

Diversity Database[™]

User Guide for Version 2

Windows and Macintosh

Diversity Database User Guide

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Diversity Database User Guide

Preface

1. About This Document

This user guide is designed to be used as a reference in your everyday use of Diversity Database™. It provides detailed information about the tools and commands of Diversity Database for the Windows and Macintosh platforms. Any platform differences in procedures and commands are noted in the text.

This guide assumes you have a working knowledge of your computer operating system and its conventions, including how to use a mouse and standard menus and commands, and how to open, save, and close files. For help with any of these techniques, see the documentation that came with your computer.

This guide uses certain text conventions to describe specific commands and functions.

Example	Indicates
File > Open	Choosing the Open command under the File menu.
Dragging	Positioning the cursor on an object and holding down the left mouse button while you move the mouse.
CTRL+S	Holding down the Control key while typing the letter s.
Right-click / Left-click / Double-click	Clicking the right mouse button / Clicking the left mouse button / Clicking the left mouse button twice.

Some of the illustrations of menus and dialog boxes found in this manual are taken from the Windows version of the software, and some are taken from the Macintosh version. Both versions of a menu or dialog box will be shown only when there is a significant difference between the two.

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2. Bio-Rad Listens

The staff at Bio-Rad are receptive to your suggestions. Many of the new features and enhancements in this version of Diversity Database are a direct result of conversations with our customers. Please let us know what you would like to see in the next version of Diversity Database by faxing, calling, or e-mailing our Technical Services staff. You can also use Solobug (installed with Diversity Database) to make software feature requests.

1. Introduction

1.1. A Brief Introduction to Diversity Database

Diversity Database™ is a powerful, flexible software package for analyzing and comparing biological samples across multiple 1-D electrophoresis gels. It combines image acquisition and analysis features with databasing functions for creating and managing large catalogs of protein and DNA samples.

Using Diversity Database's band identification tools and powerful search engine, you can quickly determine similarities among thousands of different biological samples in hundreds of gel images. You can classify your samples by protein type, enzyme, DNA probe, or any other user-defined category, and perform similarity searches within single or multiple categories. You can display your results in variety of statistical, graphical, and image reports.

The software runs in a Windows or Macintosh environment and has an easy-to-use interface with standard pull-down menus, toolbars, and keyboard commands.

Diversity Database can analyze a wide variety of biological data, including radioactive, chemiluminescent, fluorescent, and color-stained samples acquired from densitometers, phosphor imagers, fluorescent imagers, and gel documentation systems.

You can share your images among all the Discovery Series™ software, or convert them into TIFF format for compatibility with other Macintosh and Windows applications.

1.2. How Diversity Database Databases Gel Images

Every Diversity Database database starts with a single gel image. You begin building the database by identifying and matching the unique bands in that gel—these matched bands are called band types.

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Band types are used to link samples across gels. They are defined by their position and (if standards have been run) molecular weight or isoelectric point.

As you add gels to the database, you add to your collection of band types. Every band in every gel in the database will be identified as a particular band type.

Band types are grouped together into band sets; a band set includes all the band types that were created using the same enzyme, probe, etc. A database can contain multiple band sets; however, you can only perform searches and comparisons within a single band set.

You can build your database to include hundreds of images and band types. The powerful search engine within Diversity Database can perform complex searches and comparisons on hundreds of samples within minutes.

1.3. Digital Data and Signal Intensity

The Bio-Rad imaging devices supported by Diversity Database are light and/or radiation detectors that convert signals from biological samples into digital data. Diversity Database then displays the digital data on your computer screen, in the form of gray scale or color images.

A data object as displayed on the computer is composed of tiny individual screen pixels. Each pixel has an X and Y coordinate, and a value Z. The X and Y coordinates are the pixel's horizontal and vertical positions on the image, and the Z value is the signal intensity of the pixel.

Introduction

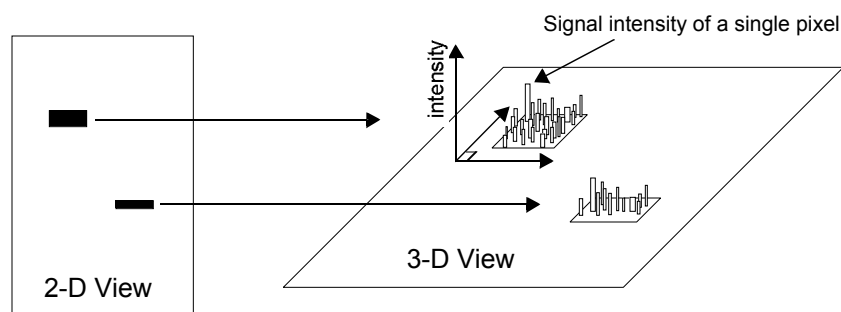


Fig.1-1. Representation of the pixels in two digitally imaged bands in a gel.

For a data object to be visible and quantifiable, the intensity of its clustered pixels must be higher than the intensity of the pixels that make up the background of the image. The total intensity of a data object is the sum of the intensities of all the pixels that make up the object. The mean intensity of a data object is the total intensity divided by the number of pixels in the object.

The units of signal intensity are Optical Density (O.D.) in the case of the GS-700 and 710 densitometers, the Gel Doc with a white light source, or the Fluor-S and Fluor-S MAX MultiImagers with white light illumination. Signal intensity is expressed in counts when using the Personal FX or FX, or in the case of the Gel Doc, Fluor-S, or Fluor-S MAX when using the UV light source.

1.4. About Gel Quality

Diversity Database is very tolerant of an assortment of electrophoretic artifacts. Lanes need not be perfectly straight or parallel. Bands need not be perfectly resolved.

However, for accurate lane-based quantitation, we suggest that bands be reasonably flat and horizontal. Lane-based quantitation involves calculating the average intensity of pixels across the band width and integrating over the band height. For the automatic band finder to function optimally, bands should be well-resolved.

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Dots that appear as halos, rings, or craters, or that are of unequal diameter, may be incorrectly quantified using the automatic functions.

1.5. Steps Involved in Diversity Database

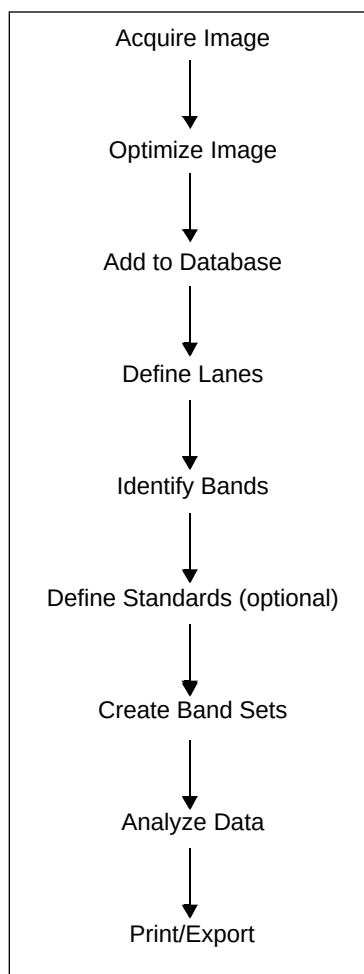


Fig.1-2. Steps in analyzing and databasing 1-D gels.

Introduction

Acquire Image

Before you can use Diversity Database to analyze a biological image, you need to capture the image into your computer. This may be done with one of the several Bio-Rad instruments supported by this software: the Molecular Imager FX and Personal Molecular Imager FX; the GS-700 and GS-710 Imaging Densitometers; the Gel Doc 1000/2000 gel documentation system; and the Fluor-S and Fluor-S MAX MultiImagers. The resulting images are stored in files on your hard disk, network file server, or removable storage media.

Optimize Image

Once you have acquired an image of your sample, you may need to reduce any noise or background density caused by film fogging or the opacity of your carrier medium. A variety of functions exist to minimize background noise while maintaining data integrity.

Create Database

After you have processed one or more related images, you can place them in a database and save the database file. Once images are stored in a database, you are ready to analyze them.

Define Lanes

After you have placed a gel image in a database, you can identify the lanes in the image. Lanes can be defined manually or automatically.

Identify Bands

Once your lanes have been defined, you are ready to identify and quantitate bands. Automatic band detection places brackets around all bands in the defined lanes. The manual tools allow you to create new bands or edit existing ones.

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Define Standards (optional)

If your gel includes molecular weight or other standards, you can enter the values into the computer and apply them on the image.

Create Band Sets

Once you have identified your experimental bands, you group them into band sets. Band sets contain the positional values of all the unique bands in your gels. They are the mechanism by which samples are linked in a database.

Data Analysis

Once your gel linked to other gels by band sets within a database file, you can use a variety of searching and population comparison tools to analyze your gel images in detail. **Diversity Database** supports both single lane and multilane sample definition as well as phylogenetic tree and 3-D principle component analysis display.

Print/Export

When your imaging and analysis are complete, you can print out your results in the form of simple images, images with overlays, reports, tables, and graphs. You can export your images and data to other systems for further analysis.

1.6. Computer Requirements for Running Diversity Database

Diversity Database will run on a PC as a Windows 95/98/NT 4.0 32-bit application or on a Macintosh as a PowerPC application.

The amount of computer memory required for using Diversity Database is mainly determined by the size of the images you will scan and analyze. Some images—particularly those scanned at high resolution—can be quite large, and your hard disk may quickly run out of space. For this reason, we recommend that you archive images on a network file server or some kind of removable storage media after your analysis.

Introduction

PC

The following are the **minimum** system requirements for installing and running Diversity Database on a PC:

Operating system:	Windows 95/98/ NT 4.0.
Processor:	Pentium 166.
RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or a network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Adaptec SCSI card with EZ-SCSI software.
Printer:	Optional.

Macintosh

Note: The default amount of memory assigned to Diversity Database on the Macintosh is 64 MB. If the total RAM in your Macintosh is 64 MB or less (the minimum recommended amount is 128 MB), you should reduce the amount of memory assigned to Diversity Database to 10 MB less than your total RAM. With the Diversity Database application icon selected, go to File > Get Info in your Finder to reduce the memory requirements for the application. See your Macintosh computer documentation for details.

The following are the **minimum** system requirements for installing and running Diversity Database on a Macintosh:

Operating system:	System 7.2 or higher.
Processor / Model:	PowerPC Macintosh 9500.
RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.

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Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Macintosh has built-in SCSI.
Printer:	Optional.

1.7. Installing Diversity Database

1.7.a. Installing Diversity Database on a PC

Before attempting to install and use Diversity Database on a PC, you should be familiar with your Windows 95/98 or Windows NT 4.0 operating system.

Diversity Database can be installed from a CD-ROM or you can download the installation program from the Internet. Make sure that Windows is up and running on your computer before attempting to install Diversity Database. We recommend that you check your hard disk with a disk utility program (such as Norton Disk Doctor) before loading Diversity Database.

Installing Diversity Database from a CD-ROM

Insert the Discovery Series™ CD-ROM into the CD-ROM drive on your computer.

Downloading from the Internet

You can download Diversity Database from the Internet using Bio-Rad's FTP site. Go to <ftp://ftp.genetics.bio-rad.com/Public/DiscoverySeries/Windows/DiversityDatabase.exe> and click on Diversity Database.exe. This will download the installer onto your computer. Then you can double-click on the Setup.exe icon on your desktop to begin running the installer.

Introduction



Fig.1-3. Setup.exe icon (Windows).

The Installation Program

The Diversity Database installation program will guide you through a series of screens. The installer will create a default directory tree under Program Files on your hard disk called Bio-Rad / The Discovery Series / Bin (you can select your own directory if you wish). The main Diversity Database program will be placed in the Bin directory. Additional directories for storing user profiles and sample images will also be created.

The installer will place a Diversity Database icon on your desktop and create a Discovery Series folder in the Programs folder on your Windows Start menu.

Finally, after installation is complete, the installer will ask you if you want to open Diversity Database.

1.7.b. Installing Diversity Database on a Macintosh

Diversity Database can be installed on a Macintosh from a CD-ROM or you can download the installation program from the Internet.

Installing Diversity Database from a CD-ROM

Insert the Discovery Series CD-ROM into the CD-ROM drive on your Macintosh.



Fig.1-4. Diversity Database installation program icon (Macintosh).

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Downloading the Installation Program from the Internet

You can download Diversity Database from the Internet using Bio-Rad's FTP site. Go to [ftp://ftp.genetics.bio-rad.com/Public/Discovery Series/Macintosh/Diversity Database](ftp://ftp.genetics.bio-rad.com/Public/DiscoverySeries/Macintosh/Diversity%20Database), and click on Install DivDB. This will download the installer and place it on your desktop. Then you can double-click on the installer icon to begin running the installer.

The Installation Program

The Diversity Database installer will guide you through a series of screens. Once installation is complete, the Diversity Database folder will appear on your desktop.

1.8. Starting Diversity Database

A hardware protection key (HPK) has been provided with your Diversity Database software.

Most computers will not need an HPK to run Diversity Database. However, if you attempt to start Diversity Database and receive an "Unable to obtain authorization" message, turn off your PC or Macintosh, attach the HPK, and restart the software.

Note: Some parallel port devices such as zip drives may be incompatible with HPKs. Please check with your peripherals vendor.

1.8.a. Attaching the Hardware Protection Key

Before attaching the HPK, you should first attempt to start Diversity Database without the HPK. If Diversity Database opens successfully, you can skip this section.

PC

The code for the PC HPK is EYYCY. The code is printed in the lower left corner of the HPK label.

Introduction

Before proceeding, please turn off your computer. If you have a printer attached to your computer's parallel port, please turn that off as well.

The HPK attaches to the parallel port on the back of your PC. If a printer cable is attached to this port, disconnect it. After you have attached the HPK, you can attach the printer cable to the key itself and restart your computer and printer.

You will also need to install the system driver that allows your Diversity Database software to "read" the HPK.

Note: Windows NT users must be in the local administrator group to install the HPK driver.

To install the driver, click on your Windows Start menu and select Programs > The Discovery Series. (This directory will not exist until you run the Diversity Database installer.) Select Install HASP HPK driver to begin installation. After the driver is installed, a message box will appear confirming the installation.

Macintosh

The code for the Macintosh HPK is QCIDIY. The code is printed in the lower left corner of the HPK label.

Before proceeding, please turn off your Macintosh.

The Macintosh HPK must be inserted in the Apple Desktop Bus (ADB) path. The ADB is located on the back of your Macintosh.



Fig.1-5. Apple Desktop Bus icon on back of Macintosh.

The HPK can be inserted at any point in the ADB path—between the computer and the keyboard, between the keyboard and the mouse, between the keyboard and the monitor, etc. After you have attached the HPK, you can restart your computer.

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1.8.b. Opening Diversity Database

PC

The installation program creates a Diversity Database application icon on your desktop. To start the program, double-click on this icon.



Fig.1-6. Diversity Database application icon.

The installer also creates a Discovery Series folder in the Programs folder on your Windows Start menu. You can start Diversity Database by double-clicking on the icon in this folder.

Macintosh

After installation, the Diversity Database folder will be open on your desktop. To start the program, double-click on the application icon inside the folder.

1.9. Registering Diversity Database

When you open Diversity Database, the Main Registration Screen will pop up welcoming you to Diversity Database.

Note: If you receive a message saying “Unable to obtain authorization,” attach the HPK if you haven’t already (refer to the previous section). If you have already attached your HPK, make sure that it is attached correctly and (if you have a PC) your HPK driver is installed. If you continue to have problems, contact Bio-Rad technical service at the numbers listed in section 1.10., Contacting Bio-Rad, below.

The type of screen you see will depend on whether you have an HPK attached or not.

Introduction

Without HPK

If you do not have an HPK attached, you can register to receive a 30-day free trial of the software.

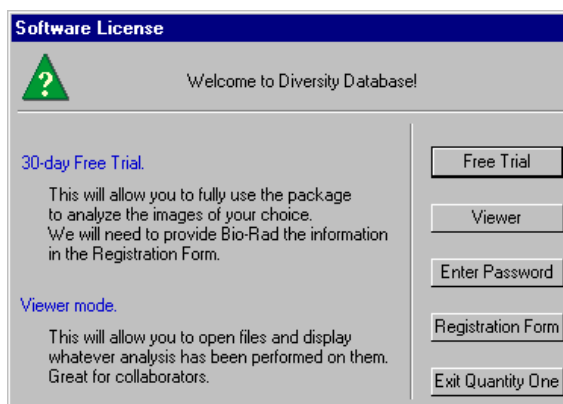


Fig.1-7. Main Registration Screen without HPK

Click on the Free Trial button. This will open the Software License Registration Form.

With HPK

If you have an HPK attached, the Main Registration Screen will reflect the fact that you receive a 30-day temporary license with your HPK ("Your license will expire on _____").

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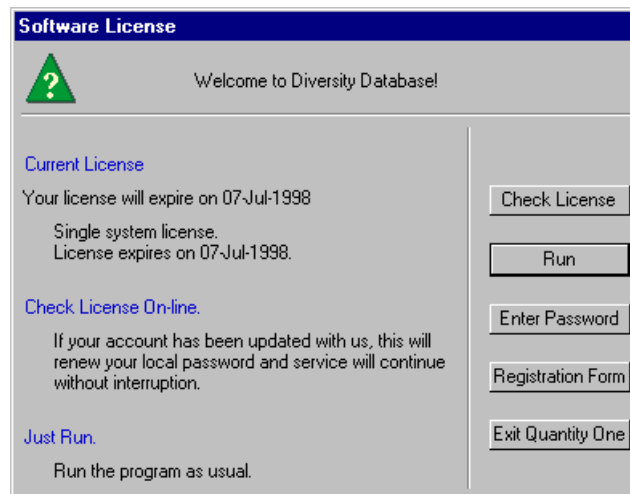


Fig.1-8. Main Registration Screen with HPK.

Click on the Run button to begin using the software.

Some time during the 30-day trial period, click on the Registration Form button to register your software (to access the Registration dialog from within Diversity Database, select Register from the Help menu).

If your 30-day period has expired, a Free Trial button will appear when you open the software. Click on this button to register your software.

This will open the Software License Registration Form.

1.9.a. Software License Registration Form

To obtain your software license, you will need to fill out the information in the Software License Registration Form.

Introduction

The screenshot shows a window titled "Software License Registration Form". Inside, there is instructional text about how to submit the form (via Internet, fax, or mail) and a list of three options. Below the text are three main sections of input fields: "Software User" (with fields for Dr./Mr./Ms., First Name, Last Name, Phone #, Fax #, and E-mail), "Company or Institution" (with fields for Company/Institution, Principal Investigator, Division/Department, Street Address, City, State or Province, Postal Code, and Country), and "Purchase Information" (with fields for Bio-Rad Representative, Purchase Order Number, Software Serial Number, and a dropdown for "What imaging device will you use?" which is currently set to "Gel Doc 2000"). To the right of the "Purchase Information" section is a "Software and System" section showing pre-filled information: Software: Diversity Database, Version: 2.0, Operating System: Windows NT, and System ID: 4SVXXLCH89. At the bottom of the window are three buttons: "Submit via Internet", "Print", and "Close".

Fig.1-9. Software License Registration Form.

Note: If you do not yet have a Purchase Order Number or Software Serial Number, you may leave these fields blank to receive a trial license.

Submitting the Registration Form by Internet

If you have Internet access on your computer (the same computer on which you loaded Diversity Database), you can register quickly and easily.

In the Software License Registration Form, click on the Submit via Internet button.

Your information will be submitted automatically over the Internet, and a password will be generated automatically and sent back to your computer. Simply continue to run Diversity Database as before.

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This password will be good for 30 days. During this period, if you have already submitted your Software Serial Number, click on Check License in the Main Registration Screen to update your license. (To access the Main Registration Screen from within Diversity Database, select Help > Register.)

If you have not yet submitted your Software Serial Number, open the form again, enter the serial number, and resubmit it over the Internet. In 1–2 days, click on the Check License button to update your license.

Submitting the Registration Form by Fax or E-mail

If you do not have Internet access, click on the Print button in the Software License Registration Form and fax the form to Bio-Rad at the number listed on the form. Alternatively, you can enter the contents of the form into an e-mail and send it to Bio-Rad at the address listed in the Registration Form.

If you include your Software Serial Number in the form, Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

If you do not include your Software Serial Number in the form, Bio-Rad will contact you in 2–3 days with a temporary 30-day license. Then, some time during the 30-day period, enter the serial number in the form and fax the revised form to Bio-Rad. Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

1.9.b. Entering a Password

If you submit your registration information over the Internet, you do not need to enter a password; it is done automatically.

If you fax your registration information, you will receive a password from Bio-Rad. You must enter this password manually.

To enter your password, click on Enter Password in the Main Registration Screen. If you are not currently in the Main Registration Screen, select Register from the Help menu in Diversity Database.

Introduction

Enter Password

To register, please contact Bio-Rad during US/Pacific business hours:
Phone: (800) 424-6723 opt 2 opt 3 (in the U.S.)
(510) 741-6576
Fax: (510) 741-5802
e-mail: LSG.TechServ.US@Bio-Rad.com (in the U.S.)
LSG.TechServ.Intl@Bio-Rad.com (International)

Please provide the information in the completed registration form.

Current license
Single system license.
No time limit.
Version limit = 6.0.

System ID
4SWXKLCB9

Password
Diversity Database OK Not OK

Enter Cancel

Fig.1-10. Enter Password screen.

In the Enter Password screen, type in your password next to the Diversity Database prompt.

Once you have typed in the correct password, the OK light next to the password field will change to green and the Enter button will activate. Click on Enter to run Diversity Database.

1.10. Contacting Bio-Rad

If you have problems with the HPK / password system, please contact Bio-Rad technical service. Technical service hours are from 8:00 a.m. to 4:00 p.m., Pacific Standard Time in the U.S.

Phone: (800) 424-6723, option 2, option 3
(510) 741-6576

Fax: (510) 741-5802

E-mail: LSG.TechServ.US@Bio-Rad.com (in the U.S.)
LSG.TechServ.Intl@Bio-Rad.com (International)

For software registration, phone:

(800) 424-6723, option 1, or (510) 741-6996

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2. General Operation

2.1. The Graphical Interface

2.1.a. Menu Bar

Diversity Database has a standard menu bar with eleven pulldown menus that contain all the major features and functions available in the software.

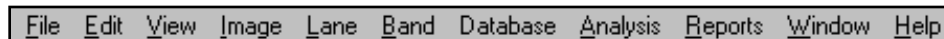


Fig.2-1. Menu Bar

The pulldown menus contain groups of related functions:

- File—General file commands (Open, Save, Print), imaging device acquisition windows (Gel Doc, GS-700, GS-710, Fluor-S, Fluor-S MAX, Personal FX, FX).
- Edit—Annotation tools, preferences, miscellaneous editing tools.
- View—General viewing tools (Zoom Box, Grab), quick analysis tools (Density in Box, Plot Density), Transform function.
- Image—Image processing tools (Crop, Flip, Subtract Background).
- Lane—Lane-finding tools, array tools.
- Band—Band-finding tools, contour tools.
- Database—Database commands, standards, band matching tools.
- Analysis—Volume tools, quantity standards, colony counting, differential display, VNTR analysis.
- Report—Analysis reports (Lanes, Matches, Volumes, VNTR).
- Window—Tile windows commands, imitate zoom.
- Help—Quick Guides, on-line Help, software registration.

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Below the menu bar is the main toolbar, containing some of the most commonly-used commands. Next to the main toolbar are the status boxes, which provide information about cursor selection and toolbar buttons.

2.1.b. Main Toolbar

The main toolbar in Diversity Database appears below the menu. It includes buttons for the main file commands (Open, Save, Print) and essential viewing tools (Zoom Box, Grab, etc.), as well as buttons that open the secondary toolbars.

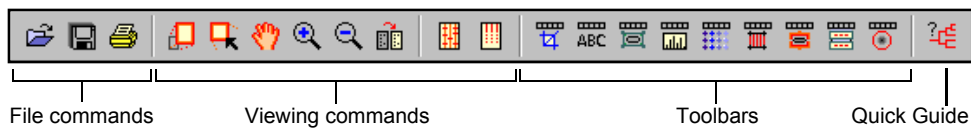


Fig.2-2. Main Toolbar

Tool Help

If you hold the cursor over a toolbar icon, the name of the command will pop up below the icon. This utility is called Tool Help. Tool Help appears on a time delay basis that can be specified in the Preferences dialog box under Edit > Preferences. You can also specify how long the Tool Help will remain displayed.

2.1.c. Status Boxes

There are two status boxes in Diversity Database. These appear to the right of the main toolbar.

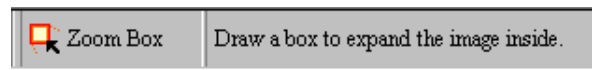


Fig.2-3. Status boxes.

The first box displays any function that is assigned to the mouse (see section 2.2., Mouse-assignable Tools). If you select a command such as Zoom Box, the

General Operation

name and icon of that command will appear in this status box and remain there until another mouse function is selected or the mouse is deassigned.

The second status box is designed to supplement Tool Help (see above). It provides additional information about the toolbar buttons. If you hold your cursor over a button, a short explanation of that command will be displayed in this second status box.

2.1.d. Secondary Toolbars

Diversity Database has nine secondary toolbars, which contain icons for groups of associated functions. You can open these toolbars from the main toolbar or from the View > Toolbars pulldown menu.

The secondary toolbars can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button on the toolbar itself.

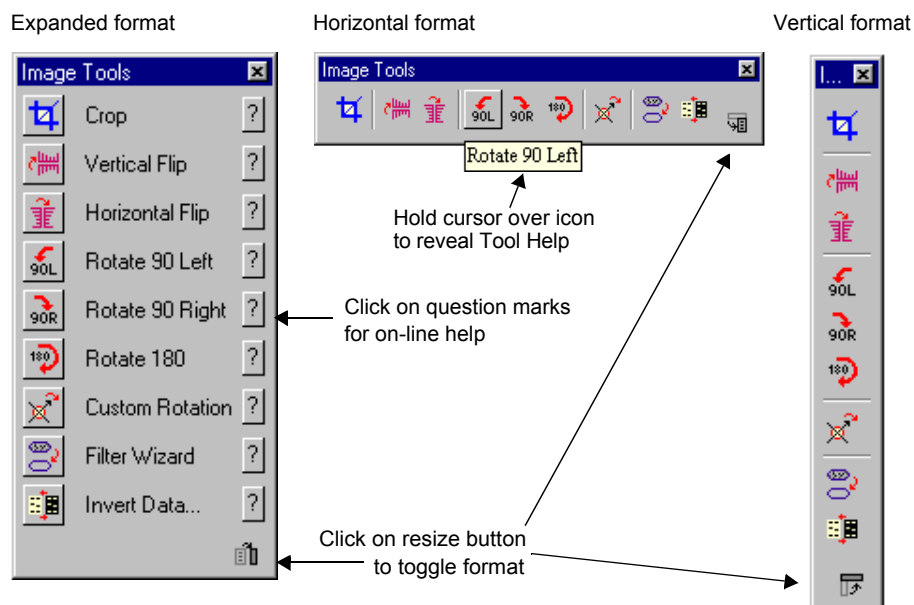


Fig.2-4. Secondary toolbar formats and features.

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The expanded toolbar format shows the name of each of the commands. Click on the “?” icon next to the name to display on-line Help for that command.

2.1.e. Quick Guides

Diversity Database has six Quick Guides to guide you through the major applications of the software. The Quick Guides are listed under the Help menu; the Diversity Database Quick Guide is also available on the main toolbar.

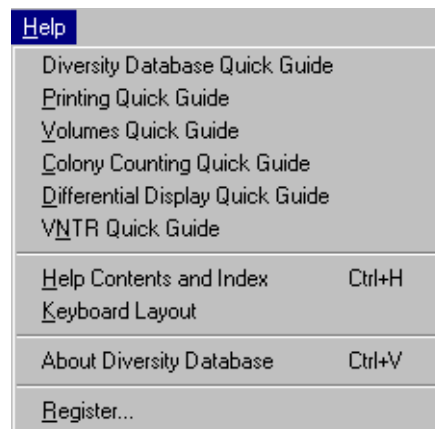


Fig.2-5. Quick Guides listed in Help menu.

The Quick Guides are similar in appearance to the secondary toolbars, but are application-specific.

Each Quick Guide contains all of the functions necessary for a particular application, from opening the image to outputting data.

General Operation

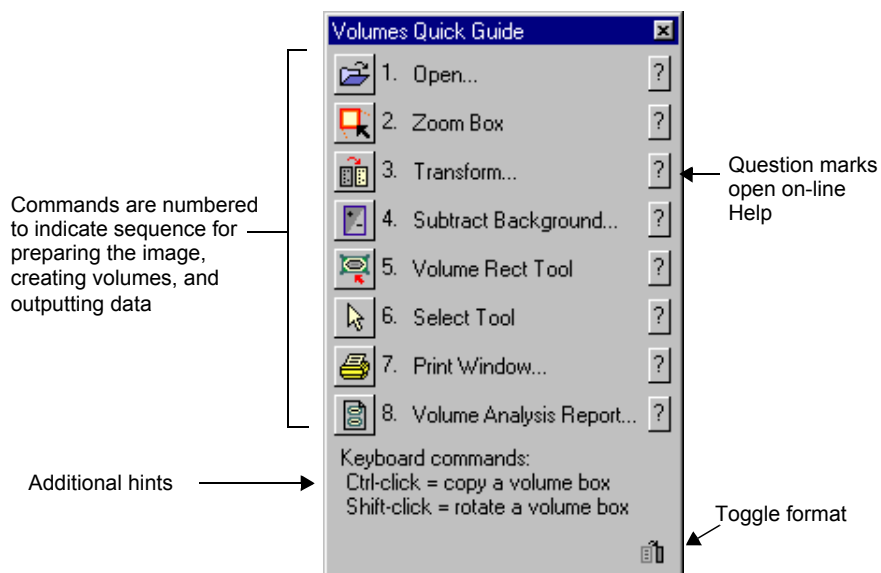


Fig.2-6. Example of a Quick Guide: Volumes

Note: You will find that as you become more familiar with the application, you will skip operations that are not needed for your particular images.

In their expanded format, the Quick Guide commands are numbered as well as named, so that the order of operation is clear. Simply follow the steps and the Quick Guide will lead you through the analysis.

As with the secondary toolbars, you can click on the “?” next to the name of a function to display the Help text.

2.2. Mouse-assignable Tools

Diversity Database has a number of commands that don’t perform actions right away, but instead “assign” a function to your mouse (e.g., Zoom Box, Add Gel, Lane Background). You first select one of these commands from a menu, toolbar, or dialog box, then execute the command by clicking or dragging on the image.

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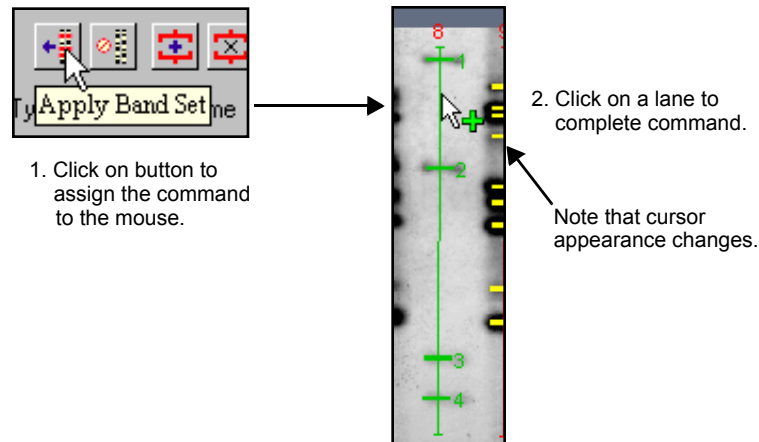


Fig.2-7. Example of a mouse-assignable tool: Apply Band Set

When a “mouse-assignable” function is selected, the cursor appearance changes. The name and icon of the function appear in the status box next to the main toolbar (see section 2.1.c).

Note: To *deassign* a function from the mouse, click on the toolbar button of the assigned function or click in the status box displaying the assigned function. You can also double-click on the Hide Overlays button.

2.3. Keyboard Shortcuts

Many of the functions in Diversity Database may be executed using keyboard shortcuts. These are listed in the Help menu under Keyboard Layout, as well as in Appendix H of this manual.

The pulldown menus also list the shortcut keys for the menu commands.

2.4. File Commands and Functions

This section describes the basic file commands and functions of Diversity Database. These are selected from the File menu.

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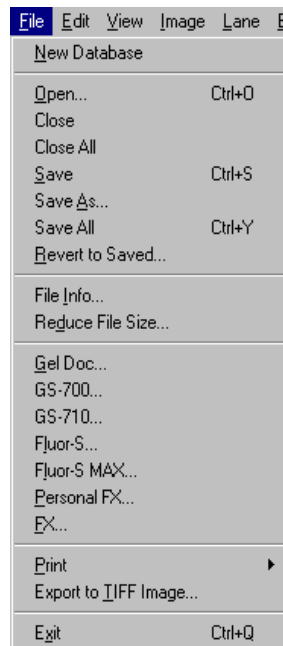


Fig.2-8. File menu.

New Database

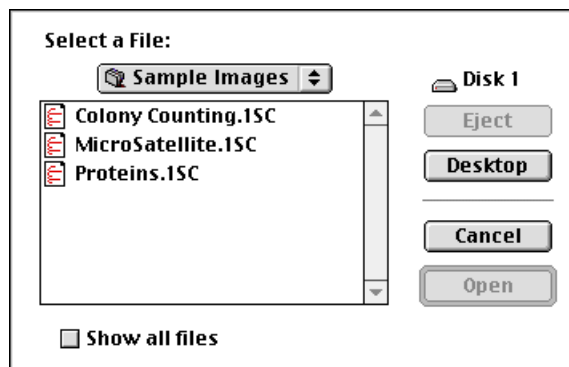
The New Database command creates a new database file and displays a new Database Summary form on-screen. The new database file will be unsaved and will not contain any gels. See Chapter 3 (tutorials) or Chapter 8 for more information on creating database files.

Open

To open a previously saved image, select Open from the File menu or main toolbar. This opens the Open dialog box.

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Macintosh version:



Windows version:

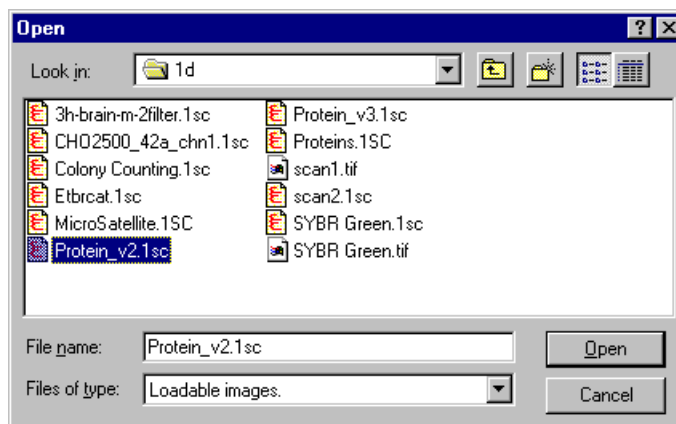


Fig.2-9. Open dialog box.

In the dialog box, open a file by selecting the file name and clicking on the Open button. To look for files in other directories, use the directory pulldown menu at the top of the dialog box.

An image created in the Windows version of Diversity Database can be opened in the Macintosh version, and visa versa. However, you must add a .1sc extension to your Macintosh files to open them in Windows.

Note: This version of Diversity Database will open images and databases created with previous versions of Diversity Database.

General Operation

Diversity Database comes with a selection of sample images and databases. These are for use with the tutorials in Chapter 3. In Windows, these files may be found in the Discovery Series/Sample Images/Div Database directory. On the Macintosh, they are stored in the Sample Images folder in the Diversity Database folder.

Opening TIFF Images

The Open command can also be used to import TIFF images from other software applications.

TIFF images do not contain all of the tagged information that would normally be included in a Discovery Series image file (e.g., scanner type, date, color, etc.). For this reason, the File Info and Scan Report commands will not display this tagged information for TIFF files.

There are many types of TIFF formats that exist on the market. Not all are supported by the Discovery Series. There are two broad categories of TIFF files that are supported:

1. 8-bit Grayscale. Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with the Discovery Series software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray. Hewlett Packard™, Microtek™, and Sharp™ each make scanners that produce compatible 8-bit grayscale images.
2. 16-bit Grayscale. Bio-Rad's Molecular Imager (storage phosphor) systems use 16-bit pixel values to describe intensity of scale. Molecular Dynamics™ imagers also use 16-bit pixel values. The Discovery Series understands these formats and can interpret images from both Bio-Rad and Molecular Dynamics storage phosphor systems.

Note: Diversity Database can import 8- and 16-bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1. 1-bit Line Art. This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as

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either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.

2. 24-bit Full Color or 256 Indexed Color. These formats are frequently used for retouching photographs and are currently unsupported in the Discovery Series, although most scanners that are capable of producing 24-bit and indexed color images will be able to produce grayscale scans as well.
3. Compressed Files. The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with the Discovery Series should be formatted with the compression option turned off.

Close

To close an image or scan, select File > Close. If the file has changed since it was last saved, a message box will ask you if you want to save the changes before closing.

Close All

File > Close All closes all open images. If you have made changes to any of the images, a message box will open for each unsaved image giving you the option of saving those changes.

Save

File > Save will save a new image or an old image to which you have made changes.

In Windows, new Diversity Database images will be given a .1sc extension when they are first saved.

Note: Save on the File menu can only be used to save images. The commands for saving databases are located on the Database Summary Form.

General Operation

Save As

File > Save As can be used to save a new image, rename an old image, or save a copy of a image in a different directory.

In Windows, new Diversity Database images will be given a .1sc extension when they are first saved.

Note: Save As on the File menu can only be used to save images. The commands for saving databases are located on the Database Summary Form.

Save All

Save All on the main toolbar or File menu saves all images that are currently open.

In Windows, new Diversity Database images will be given a .1sc extension when they are first saved.

Note: Save All can only be used to save images. The commands for saving databases are located on the Database Summary Form.

Revert to Saved

File > Revert to Saved will reload the last saved version of the image you are working on. This is a quick way to undo any alterations you may have made to an image since you last saved it.

Any changes you have made since last saving the file will be lost. (Also, all open dialog boxes will be closed.) A message box will warn you before completing the command.

File Info

File > File Info displays general information about your image, including the scan date, scan area, number of pixels in the scan, data range, and the size of the scan file. There is also a field where you can type in a file description or comments.

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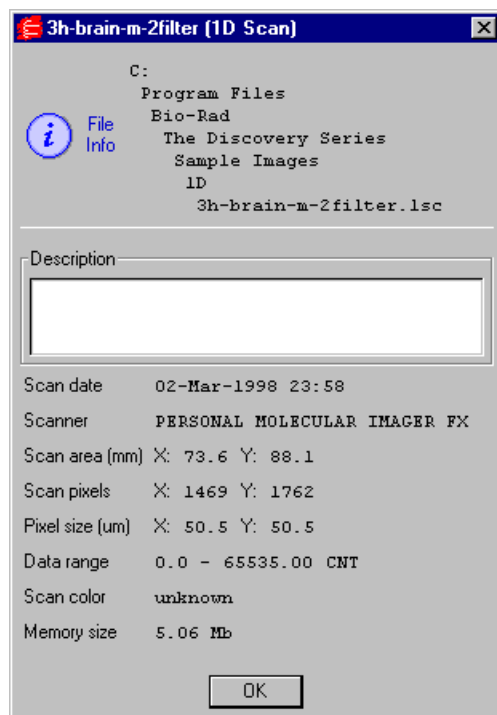


Fig.2-10. File Info box.

Reduce File Size

Image files can be quite large, and computer systems do not have unlimited memory or storage space. If you are having difficulty loading or storing a particular image, you may want to reduce the size of the file by reducing the number of pixels in the image. (You can also trim unneeded parts of an image to reduce its memory size. See section 4.6.a., Cropping Images.)

This function is comparable to scanning at a lower resolution, in that you are increasing the size of the pixels in the image, thereby reducing the total number of pixels and thus memory size.

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Note: Reducing the file size of an image will result in some loss of resolution. In most cases this will not affect quantitation. In general, as long as the pixel size remains less than 10 percent of the size of the objects in your image, changing the pixel size will not affect quantitation.

Select File > Reduce File Size to open the Reduce File Size dialog box.

The Reduce File Size dialog box shows you the size of the pixels in the image (Pixel Size: X by Y microns), the number of pixels in the image (Pixel Count: X by Y pixels), and the memory size of the image.

As you increase the size of the pixels, the pixel count will decrease, as will the memory size. You can increase the pixel size in either dimension (see the following figure for an example).

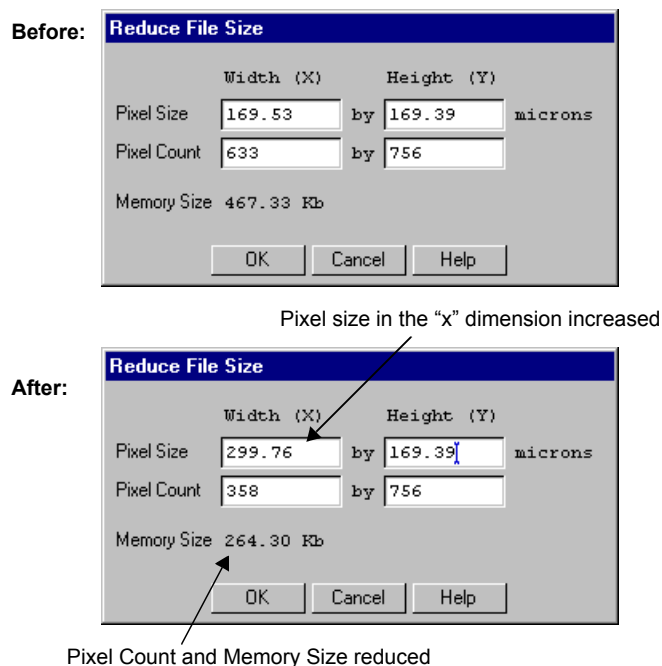


Fig.2-11. Reduce File Size dialog box, before and after pixel size increase.

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Note: Since with most 1-D gels you are more concerned with resolving bands in the vertical direction than the horizontal direction, you may want to reduce the file size by making rectangular pixels. That is, keeping the pixel size in the “y” dimension the same, while increasing the size in the “x” dimension. This lets you decrease the total number of pixels and therefore file size without sacrificing detail.

When you are finished, click on the OK button.

A pop-up box will give you the option of reducing the file size of the displayed image or making a copy of the image and then reducing the copy's size.

Reducing the file size is an irreversible process. For that reason, we suggest you make a copy of the image and reduce its file size. That way, if you lose too much resolution, you can simply delete the copy and try again. Once you are happy with the reduced image, don't forget to delete the original. The goal is to save space!

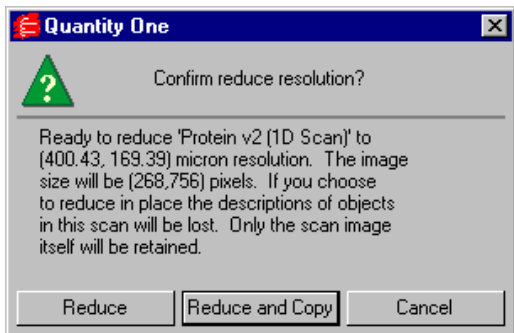


Fig.2-12. Confirm Reduce File Size pop-up box.

If you choose to make a copy of the image, you will be asked to enter a name for the new copy before the operation is performed.

Imaging Device Acquisition Windows

The File menu contains a list of Bio-Rad imaging devices supported by the Discovery Series software. These are:

General Operation

1. Gel Doc 1000 / 2000
2. GS-700 Imaging Densitometer
3. GS-710 Imaging Densitometer
4. Fluor-S MultiImager
5. Fluor-S MAX MultiImager
6. Personal Molecular Imager FX
7. Molecular Imager FX

Selecting a name in the File menu will open the acquisition window that allows you to scan using that instrument.

See the individual appendices on the imaging devices for more details.

Exit

File > Exit quits the application.

2.5. Preferences

The Preferences dialog box (accessed under the Edit menu) allows you to customize basic features of your system, such as displays and toolbars.

After you have selected your preferences, click on OK to implement any changes you make.

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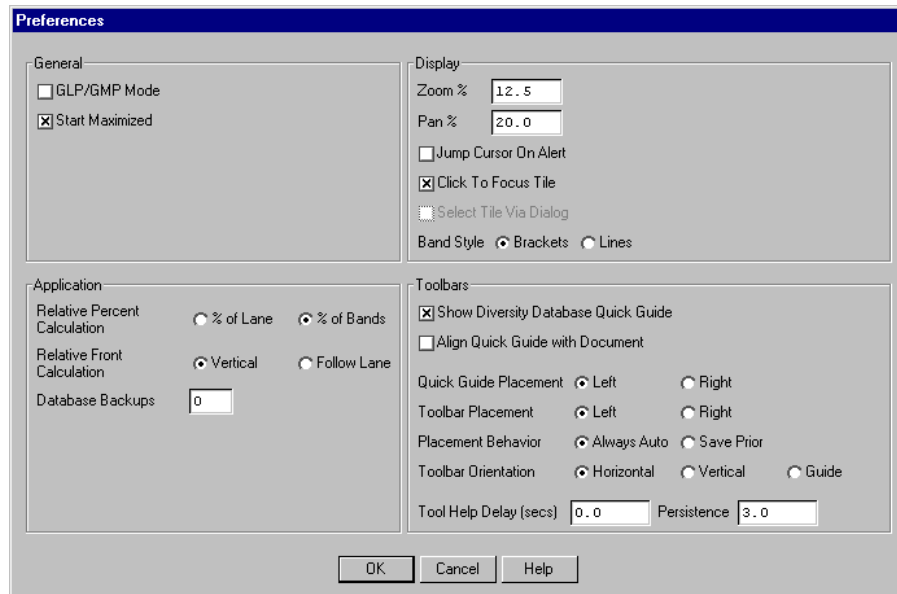


Fig.2-13. Preferences dialog box

2.5.a. General

GLP/GMP Mode

The GLP / GMP Mode checkbox allows you to prevent changes to an image that would change the raw image data. In GLP / GMP mode, the software will not allow the following operations to be performed (these are all located on the Image menu):

- Subtract Background
- Custom Rotation
- Filters
- Invert Data

If a user attempts to use any of these functions in GLP / GMP mode, he will receive a message that that function is not available.

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To set GLP / GMP mode, click on the checkbox. A dialog box will pop up asking you to enter a password.

After you enter a password and click on OK, another dialog will ask you to reenter the password for confirmation. Retype the password and click on OK again.

To disable GLP / GMP mode, click on the checkbox to deselect it. The dialog box will pop up asking you to enter the password. When you click on OK, GLP / GMP Mode will be disabled.

Start Maximized

In the Windows version, the Start Maximized checkbox determines whether Diversity Database occupies the entire computer screen when first opened. If this is unchecked, Diversity Database's menu and status bars will appear across the top of the screen and any toolbars will appear "floating" on the screen.

2.5.b. Display

The Zoom % field allows you to specify the percentage by which an image moves closer or farther away when you use the Zoom functions. This percentage is based on the size of the image.

Pan % determines the percentage by which the image moves side to side or up and down when you use the arrow keys. This percentage is based on the size of the image.

When an alert box pops up in Diversity Database, you can set your cursor to automatically go to the OK button in the box by selecting Jump Cursor on Alert.

Bands in your gel image can be marked with brackets that define the top and bottom boundaries of the band, or they can be marked with a dash at the center of the band. Indicate your preference by clicking on the Brackets or Lines button next to the Band Style prompt. (This setting can also be changed in the Band Attributes dialog box.)

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2.5.c. Application

Relative Percent Calculation

The Relative Percent Calculation option allows you to define how Relative % will be determined for all reports, histograms, and band information functions in Diversity Database: either as a percentage of the signal intensity of an entire lane or as a percentage of the signal intensity of the defined bands in a lane.

Selecting % of Lane means that the total lane intensity will equal 100 percent and the band that you select will be reported as a fraction thereof.

Selecting % of Bands means that the sum of the intensity of the defined bands in a lane will equal 100 percent, and that your band will be reported as a fraction of that sum.

If you create, adjust, or remove bands with Relative % defined as % of Bands, the Relative % of all the other bands in the lane will update to reflect changes in the total intensity of defined bands.

Relative Front Calculation

The Relative Front Calculation option lets you to define how relative fronts (Rf) will be calculated for lane and band analysis. This is useful if your gel image is curved or slanted.

Rf is calculated by dividing the distance a band has migrated from the gel's origin by the total distance between the origin and the dye front. These distances can be determined in one of two ways:

1. Drop a vertical line from the origin to the dye front (Vertical Projection);
2. Measure along the length of the lane (Follow Lane).

If a lane is straight and vertical, these methods will give identical results. However, the results will differ if a lane is curved or slanted. Choose your preferred method by clicking on one of the two buttons next to the Relative Front Calculation prompt.

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Database Backups

The Database Backups option allows you to specify the number of backup database files (0-9 copies possible) that will be created when you save your databases.

If you enter 1 in the field, whenever you save a database, a backup of that database will be saved in a Backup folder in the same directory as the database. The database backup name will be appended with a number 1 (Windows files will also have the extension .dbk).

If you enter 2, the most recent backup will be appended with the number 1 and the next most recent will be appended with the number 2. And so on.

Backups allow you to recover from mistakes or inadvertent changes to your database file. Specifying a Database Backups value of 0 means that no backup copies will be maintained.

To open a backup database in Windows, you must change the .dbk extension to .ddb using Windows Explorer.

2.5.d. Toolbars

These settings determine the behavior and positioning of the secondary toolbars and Quick Guides.

If the Show Diversity Database Quick Guide checkbox is selected, the Diversity Database Quick Guide will open automatically when you open Diversity Database.

If Align Quick Guide with Document is selected, the Quick Guides will pop up flush with the edge of your documents. Otherwise, they will appear flush with the edge of the screen.

Quick Guide Placement and Toolbar Placement determine which side of the screen the Quick Guides and toolbars will first open.

The Placement Behavior setting determines whether a Quick Guide or toolbar will always pop up in the same place and format (Always Auto), or whether they will pop up in the last location they were moved to and the last format selected (Save Prior).

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The Toolbar Orientation option buttons specify whether toolbars will first appear in a vertical, horizontal, or expanded format when you open Diversity Database.

Tool Help Delay allows you to specify the amount of time (in seconds) the cursor must remain over a toolbar icon before the Tool Help appears. First-time users may want to specify a short delay to learn the names of the toolbar functions, while experienced users can specify a longer delay once they are familiar with the icons.

Tool Help Persistence determines how long the Tool Help will linger on the screen after you move the cursor off a toolbar icon.

2.6. User Settings

If Diversity Database is on a workstation that has multiple users, each user can have his or her own set of Diversity Database preferences and settings.

In multiple-user situations, Diversity Database's preferences and settings are associated with individual user names. On a PC with Windows 95/98/NT, your user name is the name you use to log onto the computer. On a Macintosh, your user name is the Owner Name under the File Sharing (OS 8.x) or Sharing Setup (OS 7.x) control panel.

Note: If you do not log onto your PC under Windows 95/98 or do not have a Owner Name on your Macintosh, then you do not have a user name and your preferences and settings will be saved in a generic file.

3. Tutorials

This chapter provides four tutorials that are designed to guide you through the main features and functions of Diversity Database. All of the features described in this chapter are discussed in greater detail in subsequent chapters.

These lessons utilize three sample databases and six images that were installed when you installed Diversity Database. (See below for the location of these files.)

The lessons are designed to be read sequentially. However, the files created and altered in each lesson are independent of those in the other lessons. You do not have to perform the steps in Lesson 1 before you can move on to Lesson 2, and so forth.

Lesson 1 describes how to create a new database, add a gel image to it, and define lanes and identify bands in that gel.

Lesson 2 describes how to create standards and a band set and use the band set to match the bands in your image.

Lesson 3 describes how to create a band set without standards and use it to match the bands in your image.

Lesson 4 describes how to add a gel to a preexisting database and outlines some of the similarity searches you can perform on the images in your database.

Location of Sample Images and Databases

If you are using Windows, the sample databases and images are installed in the Div Database folder in the Sample Images folder in the Discovery Series directory on your hard drive (Program Files > Bio-Rad > The Discovery Series > Sample Images > Div Database).

If you are using a Macintosh, the sample databases and images are installed in the Sample Images folder inside the Diversity Database folder on your hard disk.

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3.1. Lesson 1. Creating a Database and Preparing an Image

In this lesson, you will create a new database, add a gel image to that database, and define the lanes and bands in the gel image. (See Chapters 5–7 for more detailed explanations of the commands described in this lesson.)

3.1.a. Creating a New Database

Every database starts with a good, clean gel image with well-defined bands. You should choose a gel image that has most of the bands in which you are interested in your research. The more distinct bands in your “starter” image, the less work you will have to do when you add other images to the database.

In this example, we will create a new database using Image.4 as the “starter” gel image.

1. From the main toolbar, click on Open. In the Open dialog box, select Image.4 from the Sample Images folder. The raw image will be displayed.

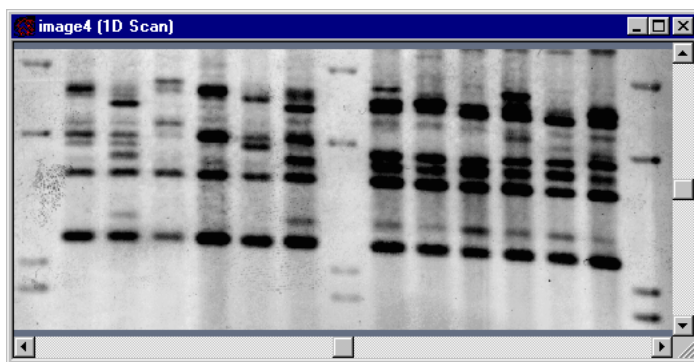


Fig. 3-1. Image.4.

2. From the File menu, select New Database. This will open the Database Summary form.

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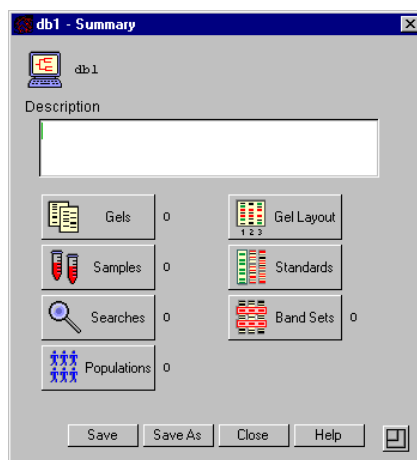


Fig. 3-2. Database Summary form.

3. In the Summary form, click on the Gels button. This will open the Gels dialog box.

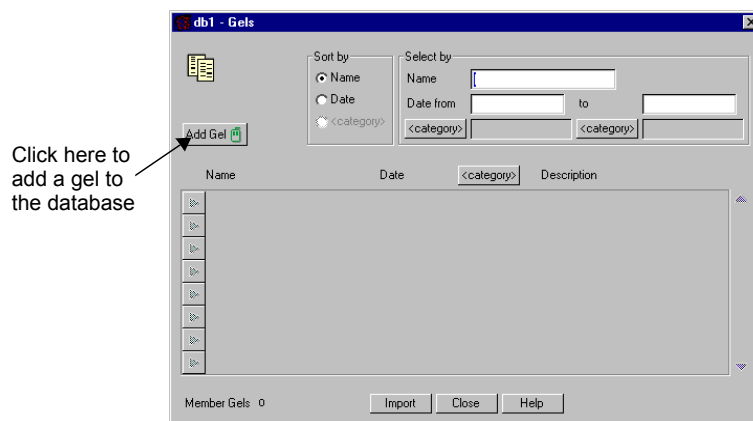


Fig. 3-3. Gels dialog box.

4. In the Gels dialog, click on the Add Gel button, then click anywhere on Image.4.

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5. Image.4 and its creation date will appear listed in the Gels dialog box. Close the dialog by clicking on the Close button.
6. In the Database Summary form, the number 1 will appear next to the Gels button, indicating that the database now contains one gel.



Fig. 3-4. Addition of the gel is reflected in the Database Summary form.

7. If you do not want to use this tutorial again, you can now save the new database. (If you do want to use this tutorial again, skip this step.) Click on the Save As button in the Database Summary form to save the newly created database. In the Save As dialog, the default file name will be db1. Click on the Save button to finish saving.
8. Finally, the Add Gel command is still assigned to your mouse. To unassign it, go the status box next to the main toolbar. The Add Gel command is displayed there. Click once in the status box to unassign the command. (All mouse-assignable functions may be unassigned in this manner.)

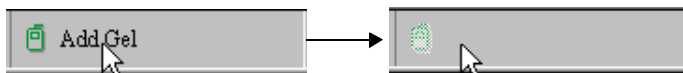


Fig. 3-5. Click in the status box to deassign Add Gel command from the mouse.

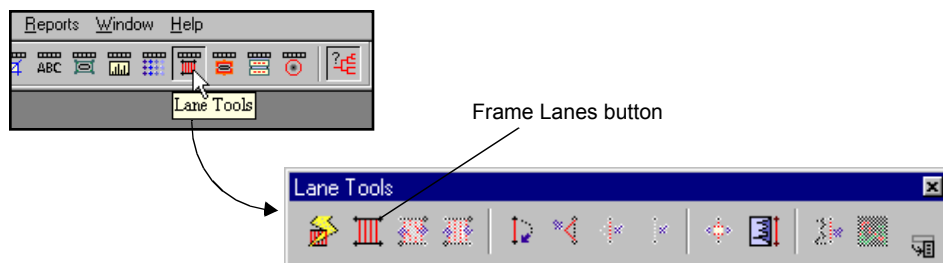
Now Image.4 has been added to db1. Every time you open the image, the database will be opened as well. Conversely, you can open the database first, then open the image from within the database. (See Chapter 8 for more details.)

3.1.b. Defining Your Lanes

Now you are ready to define the lanes in Image.4. Image.4 has 15 lanes.

1. Click on the Lane Tools button on the main toolbar. This will open the Lane Tools toolbar.

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2. On the toolbar, click on the Frame Lanes button.
3. In the pop-up box, under “Number of lanes to create?,” enter “15” and click on OK.

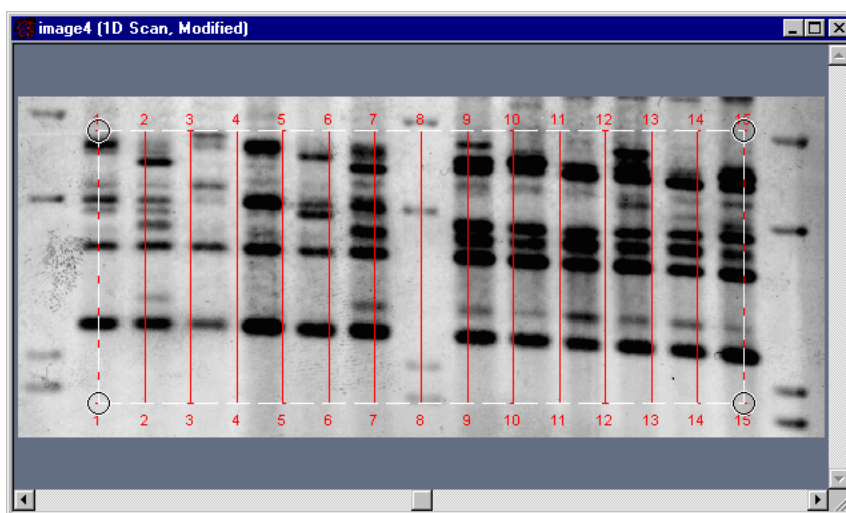


Fig. 3-6. Initial lane frame.

The lane frame will appear overlaid on the image. However, it will not be aligned with the lanes in the image. The next step is to align your lane frame.

4. On the Lane Tools toolbar, click on the Add / Adjust Anchors button. Position your cursor inside the circle in the upper left corner of the frame.

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Drag the circle so that it is centered slightly *above* the top band in the first lane of standards.

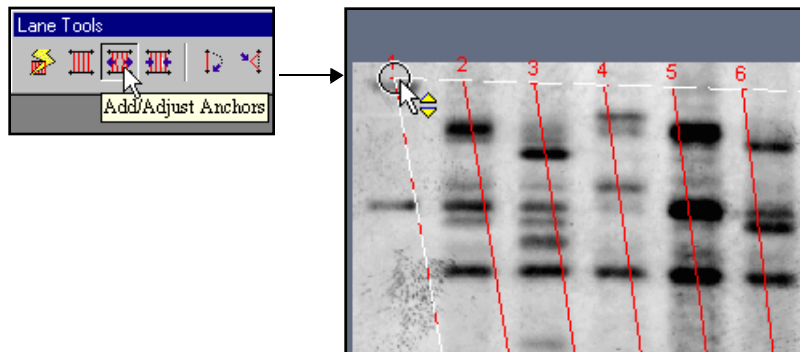


Fig. 3-7. Positioning the upper left corner of the lane frame.

5. Drag the upper right corner of the frame in the same manner, so that it is positioned above the top band in the last lane of standards on the image.
6. Drag the lower left corner of the frame so that it is positioned below the last band in the first lane of standards.
7. Drag the lower right corner of the frame so that it is positioned below the last band in the last lane of standards.

When you are finished, your frame lines should be aligned down the center of your lanes. (The remaining lane adjustment tools are described in Chapter 5.)

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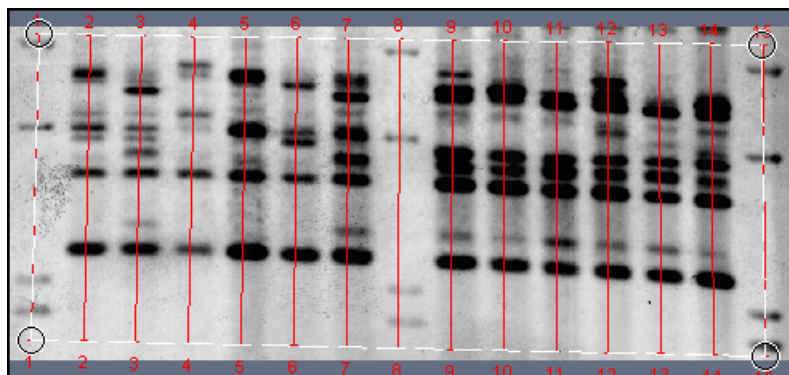


Fig. 3-8. Lane frame after alignment.

Note that when you identify the lanes on your image, the number of lanes appears in the Database Summary form, next to the Samples button. Diversity Database treats each lane as a sample, and performs similarity searches based on these samples. See Lesson 3 and Chapter 9. for more details.

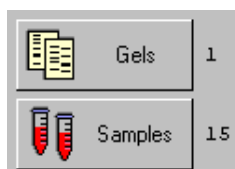


Fig. 3-9. Fifteen lanes have been added to the database.

3.1.c. Detecting Bands

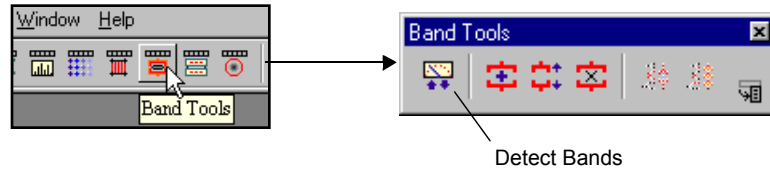
Now you are ready to detect the bands in your gel image.

Note: If your gel image were not already placed in a database, Diversity Database would prompt you to do so before you detect bands.

First, click on the Lane Tools button on the main toolbar to dismiss the lane tools.

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1. Click on the Band Tools button on the main toolbar. On the Band Tools toolbar, click on Detect Bands. This will open the Detect Bands dialog box.



2. This dialog box has a small version and an expanded version. Click on the square button in the upper left corner of the dialog box (next to the “close” button) to change the size to the expanded version.

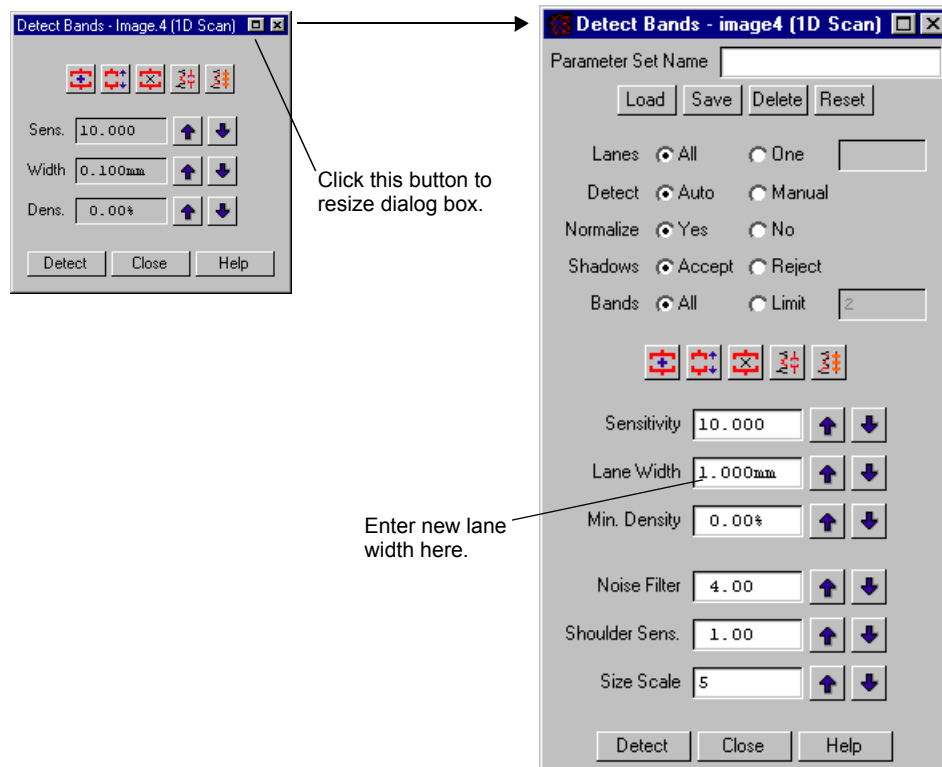


Fig. 3-10. Detecting bands.

3. In the expanded Detect Bands dialog, change the lane width by typing 2.99 into the Lane Width field and pressing the RETURN key. When you do so, your bands will be automatically detected (because the Detect: Auto option is selected in the dialog box). Click on Close to dismiss this dialog box.

Your bands will appear marked as brackets on the image. To display your bands as lines instead of brackets:

4. Select Band Attributes from the View menu. This will open the Band Attributes dialog box. At the bottom of the dialog, select Lines instead of Brackets. Click on OK to dismiss this dialog box.

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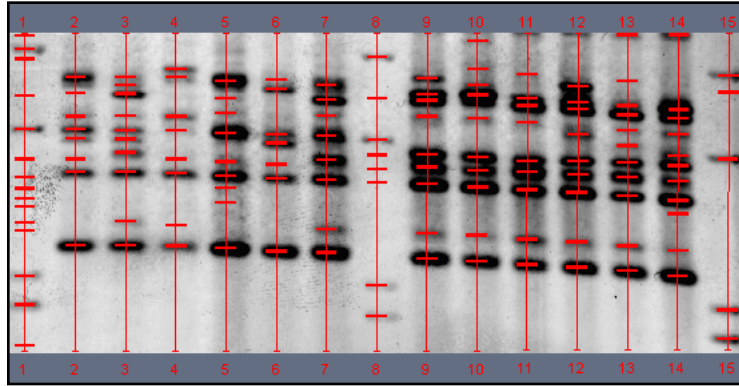


Fig. 3-11. Detected bands, displayed as lines.

Each standard lane has four bands. Looking at the standard lanes in the image (1, 8, and 15), you can see that some image noise has been mistakenly detected as bands. You will need to manually delete these bands.

5. Go to the tool palette at the center of the Detect Bands dialog box and click on the Remove Band button. Then drag a box around the cluster of misidentified bands in lane 1. When you release the mouse, the bands will be removed.

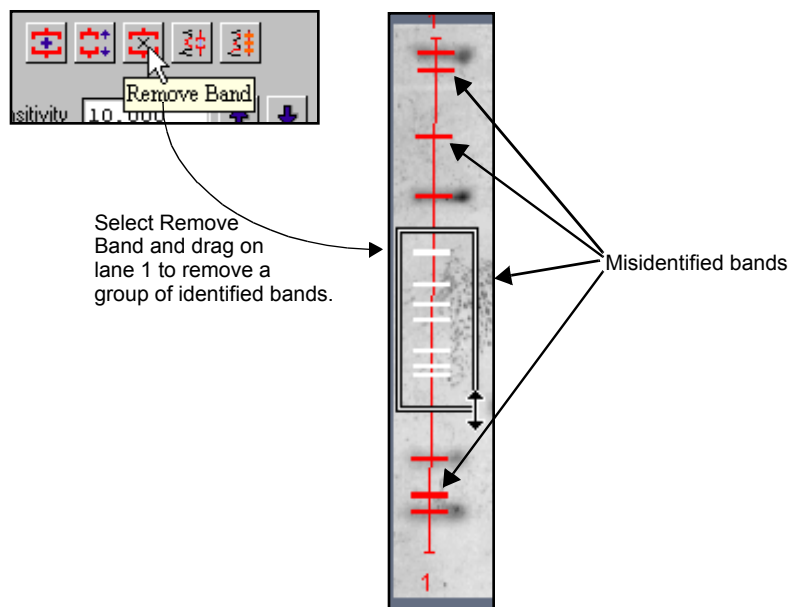


Fig. 3-12. Removing a group of misidentified bands from lane 1.

6. You can also remove bands individually by clicking on them one at a time with Remove Band selected.

To add bands, click on the Add Band button on the same tool palette, and click on a spot on the image where you want to place the band. Bands can only be added individually.

Almost of the misidentified bands in Image.4 are in the three standard lanes. When you are finished removing bands, your image should look like this:

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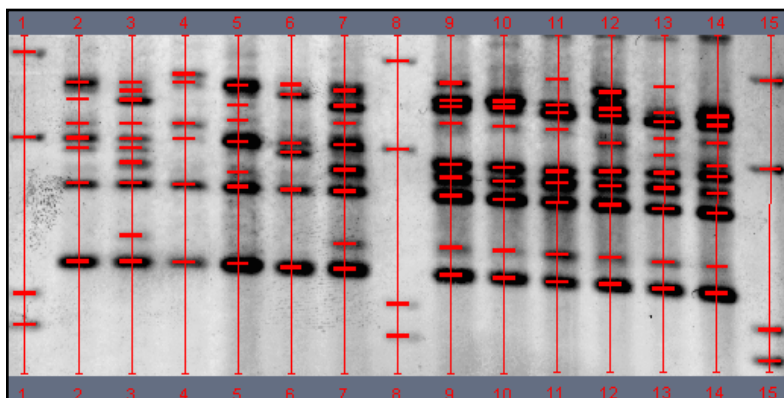


Fig. 3-13. Image with bands manually removed.

This completes Lesson 1. If you want to use this tutorial again, close the open image and database without saving. All changes you made will be lost.

3.2. Lesson 2. Creating Standards and Band Sets and Matching Bands

In this lesson, you will take an image that has already been added to a database, define the standard lanes and bands, and match all the experimental bands. (See Chapters 8–9 for a more detailed explanation of the commands described in this lesson.)

Note: If you are not using standards in your gels, you should skip to Lesson 3, Creating and Applying Band Sets Without Standards.

For this example, open Image.5 from the Sample Images folder. This image has already been placed in a database: Database.2. Therefore, the database will open with the image.

When the image first opens, select Sort and Recalculate from the Edit menu to view the current identification of the bands.

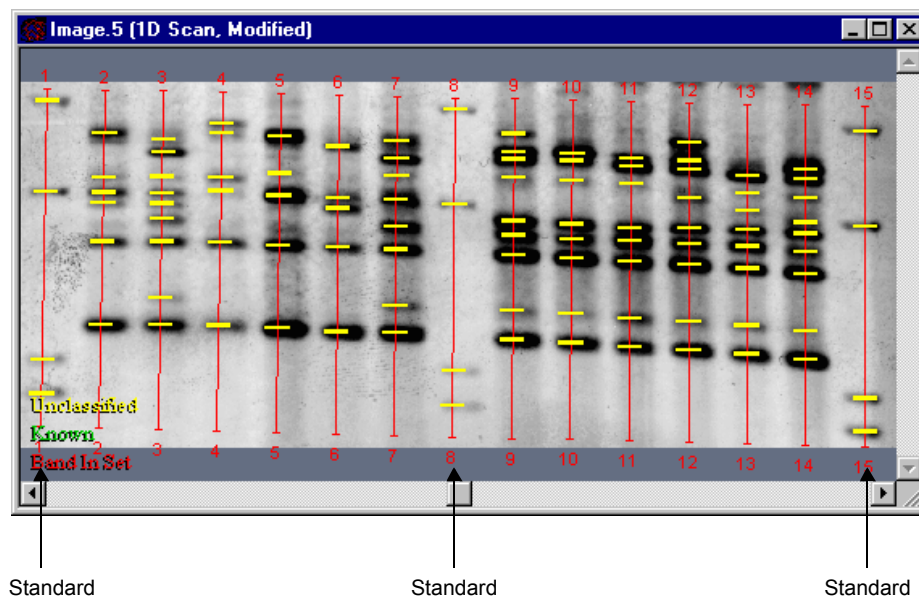


Fig. 3-14. Image.5 with standard lanes indicated.

Image.5 contains three lanes of standards (numbers 1, 8, and 15) and 12 experimental lanes. Note that the three evenly spaced standard lanes are valuable in this case because of the curvature of the gel. Also, note that all bands at this point are undefined, and therefore are marked in yellow.

3.2.a. Entering and Labeling Standards

We will start by entering the values of your standards into the computer. We will then assign these standard values to the actual standard bands in the image.

1. On the Database Summary form, click on the Standards button. This will open a small pop-up box. In that box, click on New Standards. This will open another pop-up box, in which you will select the units of your standards.

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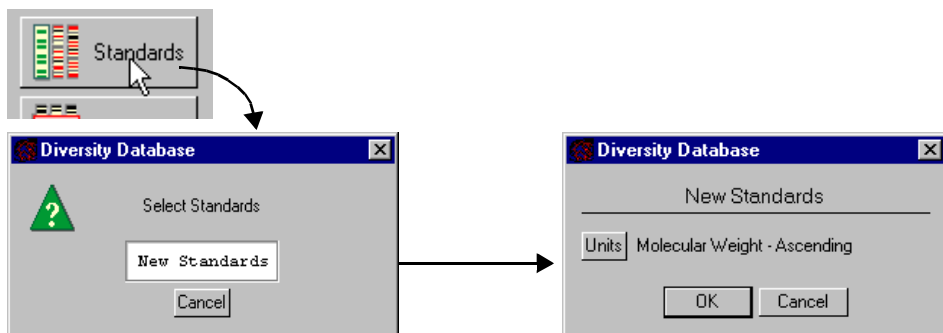


Fig. 3-15. Creating a new set of standards.

2. The default units (Molecular Weight—Ascending) are correct, so click on OK.
3. The Standards dialog box will open. Name this set of standards by typing Std 1 in the field at the top of the form.

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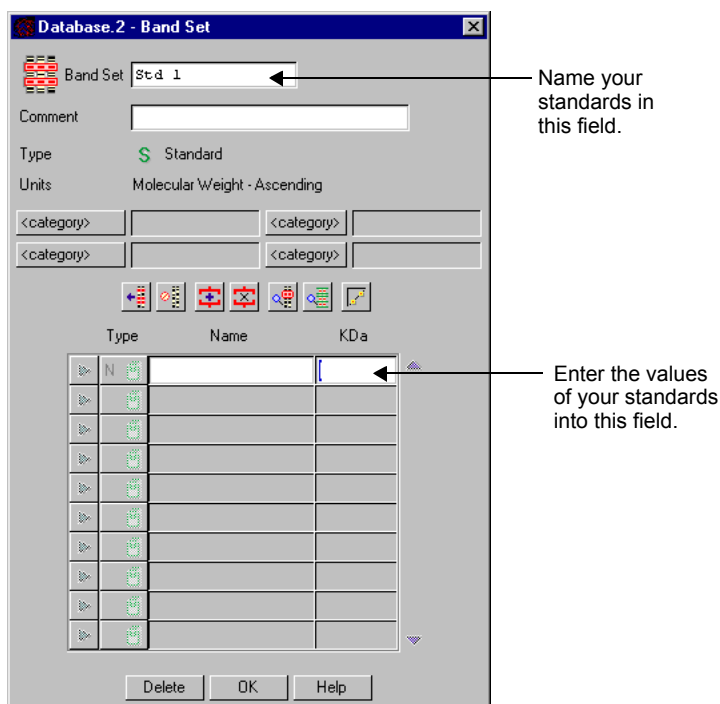


Fig. 3-16. Standards dialog box.

4. You are now ready to enter the values of your standard bands into the appropriate fields. Begin by clicking in first row in the KDa column in the center of the dialog box. Type the number 6557 into this field, and press the RETURN key.
5. In the next row in the column, type 4361 and press RETURN. Complete the column by entering 2322 and 2027 in the next two rows. (There are four standard bands in this gel.)

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Type	Name	KDa
▶ 1 		6557.00
▶ 2 		4361.00
▶ 3 		2322.00
▶ 4 		2027.00
▶ N 		

Fig. 3-17. Standard values entered into the Standard Band Set dialog box.

- Now you are ready to apply this band set to the standard bands on your image. Click on the Apply Band Set button on the tool palette in the middle of the Std 1 dialog box.



Fig. 3-18. Standard Band Set tool palette.

- Now click on the first standard lane in your gel image. The lane will turn green, indicating that all the bands in it are now known standards. Click on the remaining two standard lanes to assign your standards to those.
- Finally, click on OK in the Std 1 dialog box to save your standards and close the dialog box.

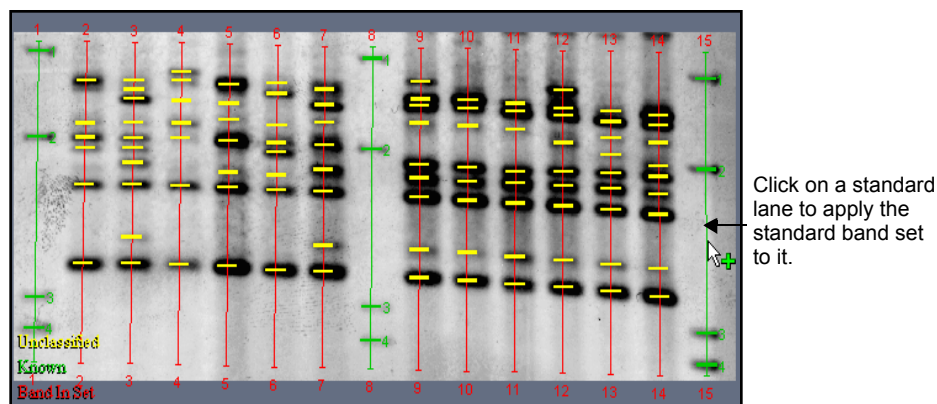


Fig. 3-19. Image.5 with standard lanes defined.

On Image.5, standard bands are now marked in green and unknown bands are marked in yellow.

3.2.b. Matching Experimental Bands

You are now ready to match your experimental bands with one another. When you match your experimental bands together, you create a “band set” containing known “band types.” This band set can then be applied to other gel images, linking them within a database. Thus, matching bands is critical to creating a database.

Your first step is to get Diversity Database to automatically match as many of the bands as possible.

Automatic Matching

You start by selecting one lane that contains most, if not all, of the experimental bands you are interested in, with minimal distortion and clean, easily identifiable bands. This will be the “first lane” in your database. You will use this lane to begin creating a band set, containing experimental band types.

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The remaining experimental bands will be matched with the band types in this first lane.

In the case of Image.5, lane 7 has eight good, distinct bands, and it is positioned next to the center standard lane, so distortion is minimal. (Lanes 3, 9, and 12 have nine bands each, but these bands aren't as clearly defined.) We will use Lane 7 as the starting point for our band set.

1. From the Database pulldown menu, select the Match command. Then click on lane 7.
2. A dialog box will pop up prompting you to select a band set. Click on New Band Set. Another dialog box will open in which you can specify the units for your band set. Molecular Weight—Ascending is what you want, so click on OK.

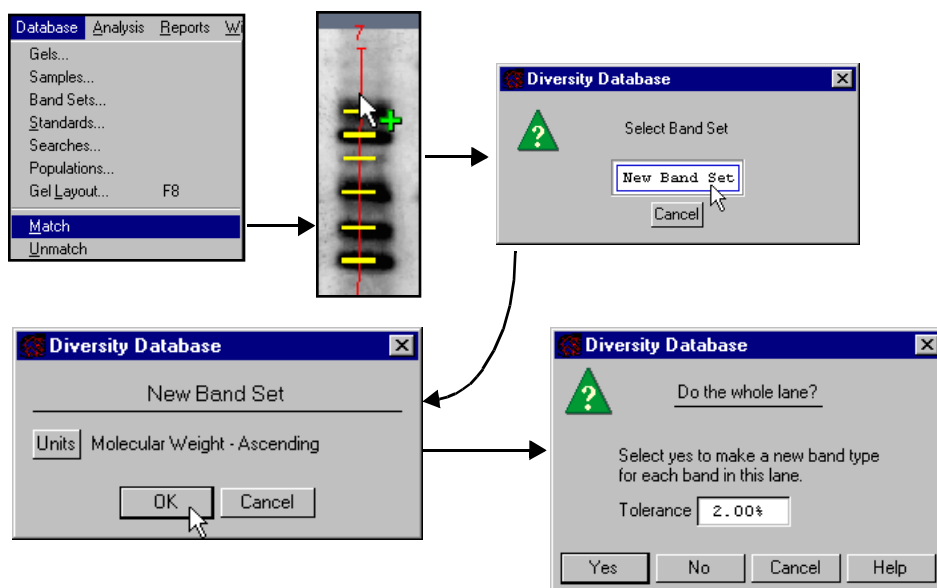


Fig. 3-20. Using lane 7 to create new band set.

3. Finally, a dialog box will ask if you want to make a new band type for each band in this lane, which you do. The Tolerance setting controls the

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automatic matching mechanism. A lower tolerance is preferred for closely spaced bands. In this gel, several bands are closely spaced, so a tolerance of 2% is good. Click on Yes.

When you click on Yes, the bands in lane 7 change to green, indicating that they are your new set of known band types. Most of the unknown bands in the image change to red. Diversity Database has automatically matched these bands, based on the band types in lane 7 and the position of the standards, as well as the tolerance setting.

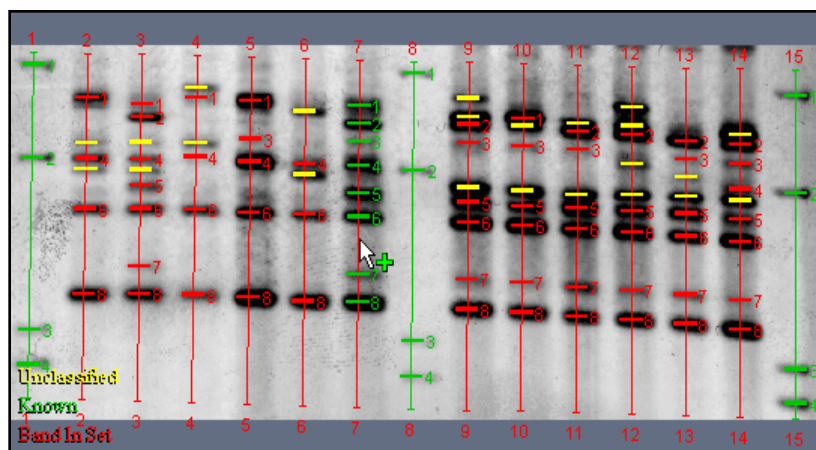


Fig. 3-21. Automatically matching experimental bands based on band types in lane 7.

You can now view the molecular weight values of the new band types you created when you clicked on lane 7.

4. Click on the Band Sets button in the Database Summary form. In the pop-up box, click on Band Set 1. Band Set 1 was created automatically when you performed steps 1–3 above.

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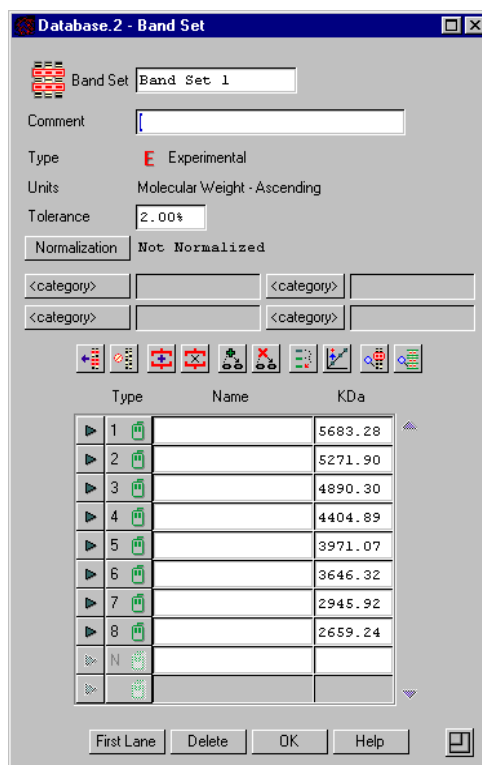


Fig. 3-22. Band Set 1, displaying the band types created from lane 7.

The eight band types correspond to the eight green bands in lane 7. The molecular weights are calculated based on your standards and Diversity Database's band modeling. Leave this dialog box open next to your image while you perform the following steps.

5. To view the band modeling lines on your image, select Show Band Models from the Database menu.

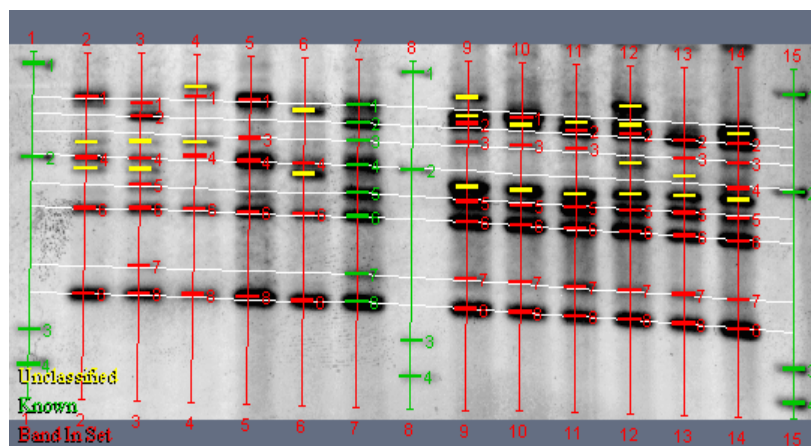


Fig. 3-23. Band modeling lines.

White lines will appear across your gel image, connecting the green bands in all three standard lanes. Other modeling lines pass through the band types in lane 7; these follow the modeling lines connecting the standards.

Note how the matching algorithm relies on the three lanes of standards to account for the curving of the gel. The modeling lines running through the experimental band types would go straight across the gel if it weren't for the changing position of the standards.

Now you must identify the remaining yellow bands as either existing band types that automatic matching missed or new band types. You do this by manual matching.

Manual Matching

To begin with, there is clearly a new band type between existing bands 4 and 5 in rows 9–14, as well as in rows 2, 3, and 6.

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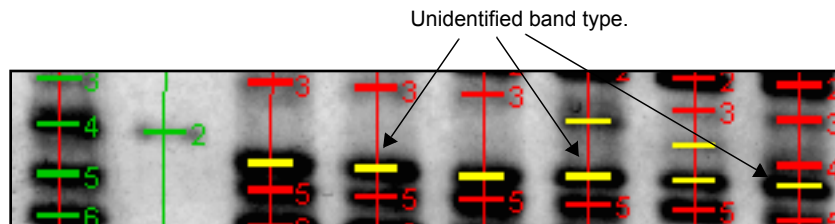


Fig. 3-24. Unidentified band type between existing bands 4 and 5.

6. To define this new band type, click on the Match button in the middle of the Band Set 1 dialog box, then click on the yellow band between bands 4 and 5 in lane 14. (When defining a new band type that appears in multiple lanes, it is a good idea to choose a yellow band next to a standard lane. This helps the matching algorithm more accurately determine the position of that band type.)

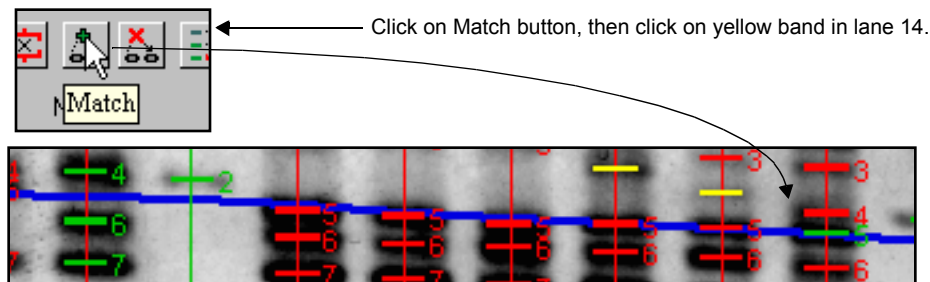


Fig. 3-25. Unidentified band type has been identified as new band type 5.

When you click on the yellow band, it changes to green and is identified as the new band type 5. Former band types 5–8 are renumbered 6–9. Also, a blue modeling line appears, showing the position of new band type 5 across the entire gel.

Now look at the top of lanes 9–12.

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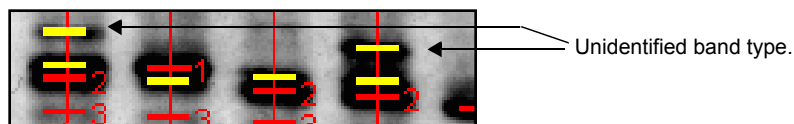


Fig. 3-26. Unidentified band type at top of lanes 9 and 12.

7. Again, click on the Match button, then click on the yellow band at the top of lane 12 (the lane nearest the standard lane). This band becomes the new band type 1, and all the other band types are renumbered accordingly. Notice that the modeling line also identifies this band type at the top of lane 4.

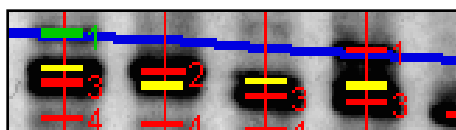


Fig. 3-27. New band type 1.

Sometimes it is not clear whether a band belongs to a new band type or to an existing one, and additional modeling is required before you can add a new type.

For example, there appears to be a new band type above band 3 in lanes 9, 12, 13, and 14. However, this has been identified as band type 2 in lane 10. Because the computer was unable to identify this band in the other lanes, band type 2 in lane 10 looks suspicious.

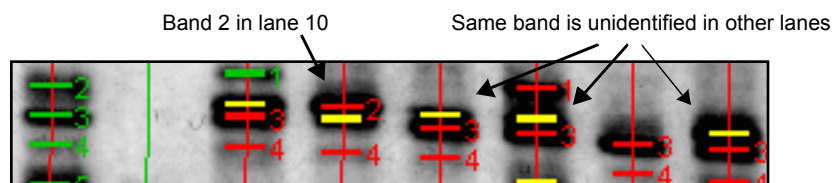


Fig. 3-28. Band 2 in lane 10 may in fact be a new band type.

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One possible solution is to improve the modeling around the mystery band. Changing the red band 3 in lane 11 to known (i.e., green) will adjust the modeling lines slightly. (Modeling lines are connected to green bands.)

8. In the Experimental Band Set dialog box, click on the button for band type 3, then click on the red band type 3 in lane 11.

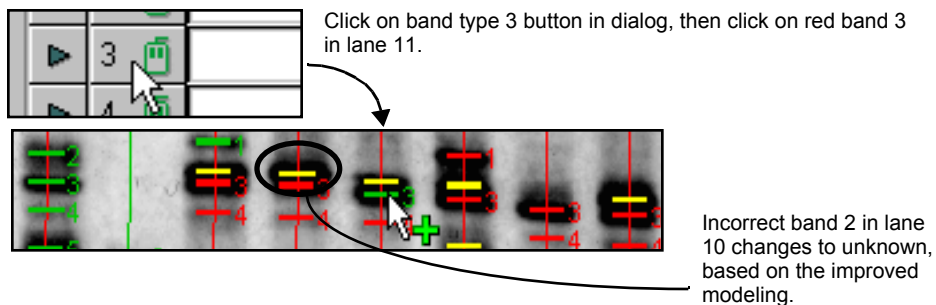


Fig. 3-29. Changing band type 3 in lane 11 to known improves the modeling.

Band 3 in lane 11 changes to green, while the incorrectly identified band 2 in lane 10 reverts to unknown, based on the improved modeling. Now you are ready to confidently identify a new band type.

9. Now once again click on the Match button in the Band Set dialog, then click on the unknown band between bands 1 and 3 in lane 9.

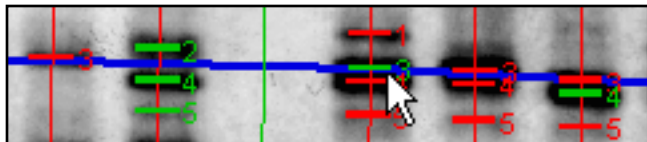


Fig. 3-30. Identifying new band type 3.

10. Finally, in lanes 2–4 and 12, there is another new band type. Click on the Match button, then click on the remaining yellow band in lane 2. This should create a new band type 6 and match the remainder of your gel.

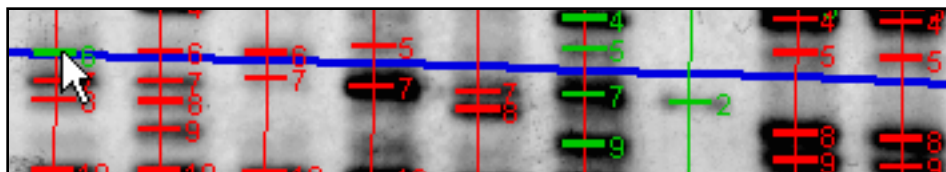


Fig. 3-31. Identifying the last yellow band in lane 2 matches the rest of the gel.

There should be no unknown bands left on your image. You end up with 12 distinct band types listed in the Band Set 2 dialog box. Click on the OK button to save this band set.

This completes Lesson 2. If you want to use this tutorial again, close the open image and database without saving. The changes you made to the files will be lost.

Important Points to Remember About Matching

Band modeling can take some practice. When modeling, remember these general points:

- After you have identified a first lane and created an initial band set, study the band modeling lines and look for patterns of identification across the lanes.
- If a band has been misidentified by auto-matching, the best method to correct this is to change the accurately modeled bands around it from matched to known (red to green) as described in step 8 above. This improves the overall modeling, and is better than trying to change the band directly, which can disrupt the modeling.
- Be careful about identifying new band types. If you see a yellow band in the same position across three or more lanes in the gel, it is probably safe to call it a new band type.
- When identifying a new band type, select a lane near a standard lane to create the new band type (as noted in step 6 above).
- If you find that previously matched bands are reverting to unknown when you try to match, you may be forcing the modeling lines out of their

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appropriate pattern. Recheck the modeling lines, and make sure they follow those connecting the standards.

3.3. Lesson 3. Creating and Applying Band Sets Without Standards

In this lesson, you will take an image that has already been added to a database and match all the experimental bands in it without using standards. (See Chapter 9 for a more detailed explanation of the commands described in this lesson.)

Note: If you are using standards in your gels, you should perform Lesson 2, Creating and Applying Band Sets Without Standards.

For this example, open Image.6. It is located in the Sample DvDB folder. This image has already been placed in a database: Database.3. Therefore, the database will open with the image.

When the image first opens, select Sort and Recalculate from the Edit menu to view the current identification of the bands.

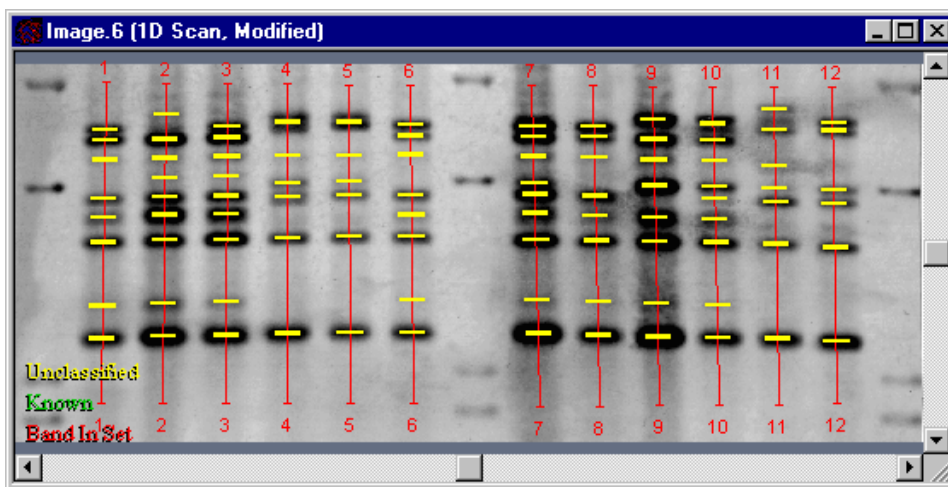


Fig. 3-32. Image.6 with 12 experimental lanes.

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Image.6 contains 12 experimental lanes. (There are standard lanes in this image; however, we will proceed as if they didn't exist.) Also, note that all bands at this point are undefined, and therefore are marked in yellow.

You start by selecting one lane that contains most, if not all, of the experimental bands you are interested in, with minimal distortion and clean, easily identifiable bands. This will be the “first lane” in your database. You will use this lane to begin creating an “experimental band set,” containing experimental band types.

The remaining experimental bands will be matched with the band types in this first lane.

In the case of Image.6, lane 2 has nine good, distinct bands, with minimal distortion. We will use Lane 2 as our first lane.

1. From the Database menu, select the Match command. Then click on lane 2.
2. A dialog box will pop up prompting you to select a band set. Click on New Band Set. Another dialog box will prompt you to specify the units for defining your band types. Click on the Units button to select from a list of units or to define your own.

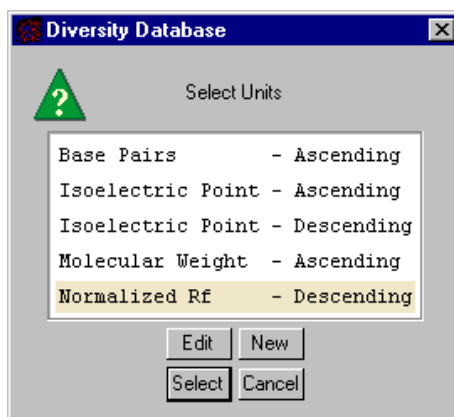


Fig. 3-33. Selecting the units for your band types.

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3. In this case, with no defined standards, Normalized Rf (Relative Front)—Descending is what we want. Normalized Rf units are based on the relative front position of the bands in the lane you use to create the band set. Select this, then click on OK in the New Band Set dialog.

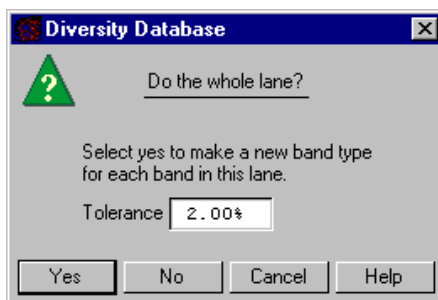


Fig. 3-34. Setting the auto-matching tolerance.

4. Finally, a dialog box will ask if you want to make a new band type for each band in this lane, which you do. The Tolerance setting controls the automatic matching mechanism. A lower tolerance is preferred for closely spaced bands. In this gel, several bands are closely spaced, so a tolerance of 2% is good. Click on Yes.

When you click on Yes, the bands in lane 2 change to green, indicating that they are your new set of known band types. Many of the unknown bands in the image change to red. Diversity Database has automatically matched these bands, based on the band types in lane 2 and the tolerance setting.

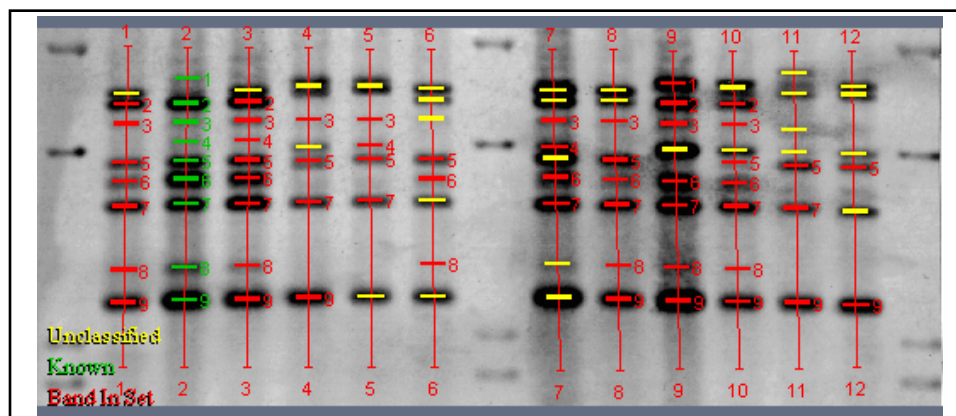


Fig. 3-35. Automatically matched experimental bands based on band types in lane 2.

You can now view the normalized Rf values of the new band types you created when you clicked on lane 2.

5. Click on the Band Sets button in the Database Summary form. In the pop-up box, click on Band Set 1 next to the red letter E. Band Set 1 was created automatically when you performed steps 1–4 above.

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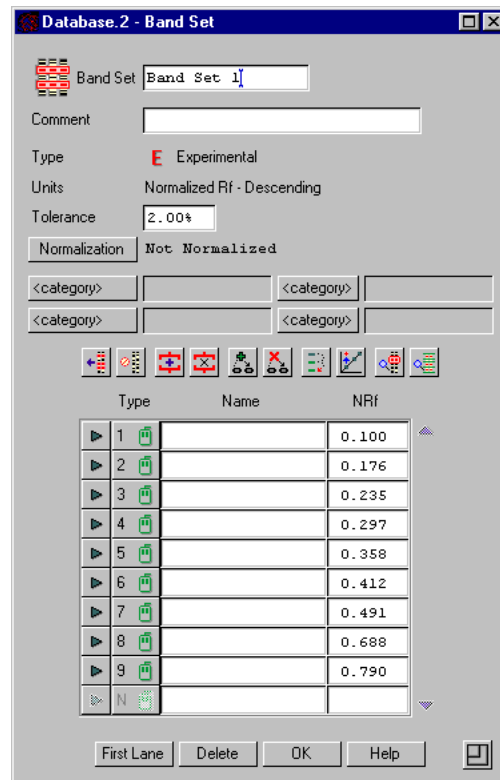


Fig. 3-36. Band Set 1, displaying the band types created from lane 2.

The nine band types correspond to the nine green bands in lane 2. The normalized Rf values (fractional distances along the length of the lane) are calculated based on Diversity Database's band modeling. Leave this dialog box open next to your image while you perform the following steps.

Now you must identify the remaining yellow bands as either existing band types that automatic matching missed or new band types. You do this by manual matching.

Manual Matching

To begin with, note that the gel is exhibiting some slight smiling, with the lanes at the ends running slightly faster than the lanes in the middle. This may account for some of the unmatched bands.

6. To compensate for the smiling, pick a band type that is present in all the lanes across the gel. Band type 7 is a good choice. Click on the band type 7 button on the band set form, then click on the middle red band in lane 7 that is in line with band type 7 and the middle yellow band in lane 12 in the same position. These bands will change to green (i.e., “known”).

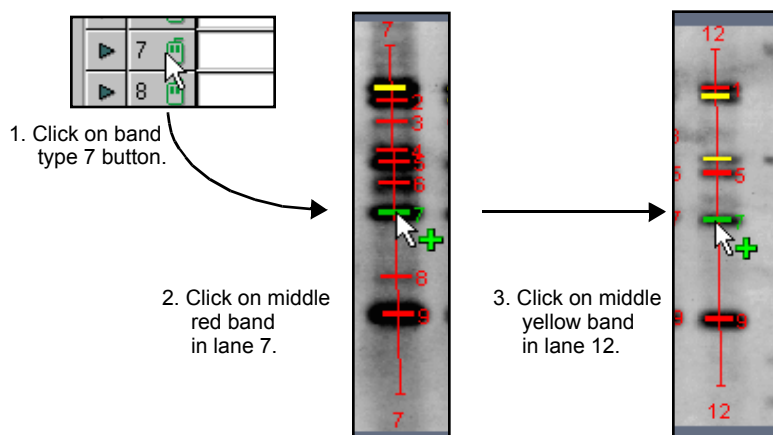


Fig. 3-37. Identifying band type 7 in lanes 7 and 12.

Note that as you define these red and yellow bands as a “known” band type 7, many of the other unknown bands are now automatically matched (marked in red). This is because the matching algorithm now “understands” that it has to compensate for the smiling, based on the position of the newly known bands.

Now we are ready to identify some new band types near the top of the image.

Note that there is clearly a new band type at the top of lane 1. Comparing lane 1 with lane 2, we can see that it falls in between existing band types 1 and 2.

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7. To create a new band type, click on the Match button in the Band Set dialog box, then click on the yellow band at the top of lane 1.

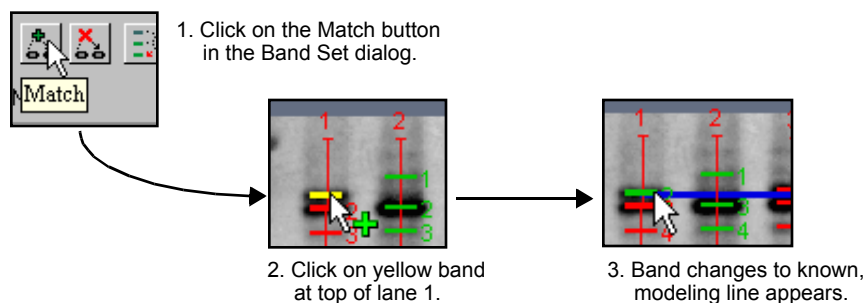


Fig. 3-38. Creating a new band type at top of lane 1.

When you create a new band type, a blue modeling line appears across the image, showing where the matching algorithm expects to find bands of the same band type.

Notice that there are a number of yellow bands at the top of lanes 7–11. Some of these may be existing band types that the matching algorithm couldn't identify. One way to clarify this situation is to convert some red (matched) bands that you know are correctly identified to green (known) bands. This improves the modeling lines (which connect green bands).

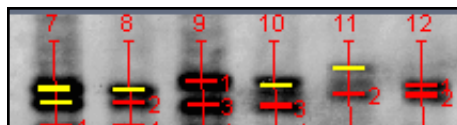


Fig. 3-39. Band 2 in lane 8 and band 1 in lanes 9 and 12 may be misidentified.

Red band type 2 in lane 8 appears to be misidentified, based on the position of band type 2 in lanes 11 and 12. Also, band type 1 in lanes 9 and 12 is suspect, based on the position of known band type 1 in lane 2.

Band type 2 in lane 11 appears to be correctly identified (based on the position of known band type 2 in lane 1), so let's change its status from red to green.

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8. Click on the band type 2 button in the Band Set dialog, then click on the red band type 2 in lane 11.

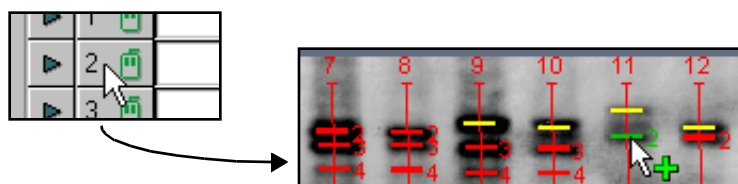


Fig. 3-40. Changing red band 2 to known correctly matches unknown bands in lane 7.

When you do this, note that red band 2 in lane 8 changes to band type 3, the yellow bands in lane 7 change to matched, and the red band 1 in lanes 9 and 12 changes to unknown.

Now we can see that the yellow bands at the tops of lanes 4, 5, 9, and 10 are probably a new band type.

9. Click on the Match button, then click on the yellow band at the top of lane 5. This identifies a new band type 2 across lanes 4, 5, 6, 9, 10 and 12.

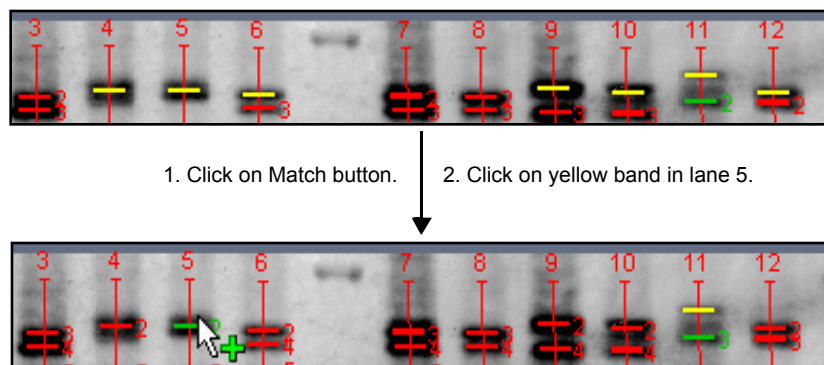


Fig. 3-41. Identifying a new band type at the top of lanes 4, 5, 6, 9, 10, and 12.

10. The yellow bands in the middle of lanes 4, 5, 7, 9, 10, 11, and 12 are also clearly a new band type. Once again, click on the Match button, then click on the yellow band in lane 5. You may get a warning message that this

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band is close to existing type 6; however, we are sure that this a new band type, so click on Continue.

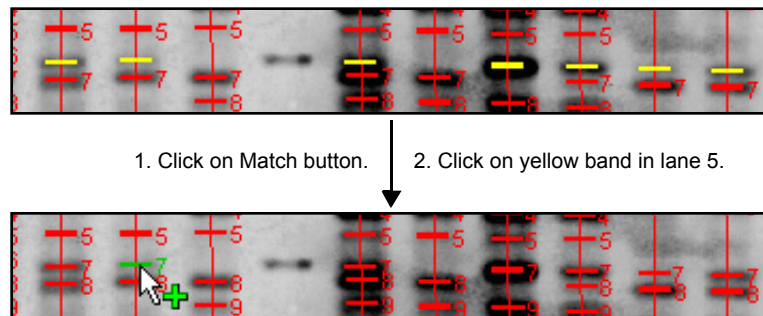


Fig. 3-42. Identifying a new band type at the middle of lanes 4, 5, 7, 9, 10, 11, and 12.

Finally, the yellow band at the top of lane 11 is clearly band type 1, and the yellow band near the bottom of lane 7 is clearly band type 11. Identify them as such by clicking on the appropriate band type button and clicking on the yellow bands.

Your final image should have all the bands identified as 12 different band types.

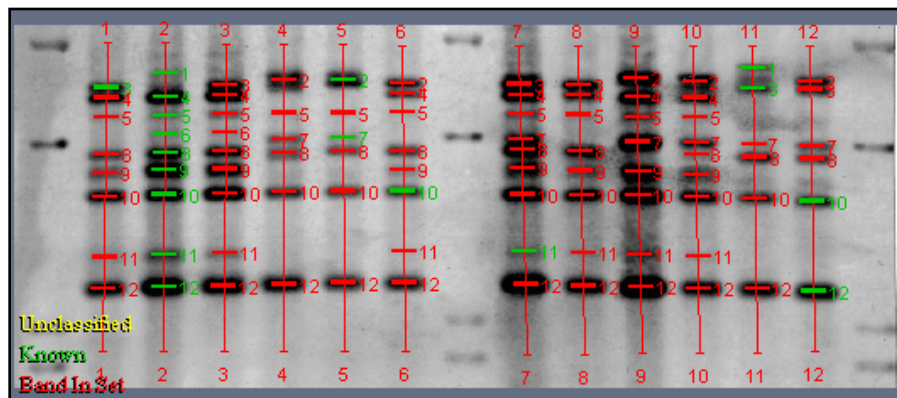


Fig. 3-43. Final Image.6 with all bands identified as 12 different band types.

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This completes Lesson 3. If you want to use this tutorial again, close the open image and database without saving. All changes you have made to the files will be lost.

Important Points to Remember About Matching

Band modeling can take some practice. When modeling, remember these general points:

- After you have identified a first lane and created an initial band set, study the band modeling lines and look for patterns of identification across the lanes.
- If a band has been misidentified by auto-matching, the best method to correct this is to change the accurately modeled bands around it from matched to known (red to green) as described in step 8 above. This improves the overall modeling, and is better than trying to change the band directly, which can disrupt the modeling.
- Be careful about identifying new band types. If you see a yellow band in the same position across three or more lanes in the gel, it is probably safe to call it a new band type.
- When identifying a new band type, select a lane near a standard lane to create the new band type (as noted in step 6 above).
- If you find that previously matched bands are reverting to unknown when you try to match, you may be forcing the modeling lines out of their appropriate pattern. Recheck the modeling lines, and make sure they follow those connecting the standards.

3.4. Lesson 4. Building and Searching the Database

In this lesson, you will add a gel image to a preexisting database and then learn some techniques for searching the database. (See Chapters 5 and 10 for a more detailed explanation of the commands described in this lesson.)

1. Start by opening Image.3. This image has lanes and bands already defined on it. To view these, click on the Show Overlays button on the main toolbar.

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2. The first step is to place this image in Database.1. Open Database.1 from the Sample Images folder. In the Summary form, click on the Gels button to open the Gels dialog box. You will notice that Image.1 and Image.2 already belong to this database.

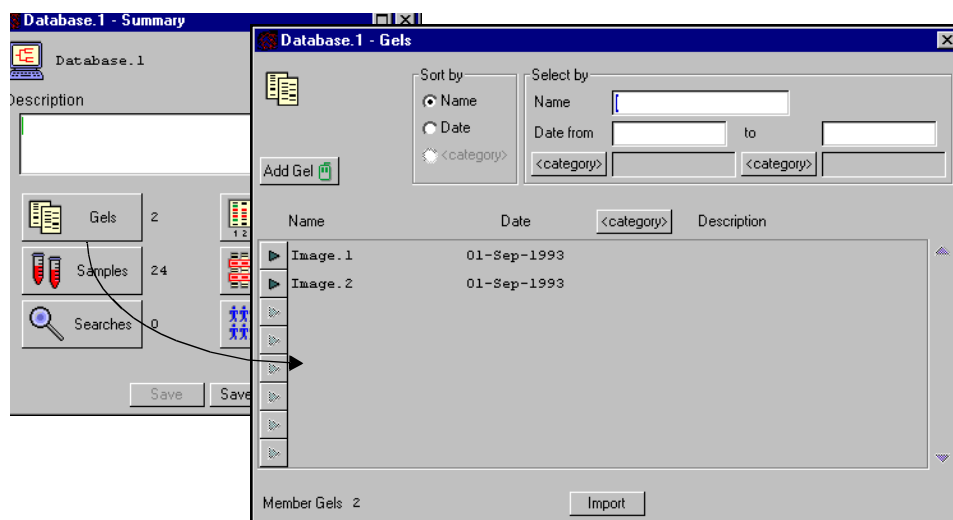


Fig. 3-44. Images 1 and 2 listed in Database.1.

3. In the Gels dialog, click on the Add Gel button, then click on Image.3. This adds Image.3 to the database.

Now you are ready to apply the band sets already entered into the database to Image.3.

4. Close the Gels dialog, and click on the Band Sets button in the Database Summary form. Select the "std" band set to open the Standard Band Set dialog.
5. In this dialog box, click on the Apply Band Set button, then click on each of the three standard lanes in Image.3. These are lanes 1, 8, and 15.

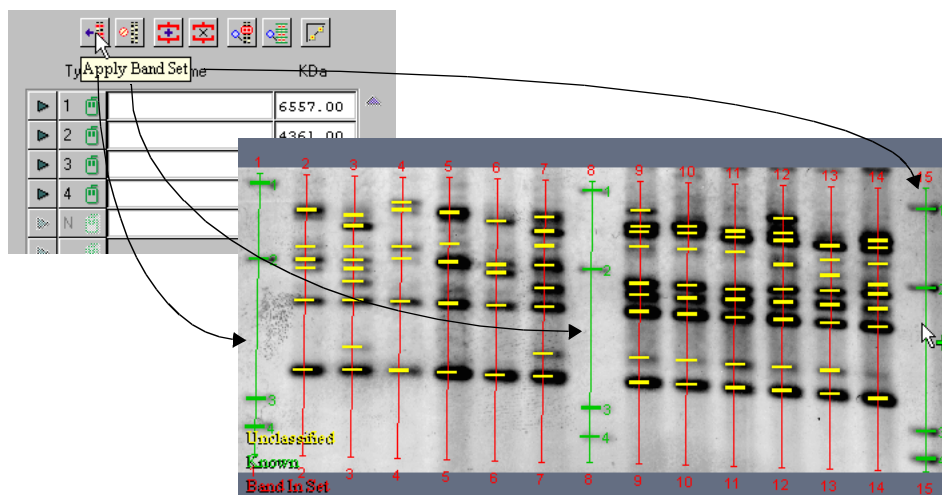


Fig. 3-45. Apply the standard band set to the standard lanes in Image.3.

6. Close the standard band set, click on the Band Sets button again, and this time select the "expt" band set to open the Experimental Band Set dialog box.

There are 13 band types listed in this dialog box. However, no lane in Image.3 contains more than nine bands. This is because band types have been added from various lanes in Images 1 and 2.

There are a number of ways to add these band types to Image.3. You can click on the Apply Band Set button in the dialog box and click on any lane in Image.3. Alternatively, you can click on one of the buttons next to a band type and apply that band to a band on the image.

For example, clicking on the band type 1 button, then clicking on the first band in lane 4 will automatically correctly assign all the band types to all the bands in the image.

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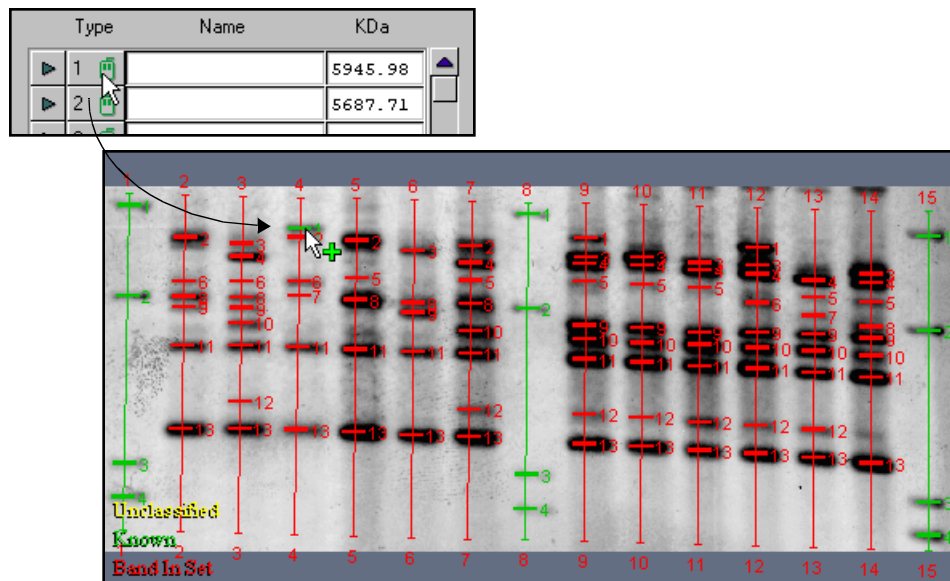


Fig. 3-46. Applying band type 1 to the first band in lane 4 matches all the bands.

However, clicking on the Apply Band Set button and applying it to lane 4 leaves some bands unmatched.

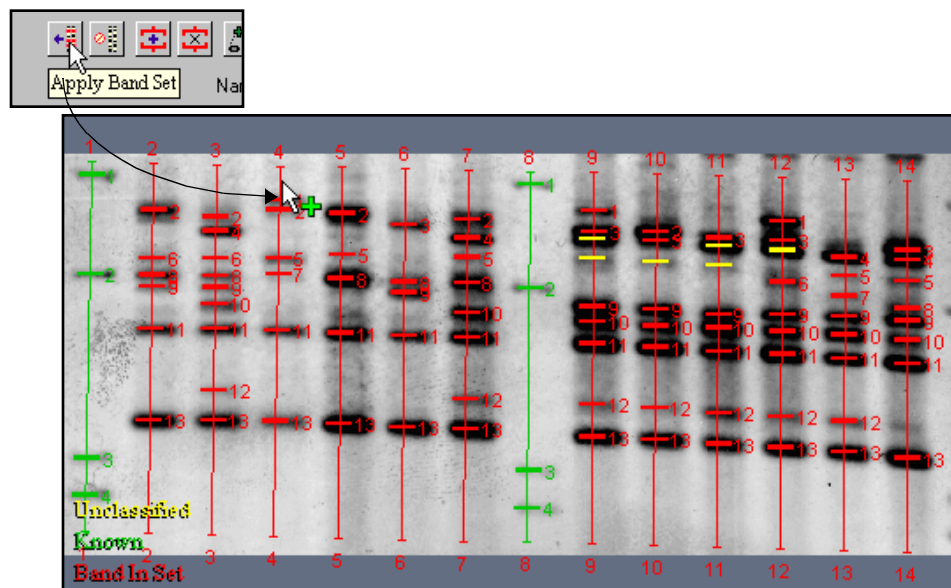


Fig. 3-47. Applying whole band set to lane 4 leaves some bands unmatched.

Applying band type 1 to the first band in lane 9 also leaves some bands unmatched, and so forth.

To summarize, there is no single, “right” way to apply an experimental band set to a new gel. If one way results in too many unmatched bands, you can “unapply” the band set by clicking on the Clear Band Set button and then clicking on each experimental lane in turn. Then you can try clicking on a different band or lane.

In most cases, further editing will be required to include unmatched bands in the band set.

3.4.a. Creating Categories and Attributes for Samples

Now that you have added Image.3 to the database and assigned the standard and experimental band sets to it, you are ready to enter some information about your samples.

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1. Click on the Samples button in the Database Summary form.

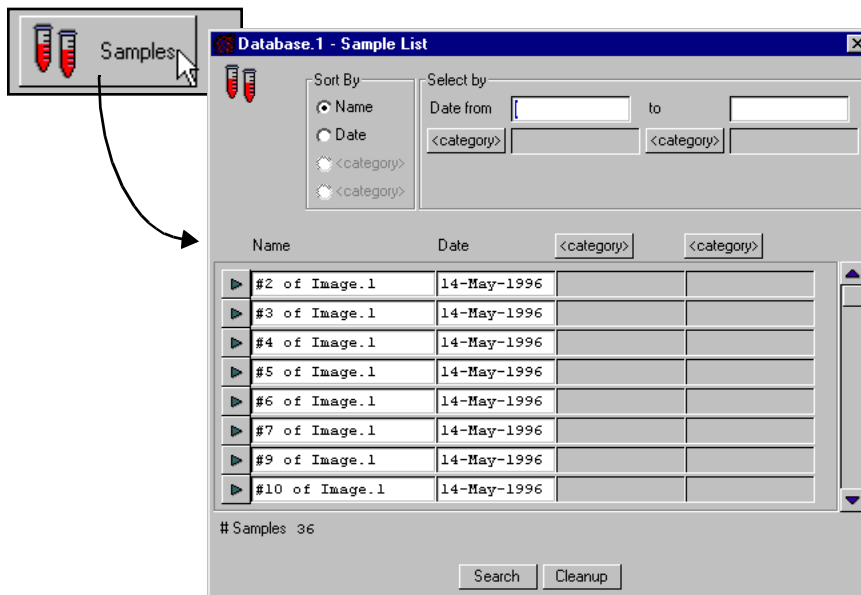


Fig. 3-48. Opening the Samples List dialog box.

The Samples List dialog box lists all the samples in the database, and allows you to rename them and sort them by name, date, and any categories you create.

For this example, you will create categories and attributes for some of your samples. We will start by creating a category called “Source,” which will include three hospitals.

2. Click on the <category> button at the top of the third column. In the pop-up box, type “Source” in the field and click on the OK button. This will change the undefined <category> button to “Source” and activate the column.

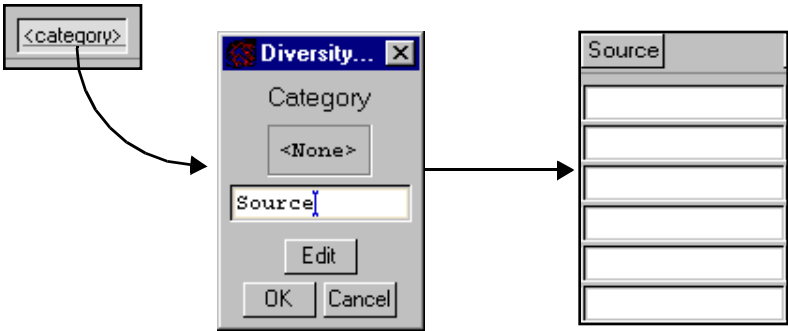


Fig. 3-49. Creating a Source category.

3. For the first sample, type “Mayfair Hosp.” in the Source column. Mayfair Hospital is the source of that sample; therefore, “Mayfair Hosp.” becomes an attribute of the category “Source.”
4. Press RETURN to move down to the next row in the same column. Type “m” and the word “Mayfair” is filled in automatically in this field. Press RETURN again and type the letter “m” in the third sample row.
5. Using the same procedure, enter “Mercy Hosp.” in the next three sample rows, and “General Hosp.” in the next three sample rows, so that your Source column appears as it does in Fig. 3-50.

	Name	Date	Source
▶	#2 of Image.1	14-May-1996	Mayfair Hosp.
▶	#3 of Image.1	14-May-1996	Mayfair Hosp.
▶	#4 of Image.1	14-May-1996	Mayfair Hosp.
▶	#5 of Image.1	14-May-1996	Mercy Hosp.
▶	#6 of Image.1	14-May-1996	Mercy Hosp.
▶	#7 of Image.1	14-May-1996	Mercy Hosp.
▶	#9 of Image.1	14-May-1996	General Hosp.
▶	#10 of Image.1	14-May-1996	General Hosp.

Fig. 3-50. Filling in sample sources.

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- Now click on the <category> button at the top of remaining column and create an “Epi Data” category. This will include some catalog numbers.
- For the first sample type “1435” into the Epi Data column. For the next sample type “8966,” and for the next sample type “4437.” Then repeat this sequence for the next three samples, and again for the following three samples. Note that you only have to type the first digit of each number to complete the sequence, because within the category the first digit of each attribute is unique.

Name	Date	Source	Epi Data
▶ #2 of Image.1	14-May-1996	Mayfair Hosp.	1435
▶ #3 of Image.1	14-May-1996	Mayfair Hosp.	8966
▶ #4 of Image.1	14-May-1996	Mayfair Hosp.	4437
▶ #5 of Image.1	14-May-1996	Mercy Hosp.	1435
▶ #6 of Image.1	14-May-1996	Mercy Hosp.	8966
▶ #7 of Image.1	14-May-1996	Mercy Hosp.	4437
▶ #9 of Image.1	14-May-1996	General Hosp.	1435
▶ #10 of Image.1	14-May-1996	General Hosp.	8966
# Samples 36			

Fig. 3-51. Completed categories and attributes.

You’ve now created and assigned categories and attributes for the first nine samples in your database. Now let’s sort your data by Epi Data number.

- At the top of the Sample List dialog box, there is a Sort By section and a Select By section. In the Select By section, click on the <category> button and select Epi Data from the pop-up box.
- Now click on the Attribute button and select 1435 from the list of attributes. Click on OK to close the Category and Attribute pop-up boxes.

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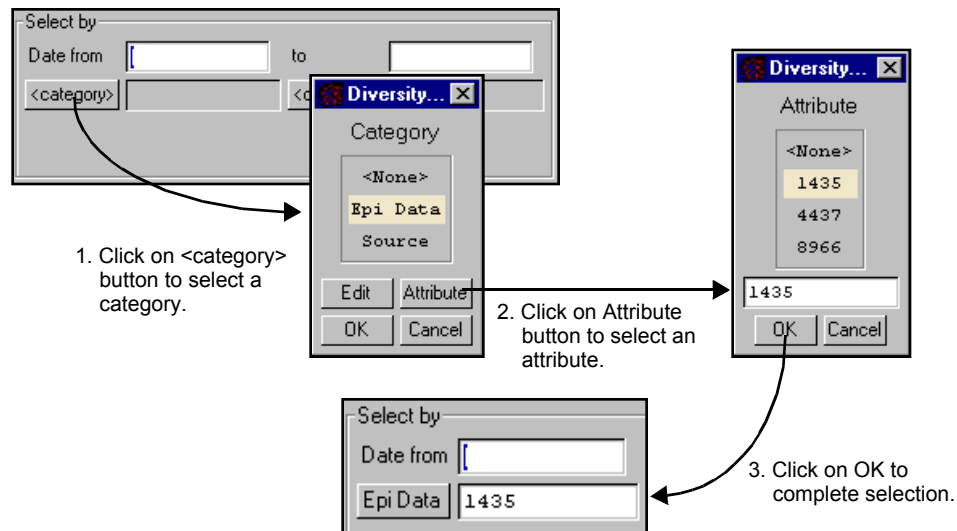


Fig. 3-52. Selecting a category and attribute.

- Now click on the Search button at the bottom of the Sample List dialog box. Only those samples with that attribute will be displayed.

	Name	Date	Source	Epi Data
▶	#2 of Image.1	14-May-1996	Mayfair Hosp.	1435
▶	#5 of Image.1	14-May-1996	Mercy Hosp.	1435
▶	#9 of Image.1	14-May-1996	General Hosp.	1435
▶				
▶				
▶				

Fig. 3-53. Samples selected on basis of Epi Data number 1435.

To return to the complete list of samples, click on the Epi Data button under Select By and select <none> from the pop-up box. Click on the Search button again, and the full list of samples will be displayed.

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3.4.b. Searching the Database

Now you are ready to search your database.

1. Click on the Searches button in the Database Summary form to open the Search form.

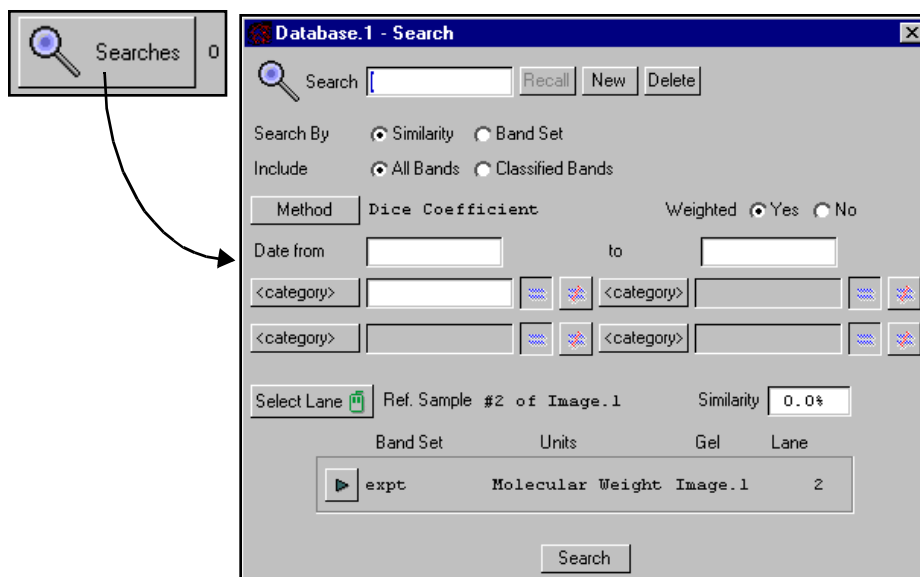


Fig. 3-54. Database search form.

As you can see from the Search form, Diversity Database offers a wide variety of search strategies. You can search by sample similarity, in which selected samples are compared to one sample, or by band set, in which the selected members of a band set are compared with one another.

For this example, we will search by sample similarity and restrict the search by category and attribute.

2. Type "Search 1" into the Search field at the top of the dialog box. This will save the settings for this particular search so that it can be performed again.

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3. Select Search By: Similarity and Include: All Bands. Classified Bands are only those bands that have been assigned to a band set; in this case, this is the same as All Bands (i.e., all the bands in all the images in the database have been assigned to a band set).
4. Click on the Method button and select Dice Coefficient, then select Weighted: Yes. (For explanations of these settings, see section 10.5.)
5. For this search, we want to compare the samples we received from Mayfair Hospital. So click on the first <category> button and in the pop-up box select Source; then click on the Attribute button and select Mayfair Hosp.

Finally, you are ready to select the lane that the other lanes in your search will be compared against.

6. First make sure that Image.1 is open. In the Search dialog box, click on the Select Lane button, then click on lane 2 in Image.1.

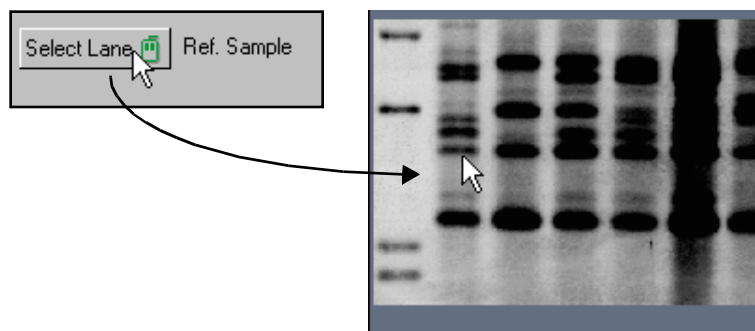


Fig. 3-55. Selecting the lane with which to perform the similarity search.

7. “Ref. Sample #2 of Image.1” will appear in the Search dialog. This lane is now the basis for your similarity search. Leave the Similarity field at 0.0%, so you are not filtering the results based on similarity. (For example, if this were set at 70%, all samples that were less than 70% similar to lane 2 of Image.1 would not be included in the search results.)

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The screenshot shows a window titled "Database.1 - Search". At the top, there is a search bar with a magnifying glass icon, containing the text "Search 1", and three buttons: "Recall", "New", and "Delete". Below this, the "Search By" section has two radio buttons: "Similarity" (selected) and "Band Set". The "Include" section has two radio buttons: "All Bands" (selected) and "Classified Bands". The "Method" section shows "Dice Coefficient" and a "Weighted" option with "Yes" (selected) and "No" radio buttons. There are two empty text boxes for "Date from" and "to". The "Source" section has a text box containing "Mayfair Hosp.", followed by an equals sign, a plus sign, and a "<category>" dropdown. Below this, there are two more "<category>" dropdowns, each preceded by a plus sign. The "Select Lane" section has a green icon, a text box containing "Ref. Sample #2 of Image.1", and a "Similarity" text box containing "0.0%". Below this is a table with four columns: "Band Set", "Units", "Gel", and "Lane". The table has one row with the values "expt", "Molecular Weight", "Image.1", and "2". At the bottom of the window is a "Search" button.

Band Set	Units	Gel	Lane
expt	Molecular Weight	Image.1	2

Fig. 3-56. Complete settings for Search 1.

- Now click on the Search button. A dialog box will pop up with the name "Population 1." Search results are called "populations." Save Population 1 by clicking on the Save button. (To reopen your saved search results, click on the Populations button in the Database Summary form.)

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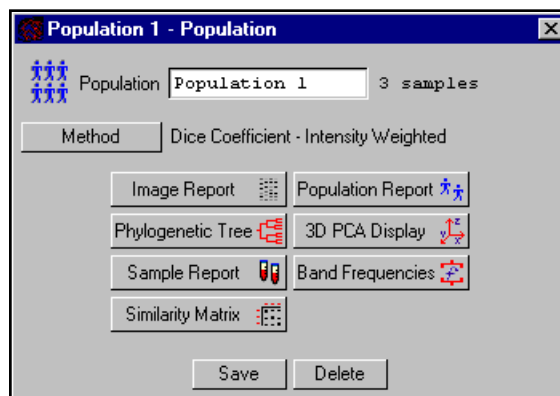


Fig. 3-57. Search results are displayed as Population 1

9. In the Population 1 dialog box, click on the Image Report button. The Report Features dialog box will open, prompting you to select the information to display in the report. Click on Report to view the report.

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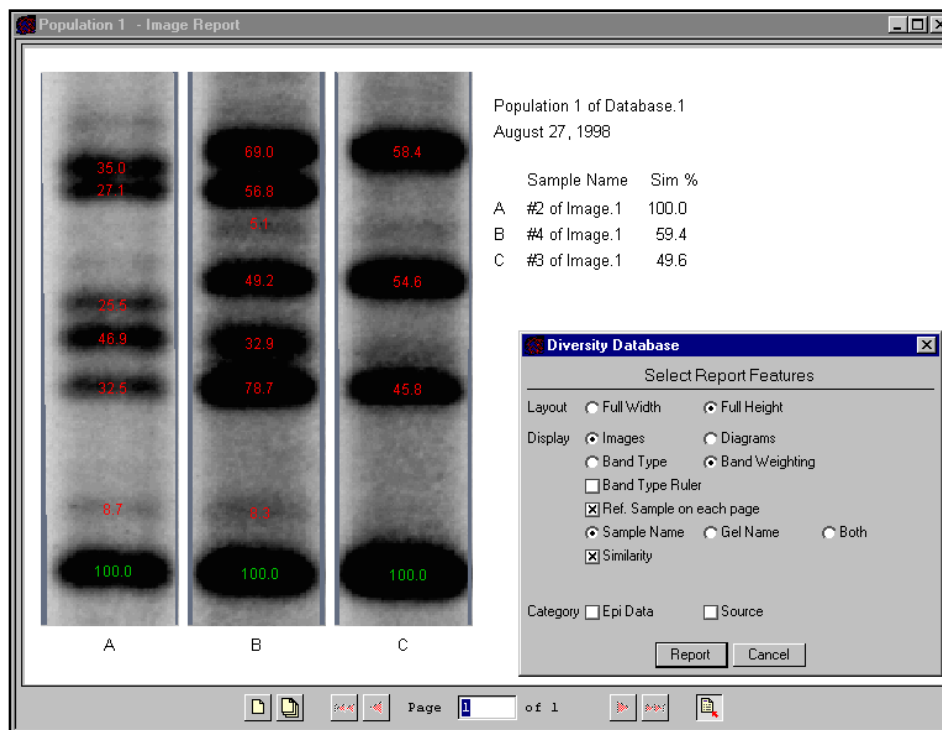


Fig. 3-58. Image Report and Report Features dialog box.

The report shows images of the three samples that came from Mayfair Hosp. The percent similarity to your reference sample (#2 of Image.1) is listed next to each sample name. The band weighting is displayed on each band on the images.

10. To reformat your report, click on the Reformat button at the bottom right of the display window. This will open the Report Features dialog box, allowing you to select different features. Click on Diagrams instead of images and check both Category boxes, then click on the Report button to display the reformatted report.

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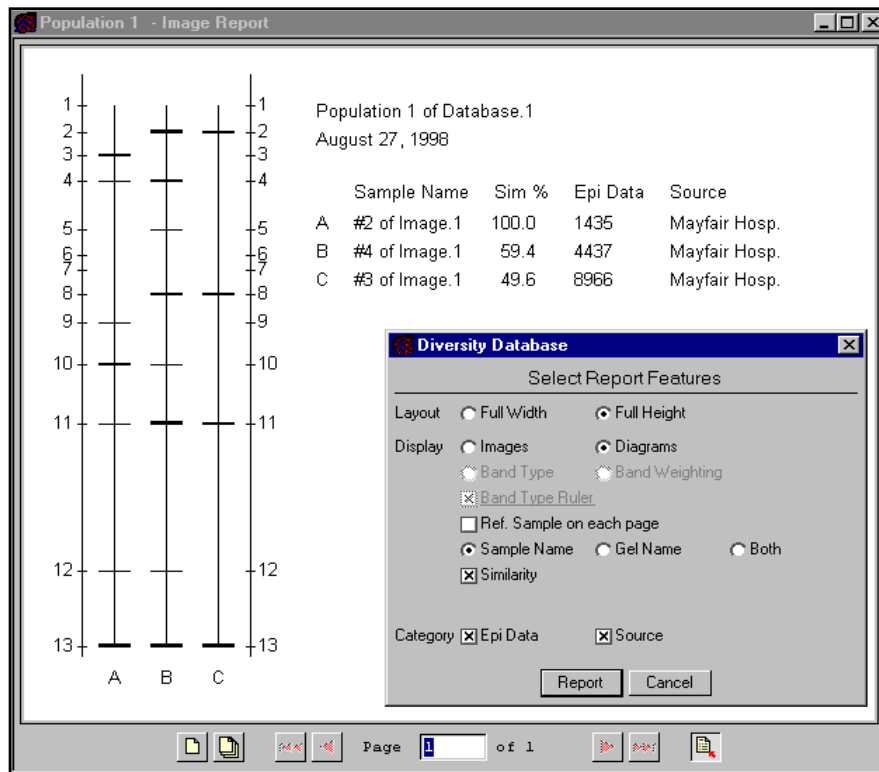


Fig. 3-59. Image Report with diagrams instead of lane images.

Let's look at another report. Close the Image Report by clicking on the Close box in the upper right corner of the window, and in the Population 1 dialog click on the Band Frequencies button.

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Population 1 - Band Frequencies

Band Frequency Report

Population 1 of Database.1

August 27, 1998

Bandset: expt Units: Molecular Weight

Normalization: Band Type 13

Samples: 3

Band #	Name	KDa	Band Frequency	
			Unweighted %	Weighted %
1		5945.99	0.0	0.0
2		5687.71	66.7	42.5
3		5477.17	33.3	11.7
4		5243.58	66.7	27.9
5		4838.65	33.3	1.7
6		4640.99	0.0	0.0
7		4520.08	0.0	0.0
8		4352.12	66.7	34.6
9		4179.56	33.3	8.5
10		3943.85	66.7	26.6
11		3628.10	100.0	52.3
12		2950.30	66.7	5.6
13		2655.65	100.0	100.0

Screen Page 1 of 1

Fig. 3-60. Band Frequencies Report.

This report lists each band type in the experimental band set for the three samples, as well as the band type name (if any) and its molecular weight. The Unweighted column shows how often a particular band type appears in the samples (e.g., band type 8 appears in two of the three samples, or 66.7 percent). The Weighted column shows the intensity of a band type as it appears in the three samples, normalized against the intensity of band type 13.

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11. Print your Band Frequencies Report by clicking on the Print All Pages button at the bottom of the report window. Click on OK in the Print dialog to complete the command.

Other reports and reporting features are described in detail in section 10.7.

12. Finally, close the report window and the Population 1 dialog box. Then close the Database Summary form. (If you want to use this tutorial again, do not save your changes.)

This completes Lesson 4. If you want to use this tutorial again, close the open images and database without saving. All changes you made to the files will be lost.

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4. Viewing and Editing Images

This chapter describes Diversity Database's viewing tools, which you can use to magnify your images and clarify your data. This chapter also describes the tools for cropping, flipping, and rotating images, eliminating and filtering noise, and annotating images.

These tools may be found on the View, Image, and Edit menus.

Note: The following chapters contain instructions for analyzing X-ray films, wet and dry gels, blots, and photographs. For the sake of simplicity, these are all referred to as “gels.”

4.1. Magnifying and Positioning Tools

The magnifying and positioning tools in Diversity Database are located on the View menu and Window menu; some of these functions are also found on the main toolbar.

These commands will only change how the image is displayed on your computer screen. *They will not change the underlying data.*

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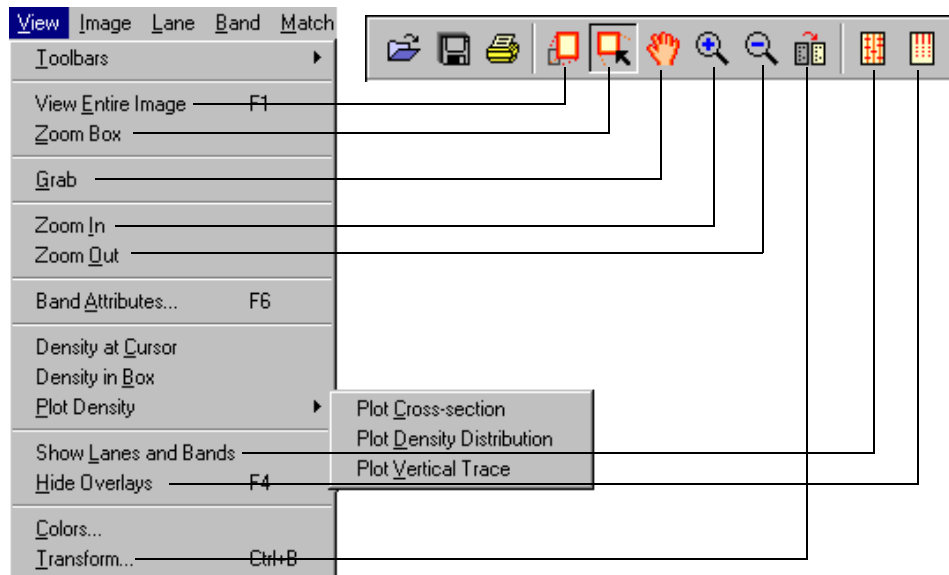


Fig.4-1. Viewing functions on View menu and main toolbar.

Zoom Box

Zoom Box allows you to select a small area of the image to magnify so that it fills the entire image window.

First, click on the Zoom Box button on your main toolbar or select View > Zoom Box. Your cursor arrow will change to a cross. Then drag the cursor across the image to enclose the area you want to magnify. The area of the image you selected will be magnified to fill the entire window.

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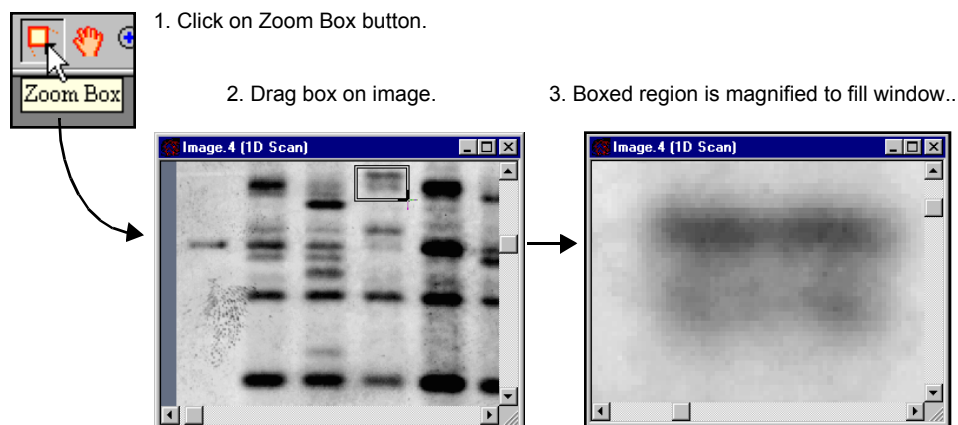


Fig.4-2. Zoom Box tool.

Zoom In/Zoom Out

These tools work like standard magnifying tools in other applications.

Click on the Zoom In or Zoom Out button on your main toolbar (or select from the View menu). Your cursor will change to a magnifying glass. Click on an area of the image to zoom in or zoom out a defined amount, determined by the setting in the Edit > Preferences dialog box.

Grab

This tool allows you to change the position of your image in the image window. Select View > Grab or click on the Grab button. Your cursor will change to a “hand” symbol. Dragging the Grab symbol on the image will move the image in any direction.

You can also move your image inside the image window by using the ARROW keys.

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View Entire Image

If you have magnified part of an image or moved part of an image out of view, View Entire Image returns to the original, full view of the image.

Imitate Zoom

If you are comparing two or more images and want to zoom in on the same area on all of them, use the Imitate Zoom command.

First zoom in on one of the images. Then, with that image window still selected (the title bar will be blue), select Imitate Zoom from the Window menu.

The zoom factor and region of the selected image will be applied to all the images.

Note: Imitate zoom will only work on comparable images with similar dimensions.

Tiling Windows

If you have more than one image open, the Tile commands on the Window menu allow you to arrange your images neatly on the screen.

Tile will resize your windows and arrange them on the screen left to right and top to bottom.

Tile Vertical will resize your windows and arrange them side-by-side on the screen.

Tile Horizontal will resize your windows and stack them top-to-bottom on the screen.

4.2. Density Tools

The Density Tools (under the View menu and on the Density Tools toolbar) are designed to provide a quick measure of the signal intensity of the data in your image.

Viewing and Editing Images

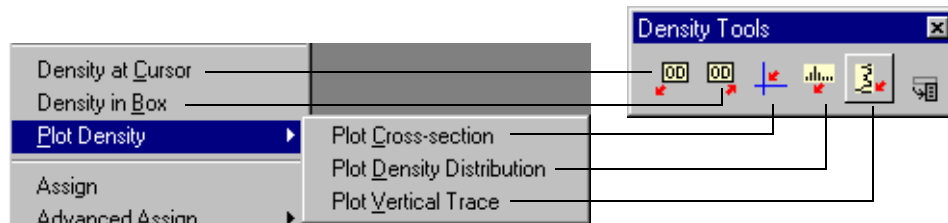


Fig.4-3. Density tools on the menu and toolbar.

Density at Cursor

Density at Cursor displays the signal intensity of the pixel on the image where you click your mouse. It also shows the average intensity for a 3 x 3 pixel box centered on that point.

To use this tool, click on the Density at Cursor button on the Density toolbar or select View > Density at Cursor. Then click on the point of interest on the image.

Density in Box

Density in Box displays the average and total intensity within a boxed region on the image. Select Density in Box from the Density toolbar or View menu, then define the region you want to measure by dragging your cursor across the image.

Plot Density Distribution

Plot Density Distribution displays a histogram of the signal intensity distribution for the part of the image displayed in the image window. The average intensity is marked in yellow on the histogram.

The histogram will appear along the right side of the image. Use the Zoom functions to display the histogram for magnified regions of your image.

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Plot Cross-section

Plot Cross-section displays an intensity trace of a cross-section of the image centered on the point where you click your mouse. The horizontal trace is displayed along the top of the image, and the vertical trace is displayed along the side of the image.

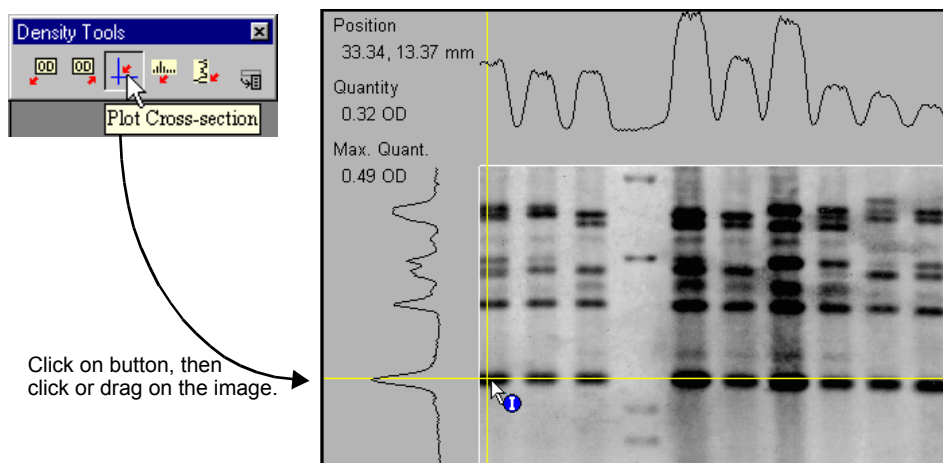


Fig.4-4. Plot Cross-section tool.

The intensity at the point you clicked on is displayed, as is the maximum intensity along the lines of the cross-section. Dragging the mouse with this function selected will continuously update the display.

Plot Vertical Trace

Plot Vertical Trace plots an intensity trace of a vertical cross-section of the image centered on the point where you click your mouse. Dragging the mouse with this function selected will continuously update the display.

Note: These traces will look different than the traces for functions like Plot Lane or Plot Band, because the sampling width for the Density traces is only one pixel.

Viewing and Editing Images

4.3. Showing and Hiding Overlays

As you create plots and traces and identify bands or spots on an image, you create a variety of image overlays. Hide Overlays on the main toolbar and View menu is useful for concealing any overlays you have created. It can also be used to erase plots, traces, and information boxes that are created by other functions.

Note: Clicking once on Hide Overlays will conceal the overlays. Clicking twice will deassign any function that has been assigned to the mouse.

View > Show Lanes and Bands reveals the lane and band overlays that have been added to an image.

4.4. Colors

View > Colors opens a dialog box in which you can adjust the colors of the image, windows, buttons, etc.

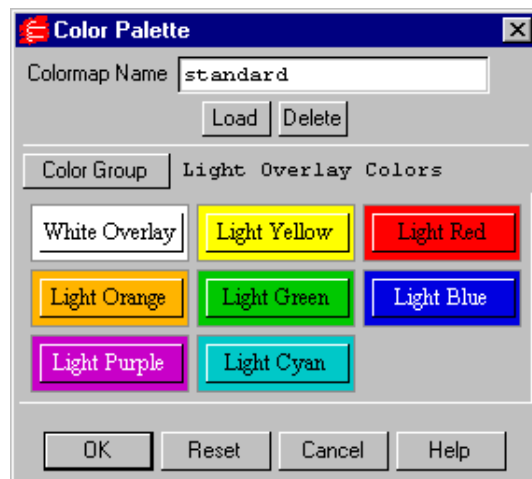


Fig.4-5. Colors dialog box.

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Selecting a Color Group

Within the dialog box, the Color Group button allows you to select the colors of a particular group of objects (e.g., pop-up boxes, image colors, etc.). Click on the button to open the list of objects you can change.

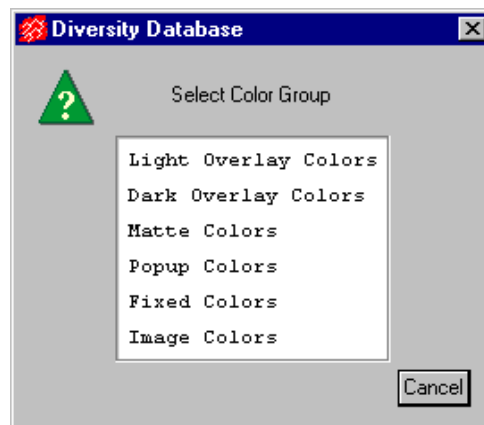


Fig.4-6. List of Color Groups.

Click on a color group in the list to select it.

Changing a Color

After you have selected the color group to change, click on the specific color button to change. The Color Edit dialog box will open, allowing you to adjust the red-green-blue (RGB) values of the color you selected.

Viewing and Editing Images

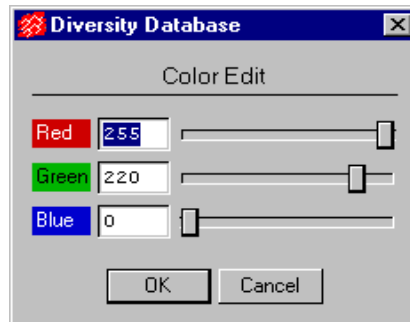


Fig.4-7. Color Edit dialog box.

Drag the sliders or enter a value in the fields. The color of the button will change with your adjustments.

Saving/Selecting a Defined Set of Colors

After you have changed the colors within color groups, you can save these settings for future use on other images. The Colormap Name field displays the name of a defined set of colors and color groups. There are several standard colormaps included with Diversity Database, or you can create your own.

To select a predefined colormap, click on the Load button.

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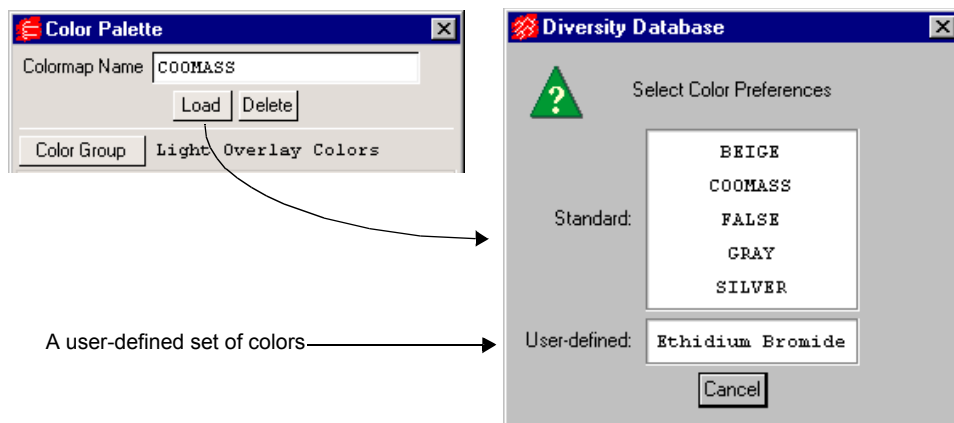


Fig.4-8. Selecting a Colormap.

From the list displayed, click on the set of colors you want to apply.

To create your own colormap, adjust the colors within the color groups as described above and type in a new colormap name. Click on OK to apply your changes.

To remove a colormap, click on the Delete button. Select the colormap to be deleted from the displayed list. A pop-up box will ask you to confirm the deletion.

If you change your mind about applying any changes you make, click on the Cancel button. If you want to return to the Standard colormap, click on the Reset button. All colors will be returned to their default values.

4.5. Transform

If features in your image are indistinct or fine details appear to be lost in background noise, you can use the Transform functions to optimize your displayed image. To open the Transform dialog box, select View > Transform.

Viewing and Editing Images

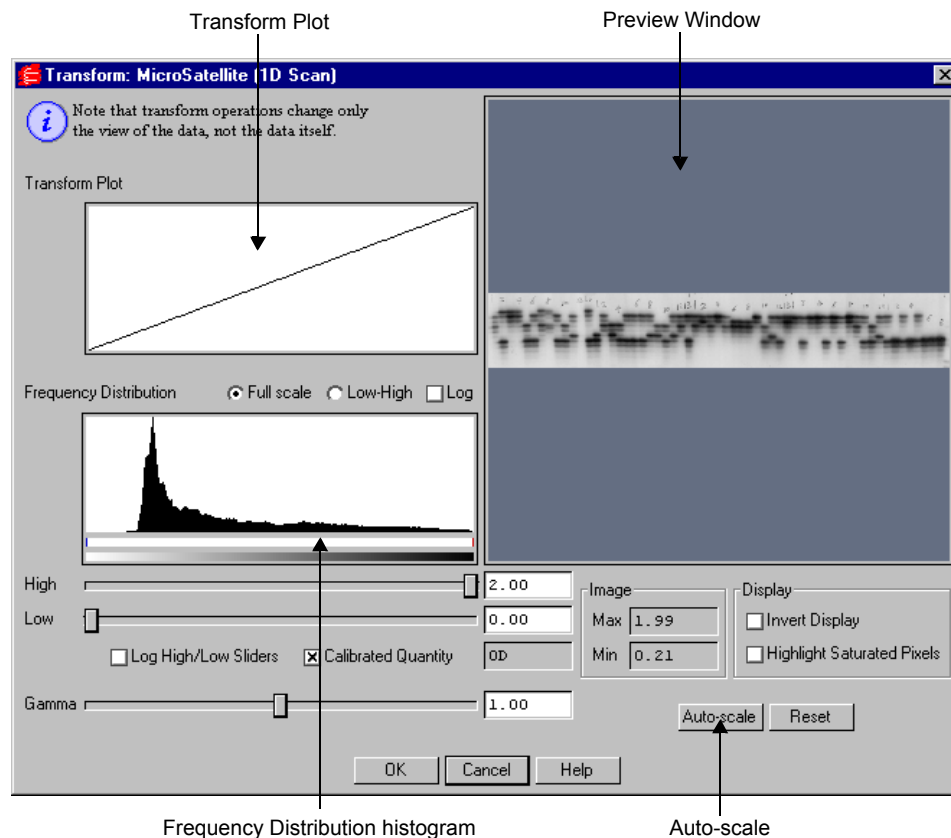


Fig.4-9. Transform dialog box.

The Transform dialog box contains a Preview Window, a Frequency Distribution histogram, a Transform Plot, and three main methods of optimizing your image: Auto-scale, High and Low sliders, and a Gamma slider. You can use these controls to adjust the way the software transforms raw data into visual data.

Note: Any changes made with the Transform controls will only affect how the image is displayed on your screen. *They will not affect the underlying data.*

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4.5.a. Subwindows in the Transform Dialog

Preview Window

The Preview Window in the Transform dialog shows a smaller view of the same image that is displayed in the main image window. Changes in the Transform controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on OK.

You can use view tools like Zoom Box and View Entire Image in the Preview Window just as you can in the main image window, to focus on particular regions of interest. You can also use the ARROW keys to move the image within the Preview Window.

Frequency Distribution Histogram

The Frequency Distribution histogram shows the total data range in the image and the amount of data at each point in the range. In a typical scan, there is a signal spike at the left (“gray”) end of the histogram due to background noise.

Transform Plot

The Transform Plot is a logarithmic representation of how the raw pixel data are mapped to the pixels of your computer screen.

4.5.b. Transform Controls

Auto-scale

Clicking on the Auto-scale button will optimize your image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). This enhances minor variations in the image, making fine details easier to see. You can then “fine-tune” the display using the High, Low, and Gamma sliders described below.

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High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

As you drag the sliders, the slider markers on the Frequency Distribution histogram will move. Everything to the left of the Low marker will be remapped to minimum intensity, while everything to the right of the High marker will be remapped to maximum intensity. Using the histogram, you can position the markers at either end of the data range in your image, and use the low slider to cut off the "spike" of background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Log High / Low Sliders changes the feedback from the slider handles, so that when you drag them, the slider markers move a shorter distance in the histogram. This allows for finer adjustments when your data is in a narrow range.

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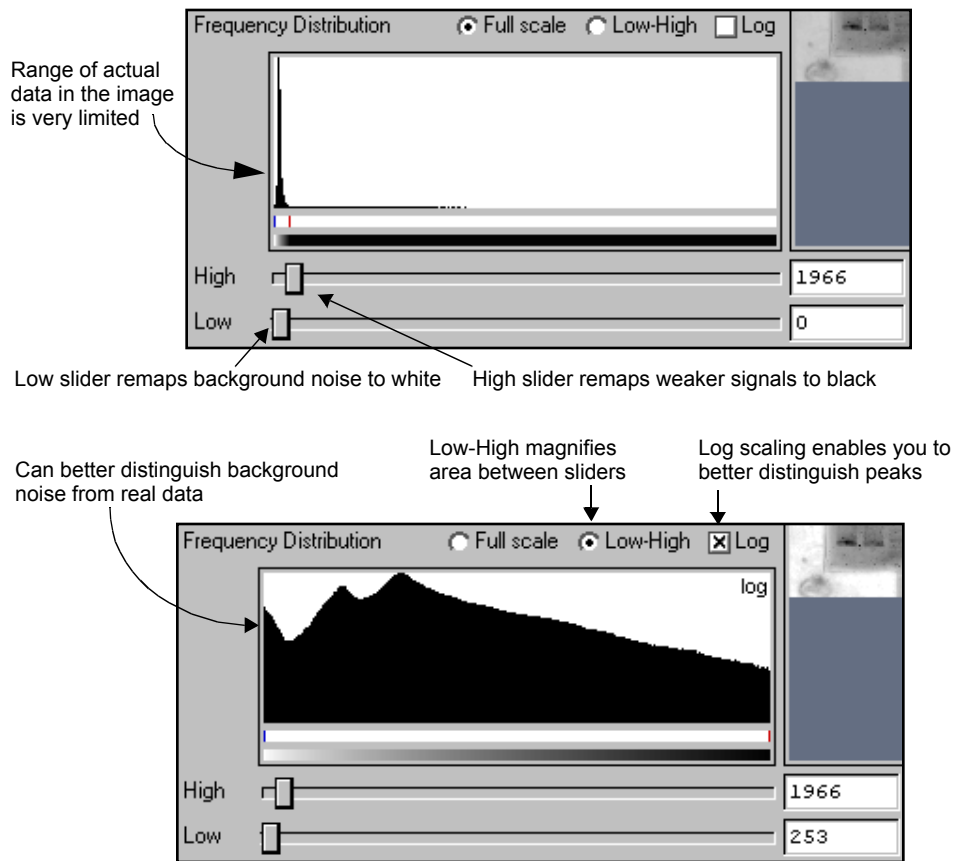


Fig.4-10. Two views of the Frequency Distribution histogram.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle expands or compresses the contrast range at the dark or light end of the range, and this is reflected in the Transform Plot and Preview Window.

Viewing and Editing Images

4.5.c. Other Features

Full Scale/Low-High

The Full Scale/Low-High option buttons adjust how the range of data in the image is displayed in the Frequency Distribution histogram and Transform Plot. They do not change how the data is displayed in the image window.

Selecting Full Scale adjusts the Frequency Distribution and Transform Plot displays so they show the full intensity range of the image.

Selecting From Low to High magnifies the range between the Low and High sliders. This makes it easier to view your data if it does not occupy the full intensity range of the image.

Log

The Log checkbox changes the way the data is displayed in the histogram so you can better discern subtle changes in signal intensity.

Image Max/Min and Units

Image Max and Min display the range of signal intensity in the image.

The image units are determined by the type of scanner used to create the image. In the case of densitometers, you can select Calibrated Quantity to display your image units in O.D.s.

Invert Display

The Invert Display checkbox flips light bands on a dark background to dark bands on a light background, and visa versa. Once, again, the actual data will not change—only the image.

Highlight Saturated Pixels

When the Highlight Saturated Pixels checkbox is selected, areas of the image with saturated signal intensity are highlighted in red.

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Reset

If at any time you want to return to an unmodified view of the scan data, click on Reset.

4.6. Resizing and Reorienting Images

Frequently, your images will require some editing prior to analysis. This section describes the features that allow you to change the size and orientation of your images.

Diversity Database's image editing tools are located on the Image menu and the Image toolbar.

Note: Many of these image editing commands are irreversible. A pop-up alert box will ask you to confirm each edit that will change your image irreversibly.

Viewing and Editing Images

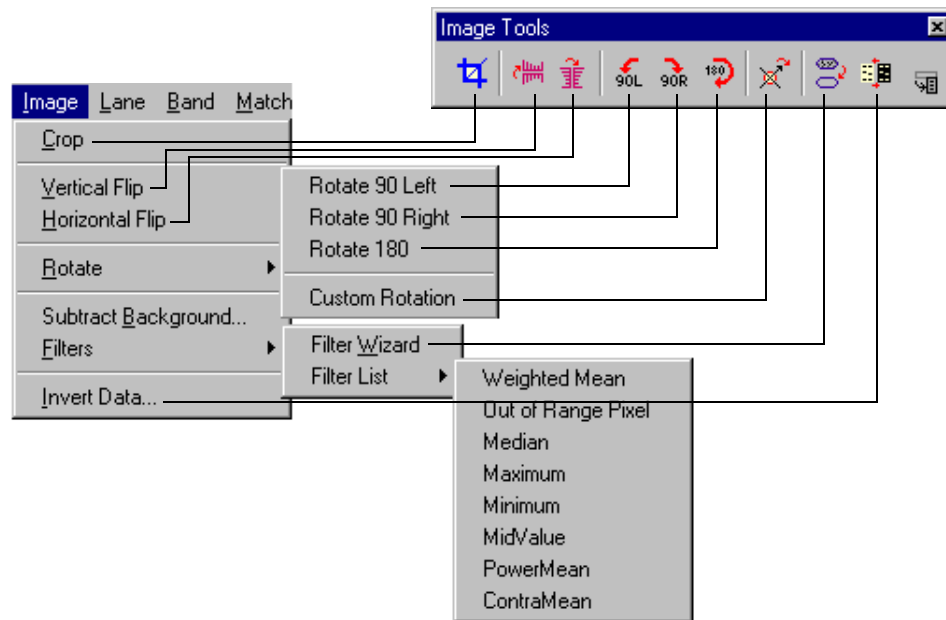


Fig.4-11. Image menu and toolbar.

4.6.a. Cropping Images

To eliminate unwanted parts of an image, you can use the Crop tool. This is also an easy way to reduce the memory requirements of an image.

To crop, click on the Crop button on the Image toolbar or select Image > Crop. Your cursor appearance will change to a Crop symbol.

Define the region to be cropped by dragging the cursor across the image, creating a box. Everything outside the box will be deleted.

Information about the physical dimensions of the crop area (given in millimeters and number of pixels) is listed at the bottom of the crop box, as is information about the memory size of the image inside the crop area.

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1. To *reposition* the crop box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.
2. To *resize* the box, position your cursor on a box border or corner. The cursor appearance will change to a bidirectional arrow. You can then drag that border or corner in or out, resizing the box.
3. To *redraw* the box, position your cursor outside the box. The cursor appearance will change back to the Crop tool, and you can draw another box, replacing the one you just drew.

Once you are satisfied with your crop box, position your cursor within the box slightly off-center. The cursor appearance will change to a scissors symbol. You can then click on the mouse to perform the crop.

A pop-up box will ask you whether you want to: (1) crop the original image, (2) save a copy of the area inside the crop box as a separate image, keeping the original image intact, or (3) cancel out of the cropping operation.

Viewing and Editing Images

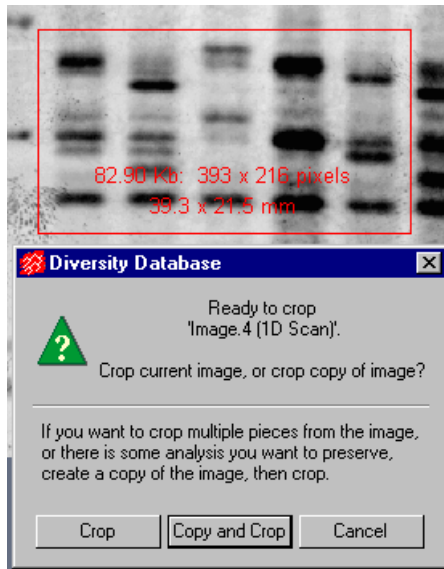


Fig.4-12. Crop box and pop-up Crop dialog.

If you select the Copy and Crop button, a dialog box will be displayed in which you can enter the name of the cropped image and its version number. Click on the OK button once you have finished.

4.6.b. Flipping and Rotating Images

If your image is not properly oriented, you can flip and/or rotate the image.

Note: These actions will erase any overlays you have created or analysis you have performed on an image. You will be asked to confirm the flip / rotation before the command is executed.

Flipping

To flip the image right-to-left, select Horizontal Flip from the Image menu or toolbar. To flip the image top-to-bottom, select Vertical Flip.

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90° Rotations

Select Rotate 90 Left, Rotate 90 Right, or Rotate 180 from the Image > Rotate menu or Image toolbar to perform the specified rotation. You will be asked to confirm your choice before the command is executed.

Note: Any analysis performed on the image cannot be rotated and will be lost.

Custom Rotation

If you need to rotate your image in increments other than 90°, you can use the Custom Rotation command.

Select Custom Rotation from the Image > Rotate menu or Image toolbar. A green “plus” sign will appear next to your cursor. Click on the image you want to rotate and a circular overlay with an orange arrow will appear. A small dialog box also will open, indicating the angle of rotation in degrees and radians.

Viewing and Editing Images

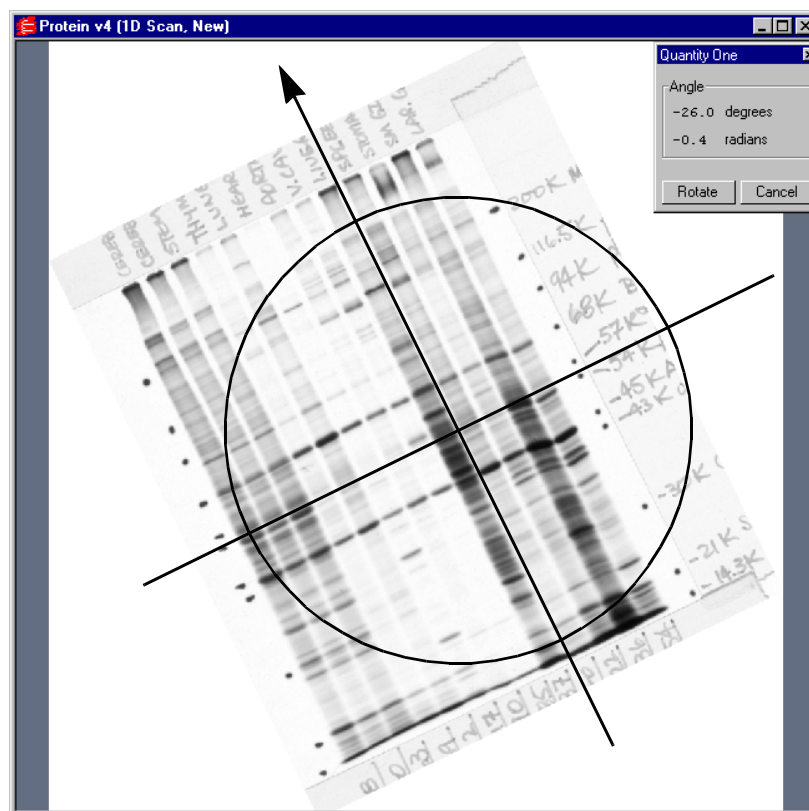


Fig.4-13. Custom rotation; the arrow points in the direction of the new top of the image.

To perform the rotation, position your cursor on the arrowhead and drag. As you drag, the arrow will rotate and the angle in the dialog box will change. Position the arrow so that it points in the direction of the new top of the image. You can “fine-tune” your rotation as much as you like.

Note: If you want to center your arrow on a particular point on the image (e.g., to align along a particular lane), you can use the F3 key command. Position your cursor on the point on the image you want to center on, and press the F3 key. The image will move so that the center of the arrow and your cursor point are aligned.

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To complete the rotation, click on the Rotate button in the small dialog box. Another, smaller image window will open containing the rotated image. You will then have the option of renaming your new image and changing the version number.

If you are not satisfied with your rotated image, simply delete it and start over.

Note: Because an image is composed of square or rectangular pixels, Custom Rotation has to perform some minor smoothing on the image to turn it at a non-90° angle. In addition, any analysis performed on the image cannot be rotated and will be lost.

4.7. Subtracting Background Noise from Entire Images

Diversity Database has a number of tools for subtracting background noise from images to improve the clarity of your data. This section describes background subtraction for the entire image.

Diversity Database also has background subtraction tools for individual lanes (see section 11.2.) and for volumes analysis (see section 14.3.).

Whole-image background subtraction is useful for reducing background noise resulting from the opacity of the carrier medium (film, gel matrix, or blot matrix) or film fogging.

Note: Whole-image background subtraction is an irreversible process. You will be asked if you want to make a copy of the image before completing the operation.

To open the dialog box for performing whole-image background subtraction, select Image > Subtract Background.

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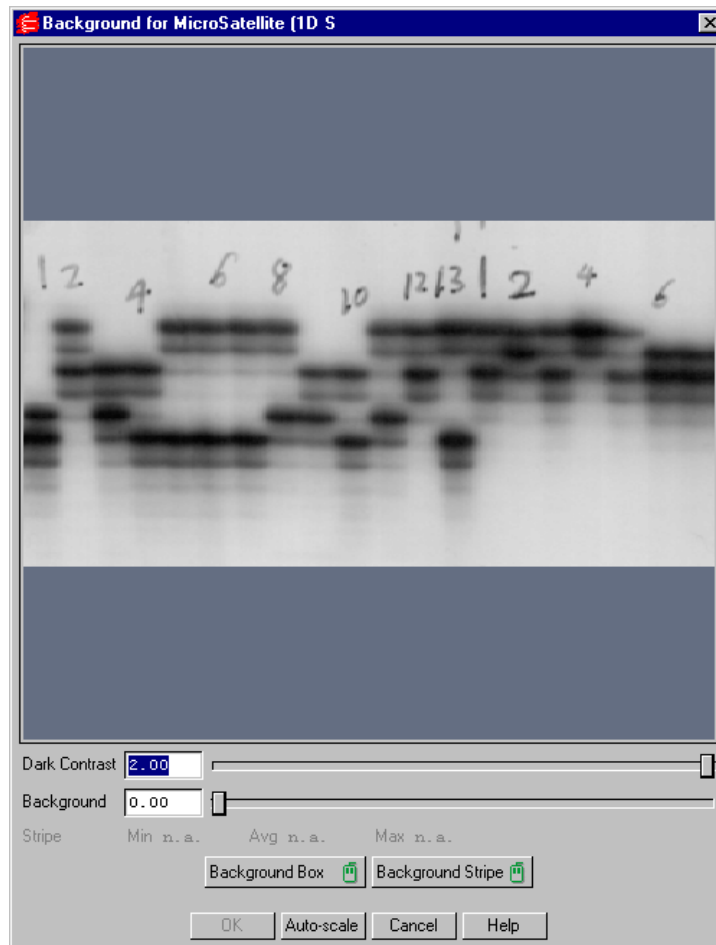


Fig.4-14. Subtract Background dialog box.

The Subtract Background dialog box has a Preview Window, which contains a smaller view of the same image that is displayed in the main image window. Changes in the subtract background controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on OK.

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You can use view tools like Zoom Box and View Entire Image in the Preview Window just as you can in the main image window, to focus on particular regions of interest. You can also use the ARROW keys to move the image within the Preview Window.

A description of each of the available features in the Subtract Background dialog is given below.

Auto-scale

Clicking on the Auto-scale button will automatically adjust your Dark Contrast and Background settings to optimal levels. You can then “fine tune” these settings using the other controls.

Dark Contrast Slider

Adjusting the Dark Contrast prior to background subtraction allows you to make fine distinctions in the amount of background noise present in your image. Faint artifacts that may not be obvious to the unaided eye are revealed by lowering the Dark Contrast level to close to the subtraction levels.

The Dark Contrast slider is similar to the High slider in the Transform dialog box. Dragging the slider handle to the left will make faint signals appear stronger. You can also move the slider incrementally by clicking on the slider bar, or you can type a value into the field next to the slider.

Note: Adjusting the Dark Contrast slider by itself does not eliminate background noise; therefore, the OK button will not activate if you only adjust this slider. If you only want to adjust the display contrast, and *not* subtract background, use the View > Transform function.

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Example A: without adjustment



Example B: with adjustment

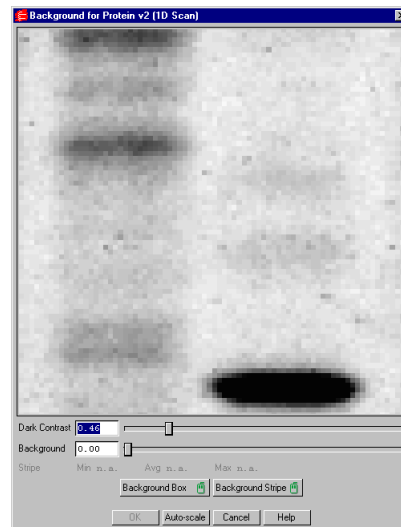


Fig.4-15. Images without and with Dark Contrast adjustment. Note that in Example A, with no Dark Contrast adjustment (3.00 O.D.), the surrounding medium appears free of background noise and image artifacts, but in Example B, with adjustment (0.46 O.D.), artifacts in the image become immediately obvious, indicating that the subtraction levels were not sufficient to remove all background artifacts from the gel image.

Background Slider

To manually adjust the background subtraction level for your image, drag the Background slider to the right. You can also move the slider incrementally by clicking on the slider bar, or you can type a specific background value into the field next to the bar.

Objects with signal intensities lower than the subtraction level will be eliminated from the image when you click on OK.

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Background Box

This method of subtraction is best suited for images with uniform backgrounds.

Click on the Background Box button, then drag on the image, defining an area that is representative of the background for your entire gel image. The average intensity of the pixels within the box will be used as the background level to be subtracted from your image.

Background Stripe

This function is useful when the image background is horizontally uniform but changes from the top to the bottom of the image, such as on a gradient gel.

To subtract a background gradient, click on Background Stripe, then drag on a background region to create a lane-like box down the length of the image. The average intensity of the pixels in every horizontal pixel row within the stripe will be subtracted from that entire pixel row. This way, if your image has more background noise at the bottom than at the top, more background will be removed from the lower regions of the image.

Note: Make sure that your background stripe box stretches down the entire length of the part of the image you are interested in. If your background stripe is shorter than your image length, the software will take the last background value at top end of the stripe and subtract it from the upper part of the image, and take the last background value at the bottom of the stripe and subtract it from the lower part of the image.

Completing the Subtraction

When you are happy with the background subtraction shown in the preview image, click on the OK button at the bottom of the dialog box.

Because whole-image background subtraction is an irreversible process, a pop-up box will give you the option of subtracting from the original image, creating a copy of the image to subtract from, or cancelling out of this operation.

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If you choose to copy and subtract, you will be asked to enter a new name and /or version number for the new copy before the operation is performed. Once this information has been entered, the background subtraction will be performed.

4.8. Filtering Images

Filtering is a process that removes small noise features on an image while leaving larger features (like data) relatively unaffected. Diversity Database has a wide range of filters for removing different types of noise from images. Depending on the nature of your data, you will probably need to use only one or two of the available filters. However, you should experiment with several different filters before selecting the ones that work best for your images.

Note: Since filtering is an irreversible process, you will be asked if you want to create a copy of the original image before you filter. If you are experimenting with various filters, you should create copies of your image and compare them side-by-side. If you filter the original image and save it, *you cannot return to the original, unfiltered state.*

4.8.a. Using the Filter Wizard

The Filter Wizard is designed to guide you through the filter selection process. First, you will identify the type of noise in your image. Next, you will select the size of the filter to use on that noise. Finally, you will filter the image.

To open the Wizard, select Image > Filters > Filter Wizard or click on the Filter button on the Image Tools toolbar.

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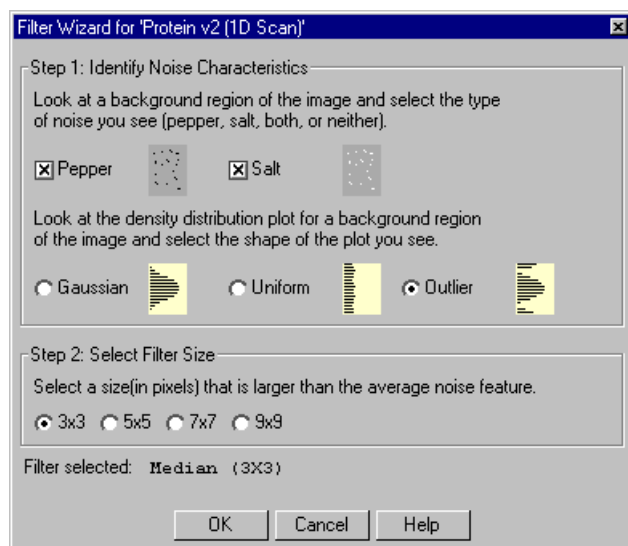


Fig.4-16. Filter Wizard dialog box.

When the Wizard dialog box opens, a vertical plot of the density distribution will pop up next to the image, showing the data range in the image and the amount of signal at each point in the range. (This is the View > Plot Density > Plot Vertical Trace histogram.)

This histogram can be used in conjunction with the Wizard to identify the type of noise in your image.

Step I: Identify Noise Characteristics

The first step in the Wizard is to identify the type of noise in your image. The Wizard allows you to select from a variety of different noise types and combinations.

To determine the type of noise in your image, it is a good idea to first magnify a background region of the image (i.e., a region where there is no data) using the Zoom Box command on the main toolbar. Note that as you magnify a region, the density distribution histogram changes to reflect only the signal within the region you have magnified.

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After examining the magnified region, select one, both, or neither of the check boxes described below.

- **Salt.** This type of noise appears as specks that are lighter than the surrounding background. The density distribution histogram of this type of noise displays noise peaks at the high end of the range. This type of noise is common in electronic cameras with malfunctioning pixels. It can also be caused by dust or lint in the imaging optics or scratches on photographic film. Salt is a type of outlier noise (see below).
- **Pepper.** This type of noise appears as specks that are darker than the surrounding background. The distribution histogram of this type of noise displays noise peaks at the low end of the range. Its causes are similar to those of salt noise. Pepper is a type of outlier noise (see below).

Next, select one of the following option buttons to describe additional features of your noise.

- **Gaussian.** The distribution histogram of this type of noise has a Gaussian profile, usually at the bottom of the data range. This type of noise is usually an electronic artifact created by cameras and sensors, or by a combination of independent unknown noise sources.
- **Uniform noise.** This type of noise appears in the histogram as a uniform layer of noise across the data range of the image.
- **Outlier noise.** This category of noise includes salt and pepper noise (see above). The distribution histogram of this type of noise displays noise peaks at the high and low ends of the range.

After you have identified the type of noise, go to Step 2.

Step 2: Select Filter Size

Image noise is filtered by means of a filtering window (or kernel), which is measured in pixels. This filtering window slides across the image, processing the pixels within it.

The available filter dimensions range from 3 × 3 pixels to 9 × 9 pixels. To select an appropriate size, magnify a background region of your image so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than your data features.

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Note: A smaller filter will alter your image less than a larger filter. Large filters can result in better suppression of noise, but can also blur desirable features in the image.

Step 3: Begin Filtering

After you have completed your selections, the filter name and size will be displayed at the bottom of the Filter Wizard dialog box.

To begin filtering, click on the OK button. Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to Copy and Filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

4.8.b. Selecting a Filter Directly

If you already know the type and size of filter you want, you can select it directly by selecting Image > Filters > Filter List. The pulldown menu includes all the filters available in the Discovery Series.

The types of filters in the Discovery Series are:

- **Weighted Mean.** This filter is useful for reducing Gaussian noise. It calculates the weighted mean of the pixels within the filtering window and uses it to replace the value of the pixel being processed.
- **Out of Range Pixel.** This filter is useful for suppressing salt-and-pepper noise; its effect on Gaussian noise is minimal. This filter calculates the mean of the pixel values in the filtering window, including the pixel being processed. If the difference between the mean and the individual pixel value is above a certain threshold, then the individual value is replaced by the mean.
- **Median.** Also useful for suppressing salt-and-pepper noise, this filter calculates the median value of the pixels within the filtering window and uses it to replace the value of the pixel being processed. The median filter produces very little blurring if a small-sized window is selected.

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- **Maximum.** This filter is useful for eliminating pepper noise in an image (it worsens the effect of salt noise). It replaces the value of the pixel being processed with the maximum value of the pixels within the filtering window.
- **Minimum.** This filter replaces the value of the pixel being processed with the minimum pixel value within the filtering window. This filter is useful for eliminating salt noise in an image (it worsens the effect of pepper).
- **MidValue.** This filter is useful for suppressing uniform noise within an image; however, it worsens the effect of pepper and salt. This filter replaces the value of the pixel being processed with the mean of the maximum and minimum pixel values within the filtering window.
- **Power Mean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise). It replaces the value of the pixel being processed with the power mean of the pixel values within the filtering window.
- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt). It replaces the value of the pixel being processed with the contra-harmonic mean of the pixel values within the filtering window.

To begin filtering, select a filter type from the pull-down list. A pop-up box will ask you to select a filter size.

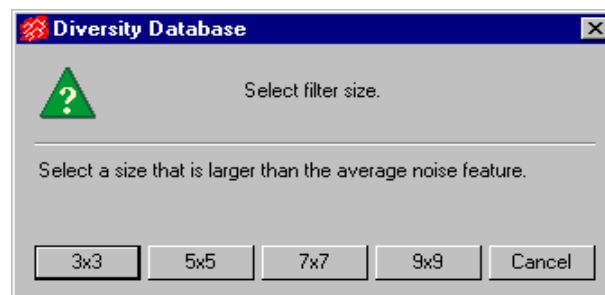


Fig.4-17. Selecting a filter size.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. (See the previous section for guidance on selecting a size.)

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Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to copy and filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

4.9. Invert Data

The Invert checkbox in the Transform dialog box (section 4.6.b) inverts only the appearance of the image. In some cases, however, you may want to invert the actual image data.

If your image has light bands or spots on a dark background (i.e., the signal intensity of the background is greater than the signal intensity of the sample), you will need to use the Invert Data function before you can analyze the image.

This function is reversible, so if you change your mind, you can always switch it back.

To invert your image data, select Image > Invert Data. You may need to use the Transform function to adjust the appearance of your inverted image.

4.10. Annotating Images

If you want to create and display textual notes directly on your image, select Annotation Tools from the Edit menu or main toolbar.

This will open the Annotation Tools toolbar.

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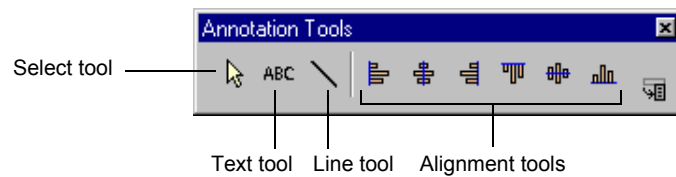


Fig.4-18. Annotation Tools toolbar.

To create an annotation, click on the Text Tool, then click on the image at the spot where you want the annotation to appear. This opens the Text Annotation Properties dialog box.

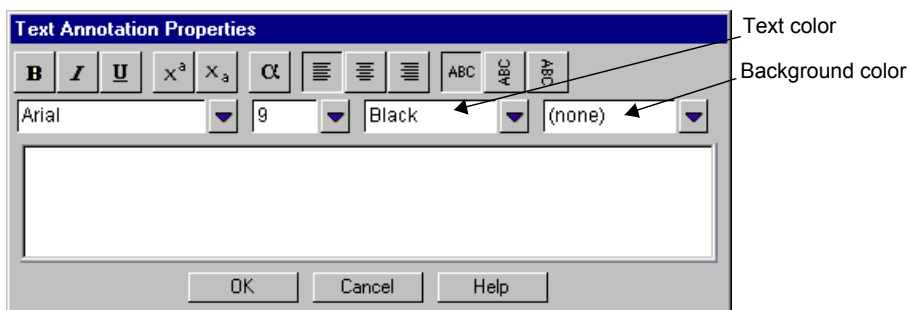


Fig.4-19. Text Annotation Properties dialog box.

To enter your annotation, simply begin typing in the main text field. The buttons in the dialog allow you to select the properties of your text, including format, alignment, and justification.

The pull-down boxes (from left to right) allow you to select the font style, font size, color of the text, and color of the background within the text box.

Once you have typed your annotation text, click on OK.

When you exit the Text Annotation Properties dialog by clicking OK, the annotation you just created will appear on the image at the spot where you originally clicked.

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Editing, Positioning, and Copying Individual Annotations

You must have either the Select Tool or the Text Tool assigned to your mouse before performing the following commands. To assign one of these functions, click on the appropriate button on the Annotation Tools toolbar.

- To edit an annotation, double-click on it to open the Text Annotation Properties dialog. The existing text will be displayed and can be edited.
- To move an annotation, position your cursor on it and drag.
- To copy, hold down the CONTROL key while you drag the annotation.
- To delete, click on the annotation once to select it, then press the DELETE key.

Line Tool

You can use the Line Tool on the Annotation Tools toolbar to draw a line between an annotation and an image feature, or between any two points of interest on your image.

Click on the Line Tool button, then drag on your image to create the line.

- To resize or adjust a line, position your cursor on one end of the line (marked by a circle) and drag.
- To move a line, position your cursor on the middle of the line and drag.
- To copy, hold down the CONTROL key while you drag the line.
- To add arrowheads to a line, double-click on the middle of the line. A dialog box will pop up with options to add arrowheads to one or both ends of the line.
- To delete a line, click on it once to select it, then press the DELETE key.

Working with Multiple Annotations/Lines

To edit multiple annotations and / or lines, first click on the Select Tool button on your Annotation Tools toolbar. Then either drag a box around the objects you want to select or hold down the SHIFT key while you click on individual objects.

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Note: When dragging a box around objects to be selected, make sure that the box entirely surrounds all the objects.

Each object in the selected group will have a green border.

You can drag the selected objects as a group, or click on the Alignment buttons on the Annotation Tools to align them.

Holding down the CONTROL key while you drag will duplicate all of the objects you have selected. Pressing the DELETE key will delete them.

You can create as many annotations and / or lines as you want.

Viewing Previously Created Annotations

To display previously created annotations after opening an image, select Annotation Tools from the Image menu.

If you have concealed all your overlays using Hide Overlays, clicking on any of the buttons on the Annotation Tools toolbar will display the hidden annotations.

4.11. Erasing All Analysis from an Image

If you want to delete all analysis that has been performed on an image (including lanes, bands, volumes, annotations, etc.), you can use the Clear Analysis command.

Select Clear Analysis from the Edit menu. Since this process is irreversible, Diversity Database will ask you for confirmation before completing the command.

4.12. Sort and Recalculate

If the numbering of your lanes or bands (see Chapters 6 and 7) is incorrect because of an addition or deletion, you can renumber them using the Sort and Recalculate function.

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Select Sort and Recalculate from the Edit menu. The lanes and bands in your image will be displayed, correctly numbered and labeled.

5. Creating a New Database

Database files store all the information saved in your gel images and allow you to perform comparative analysis on lanes, bands, and multilane samples within a single gel, a user-defined set of gels, or all the gels in your database. Database files can also be used as flexible gel cataloging systems, allowing you to store up to 50,000 lanes in a single database and perform similarity searches on those lanes in under 30 seconds.

This chapter describes the steps for creating a new database and entering gels into the database. Chapter 10 describes the other database features in greater detail.

5.1. Creating a New Database

Select New Database from the File menu. This creates a new database file and displays a new Database Summary form.

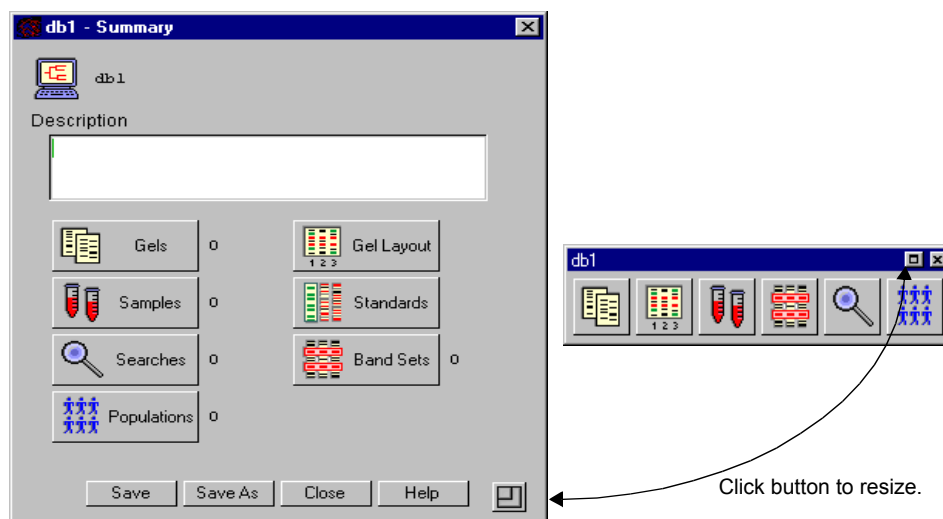


Fig.5-1. Database Summary form and smaller palette version.

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The new database file will be unsaved and will not contain any gels.

5.2. Database Summary Form

The Database Summary form is the on-screen representation of the database file. It provides all the tools you need to build, edit, and use the database.

From the Summary form, you can open seven main dialog boxes: the Gels list, the Gel Layout form, the Samples list, the Standards form, the Band Sets form, the Searches form, and the Populations form. The Gels list is described below. The other dialog boxes are discussed in subsequent chapters.

With the Database Summary form displayed, you can save or rename your database and write a brief description of it. The Summary form will be displayed for as long as the database file is loaded.

Closing the form will close the database file and each member gel. Experienced users may prefer to use the palette version of the Summary form, which is accessible by clicking on the resize button in the lower right corner of the form.

The numbers next to the buttons on the Summary form indicate the number of members of each type included in the database. For example, if there are three gels included in the database, the number “3” will appear next to the Gels button. In a new database, the number “0” appears next to each button.

5.3. Adding a Gel Image to the Database

To add a gel to the database, click on the Gels button in the Summary form. This opens the Gels dialog box.

Creating a Database

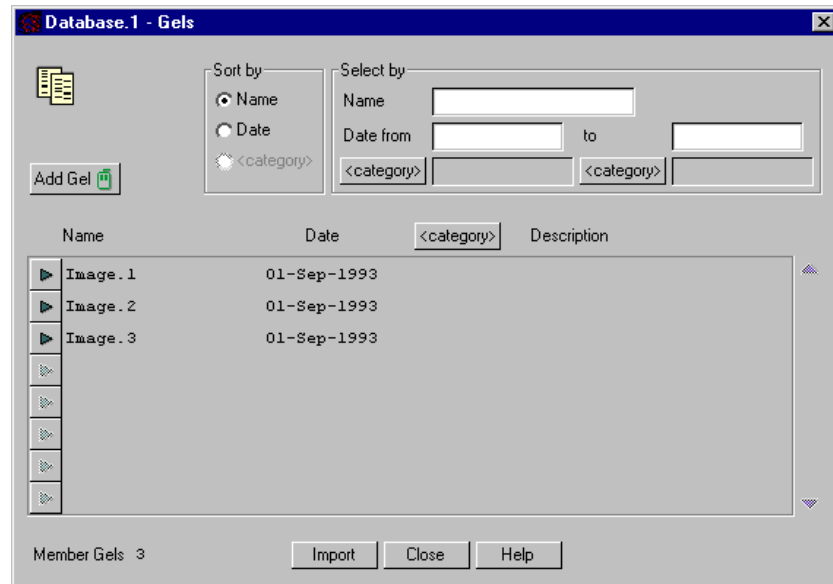


Fig.5-2. Gels dialog box, with three gels listed.

All gel images included in the database are listed here. The name of each gel is accompanied by its scan date and any user-defined category or descriptive information.

To add a new gel image to the database, click on the Add Gel button, open the image that you want to add (using the File > Open command), and click anywhere in the gel window. The name of the gel you selected will appear in the list of gels.

When you are finished adding gels to the database, click on the Close button in the upper right corner of the Gels dialog box.

5.4. Editing a Gel in the Database

To edit a gel that has been included in the database, once again click on the Gels button in the Database Summary form. In the Gels dialog box, click on the triangular button to the left of the gel name. This will open a pop-up box.

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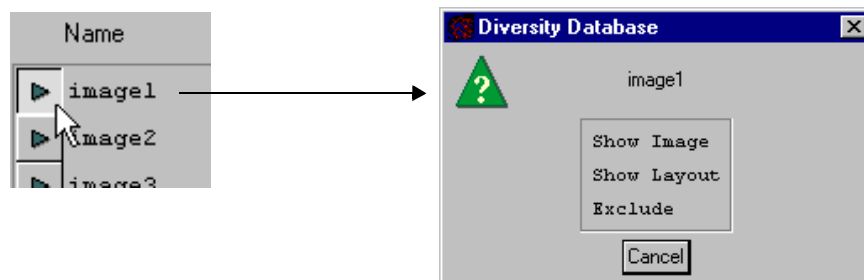


Fig.5-3. Gels pop-up box.

In the pop-up box, you can open the image by selecting Show Image. You can then perform all normal image formatting functions on the image.

You can open the Gel Layout form for that image by clicking on Show Layout. See section 10.4. for more information about the Gel Layout form.

To remove the gel image from the database, click on Exclude. Excluding a gel will delete all information about the gel from the database; however, it will not affect any image formatting.

When you are finished opening or removing images from the database, click on the Close button in the upper right corner of the Gels dialog box.

5.5. Saving/Closing a Database

To save the database, click on the Save or Save As button at the bottom of the Database Summary form.

Note: Save, Save As, and Save All on the File menu are not used to save databases—only images.

In Windows, database files are given the extension .ddb.

If you elect to make backup copies of your databases (specified in Edit > Preferences), you will make a new backup every time you save the database. These backups are stored in a Backup folder in the same directory as your database. See section 2.5.c., Application Preferences, for more information.

Creating a Database

Saving an Existing Database under a Different Name/Location

You can use Save As on the Database Summary Form to save an existing database under a new name or location.

Click on Save As, and change the name/location in the dialog box.

Note that when you finish saving, the old database will be changed to a backup database (.dbk extension in Windows). See section 2.5.c., Application Preferences, for more information about backups.

Closing a Database

To close a database, select Close from the File menu or click on the Close box in the upper right corner of the Summary form. You will be prompted to save any unsaved changes you may have made to the database or constituent images.

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6. Lanes

Once you have added a gel image to a database, you are ready to identify the lanes and bands in that image for cross-referencing using the database.

The next three chapters describe the steps for preparing a gel image to be linked within a database. First, you must define the lanes in the image.

You can define lanes individually or as part of a frame. The functions for doing this are under the Lane menu and on the Lane toolbar.

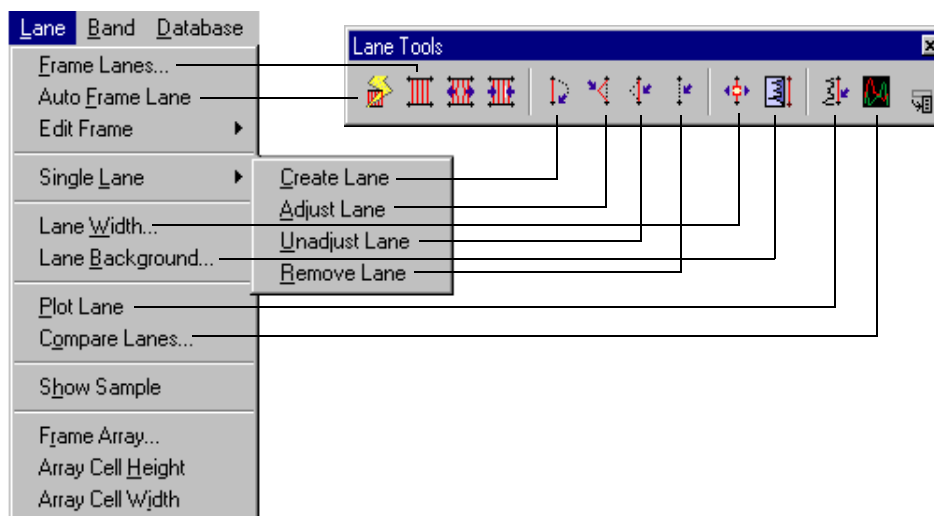


Fig.6-1. Lane menu and toolbar.

6.1. Creating a Lane Frame

The fastest way to define all of the lanes in your image is to create a lane frame using the Auto Frame Lanes command. If Auto Frame Lanes does not work well on your images, you can create and place a lane frame manually.

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Auto Frame Lanes

Note: Auto Frame Lanes works best with images with large numbers of clearly defined lanes and bands. Also, the lanes should be reasonably vertical.

Select Lane > Auto Frame Lanes or click on the button on the Lane Tools toolbar. The lane-finding program will automatically detect the lanes in your image and place a grid over them.

If Auto Frame Lanes detects too few or too many lanes, you can add or delete lanes using the single lane commands described below.

If Auto Frame Lanes does not work on the image, you will be prompted to create a lane frame manually.

If you are not satisfied with the frame created by Auto Frame Lanes, select Edit > Clear Analysis and use the manual Frame Lanes command.

Creating a Lane Frame Manually

Lane > Frame Lanes opens a dialog box in which you can type in the number of lanes in your gel.

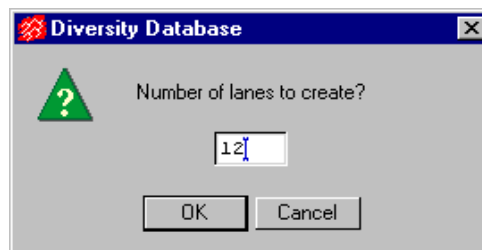


Fig.6-2. Frame Lanes dialog box.

Click on the OK button to complete the action.

Lanes

Features of a Lane Frame

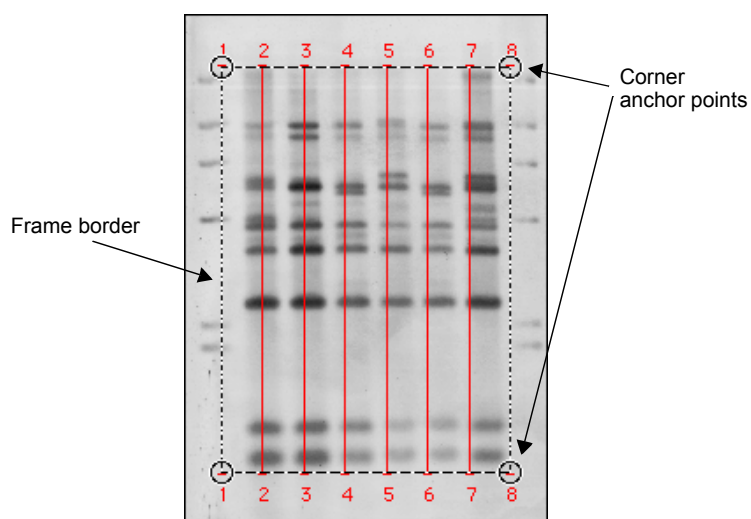


Fig.6-3. Features of a lane frame.

The lane frame overlay contains individual lanes numbered sequentially from left to right. The borders of the frame are marked with dashed lines, the interior lanes are solid red lines, and each anchor point (corner) is marked with a circle.

6.1.a. Editing the Frame

If your frame does not follow the lanes on your image to your satisfaction, you can adjust it using the frame editing commands. These are located on the Lane > Edit Frame submenu.

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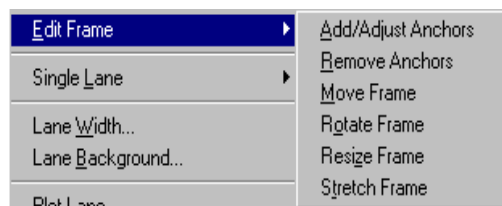


Fig.6-4. Edit Frame submenu.

Adjusting the Entire Frame

The following commands will adjust the entire frame:

- To stretch the frame (e.g., if you want to include additional bands at the top or bottom of the image), select Stretch Frame from the submenu and drag an anchor point in or out. The opposite anchor point will remain fixed while the frame expands or contracts.
- To move the entire frame to a new position, select Move Frame from the Lane > Edit Frame submenu and drag an anchor point. The entire frame will move.
- To rotate the frame, select Rotate Frame from the submenu and drag an anchor point. The entire frame will rotate.
- To resize the frame, select Resize Frame from the submenu and drag an anchor point in or out. The frame will expand or contract from the center.

Adjusting/Adding Individual Anchors

After you have created a frame, the Adjust Anchors function will automatically be assigned to your mouse.

To adjust an individual corner anchor point of a frame, select Adjust Anchors (if it is not already assigned to your mouse) and drag the anchor point. This will move both the anchor point and attached frame lines.

If the lanes are not straight or if the dye front smiles, you can create additional anchor points within the frame to change the shapes of individual lines.

Lanes

Still using Adjust Anchors, click anywhere on the dashed lines of the frame border. This creates additional anchor points, both where you clicked and on the other side of the frame. This also creates another dashed line across the frame

You can then drag the new anchor points just as you did the corner anchor points, thereby “bending” the frame. Other internal anchor points can be placed along the new dashed line if further adjustments to the lane frame are required.

You can create as many additional anchors as necessary for the lines of the frame to correspond to the lanes on the gel image. If you change your mind about the position of an anchor, you can reposition it by simply dragging it.

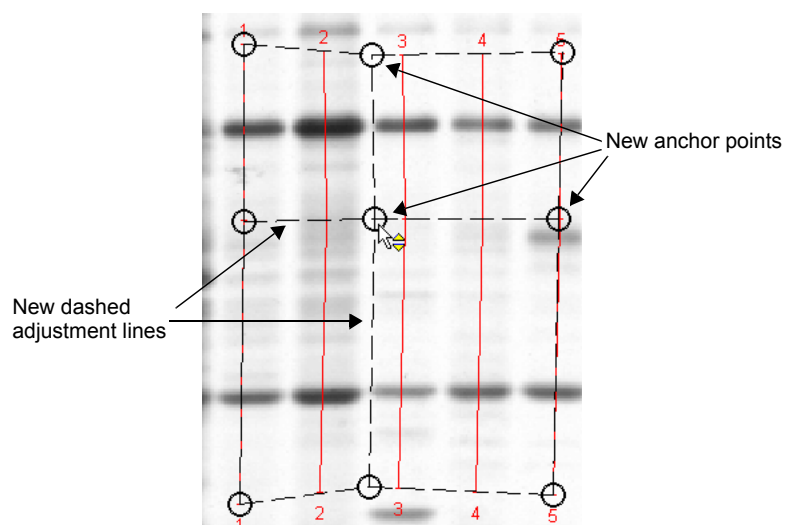


Fig.6-5. Adjusting the anchor points of the lane frame.

Removing/Unadjusting Anchor Points

If you decide to remove an anchor point, select Unadjust Anchors from the Edit Frame submenu or toolbar and click on the anchor you want to remove. The anchor will disappear and the lanes will “unadjust,” reflecting the removal of the anchor.

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6.2. Defining a Single Lane

You can define individual lanes using the Single Lane functions. These are located on the Single Lane submenu of the Lane menu or on the Lane Tools toolbar.

Note: You can use the single lane commands on lanes within a frame; however, the lane will be detached from the frame before the command is executed.

To mark an individual lane, select Create Lane, then drag a line from the top to the bottom of the desired lane.

Repeat this procedure to manually mark all the lanes you want to identify on your gel image.

Note: If while defining lanes your lane numbering gets out of sequence, click on Edit > Sort and Recalculate to renumber the lanes.

6.2.a. Adjusting Single Lanes

You can adjust the position of any individual lane line you create by using the Adjust Lane command.

Select Adjust Lane, then either drag one of the existing anchor points or click anywhere on the lane to create a new anchor point and drag it to the desired position. Repeat this procedure as many times as necessary until the lane line accurately follows the lane on the gel image.

6.2.b. Unadjusting Single Lanes

You can undo any of your lane adjustments with the Unadjust Lane command. Select Unadjust Lane from the menu or toolbar, then click on an anchor point you created to remove it. If you remove the anchor points at either end of the lane line, you will delete the line from the image.

6.3. Deleting Lanes

You can delete both single lane lines and lanes from a lane frame.

Lanes

Select Remove Lane from the Lane menu or toolbar and click on the lane you want to remove. A pop-up box will ask you to confirm the deletion.

Note: If you delete a lane from a group of lanes or a frame, click on Edit > Sort and Recalculate to renumber the remaining lanes.

6.4. Lane Sampling Width

The Lane Sampling Width command allows you to adjust the width of a single lane (in millimeters). Only the pixels within the sampling width are used for lane profiling and data quantitation purposes.

Select Lane Sampling Width from the Lane menu or toolbar, then click on a lane. This will open the Sampling Width dialog box, displaying the current width of the lane you clicked on.

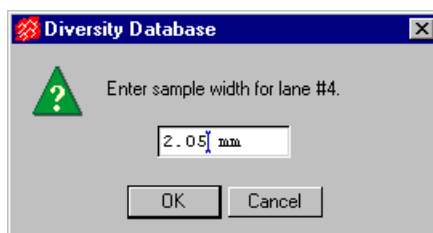


Fig.6-6. Sampling Width dialog box.

Enter a new width in millimeters and click on the OK button.

You can adjust the sampling width of all the lanes in your image from within the Detect Bands dialog box. See section 7.1., How Bands Are Identified and Quantitated for a full discussion of the effect of sampling width on band quantitation.

6.5. Creating a Lane Profile

After you have defined a lane, you can create an intensity profile of that lane using the Plot Lane function. A lane profile provides a quick visualization of

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the intensity of your sample data, and is also useful for determining the amount of background noise in your image.

Select Plot Lane from the Lane menu or toolbar, then click on a lane. A profile window will pop up.

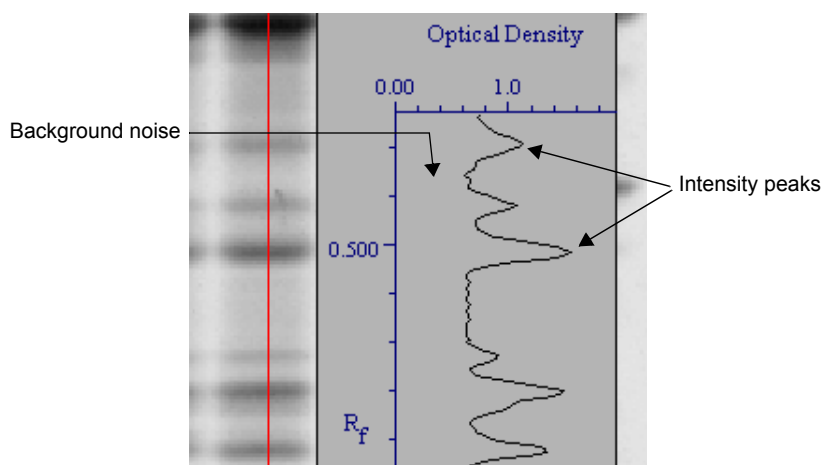


Fig.6-7. Profile of a defined lane.

To profile another lane, simply click on the lane with the Plot Lane command still assigned to the mouse.

The profile is generated by calculating the average intensity of the pixels in every horizontal scan line along the defined lane.

The profile shows the data objects in the image, represented by the intensity peaks in the profile, as well as the background noise in the image, represented by the region underneath the intensity peaks. The number of pixels used in this calculation depends on the lane sampling width (see previous section).

To close the lane profile window, click on Hide Overlays on the main toolbar.

6.6. Lane-Based Background Subtraction

After defining your lanes, we strongly recommend that you perform lane-based background subtraction. This is the best method for removing background noise from your lanes.

Select Lane Background from the Lane menu or toolbar. When you click on a lane, a lane profile will be displayed and the Lane Background dialog box will open.

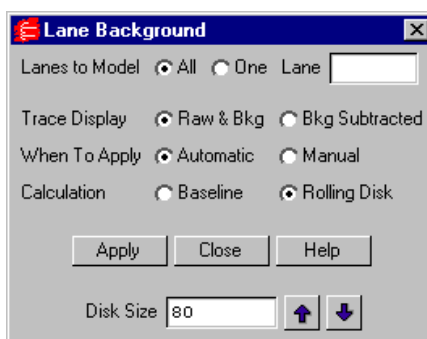


Fig.6-8. Lane Background dialog box.

The following information can be specified in the dialog.

Lanes to Model

If you want to apply the background subtraction to all the lanes of your image, click on the All button. Otherwise, select One and enter the number of the lane you want to perform the operation on.

Trace Display

The profile trace of the lane can be displayed in one of two ways. Selecting the Raw & Bkg option (with the functions Automatic and Rolling Disk activated) will display the original “raw” trace of the lane in white with an orange line representing the background noise beneath the peaks of the trace.

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Clicking on the Bkg Subtracted button will display the lane trace as if all the background noise were actually removed. This display better enables you to visually compare the relative intensities of the bands in the lane.

When to Apply

As you change either the calculation method or the rolling disk size, the new settings can be applied to the image automatically every time a change is made by selecting the Automatic option.

Alternatively, you can apply each new setting manually by selecting the Manual option and clicking on the Apply button whenever the settings are changed.

Calculation Method

If you select Baseline, background noise will not be subtracted from your lanes. Selecting Rolling Disk will allow you to use the Rolling Disk method of background subtraction.

“Rolling disk” refers to a hypothetical disk that follows the contour of the lane’s profile trace, removing different intensities along the length of the lane.

The size of the disk determines how much background will be subtracted. A large disk will follow the profile trace less closely, touching fewer points along the trace and identifying less background. A smaller disk will more closely follow the profile trace, thus identifying more background.

If you select Rolling Disk, the dialog box will expand to reveal a field showing the size of the rolling disk. The Disk Size is the disk radius in pixels.

You can type in a numerical value for the disk size or you can use the arrows to change the value in 10 percent increments.

A disk radius that is too large will result in slow execution and poor removal of background. A disk radius that is too small may subtract nonbackground intensity. Typical values range from 50 to 150.

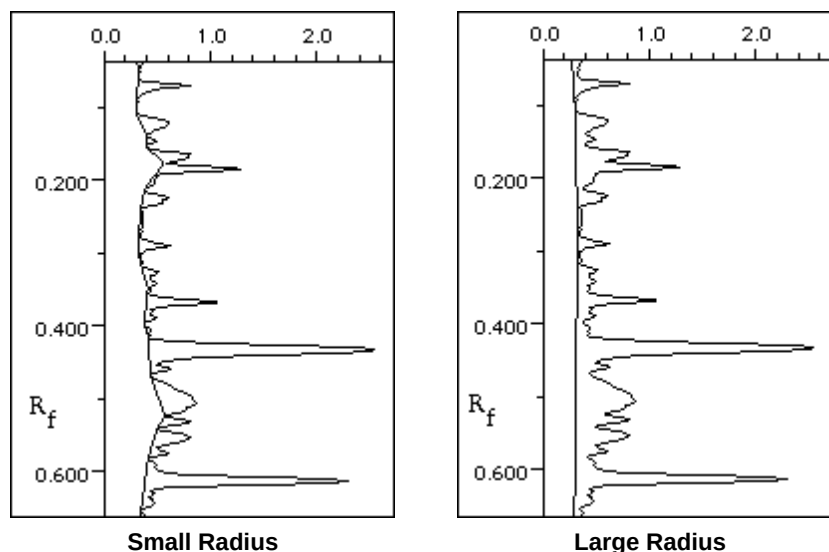


Fig.6-9. Examples of the background trace for small and large rolling disks. The small disk follows the profile trace more closely, resulting in more background subtraction.

Try different disk sizes to find the one that gives you the desired background noise trace.

When you apply the Rolling Disk background subtraction, the lane trace display will change but the image will not reflect the change in background intensity.

6.7. Compare Lanes

The Compare Lanes command opens a multicolored graph that allows you to superimpose the intensity profiles of any number of lanes present on any number of open images. Once a lane is displayed, Compare Lanes can zoom in on particular regions of interest and display them in greater detail.

Select Compare Lanes from either the Lane menu or toolbar, then click on the first lane you want to display.

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Note: Your image must have defined lanes for this command to work.

After the Compare Lanes window has opened, you can add lanes by clicking on the lanes you want to include.

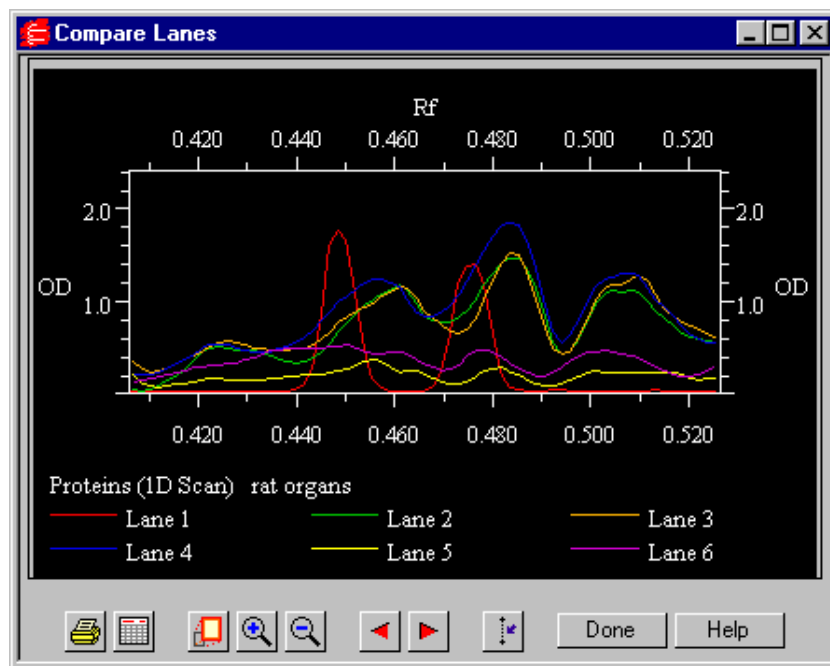


Fig.6-10. The Compare Lanes dialog box.

Each lane you select will be profiled in one of eight colors, and will be identified by color in a legend underneath the graph. If you select more than eight lanes to display, the colors will repeat themselves, but each lane will still be identified underneath the trace display. There is no absolute limit to the number of lanes that can be displayed simultaneously.

The X axis of the graph is Rf values and the Y axis is pixel intensity values. Compare Lanes automatically “best fits” lanes within the display window to maximize the range of intensity values included in the graph. Rf values are displayed from 0.0 to 1.0.

Lanes

Magnifying Within Compare Lanes

The Zoom In and Zoom Out buttons in the dialog can be used to zoom in and out within the display to magnify regions of interest. Dragging the cursor horizontally across the graph changes the cursor appearance to a bidirectional arrow and magnifies the range you defined by dragging.

Note: The magnifying functions in Compare Lanes only magnify the profile in the direction of the X axis. Therefore, the profile will appear to “stretch” without increasing in height.

The Full View button returns the graph to its complete, unadjusted appearance.

If you have zoomed in on part of the graph, the Left and Right scroll buttons in the dialog can be used to pan left or right.

Function Buttons

The Remove Lane button displays a dialog box that allows you to remove any lane profile from the graph. If you remove a lane, the color designations of the remaining lanes will change. Check the lane legend for an updated color code.

Click on the Print button to print a copy of the Compare Lanes display.

Click on the Export button to export the data in your graph to Excel or 1-2-3 spreadsheets.

6.8. Show Sample

You can display any database information you may have entered about your lanes using the Show Sample command.

Select Lane > Show Sample, then click on a lane. The Sample Detail dialog box for that lane will open, displaying all the database information that has been entered about the sample in that lane.

See section 10.3. for more information about samples and the Sample Detail dialog box.

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7. Bands

Once you have defined the lanes in your gel image, you can automatically identify and quantitate the bands in those lanes using a set of adjustable parameters.

The tools for band detection are located on the Band menu and on the Band Tools toolbar.

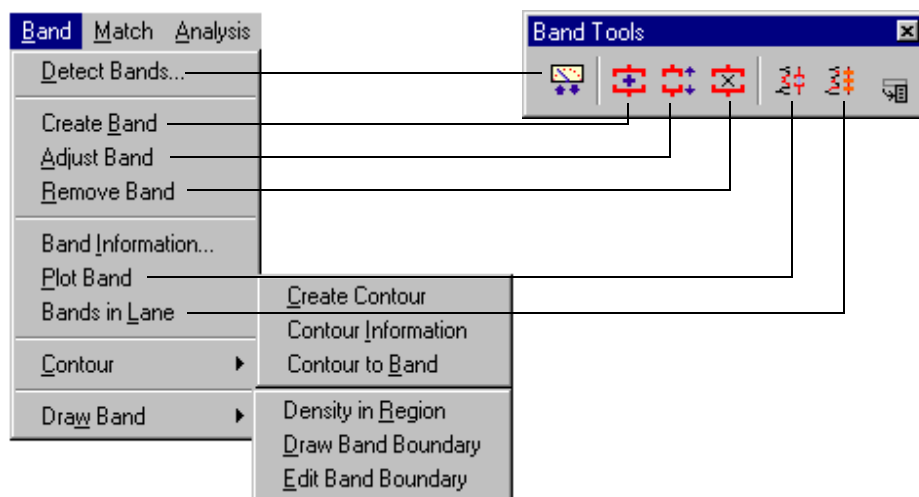


Fig.7-1. Band menu and Band Tools toolbar.

7.1. How Bands Are Identified and Quantitated

You can automatically identify all the bands in your image using the Detect Bands command, or you can mark them individually using the Create Band command.

Each identified band is defined by brackets above and below the band. You can see these if you select the Brackets option button in the Band Attributes

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dialog box. (Select View > Band Attributes.) These brackets mark the boundaries of the band.

The width of each set of brackets is set by you before the bands are detected. This is the lane sampling width. The height of each set of brackets is determined automatically, using a band-finding formula together with parameters that you select. You can also adjust the bracket height using the Adjust Band command.

When a band is quantitated, the average intensity value of each horizontal row of pixels within the brackets is calculated. Next, the number of pixel rows between the top and bottom brackets is determined. Taken together, these result in an intensity profile for the band.

Finally, the area under the profile curve to the baseline is integrated, resulting in units of intensity x millimeters. This is the “trace quantity” of the band.

Bands

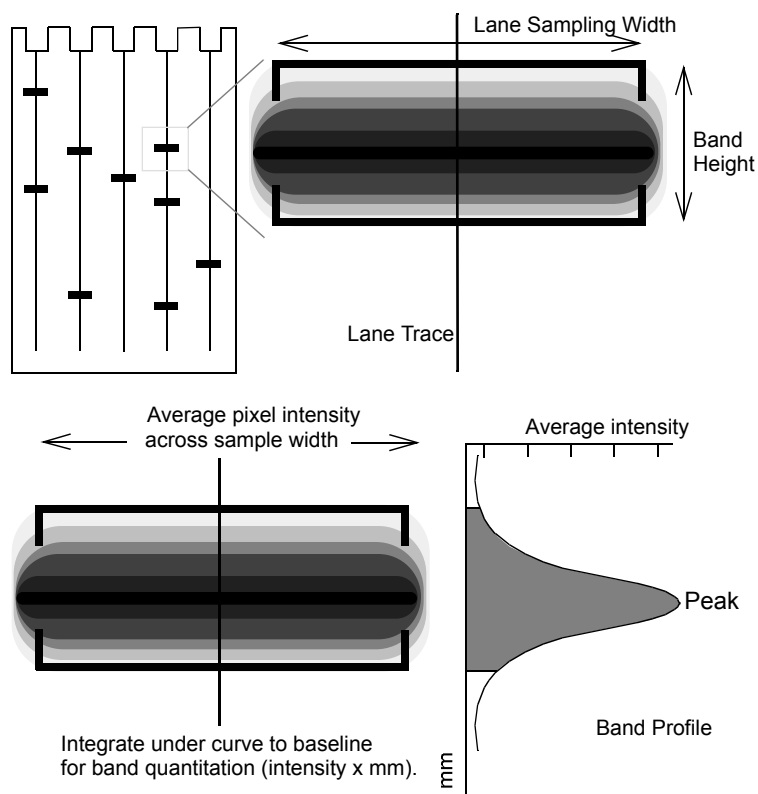


Fig.7-2. Illustration of bracket quantitation.

7.2. Automatically Identifying All Bands

To identify the bands in your image automatically, open the Detect Bands dialog box (Select Band > Detect Bands or click on the button on the Band Tools toolbar).

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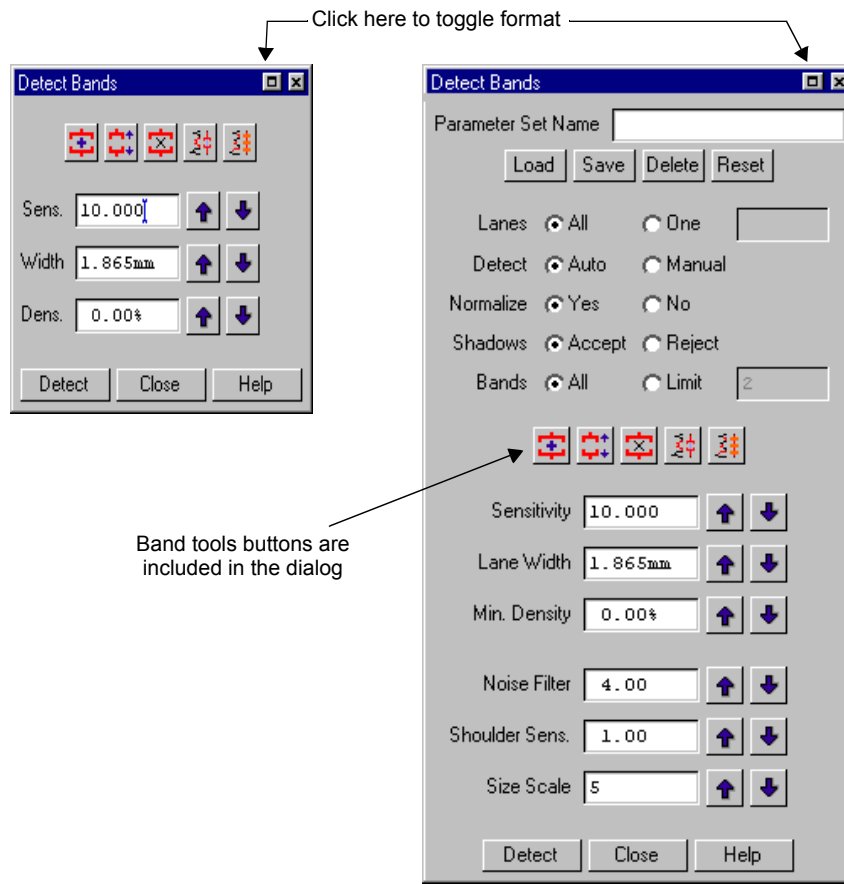


Fig.7-3. Detect Bands dialog box, short and long formats, with default values.

The Detect Bands function will find all of the bands in your defined lanes, based on parameters that you select.

Note: If you have already manually identified bands (using Create Band, Adjust Band, etc.), the Detect Bands function will overwrite your manual detection. For this reason, it is advisable to use Detect Bands first, and then manually add, adjust, or remove bands as needed.

Bands

The Detect Bands dialog box has both a short format and a long format, which you can toggle between by clicking the toggle box in the upper right corner of the dialog.

The long format allows you to set more of the parameters that affect band detection. Also in the long format, the detection parameters you select can be saved for use on future gel images.

To change any of the parameter settings, you can either type in a new value or use the arrows to increase or decrease the setting by 10 percent. Since the properties of your gel images are unique, we suggest that you experiment with various settings to find those best suited to your images.

7.2.a. Setting the Detection Parameters

Lanes to Detect

You can specify whether you want to detect the bands in all the lanes or in a single lane by selecting All or One next to the Lanes prompt. If you choose One, type the lane number in the field next to the One button.

Note: It may be necessary to detect one lane individually if the detection parameters applied to the whole image are not optimal for that lane (e.g., faint bands are not being detected or high background in that lane is being detected as bands). You can use the Detect Bands function to redetect bands in that lane using a different set of detection parameters.

When to Detect

If you select Auto next to the Detect prompt, band detection will occur immediately each time you change a detection parameter. You will not need to click on the Detect button located at the bottom of the form.

If you want to change more than one parameter before detecting, choose Manual. With the Manual option, you can change parameter settings first, and then apply them by clicking the Detect button.

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Normalization

Normalization is a way for band detection to compensate for differences in the darkness of the lanes on a gel image. *It does not normalize for band quantitation.*

If you select the No button next to the Normalize prompt, the band detection parameters are taken as absolute values and are applied in the same way to every lane on the image.

If you select Yes, the parameters are automatically adjusted according to the darkness of the lane. The darkness of the lane is determined by the darkest band in the lane. For example, suppose that all but one of the lanes on an image contain bands with an intensity of up to 50,000 counts. In the one light lane, the darkest band is only 25,000 counts. If normalization is turned on, band detection will be twice as sensitive when processing the light lane, improving the detection of faint bands.

Shadow Rejection

“Shadow bands” are common gel artifacts. Shadow bands are spaced at tandem repeat intervals and decrease in intensity as they progress further from a real band. The Shadows parameter is designed to limit the detection of shadow bands (see also Band Limit, below).

If you select Reject next to the Shadows prompt, the Detect Bands function will only find a band if it is darker than the one above it or if it is spaced further than one tandem repeat unit from the previous band. This greatly reduces the number of shadows identified as real bands.

If you select Accept next to the Shadows prompt, Detect Bands will not filter for shadows.

Band Limit

If you know that all the lanes in your gel image contain a specific number of bands, you can click on the Limit button next to the Bands prompt and type in the number of bands that you know are present. Only that number of bands will be detected in each lane, reducing the need for later editing.

Bands

Sensitivity

The Sensitivity setting determines the minimum signal intensity in the image that will be defined as a band. The higher the sensitivity value, the more bands will be detected.

If the sensitivity is set too high, background noise will be erroneously detected as bands. If the setting is too low, real bands may be missed.

The default sensitivity setting is 10.00. If your gel image has faint bands (e.g., O.D. < 0.05, counts < 2,000), you may want to increase this value to 20.00.

Lane Width

The Lane Width determines the width along the lane lines that will be sampled for quantitative data. (You can also set the lane width for individual lanes using the Lane > Lane Width command.)

When a band is quantitated, an average intensity value for each horizontal row of pixels within the band brackets is calculated. You determine the number pixels in a row when you enter the Lane Width. The wider you set the Lane Width, the more pixels are included in the intensity average.

A Lane Width of two-thirds the actual lane widths in your gel is optimal for allowing high sensitivity with few false bands.

You should be careful not include background in the Lane Width, since this will decrease the average pixel values used in band quantitation. If you need to choose between including background versus leaving out part of the band, it is better to opt for the latter. Since you are taking the average intensity of all the pixels in a row, it's not critical if a few pixels are omitted from the band.

Minimum Density

When Normalization is turned off, this is the lowest signal intensity value that will be counted as a band.

Before entering a value for Min. Density, use the Lane > Plot Lane command to plot a trace of a lane that includes some faint bands. Then enter a value that is lower than the intensity of the peak of a faint band but is still above the background.

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If faint bands are still being missed after applying this parameter, double-check the intensity of faint bands on the lane trace and / or consider increasing the Sensitivity setting.

If Normalize is turned on, Min. Density is converted from an absolute value to a percentage. This percentage is the fraction of the signal intensity of the darkest band in the lane that will be detected as a band. For example, if the darkest band in a lane is 50,000 counts and the Min. Density is set to 25,000 counts, when you turn on Normalize, the Min. Density will switch to 50%. In other words, for a band to be detected it must be at least half as dark as the darkest band in the lane.

Noise Filter

The Noise Filter can be used to minimize the number of small fluctuations in the image (i.e., noise) that are called bands while still recognizing larger features (i.e., real bands). This distinction becomes increasingly critical the higher the Sensitivity parameter is set.

The Noise Filter value refers to the size of the filter in pixels (e.g., a value of 2.50 equals a filter size of 2.50 x 2.50 pixels). Features smaller than the filter size will not be recognized as bands. Entering a noise filter size of zero turns it off completely. The default value is 4.00.

If band detection calls a doublet a single band, decrease the Noise Filter setting and / or increase the Sensitivity.

You can also try decreasing the Size Scale parameter instead of the Noise Filter to improve the detection of closely-spaced bands. Be wary of decreasing both the Noise Filter and the Size Scale, as this may result in the fuzziness around bands being mistakenly detected as separate bands.

Shoulder Sensitivity

Normally, band detection tries to distinguish shoulders as separate bands. When looking at a lane trace, these bands appear as flat or gently sloping abutments to darker, better-defined bands (i.e., there is no dip on the trace between the two bands).

Bands

Increasing the Shoulder Sensitivity value will result in more shoulders being detected as bands. Changing this setting to zero will result in no shoulders being recognized as separate bands.

If band detection calls a doublet a single band, check the lane trace to see if there is a dip between the peaks of the two bands. If there is no dip, increasing the Shoulder Sensitivity value will help resolve the two bands.

Size Scale

The Size Scale field helps distinguish between trends in signal intensity and random intensity fluctuations. It is the number of pixels in a vertical column that are taken together to determine whether a band is present.

The Size Scale parameter is similar to the Noise Filter in that it uses the size of objects in the image to determine the nature of those objects. The default Size Scale setting of 5 pixels is optimal for most gel images. It can be set to any whole number greater than or equal to 3.

If a gel image has high levels of background noise, a larger Size Scale may be preferable. When noise permits, small values are preferred so that small features will be detected.

You may also choose to increase the Size Scale if your gel only has a small number of thick bands and it is scanned at high resolution.

7.2.b. Band Detection Parameter Sets

Saving Parameters

Once you have found a set of parameters that are optimal for detecting bands in your image, you can save them for future use on similar images. The Save button at the top of the Detect Bands dialog box allows you to do so.

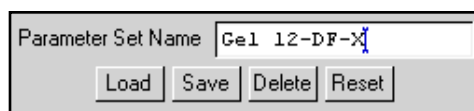


Fig.7-4. Parameter set controls in the Detect Bands dialog box.

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If you have not already entered a name for your parameter set in the Parameter Set Name field, you will be prompted to do so.

Loading Parameters

To load a previously saved set of parameters, click on the Load button.

Select the parameter set that you wish to load from the list. The values of that parameter set will be displayed in the Detect Bands form.

Deleting Parameters

To remove a set of parameters from your list, first load it, then press the Delete button. A dialog box will ask you to confirm the deletion.

Resetting Parameters

To return all detection parameters to their default values, click on the Reset button at the top of the form. Saved detection parameter files are not affected by this action, and can be reloaded at any time.

7.3. Band Attributes

Once you have identified bands in your image, you can display information about them using the Band Attributes command.

Select View > Band Attributes to open the dialog box.

Bands

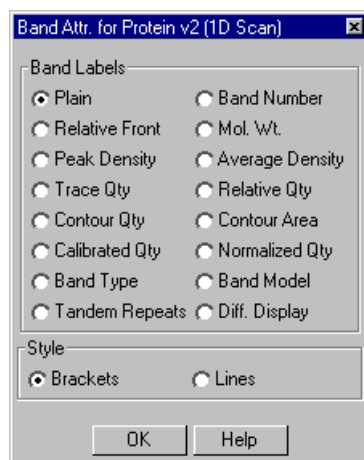


Fig.7-5. Band Attributes dialog box.

7.3.a. Band Labels

Click on any of the buttons in the Band Labels group to display that particular attribute next to each defined band on your image.

Each band label is summarized below.

- **Plain**— No attribute is displayed.
- **Relative front**— The distance of a band from the top of a defined lane to the bottom, divided by the total length of the lane. This distance can be determined either by measuring a vertical line from the top of the lane to bottom or (if the lane is curved) by measuring along the length of the lane. Select Edit > Preferences to set the preferred measuring method.
- **Peak density** —The intensity of a band as determined by its peak intensity value.
- **Trace qty**—The quantity of a band as measured by the area under its intensity profile curve.
- **Contour qty** —The quantity of a band that has been identified using the Contour or Draw Band tools. It is the sum of the intensities of all the pixels within the band boundary multiplied by the area of each pixel.

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Units are intensity x mm².

- **Calibrated qty** —The trace quantity of a band that has been calibrated using quantity standards.
- **Band type** —The band type number of a particular band that has been matched and placed in a band set.
- **Tandem repeats** —The number of repeated base-pair units in a band that has been analyzed using the VNTR Calculations function.
- **Band number** —The sequential number of a band in lane, as counted from the top of the lane.
- **Molecular weight** —The molecular weight of a band in a lane, based on band position and the defined molecular weight standards in the gel.
- **Average density** —The total intensity of the rows of pixels used to generate the profile of a band, divided by the number of rows.
- **Relative qty** —The trace quantity of a particular band expressed as a percentage of the total quantity of all the bands in the lane. Alternatively, the trace quantity of a particular band expressed as a percentage of the total intensity of the entire lane. The calculation method (% of Lane / % of Bands) may be selected in Edit > Preferences.
- **Contour area** —The area (in mm²) inside the boundary of a band that has been identified using the Contour or Draw Band tools.
- **Normalized qty** —The trace quantity of a particular band expressed as a percentage of the quantity of a selected band type that is present in the same lane.
- **Band model** —Displays the modeling lines used by the matching algorithm to match bands.
- **Differential Display** —If your band types have been normalized, this displays trends in increasing or decreasing expression of a band type across a gel based on its normalized quantity.

7.3.b. Displaying Bands as Brackets or Lines

The options in the Style group of the Band Attributes dialog allow you to choose how your defined bands are marked—as brackets surrounding each band, or as a single line at the center of each band. Lines are usually easier to

Bands

read than brackets in images with closely packed bands, while brackets are better for displaying the boundaries of bands.

Note: When editing individual bands (see the following section), it is useful to display your bands as brackets. When you edit bands in brackets mode, a trace diagram will be displayed to help you determine band boundaries.

The boundaries of a band can be defined as either the troughs between peaks on the trace or as the inflection points on the curves defining the peaks. Most people looking at a gel will define a band by its inflection points, which correspond to the visual limits of the band. When using the trace profile of a gel image, they tend to include much more of the band's tail.

7.4. Identifying and Editing Individual Bands

To manually identify and edit individual bands in your image, you can use the Create, Adjust, and Remove Band commands on the Band menu and Band Tools toolbar. (These are also available as buttons in the middle of the Detect Bands dialog box.)



Fig.7-6. Create, Adjust, and Remove Bands buttons.

7.4.a. Identifying Individual Bands

If your bands are very closely spaced, marking them in brackets mode gives you greater control for more precise band definition. It also allows you to define overlapping bands.

In Brackets Mode

With your bands displayed as brackets, first select Create Band from the menu or toolbar, then click on either the top or bottom boundary of the band of interest. An intensity trace of the lane will pop up next to the band.

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Drag the cursor until the area of the band that you want to define—represented by a peak on the intensity trace—has been completely enclosed. As you drag, lines extending from the band boundaries on the image to the corresponding boundaries on the intensity trace will appear.

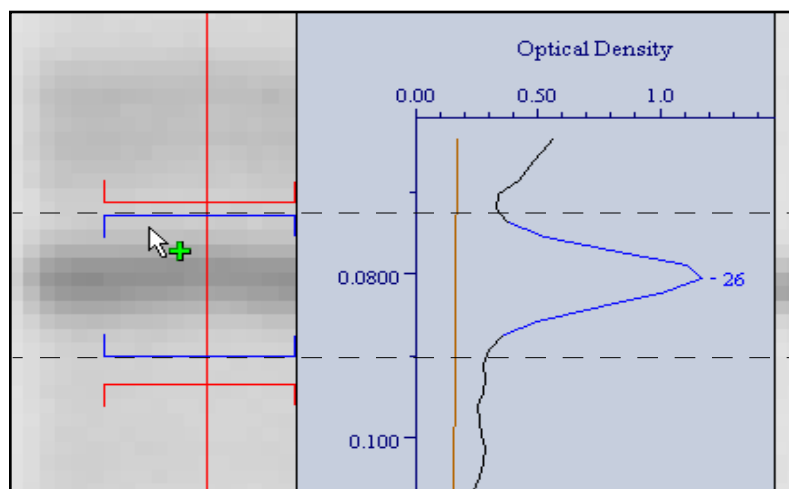


Fig.7-7. Creating a band in brackets mode.

When you release the mouse button, brackets will appear enclosing the region that you have called a band on the image. The corresponding section of the intensity trace will be highlighted.

In Lines Mode

With your bands displayed as lines, first select Create Band from the menu or toolbar, then click on the center of the band of interest. The band-finding mechanism will determine the peak and shape of the band, and a line will appear at the center of the band at that point on the lane.

Bands

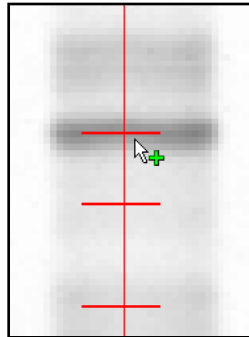


Fig.7-8. Creating a band in lines mode.

In lines mode, no intensity trace will pop up when you click on the image.

Note: After identifying several bands, be sure to renumber the bands in your image by selecting Sort and Recalculate from the Edit menu.

7.4.b. Adjusting Bands

After you identify the bands on your image—either automatically or manually—you may want to go back and change the placement of the band boundaries.

To reposition a band's boundaries, select Adjust Band from the menu or toolbar.

If you are in brackets mode, click on the upper or lower bracket of the band that you want to adjust. If you are in lines mode, click near the band you want to adjust.

If it is not already shown, an intensity trace will be displayed. The band being adjusted will be highlighted, and lines extending from the band boundaries on the image to the corresponding boundaries on the intensity trace will appear.

Drag the band boundary to the desired location, then release the mouse button.

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7.4.c. Deleting Bands

You can use Remove Band to delete bands from lanes.

Select Remove Band from the menu or toolbar, point the cursor at the unwanted band, and click the mouse button.

If your bands are displayed as brackets, a trace of the band will be displayed and a pop-up box will ask you to confirm removal of the band. Click on Yes to delete the band or No to cancel the procedure.

If your bands are displayed as lines, the band will simply be deleted.

Note: If your bands are displayed as lines, you can delete more than one band at a time. With Remove Band assigned to your mouse, drag a box around the bands to be removed. The bands in the box will be deleted when you release the mouse button.

If you delete a band, the region will no longer be counted as a band, but its intensity will still contribute to the total lane intensity.

Note: After removing bands, be sure to renumber the bands in your image by selecting Sort and Recalculate from the Edit menu.

7.5. Plotting Traces of Bands



Fig.7-9. Plot Band and Bands in Lane buttons.

Band > Plot Band (also on the Band Tools toolbar) displays an intensity trace of any band that you select on the image. First select the command, then click on the band of interest.

Band > Bands in Lane (also on the Band Tools toolbar) displays an intensity trace of any lane you select and highlights all the defined bands in that lane. First select the command, then click on the lane of interest.

Bands

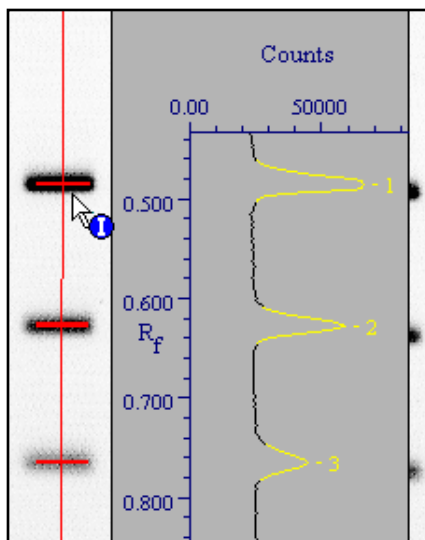


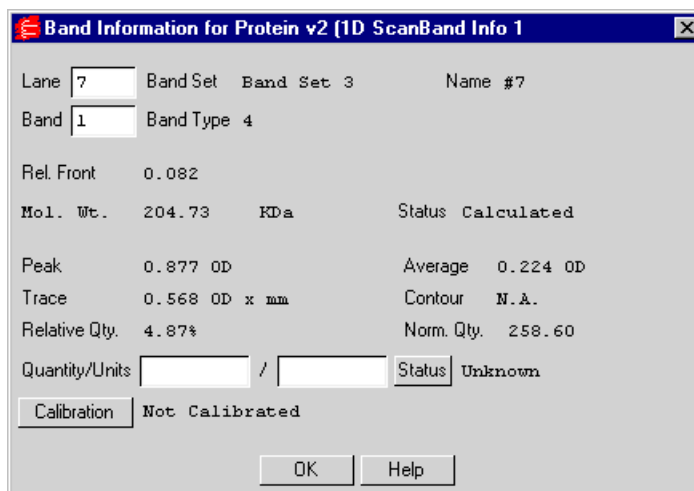
Fig.7-10. Bands in Lane command.

7.6. Displaying Band Information

You can display and enter information about individual bands using the Band Information dialog box. To open this dialog, select Band Information from the Band menu and click on a band of interest.

An intensity profile of the band's lane will be displayed, the selected band will be highlighted, and the dialog box will be displayed.

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The dialog box titled "Band Information for Protein v2 (1D ScanBand Info 1)" contains the following fields and buttons:

Lane	7	Band Set	Band Set 3	Name	#7
Band	1	Band Type	4		
Rel. Front	0.082				
Mol. Wt.	204.73	KDa	Status	Calculated	
Peak	0.877 OD	Average	0.224 OD		
Trace	0.568 OD x mm	Contour	N.A.		
Relative Qty.	4.87%	Norm. Qty.	258.60		
Quantity/Units		/		Status	Unknown
Calibration		Not Calibrated			

Buttons: OK, Help

Fig.7-11. Band Information dialog box.

The Lane and Band fields at the top of the form allow you to type in the numbers of other bands on your gel image. The information displayed in the Band Information dialog box will update to reflect the new band, as will the trace overlay and band highlighting in the image window.

If the band is part of a Band Set and is of a known Band Type (see Chapter 12), this will be noted in the form.

If the quantity of the band you have selected is known, you can enter it next to tQuantity Standard's Quantity / Units prompt.

If you want to calibrate that band against known quantities, you can do so by clicking on the Calibration button. You will be asked to select the calibration curve you want to use. (See section 11.2.a. for information on calibration curves.)

7.7. Identifying Irregularly Shaped Bands

If the bands on your gel image are irregularly shaped, you can use the contour or drawing features to define them. These functions give you more control over defining your bands than either Detect Bands or Create Bands.

Note: The contour functions will create contours semi-automatically, whereas the drawing functions allow you create bands entirely manually. At least one lane must be defined on your image before you can create a band with the contour or draw band functions.

Contoured or hand-drawn bands are quantitated based on the signal intensity of all the pixels within the band boundary, using the following formula:

Quantity = Sum of (intensity of pixel x pixel size) for all the pixels in the boundary.

Diversity Database first multiplies the intensity of a pixel by the area of the pixel. This is done for all the pixels that are within the contour or drawn boundary. The area of the pixel is determined by the scan resolution of the image.

The resulting values have units of intensity x mm².

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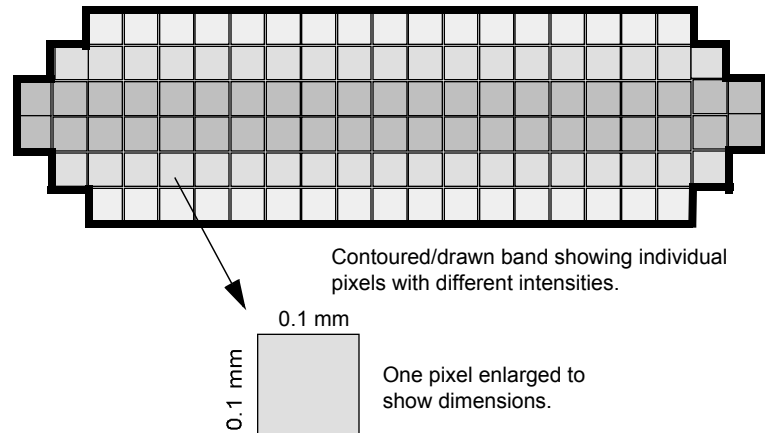


Fig.7-12. A contoured/drawn band scanned at 100 x 100 microns (micrometers).

The functions needed to contour and draw bands are found on the Band > Contour submenu and the Contour Tools toolbar.

Before using any of these functions, we suggest that you magnify the image so that the individual pixels in the band are clearly visible. This allows you to position the cursor more accurately.

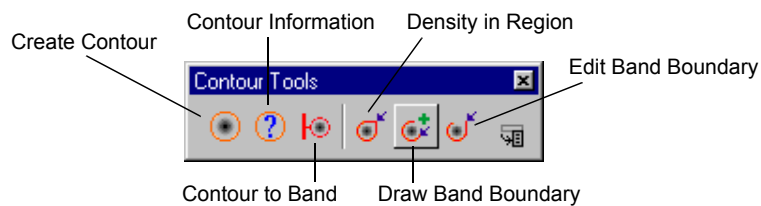


Fig.7-13. Contour Tools toolbar.

7.7.a. Contouring Bands

Creating Contours

Select Create Contour from the Band > Contour submenu or Contour Tools toolbar and click on a pixel at the edge of the band. This will display a contour that encloses pixels whose intensity is equal to or greater than that of the pixel at the cursor.

Contour encloses pixels with intensities greater than or equal to intensity of pixel at cursor

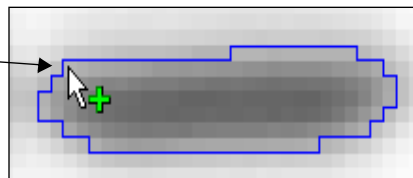


Fig.7-14. Creating a Contour

If the contour does not encircle the band, reposition the cursor and click again. A new contour will be drawn in place of the old.

Contour Information

To display information about the contour, select Contour Information while the contour is highlighted. A pop-up box will display the area, total intensity, and average intensity of the contour.

Converting a Contour into a Band

When you are satisfied with the contour, you can redefine the contour as a band with the function Contour to Band.

Note: Before you can convert a contour into a band, you must define at least one lane on your gel image. You can define a lane using the Create Lane tool while the contour is currently displayed.

Selecting Contour to Band from the menu or toolbar converts any currently displayed contour into a band and assigns that band to the nearest lane. The

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contour boundary will change color from yellow to red, and a band will appear on the nearest lane.

Once a contour is converted, all the functions in the Bands menu can be applied to it.

7.7.b. Calculating the Density in an Arbitrary Region

Use Density in Region to get intensity and area information about any region on an image.

Select Density in Region from the Band > Draw Band menu or Contour Tools toolbar. The cursor will change to an “information cursor.” Drag the cursor as if it were a pen to enclose a region of interest.

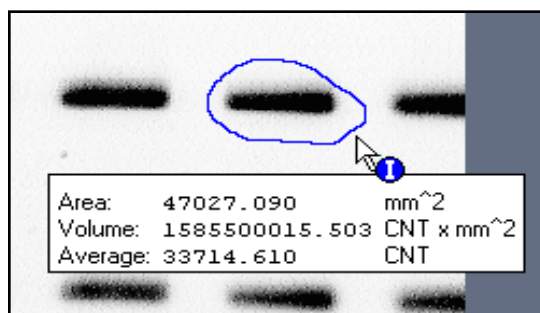


Fig.7-15. Density in Region tool.

When you close the border, it will change color and a pop-up box with information about the enclosed area will be displayed.

For very small regions, you will have to magnify the region before using this command.

7.7.c. Drawing Bands

Diversity Database has tools for drawing bands directly, without having to detect them first.

Bands

Note: You must have a lane defined on your image before using the draw tools. The drawing tools work best if you first use the Zoom commands to magnify the region you want to draw in.

Drawing Band Boundaries

If you want to draw the boundary of a band manually, you can do so using Draw Band Boundary.

Select Draw Band Boundary from the Band > Draw Band menu or Contour toolbar. Your cursor will change to a pencil symbol.

Drag the cursor around the perimeter of the region you want to define as a band. A line will appear, defining the boundary.

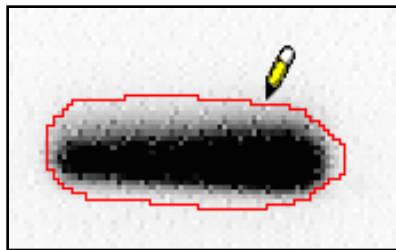


Fig.7-16. Draw Band Boundary tool.

If you make a mistake and need to retrace part of the band boundary, simply backtrack with the cursor; the previous path will be erased and you can redraw it.

Note: It is important to magnify the image before drawing a boundary. If you try to draw a very small boundary, the software will think that you are backtracking and erase the boundary.

When the cursor crosses over the line you are drawing, the drawn band is complete and the color of the line will change to indicate that it is a boundary.

If you keep drawing, each time the line crosses itself a new band will be created, replacing the old band.

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Editing Band Boundaries

If your drawn band needs minor adjustments, you can fix it using Edit Band Boundary.

Select Edit Band Boundary from the Band > Draw Band menu or Contour toolbar. Your cursor will change to a pencil symbol.

Drag the cursor across the previously defined boundary; a line will appear. When you recross the old boundary, the line will change colors and the new boundary will be created.

8. Standards

If you are using molecular weight, isoelectric point, or other positional standards on your gels, this chapter will describe how to define them on your images. If you are not using these standards, you can proceed to the next chapter, Creating Band Sets.

Note: Multiple standard lanes in your gels will facilitate the band matching process outlined in the following chapter. However, they are not required.

For additional guidance on creating and using standards, see Lesson 2 in Chapter 3., Tutorials.

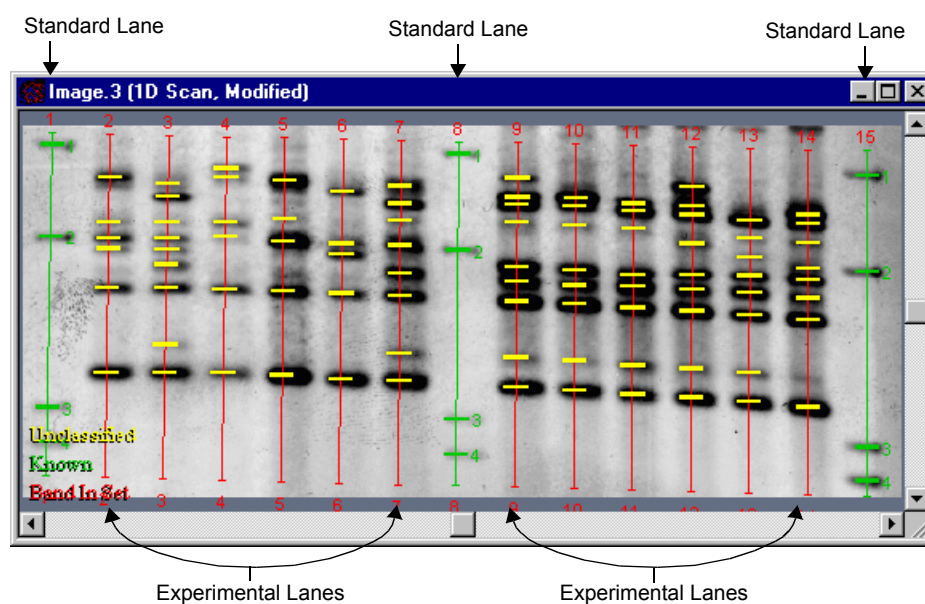


Fig.8-1. Standard and experimental lanes and bands.

The image in Figure 8-1 contains 15 lanes: 3 standard lanes (lanes 1, 8, and 15) and 12 experimentals. The standard bands have already been defined in this

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image and are marked in green, while the experimental bands are marked in yellow.

8.1. Creating a Set of Standards

Diversity Database includes sets of predefined Bio-Rad molecular weight / base pair standards. If you are not using these predefined standards, you will have to create your own set of known standard values.

To create your own standards, choose Standards from the Database Summary form or Database menu. A series of dialog boxes will guide you through the selection process.

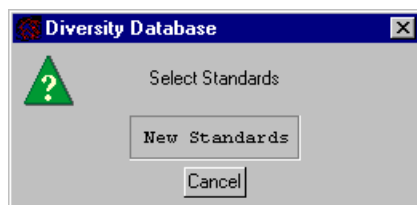


Fig.8-2. Selecting a set of standards.

The first dialog box will ask you to select new standards or load previously created standards (if any have been saved).

Select New Standards to create a new set of standards. A dialog box will pop up in which you can specify the units.

Standards

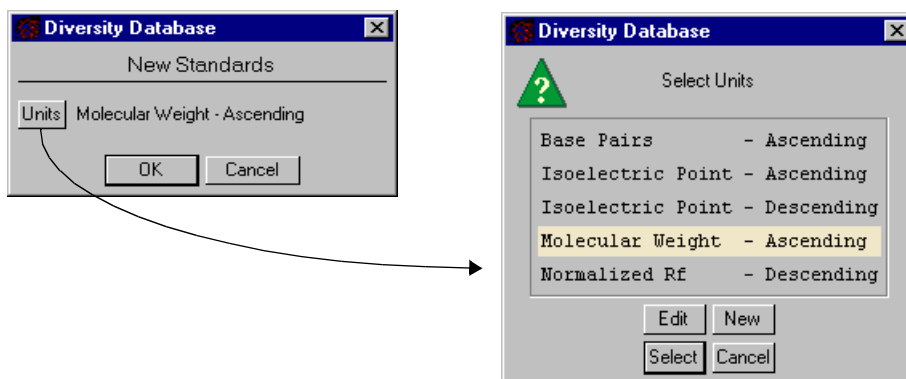


Fig.8-3. Specifying units.

Click on the Units button to specify your units. This opens a dialog box in which you can select from a list that includes Base Pairs, Isoelectric Point, Normalized Rf, and Molecular Weight. You can specify a set of units not on the list by clicking on the New button. If you have already defined a new set of units, you can edit them by highlighting them on the list and clicking on Edit.

Note: “Ascending” means that bands of higher molecular weight or isoelectric point are at the top of the gel image, and bands of lower molecular weight or isoelectric point are at the bottom of the gel image.

After you select your units, the main Standards form will appear. Here you can enter values for your standards, apply them to the bands on your image, and save them as a set for future use.

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db - Band Set

Band Set: std

Comment:

Type: S Standard

Units: Molecular Weight - Ascending

<category> <category>

<category> <category>

Icons: Add, Delete, Edit, Find, Print

Type	Name	KDa
▶ 1		6557.00
▶ 2		4361.00
▶ 3		2322.00
▶ 4		2027.00
▶ N		
▶		
▶		
▶		
▶		
▶		

Buttons: Delete, OK, Help

Fig.8-4. Standards form.

The Standards form has fields for naming your standards, categorizing the standards, and entering values of your standard bands. It also has tools for applying the standards values to the lanes in your gel image.

Naming Your Standards

The Standards form will open with a default name for your standards. You can type a new name for the standards in the text box at the top of the dialog.

Standards

You can enter additional information about the standards in the text field next to the Comment prompt.

8.1.a. Entering and Applying Standard Values

Entering Standard Values

After you opened a Standards form, the next step is to enter values for your standard bands.

In the middle of the Standards form is a table with a scroll bar on the right side. In this table, there are three columns labeled Type, Name, and the units you previously selected (e.g., KDa, pI, Rf).

The screenshot shows a table with three columns: Type, Name, and KDa. The table has 10 rows, numbered 1 to 10. Each row has a green trash icon in the Type column. The KDa column contains values: 200.000, 116.500, 94.000*, 68.000*, 57.000*, 54.000*, 45.000*, 43.000*, 30.000*, and 21.000*. Annotations include: 'Type in a name for the standard here' pointing to the Name column; 'Click here to delete this value and renumber the remaining standards' pointing to the trash icon in row 7; and 'Enter the standard value here' pointing to the KDa column.

Type	Name	KDa
▶ 1		200.000
▶ 2		116.500
▶ 3		94.000*
▶ 4		68.000*
▶ 5		57.000*
▶ 6		54.000*
▶ 7		45.000*
▶ 8		43.000*
▶ 9		30.000*
▶ 10		21.000*

Fig.8-5. Entering the values of your standards.

In the “units” column, type a value for the first standard band and press RETURN. The cursor will skip to the field below, and you can enter a value for the second standard band. Repeat this process until all your standard values have been entered.

If you accidentally skip a standard value, enter it in the next available row and press RETURN. The values will sort themselves into descending order.

You can enter a name for each standard band in the Name column. This will appear in subsequent reports and print-outs.

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To remove a standard value from the file, click on the arrow button under the Type header. A pop-up box will appear. Select Delete to remove the standard value from the template. The remaining standards will be renumbered.

Applying Standard Values to Lanes

The values you have entered in your Standards form can now be applied to a lane or lanes in your gel image. This step will also automatically calculate the values of bands in your nonstandard lanes, based on the standards.

Note: The values in nonstandard lanes are calculated based on their proximity to the standard lanes that you assign. Generally, the greater the number of evenly spaced standard lanes in a gel image, the higher the accuracy of those calculations. We recommend a minimum of two standard lanes per gel.

The Apply Band Set button (in the Standards form) applies all the standard values listed in the dialog box to the standard lane(s) in your image. Click on the button, then click on a lane.

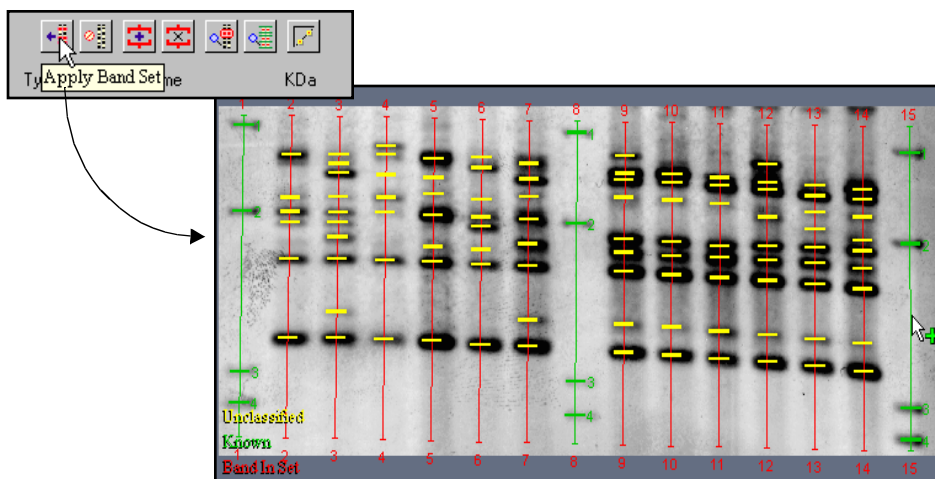


Fig.8-6. Click on the Apply Band Set button, then click on the lanes containing standards.

Standards

The band values you have entered will be applied to the lane you select. Click on any remaining standard lanes to apply the same values to them.

You can also click on the numbered button next to a band value to apply that value to a particular band in a lane on your image. Click on the button, then click on the band to be assigned the standard value.

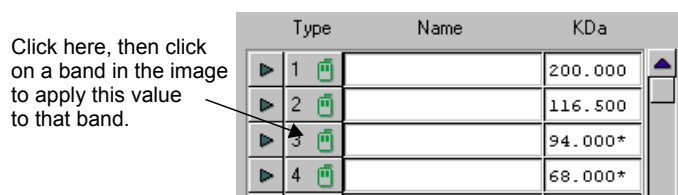


Fig.8-7. Using the numbered buttons to apply standards.

The subsequent bands in the lane will be numbered sequentially based on the initial assignment.

After you have applied your standard values to a lane, the bands in that lane will change color to green, indicating that they are now standards. The values of bands in nonstandard lanes will be calculated automatically based on your standard lane.

To view the new values of all the bands in your gel image, select Band Attributes from the View menu, then select the value to be displayed.

Removing Standard Values from Lanes

The Clear Band Set button removes all the standard values from the lane(s) you select. Click on the button, then click on the lane or lanes from which you want to delete the standard values.

Showing the Standard Modeling Lines

After you have applied your standard values, you can show the standard modeling lines across the gel image by clicking on the Show Modeling Lines button. Bands that fall along these lines will have the same values as the standards.

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8.1.b. Saving, Opening, and Deleting Standards

Values in the Standards form are automatically saved when you enter them. There is no separate “save” step.

To open a set of standards you’ve created, click on the Standards button in the Database Summary form and select the name of the standards.

To delete a set of standards, first open it, then click on the Delete button at the bottom of the dialog box. A pop-up box will ask you to confirm that you want to proceed with the deletion before completing the action.

8.2. Showing the Standard Curve

After you have defined your standards, you can display a regression curve based on those standards for any lane in your gel image.

Click on the Standard Curve button in the Standards form, then click on a lane to show the curve for that lane.

Standards

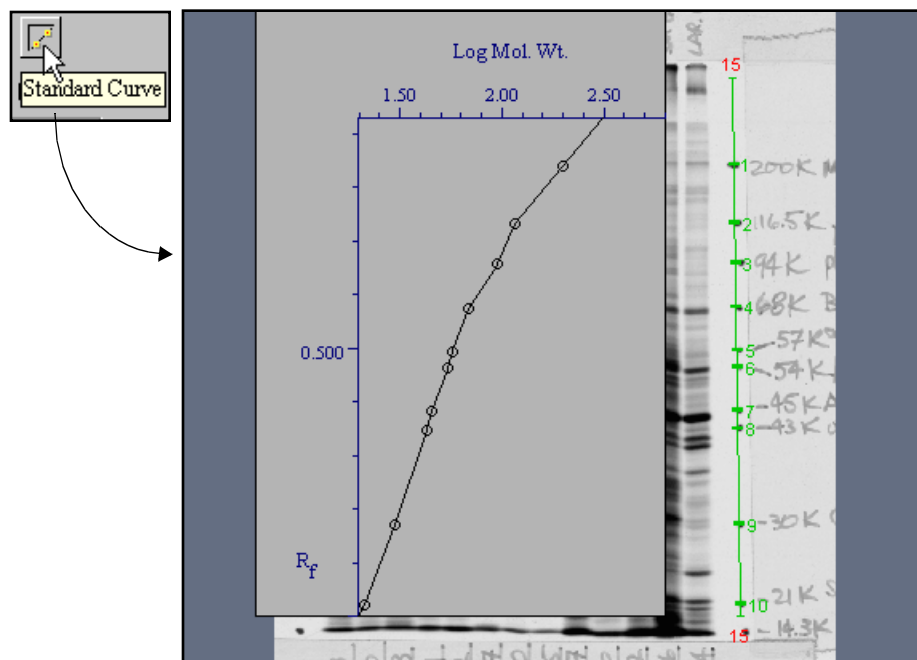


Fig.8-8. Example of a standard curve.

The X axis is the log of the standard value and the Y axis is the R_f value.

Clicking on a lane that includes your standards will display the curve with the standards marked as circles.

8.3. Adding Standards to Other Gels in a Database

After you have created a set of standards, you can apply it to all the other gels in the database that use the same standards.

First, open the gel in the database to which you want to apply your standards. (See section 5.4., Editing a Gel in the Database, for instructions.)

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Next, open your standards by clicking on the Standards button in the Database Summary form and selecting the name of the standards.

Click on the Apply button in the Standards form, then click on the lane(s) in the gel containing the standards. (See Applying Standard Values to Lanes, above.) The standard values will be applied to those lanes.

9. Creating Band Sets

Once you have included at least one good, clean gel image in a database (Chapter 5) and defined all the lanes and bands in that image (Chapters 6–7), you are ready to start building your database. If you have run standards on your gels, you should define them (Chapter 8) before proceeding.

Note: When building a database, you should start with a gel image that has clean, well-defined bands and includes most of the bands you are interested in for your research. Also, multiple lanes of standards will facilitate the matching program; however, they are not required (see section 9.6., Tips for Gels Without Standards).

The first step is to match the bands you want to compare together and create a **band set**. Band sets are the mechanism by which Diversity Database compares samples in databases. Lanes which are similarly detected (i.e., same enzyme, probe, etc.) should be placed in the same band set. All samples that you want to link in a database must belong to the same band set.

This chapter describes how to create a band set and apply it to samples to be linked in a database.

Note: See also Lesson 2 (creating band sets with standards) and Lesson 3 (creating band sets without standards) in Chapter 3, Tutorials.

Overview

A band set is made up of **band types**. A band type is a classified band that appears in a lane or across several lanes (see Fig. 6-1). It has a unique position and (if standards have been run) molecular weight/isoelectric point. When you add a band type to a band set, you are recording its position and standard value in the database. Comparisons are based on different lanes (i.e., samples) sharing the same classified bands (i.e., band types).

First you identify all the band types in a representative gel image. This collection of band types becomes your band set. You then apply this band set to other, related gel images, matching the band types in the set to the bands in those images. You can also add to the band set by identifying new band types

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in the samples in those images. Finally, you are ready to compare samples across gel images using the database.

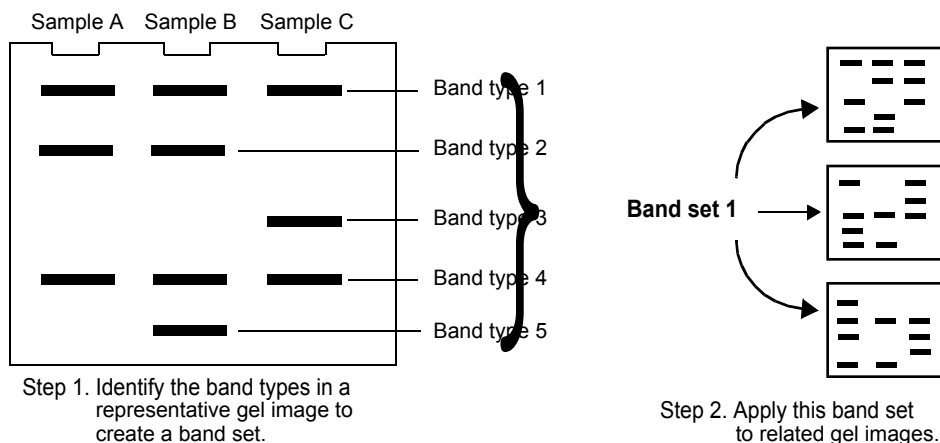


Fig.9-1. In this example, the bands in samples A, B, and C are used to create band set 1, which is then applied to three other gels.

9.1. Workflow for Creating a New Band Set

Note: When you are creating a new band set, you should start with a gel that has most of the band types in which you are interested in your research. The more band types in your initial gel, the fewer you will have to identify in later gels.

To create a new band set, you will perform the following steps (these are described in greater detail in subsequent sections):

1. Select a lane to use as the basis for your band set. This is your “first” lane; each band in this lane will become a new band type. Therefore, this lane should include as many clearly defined, unique bands in your gel as possible. (In the example in Fig. 6-1, you would choose sample B as your first lane, because it contains four distinct bands.)

Creating Band Sets

2. Match the bands in your other lanes with the bands in your first lane to create your band types. (In Fig. 6-1, the first band in each lane has been matched to create band type 1.)
3. Manually match any bands that auto-matching has missed. If, after careful matching, any bands cannot be matched with those in your first lane, use them to create new band types.
4. Once all the bands in the gel have been identified as band types, you are ready to use the gel and your newly created band set to create a database.

Note that bands in lanes that are similarly detected (same enzyme and probe, etc.) should be placed in the same band set. Bands in lanes that are *not* similarly detected belong in *separate* band sets.

Once created, a band set can be applied to new gels as you enter them into the database with a minimum of editing. This will be discussed in detail later.

9.2. Creating a Band Set

To create a band set, make sure your Database Summary form is open. Click on the Band Set button in the Database Summary form.

A dialog box will pop up prompting you to select a band set. Click on New Band Set. Another dialog box will open in which you can specify the units for your band set. Click on the Units button to select from a list of predefined units or to define units of your own.

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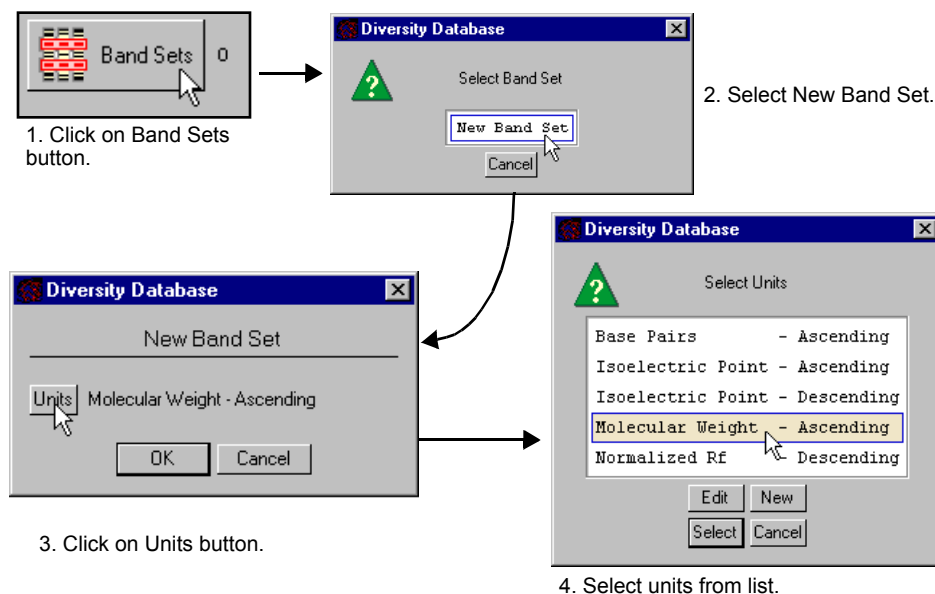


Fig. 9-2. Creating a new band set, steps 1–4.

Click on a unit name once to highlight it, then again to select it. Alternatively, click on the New button to define new units.

If you are using molecular weight or other standards, select the appropriate units. If you are not using standards in your gels, select Normalized Rf—Descending. Normalized Rf units are positional units based on the relative fronts of the bands in the sample lane from which you create the band set. Normalized Rf values are fractional distances along the length of the lane (total length of lane = 1.0).

Note: “Ascending” in the units name means that higher values (i.e., greater molecular weights, more base pairs) are near the top of the gel image.

Clicking on New will open the Edit Unit dialog box, in which you can define your new units.

Creating Band Sets

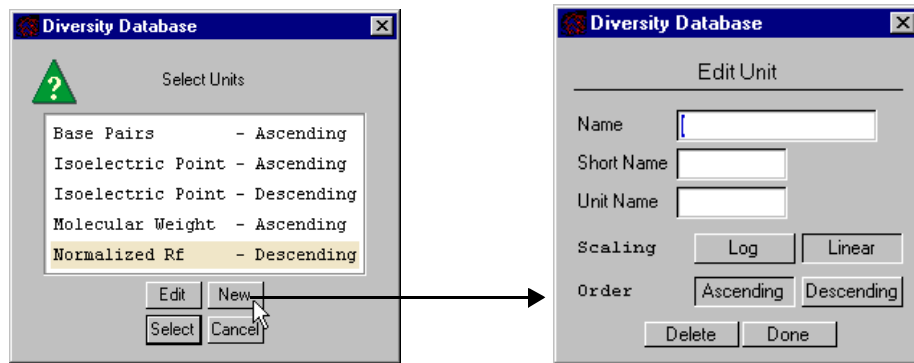


Fig.9-3. Defining new units.

When you click on Done in the Edit Unit dialog, the units you just created will be listed with the predefined units. To delete any units you have created, select the unit name from the list, click on Edit, then click on Delete in the Edit Unit dialog box. (You cannot edit or delete predefined units.)

Once you have selected your units, click on the OK button in the New Band Set dialog box. This will open a new Band Set form.

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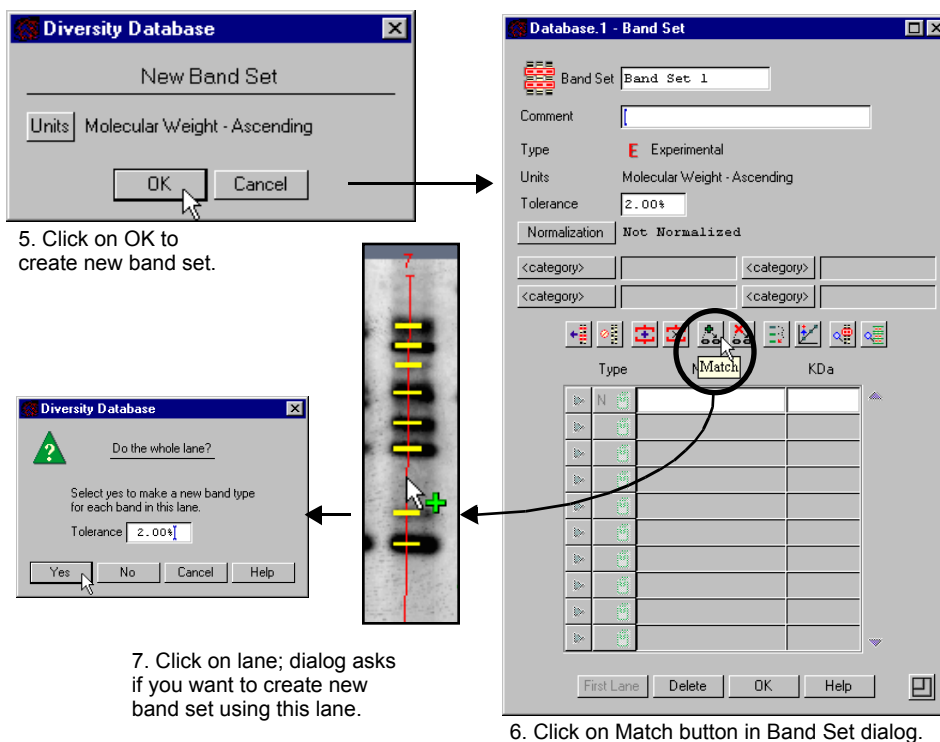


Fig.9-4. Creating a band set, steps 5–7.

In the Band Set form, click on the Match button at the center of the tool palette, then click on the lane containing the bands you want to use to start your band set.

A dialog box will ask if you want to make a new band type for each band in this lane, which you do. The tolerance setting controls the automatic matching mechanism. A lower tolerance (<2.50) is preferred for closely spaced bands, while a higher tolerance (>3.50) is optimal for widely spaced bands.

Creating Band Sets

When you click on Yes, the bands in the lane you clicked on will change to green, indicating that they are your new set of known band types. The values of these known band types will appear in the Band Set form.

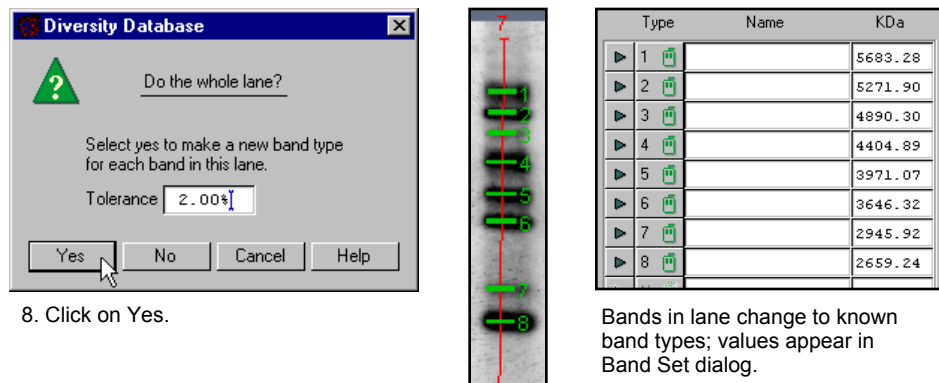


Fig.9-5. Creating a new band set, step 8.

Most of the unknown bands in the image will change to red, indicating that they have been matched with the known band types. Diversity Database has automatically matched these bands, based on the band types you just created and the position of the standards (if any), as well as the tolerance setting.

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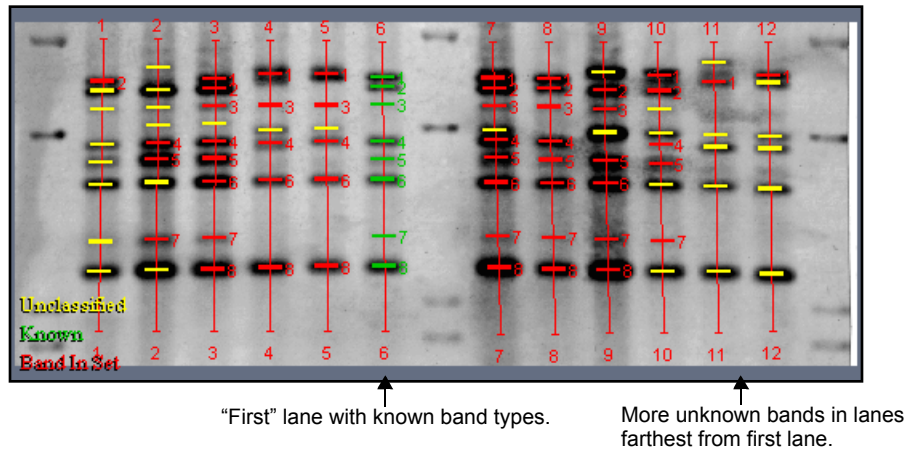


Fig.9-6. Gel image after auto-matching.

Yellow bands are bands that the software cannot accurately match. The matching algorithm is deliberately conservative to avoid incorrect labeling, so a number of yellow bands may appear on the image, particularly in the lanes farthest from the first lane. Because standards facilitate matching, gels without standards will have more yellow bands than gels with standards.

To summarize:

- Green bands are known band types, based on your identification.
- Red bands are bands that have been automatically matched with the known band types.
- Yellow bands are unknown and unclassified.

Displaying Band Types and Modeling Lines

To display the band type numbers, select Show Band Types from the Database menu. This will display all the currently assigned band types.

Band type modeling lines reveal the path along the gel image that the software uses to match bands of the same type in different lanes. These lines are based on the identified bands and the matching tolerance.

Creating Band Sets

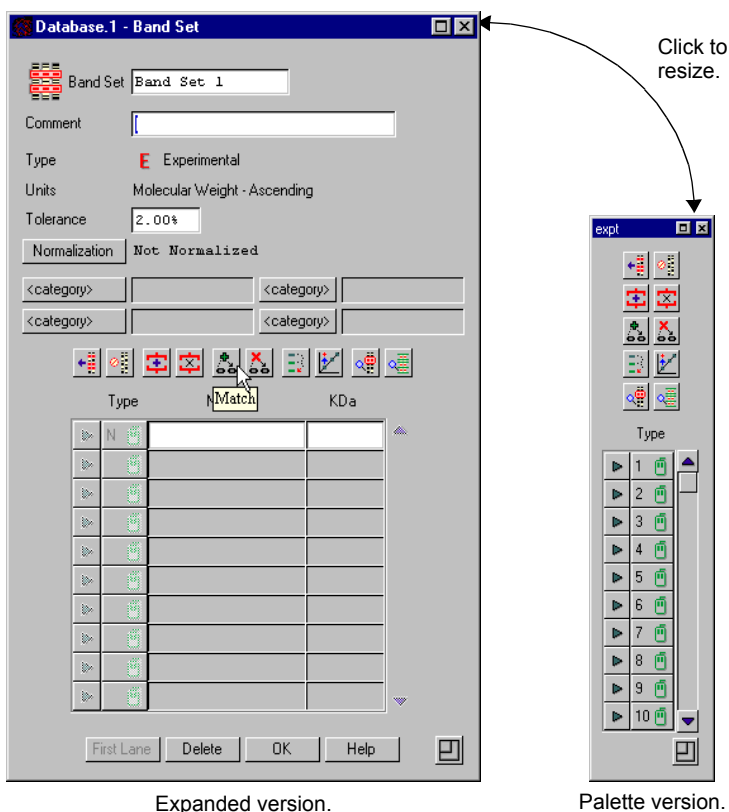
To display the band type modeling lines, select Show Band Models from the Database menu. The currently assigned band types will appear, as will the modeling lines connecting the band types.

9.3. The Band Set Form

The Band Set form contains the values of all the bands in your band set, a tool palette with the functions needed for band matching, and other information.

The dialog box will open with a default name for the band set (e.g., Band Set 1). You can type a new name in the Band Set field at the top of the dialog, and add any comments or category / attribute information you want to associate with the band set.

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Expanded version.

Palette version.

Fig.9-7. Band set form.

The band set units are displayed in the top half of the form, as is the matching Tolerance field and Normalization button. Tolerance is the minimum spacing between band types that you specified when you created the band set. The Normalization button allows you to pick a specific band type to normalize your gel against (see section 11.3., Differential Display, for more information on band normalization).

Clicking on the resize button will reconfigure the band set form to its smaller, palette version, which displays only the tool icons and band type buttons.

Creating Band Sets

List of Band Types

The values of the individual band types are listed within the dialog box in a table format. This is similar to the table of standard values in the Standards form.

Type	Name	KDa
▶ 1 		500.000
▶ 2 		400.000
▶ 3 		360.000
▶ 4 		300.000
▶ 5 		163.319
▶ 6 		148.085
▶ 7 		127.857
▶ 8 		113.739
▶ 9 		107.958
▶ 10 		102.472

Fig.9-8. Applying and editing band type values.

If you change any of the band type values in the list, that will be reflected in the band type modeling on your image.

You can assign a particular band type to a band on your image. Click on the numbered button in the Type column to assign that band type to the mouse, then click on a band on your image to change that band to the assigned band type.

You can enter names for your band types in the Name column. These will appear in subsequent reports and print-outs.

To remove a band type from the set, click on the arrow icon associated with the particular band type. A pop-up box will ask for confirmation, and the remaining band types will be renumbered.

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Band Set Toolbar

The Band Set form also contains a toolbar that contains all the commands needed for matching.

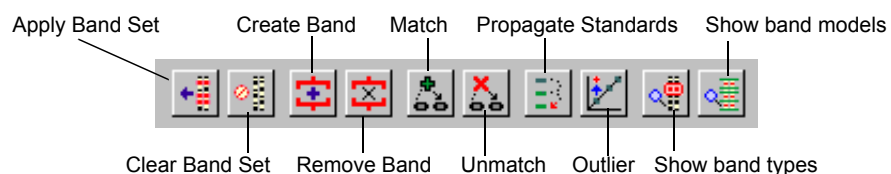


Fig.9-9. Band set toolbar.

Apply Band Set applies the band set to any lane in your gel image. Click on the Apply Band Set button, then click on the lane. The bands in that lane that can be matched will change to red.

Clear Band Set removes the band set modeling from any lane in your gel image. Click on the Clear Band Set button, then click on the lane to be cleared. The bands in the lane will change to yellow.

Create Band will mark a band where none was defined before. Click on Create Band, then click on the band in the image.

Remove Band will remove a defined band. Click on Remove Band, then click on the defined band to unmark it.

Match will define a band as a new band type. Click on the button, then click on the unknown (yellow) band to create a new band type.

Unmatch will change a band's definition from matched or known to unknown. Click on this button, then click on the matched (red) or known (green) band to change it to unknown.

Propagate Standards applies all the known band types to the bands in a lane, based on a few manually-assigned bands and the band set model. Click on this button, then click on the lane. The bands in that lane that can be matched will change to green to indicate their known status.

Outlier excludes a known (green) band from the band set model.

Creating Band Sets

Show Band Types displays all the red, green, and yellow bands on the image, with the band type numbers next to the matched bands.

Show Band Models displays the band set modeling lines across the gel image.

9.4. Manually Matching Bands

Once you have created a band set, you need to manually match the remaining yellow bands in your image using the match tools in the Band Set form or on the Database menu.

Note: See Lessons 2 and 3 in Chapter 3 for step-by-step directions on using the following commands.

You can use these tools to manually change the band type of a band, create new band types, unmatch bands, or identify bands as outliers from your band set.

- To change a band to a particular band type, first click on the band type number button in the Band Set form. This band type will be assigned to your mouse, and the status box for the cursor will change to the band type number. Then click on the red or yellow band you want to change. The band will appear green (known) and the modeling line will change to reflect the assignment of the band type.
- To change the status of a matched (red) band to known (green), click on the Match button and double-click on the red band.
- To create a new band type from an unknown (yellow) band, select Match and click on the unknown band.
- To create a new band type from a matched (red) band, select Match, hold down the SHIFT key, and click on the red band.
- To change a band type to unknown (yellow), select Unmatch and click on that band.
- To not include a red or green band in band type modeling, select Outlier and click on that band. An X will appear through the band, indicating that the software is ignoring it when modeling band types across the gel image.

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General Hints on Adding Band Types

Band types should only be added to areas of the gel that are modeled well. The modeling lines are designed to give you guidance on adding new band types.

If the gel has standard lanes, then the best choice is to define a new band type in a lane next to a standard lane. Otherwise, find a band that is in an area of the gel where its neighboring bands are assigned correctly, and the white band set lines intersect these bands.

Completing the Band Set

You should perform manual matching until all the bands in your image are assigned to band types and your band set is complete. Your gel should have no yellow (unknown) bands visible.

On well-modeled gels, the modeling lines will intersect at or near the middle of the band types across the gel. If your gel contains serious distortions, and the modeling lines are not even, you should consider using a different gel from which to build your band set. Poor modeling on your first gel will influence the modeling on all the subsequent gels in your database, so take your time while creating the band set, and if necessary choose a different gel on which to base the modeling.

Creating Band Sets

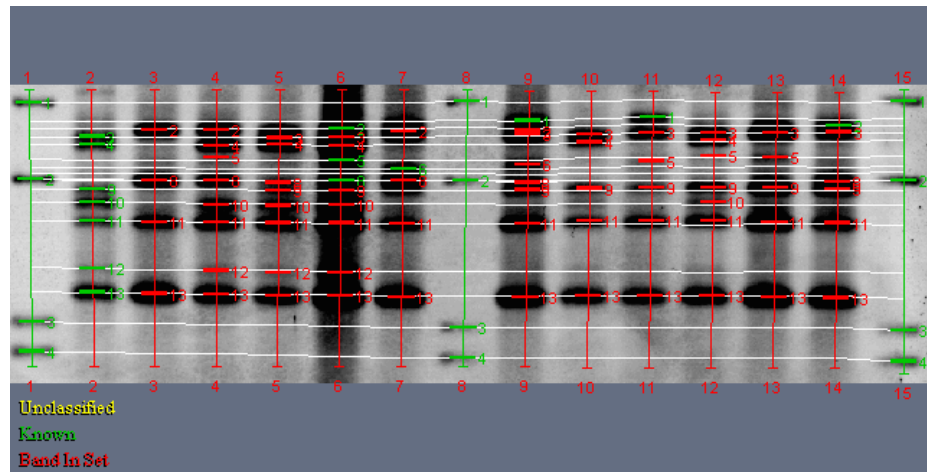


Fig.9-10. Well-modeled gel.

9.5. Applying an Existing Band Set to a New Gel

Once your band set is complete and you've saved the database, you are ready to apply your new band set to any number of other gels in the database.

Note: See Lesson 4 in Chapter 3 for step-by-step instructions on applying an existing band set.

To do so, first open a gel to which you want to apply the existing band set. If your gels have standard lanes, apply the standard set to those lanes (see section 8.3. for details).

Click on the Band Sets button in the Database Summary form and select the name of the band set that you want to apply.

There are a number of ways to add a band set to an image. You can click on the Apply Band Set button in the Band Set form, then click on any lane in the image. Alternatively, you can click on one of the buttons next to a band type and apply that band to a band on the image.

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Diversity Database will automatically assign band types to each of the unassigned lanes in your gel, based on the modeling information recorded when you created the band set.

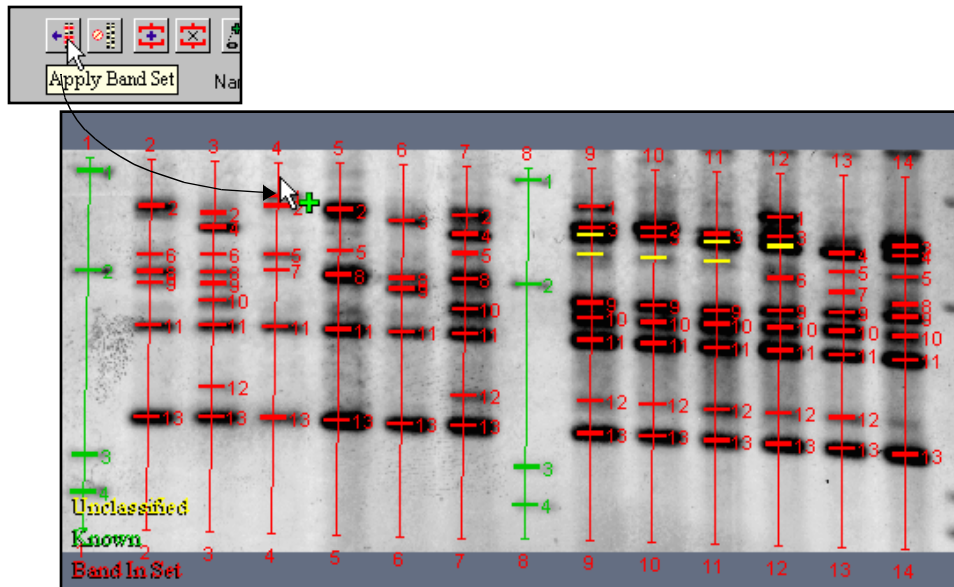


Fig.9-11. Applying a band set to gel with standards.

Note that one method of assigning band sets may leave more unmatched (yellow) bands in the image than another.

There is no single, “right” way to apply a band set to a new gel. If one way results in too many unmatched bands, you can “unapply” the band set by clicking on the Clear Band Set button and then clicking on each experimental lane in turn. Then you can try clicking on a different band or lane.

In most cases, further editing will be required to include unmatched bands in the band set.

9.6. Tips for Gels Without Standards

Ideally, if you are not using standards in your gels, you will have more than one lane containing a reference sample in your gel. After you've used this reference sample to create a band set, you should apply that band set to other reference sample lanes using the Propagate Standards button (in the Band Set form).

Propagate Standards is a feature that not only simplifies assigning band types, it allows the software to do some optimizations that will significantly speed up modeling.

Choose a reference sample lane that you want to apply the band set to. Start by assigning one or two band types in the lane that are easy to identify, then click on the Propagate Standards button and click on the lane to assign the remaining band types to the lane.

Propagate Standards assigns the band types one at a time. The command will assign a type to a band only if it is very sure that it is the correct assignment.

Once your reference sample lanes have been modeled, add any unknown band types to the gel using the methods outlined in Lesson 3 of Chapter 3, as well as in the sections above.

9.7. Displaying Graphs of Match Data

You can display graphs of different kinds of data associated with your band types. The commands for displaying these are located on the Database > Match Graphs submenu.

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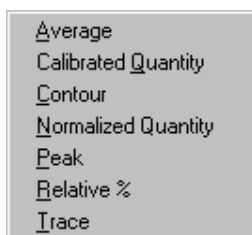


Fig.9-12. Match Graphs submenu.

From the Match Graphs submenu, select the graph you want to display, then click on a band type. The bands of that type in the image will be displayed, as will the histogram that you chose.

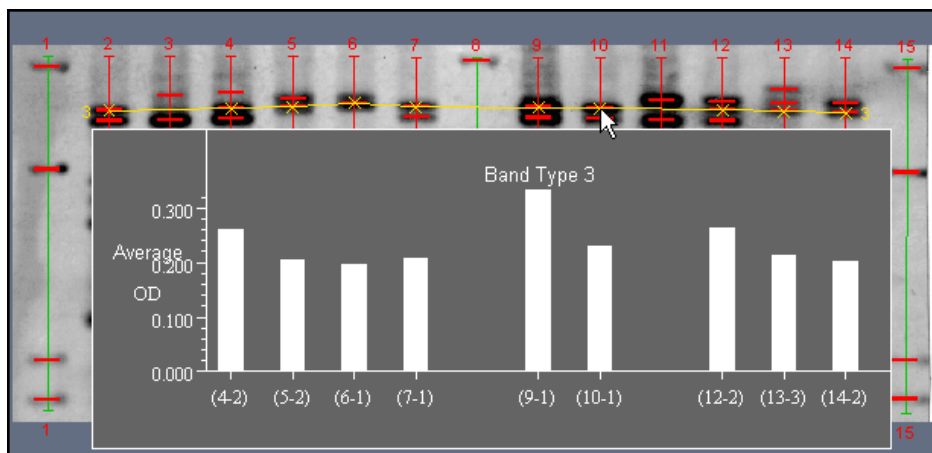


Fig.9-13. Example of a match graph.

- **Average** displays a histogram of the average densities of the bands in a band type group.
- **Calibrated Quantity** displays a histogram of the absolute quantities of bands in a band type group, based on a calibration curve (for more information on calculating absolute band quantities, refer to section 11.2.).

Creating Band Sets

- **Contour** displays a histogram of the density of each contoured band in a band type group. This function is only valid for contoured bands.
- **Normalized Quantity** displays a histogram of the normalized quantity of the bands in a band type group.
- **Peak** displays a histogram of the peak densities of the bands in a band type group.
- **Relative %** displays a histogram in which each bar represents the percentage that the bands in a band type group represent out of either the total density of their respective lanes, or the sum density of the bands in their lanes (see section 2.5.c for more information on Relative % calculations).
- **Trace** displays a histogram of the integrated densities of the bracketed bands in a band type group.

Each bar of the histogram will be labeled on the X axis with the lane number, and (where space permits) the band number. The Y axis is labeled with quantitative values.

The histogram display reserves space on the left side to label the Y axis and show its units. If the bands in the band type group span the entire window from the left edge to the right, the histogram will not include bars for the bands in the left-most lanes because of the space required for labeling the axis.

To avoid this problem, decrease the magnification of the image using Zoom Out until there is blank space between the left side of the window and the first band of the band type group.

Redisplay the histogram. If one or more bars are still not included, continue decreasing the magnification until you can see a bar lining up with each lane that contains a band in the band type group in which you are interested.

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10. Using the Database

After you have applied the same band set to several gels in a database, you are ready to compare the samples in those gels using the database.

Note: Basic database commands, such as creating, saving, and adding gels to a database, are described in Chapter 5.

Database files store all the information saved in your gels and allow you to perform comparative analysis on lanes, matched bands, or multilane samples within a single gel, a user-defined set of gels, or all the gels in your database. Database files can also be used as flexible gel cataloging systems, allowing you to store up to 50,000 lanes in a single database and perform similarity searches on those lanes in under 30 seconds.

How Databases Are Structured

Samples in databases are linked by band sets (see the previous chapter). When you search a database, you are searching all the samples that belong to the same band set.

Note: Typically, a sample is a lane on one of your gels, although Diversity Database can handle multiple lanes per sample (multiple probes), provided that each lane belongs to a different band set. A database sample can include as many lanes as necessary to define your biological sample.

Opening the Database

To begin working with a database you have created, open the database directly (File > Open) or open a gel image that is included in that database. (The Database Summary form will open with the gel image.)

The Database Summary form is the on-screen representation of the database file. Using the Summary form, you can open six main database dialog boxes: the Gels list, the Gel Layout form, the Samples list, the Band Sets form, the Searches form, and the Populations form.

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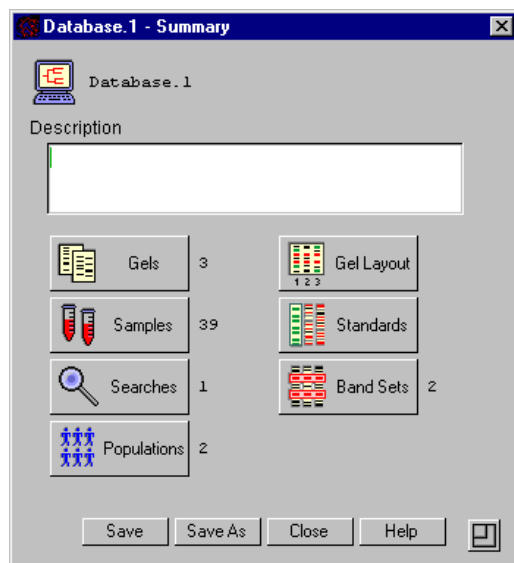


Fig.10-1. Database Summary form.

The numbers next to the buttons on the Summary form indicate the number of members of each type included in the database. For example, a “3” next to the Gels button would indicate that the database contains three gels, and the number “39” next to the Samples button would indicate that those three gels contain 39 individual samples.

10.1. The Gels List

Clicking on the first button on the Summary form displays the Gels list. All gels currently included in the database will be listed there.

To add a new gel to the database, click on the Add Gel button, open the gel image that you want to add, and click on the image. The name of the gel you selected will appear in the Gels list.

The name of each gel is accompanied by its scan date and any user-defined category or descriptive information.

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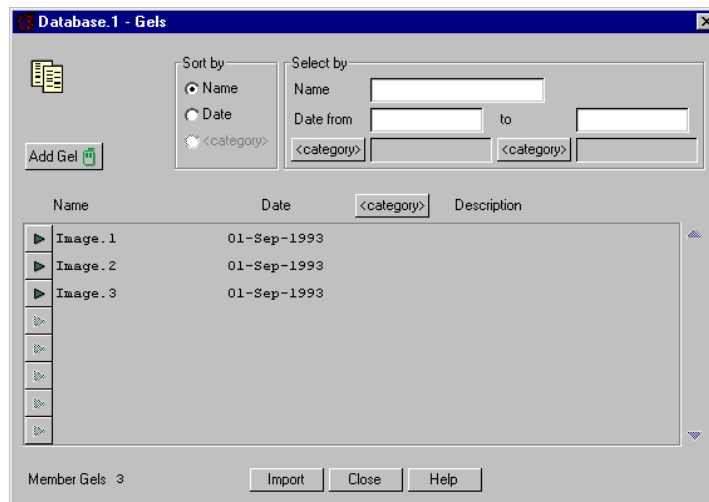


Fig.10-2. The Gels list.

Use the triangular “pop-up” icons on the left of the gel name to open the gel image, open the Gel Layout form for that gel, or to exclude the gel from the database. Excluding an image will delete all the sample and band set information about that image from the database. All matching done on the image will be lost as well.

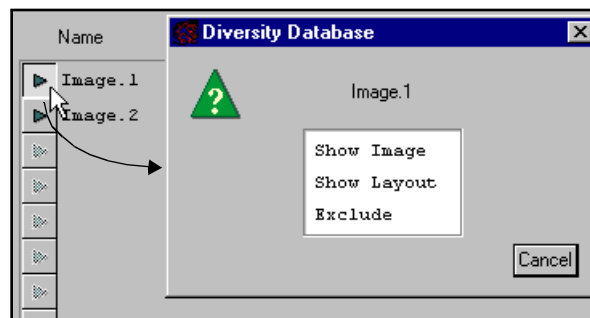


Fig.10-3. The **Gels** pop-up form.

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Importing Band Values

You can import text files containing the positional, molecular weight, or other values of bands, and Diversity Database can perform similarity searches on these values just as it can on the values of bands in images.

Click on the Import button in the Gels list dialog box. This will open a version of the standard Open dialog in which you can select the text file you want to import. When you select the file, the Import dialog box will open.

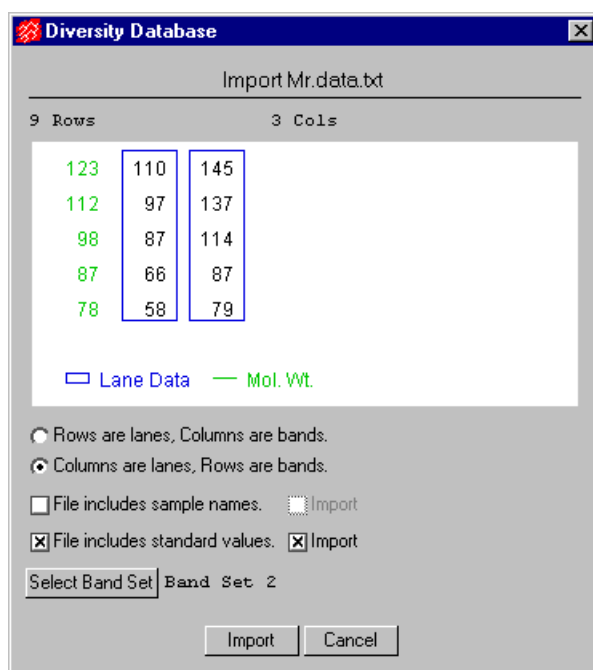


Fig.10-4. Import dialog box.

In this dialog, specify whether the rows of numbers are lanes or the columns of numbers are lanes.

If the first value in each row or column is the name of the sample, click on the “File includes sample names” checkbox. Specify whether you want to import these names by clicking on the Import checkbox to the right of this checkbox.

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(These names can then be used in any reports or sample descriptions you generate.)

If the first row or column contains Standard values, click on the “File includes standard values” checkbox. Specify whether you want to import these values by clicking on the Import checkbox to the right of this checkbox.

Finally, select the band set that the imported values will be placed in. You can create a new band set (default) or select an existing band set by clicking on the Select Band Set button.

The file name will be added to the Gels list and the band set you specified will either be added or updated. You can now perform searches on those values using the Search function.

Sorting and Selecting Gels

A database may include hundreds of gels. Two facilities exist for controlling the display of gels within the Gels list: the Sort by and Select by fields.

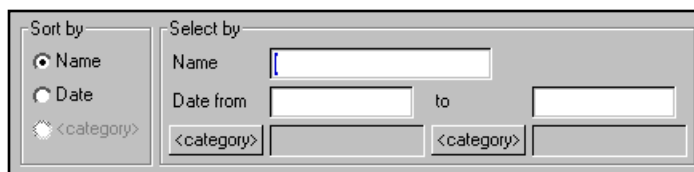
The image shows a software interface with two main sections. The 'Sort by' section on the left contains three radio buttons: 'Name' (which is selected), 'Date', and '<category>'. The 'Select by' section on the right contains a text input field with 'Name' entered, a 'Date from' field followed by 'to' and another empty field, and two '<category>' dropdown menus.

Fig.10-5. Sort by and Select by fields in the Gels list.

The Sort By field allows you to sort all the gels currently displayed in the Gels list by gel name, gel run date, or user-defined category / attribute. The Select By field allows you to restrict the field of displayed gels by name, run date, or user-defined category / attribute.

To create categories and attributes to sort and select by, see the following section.

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10.2. Categories and Attributes

User-defined categories and attributes are used throughout Diversity Database to sort and search by. They can be used to link entire gels, individual samples in gels, or band sets.

You can create categories and attributes to define any characteristics of your gel, samples, or band sets that would be useful to sort by. Typical categories might be “Enzyme,” “Primer,” “Probe,” “Type,” “Who,” etc.

To define a new category and attribute, click on a <category> button in a Sample Detail form, Gel Layout form, Band Set dialog, etc. This will open a Category pop-up box.

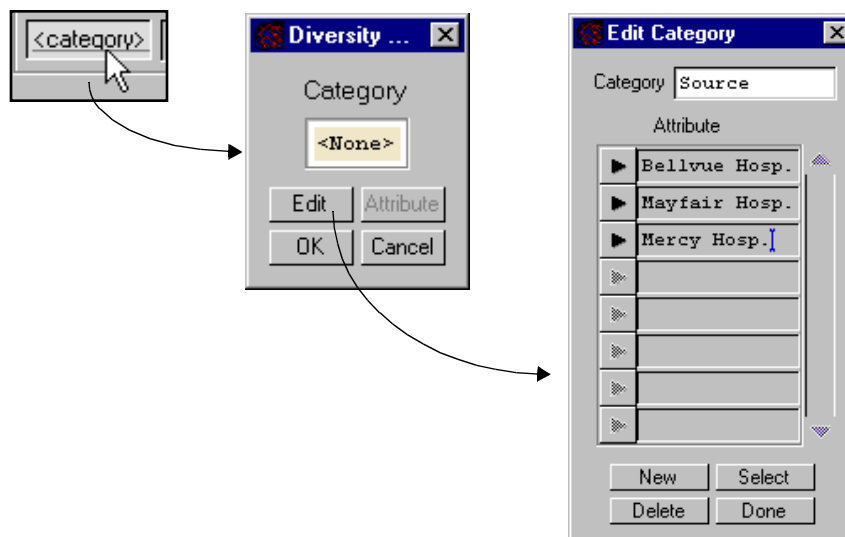


Fig.10-6. Creating a new category and attributes.

To create a category, click on the Edit button. This will open the Edit Category box.

Type the name of the new category next to the Category prompt (e.g., “Source”) and list the attributes of that category in the Attributes fields (e.g., different hospital names). The attributes will automatically be sorted

Using the Database

alphabetically within the Attributes field. When you are finished, click on Done.

To apply a category and attribute you've created, click on the <category> button in the original form again, select the category name from the pop-up list, click on the Attribute button, and select the attribute from the pop-up list.

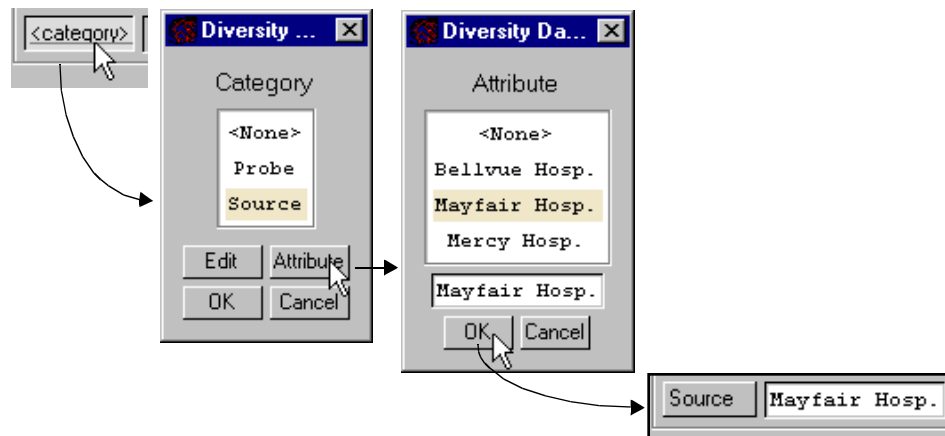


Fig.10-7. Selecting a category and attribute.

Clicking on OK in the Attributes list will specify the category and attribute in place of the <category> button in the original form.

10.3. The Samples List

Each sample in the database can be reviewed and edited from the Samples list.

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Name	Date	<category>	<category>
▶ #2 of Image.1	14-May-1996		
▶ #3 of Image.1	14-May-1996		
▶ #4 of Image.1	14-May-1996		
▶ #5 of Image.1	14-May-1996		
▶ #6 of Image.1	14-May-1996		
▶ #7 of Image.1	14-May-1996		
▶ #9 of Image.1	14-May-1996		
▶ #10 of Image.1	14-May-1996		

Fig.10-8. The Samples list.

This form also has Sort by and Select by fields (see the previous sections) that allow you to display specific groups of samples. The Sort by buttons will sort immediately; the Select by options are completed by clicking on the Select button at the bottom of the form.

Sample Cleanup

Samples are automatically entered into the database as you add gels to it, and samples can be added by typing directly into the Names list, but the list can grow out of date if gels are unlinked from the database.

Click on the Cleanup button if you have extraneous samples in your list, and Diversity Database will remove all samples that are not linked to a current gel in the database.

Using the Database

Individual Sample Info

Clicking on the pop-up button to the left of a sample name accesses a pop-up box that allows you to display the sample in its own image window or delete the sample from the database.

This pop-up box also accesses the Sample Detail form. (The Sample Detail form may also be accessed by selecting Show Sample from the Lane menu and clicking on a lane.)

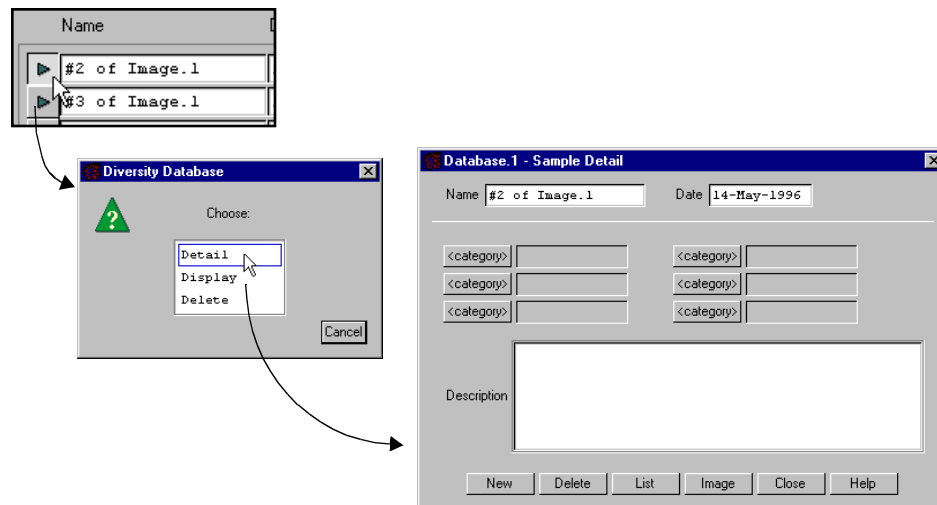


Fig.10-9. Viewing a sample detail.

Sample Detail Form

The Sample Detail form contains extensive information about a particular sample. In this form:

- Click on a <category> button to link this sample to an existing category or create a new category for it.
- Click on the New button to create a new, empty Sample Detail form.
- Click on the Delete button to remove the current sample from the Sample list.

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- Clicking on the List button to go back to the Sample list.
- Click on the Image button to open an image window of your sample lane.

10.4. The Gel Layout Form

Information about all the samples and band sets in a particular gel image can be accessed through the Gel Layout form. This form also has tools for directly assigning existing band sets to lanes in a gel image and creating templates for gels.

To open the Gel Layout form, open a gel image and click on the Gel Layout button in the Database Summary form (or select Gel Layout from the Database menu).

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Image.2 - Layout

Name: Image.2
 Description:
 <category>
 <category>

Date Run: 01-Sep-1993 #Lanes: 15 Width: 2.99mm
 Background Subtract: ☐ On ☒ Off ☐ Lane Disk Size:

Lane #	Band Set	Sample	<category>	<category>
1	std			
2	expt	#2 of Image.2		
3	expt	#3 of Image.2		
4	expt	#4 of Image.2		
5	expt	#5 of Image.2		
6	expt	#6 of Image.2		
7	expt	#7 of Image.2		
8	std			

Template Image Report
 OK Cancel Help

Fig.10-10. The Gel Layout form.

The following information can be specified for the entire gel:

- Categories and attributes information (if any) for the whole gel.
- A brief description of the gel.
- Gel run date.
- Number of lanes in the gel.
- Lane width.
- Lane-based background subtraction (if any) and disk size.

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Each lane in your gel has its own line on the Gel Layout form. The following information may be specified for a lane:

- Type of lane (standard or experimental).
- Lane number.
- Band set for that lane.
- Sample name.
- Categories and attributes information for individual lanes.

Gel Lanes

The pop-up buttons on the left side of the form offer a number of choices pertaining to the individual lanes on your gel. Clicking on one will open the Lane Choices pop-up box.

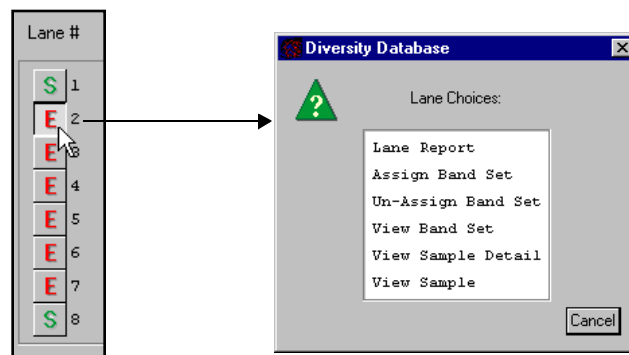


Fig.10-11. Lane Choices pop-up box.

Lane Report will open a customizable lane report form that can be printed or exported.

Assign Band Set allows you to select the specific band set that is to be applied to that lane.

Un-Assign Band Set allows you to remove the band set that is currently applied to that lane.

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View Band Set displays the Band Set form for the band set that is currently applied to that lane.

View Sample Detail displays the Sample Detail form for the current lane.

View Sample opens an image window containing the current sample and displays it with the current band types overlaid on the image.

Creating a Gel Template

If you commonly run gels using the same band sets in the same lanes, you should consider creating a template from one of your gels and applying it to new gels as you enter them into the database. Templates store the information that you enter for one image and allow you to apply it to another.

To create a template, open the Gel Layout form for a gel that you consider to be typical of your research and click on the template button. Select Make New from the pop-up box and your new gel template will appear, based on the gel you selected.

To apply a template you have created to a different gel, open the Gel Layout form for the new gel, click on the Template button, and select Apply Template from the Template Options list. A pop-up box will list the available templates; select the one you want to apply.

Edit Template allows you to make changes to an existing template.

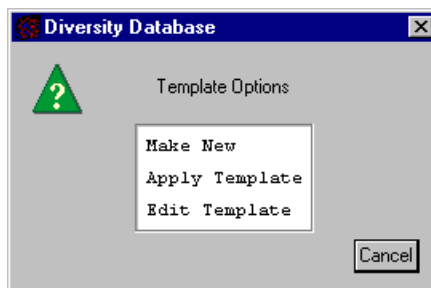


Fig.10-12. Template Options box.

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10.5. Searches

You can use the Search form to search for samples in your database based on a wide range of user-defined criteria. Searches use one of two primary search strategies: lane similarity or band set membership. In either case, for samples to be linked in the database, they must use the same band set.

Click on the Searches button in the Database Summary form to open the Search form.

Database.1 - Search

Search Recall New Delete

Search by ☒ Similarity ☐ Band Set

Include ☒ All Bands ☐ Classified Bands

Method Weighted ☒ Yes ☐ No

Date from to

<category> <category>

<category> <category>

Select Lane Ref. Sample #9 of Image.1 Similarity

Band Set	Units	Gel	Lane
play icon expt	Molecular Weight	Image.1	9

Search Close Help

Fig.10-13. The Search form.

To create a new search, specify Similarity or Band Set next to the Search by prompt, and All Bands or Classified Bands next to the Include prompt.

Using the Database

10.5.a. Similarity Searches

Similarity searches allow you to select a lane in a gel and specify the degree of similarity by which other lanes must match the lane you chose (i.e., at least 75% similar, 85% similar, etc.).

Click on the Similarity option button at the top of the form, then click on the Select Lane button and click on the lane in the image that will serve as your comparison standard.

10.5.b. Band Set Searches

Band set searches allow you to specify a particular band set and search for all samples belonging to that band set that conform to the date and category / attribute information that you specify in the middle of the form.

Select the Band Set option button, click on Select Band Set, and select the band set you want to search by from the pop-up box. Then specify the category / date information for limiting your search.

10.5.c. Search Parameters/Methods

Including All Bands in a search means that all the bands included in a sample, including unknown bands that don't belong to any band type, will be used for the purposes of comparison. Including Classified Bands only will ensure that Diversity Database ignores bands that don't belong to band types when making its comparisons.

The method of searching can be either Dice Coefficient or Jaccard Coefficient (see the following section for a detailed description of each method). Either can be specified by clicking on the Method button.

Selecting Yes or No next to the Weighted prompt will indicate whether Diversity Database should use band density as a similarity consideration when comparing band types.

10.5.d. Performing the Search

Once you have specified your search parameters and method, click on the Search button at the bottom of the form. If you want to save your search parameters for future use, enter a name next to the Search prompt at the top

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of the form and your “search” will be saved. To load an existing search, click on the Recall button and select the search from the list that appears. To remove a search you have saved, click on the Delete button.

10.6. Populations

The result of a search is a set of comparable samples which Diversity Database calls a “population.” After a search is completed, the Populations form opens, giving you access to your search results.

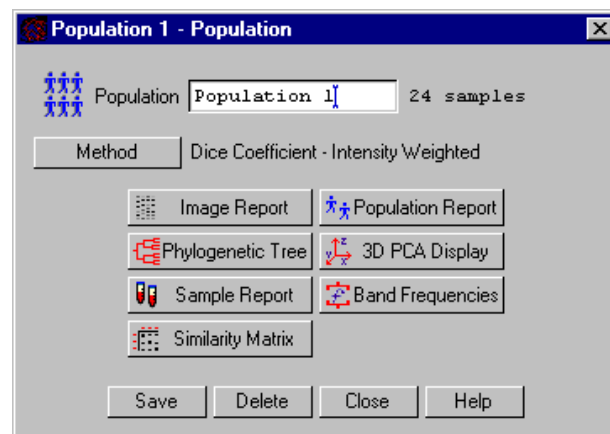


Fig.10-14. Populations form.

The Populations form contains buttons that will display your search results in a variety of reports and graphs. The available reports and graphs are: Image Report, Population Report, Phylogenetic Tree, 3-D Principle Component Analysis Display, Sample Report, Band Frequencies report, and Similarity Matrix.

Before displaying your reports / graphs, select a comparison method by clicking on the Method button and choosing Dice Coefficient, Jaccard Coefficient, or Euclidean Squared Distance.

Using the Database

10.6.a. Population Comparison Methods

Diversity Database offers three different methods for computing similarity values between samples. For simplicity, this discussion will assume one lane for each sample, and that the search was performed using the Classified Bands option.

To compute a similarity value, a vector is constructed that represents the bands found in the lane. The sample vector that is constructed depends on the type of search that was made. If the search was done on Classified Bands only, then the vector S contains B elements ($S = [s_1, s_2, s_3, \dots, s_B]$), where B is the number of band types in the lane's band set. The values for $s_1, s_2, s_3, \dots, s_B$ have the following values:

Weighting Off Search

$s_i = 1$ if the i 'th band type is found in the lane.

$s_i = 0$ if it is not found.

Weighting On Search

If the band set has a normalizing band type, then:

s_i = The normalized density of the band assigned to the i 'th band type.

$s_i = 0$ if the lane does not have a band assigned to the i 'th type.

Otherwise:

s_i = The quantity of the band assigned to the i 'th band type.

$s_i = 0$ if the lane does not have a band assigned to the i 'th type.

Let S and T be the vectors representing two samples with lanes that are in the same band set. The following similarity and distance calculation methods are available:

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Dice Coefficient

$$sim = 200 \times \frac{\sum_{i=1}^B \hat{A} \text{Min}(s_i, t_i)}{\sum_{i=1}^B (\hat{A}(s_i + t_i))}$$

$$dist = 100 - sim$$

Jaccard Coefficient

$$sim = 100 \times \frac{\sum_{i=1}^B \hat{A} \text{Min}(s_i, t_i)}{\sum_{i=1}^B \hat{A}(s_i + t_i) - \sum_{i=1}^B \hat{A} \text{Min}(s_i, t_i)}$$

$$dist = 100 - sim$$

Euclidean Distance Squared

This metric is only available from the Populations form.

$$sim = \sum_{i=1}^B \hat{A} (s_i - t_i)^2$$

$$dist = 100 \times \frac{B - dist}{B}$$

10.7. Population Reports and Graphs

The following reports and graphs are accessed from the Populations form.

Using the Database

10.7.a. Image Report

The Image Report displays a series of lane images of your population, sorted in order of decreasing similarity from the reference sample for similarity-searched populations, and alphabetically for band set-searched populations.

When you click on the Image Report button, a dialog box will pop up in which you can select your report features.

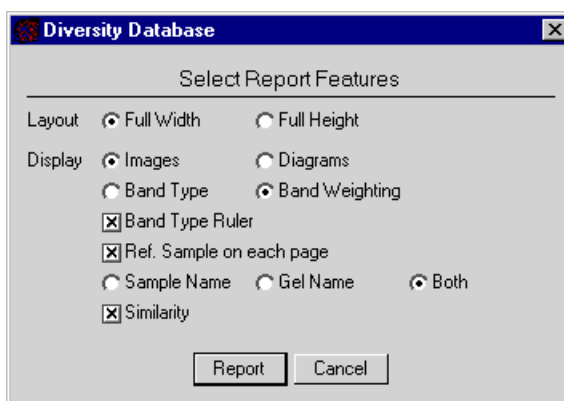


Fig.10-15. Image Report options.

If you select the Images option button, the lanes will be displayed as images. If you select Diagrams, schematic representations of the lanes will be displayed.

Select the data you want to display in the report by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

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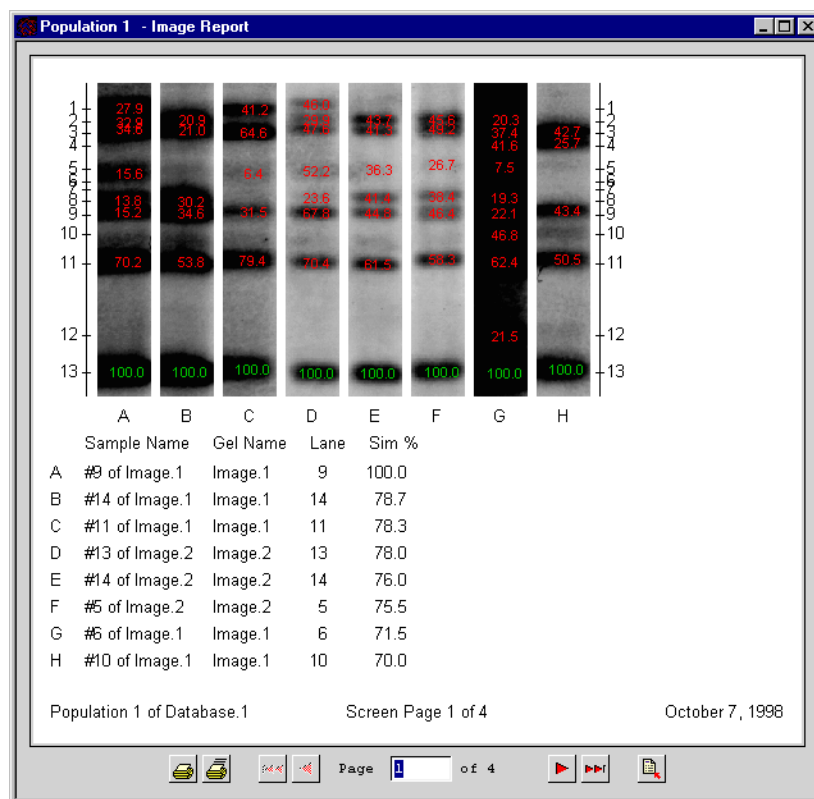


Fig.10-16. Image Report window.

The Image Report is displayed in a standard report window (see section 12.1. for information about report window options).

The Image Report window has a special feature that allows you to click on the image of a gel lane and open that lane's gel directly. This is useful if you want to edit a lane based on what you see in the report (an unassigned or incorrect band type, etc.).

Once opened, single-lane windows can be expanded, saved, etc. They are identical to normal gel image windows. Multiple-lane samples, however,

Using the Database

open special sample windows that are uneditable and unsavable, but do display all the lanes from your sample.

The Print Report dialog for image reports contains special fields for entering a title for your report.

10.7.b. Population Report

The Population Report displays all the members of your population, sorted in order of decreasing similarity from the reference sample for similarity-searched populations, and alphabetically for band set-searched populations.

When you click on the Population Report button, a dialog box will pop up in which you can select your report features.

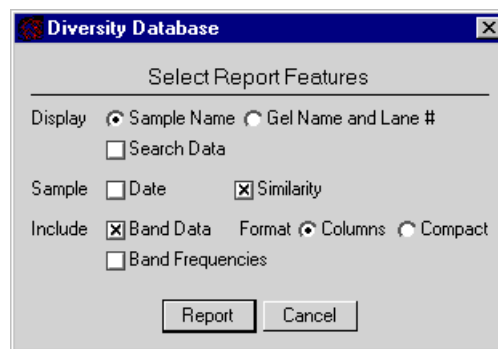


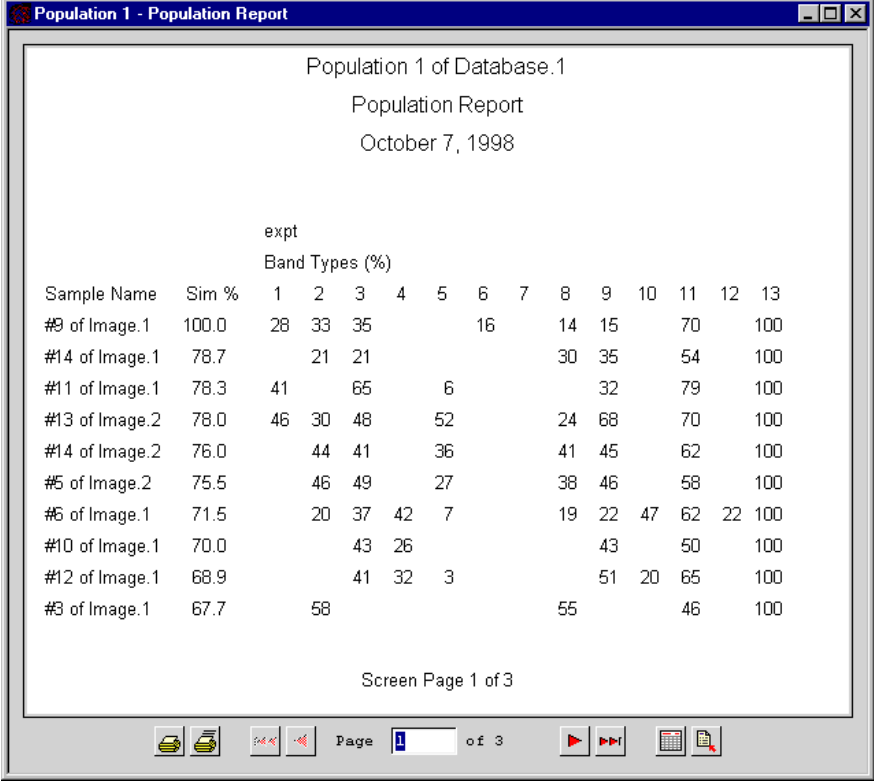
Fig.10-17. Population Report options.

Select the data you want to display in the report by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

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Population 1 of Database.1
Population Report
October 7, 1998

expt
Band Types (%)

Sample Name	Sim %	1	2	3	4	5	6	7	8	9	10	11	12	13
#9 of Image.1	100.0	28	33	35			16		14	15		70		100
#14 of Image.1	78.7		21	21					30	35		54		100
#11 of Image.1	78.3	41		65		6				32		79		100
#13 of Image.2	78.0	46	30	48		52			24	68		70		100
#14 of Image.2	76.0		44	41		36			41	45		62		100
#5 of Image.2	75.5		46	49		27			38	46		58		100
#6 of Image.1	71.5		20	37	42	7			19	22	47	62	22	100
#10 of Image.1	70.0			43	26					43		50		100
#12 of Image.1	68.9			41	32	3				51	20	65		100
#3 of Image.1	67.7		58						55			46		100

Screen Page 1 of 3

Fig.10-18. Population Report window.

The Population Report is displayed in a standard report window (see section 12.1. for information about report window options).

10.7.c. Sample Report

The Sample Report displays the samples in your population, sorted in order of decreasing similarity from the reference sample for similarity-searched populations, and alphabetically for band set-searched populations. When you click on the Sample Report button, the report will be generated.

Using the Database

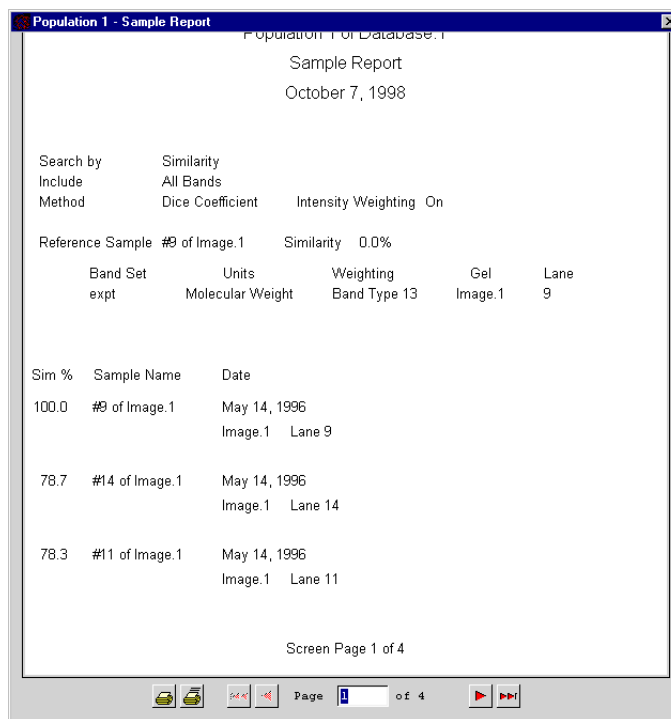


Fig.10-19. Sample Report.

The Sample Report is displayed in a standard report window (see section 12.1. for information about standard report window options).

10.7.d. Band Frequencies Report

The Band Frequencies report displays the weighted and unweighted band frequency data for each of the band types in your population. When you click on the Band Frequencies button, the report will be generated.

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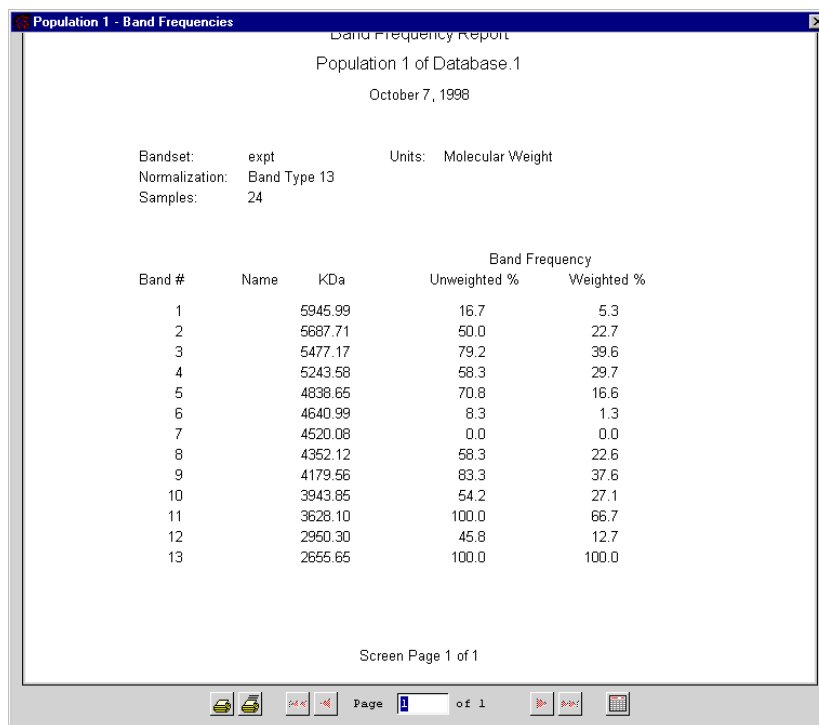


Fig.10-20. Band Frequencies report.

The Band Frequencies report is displayed in a standard report window (see section 12.1. for information about standard report window options).

10.7.e. Similarity Matrix

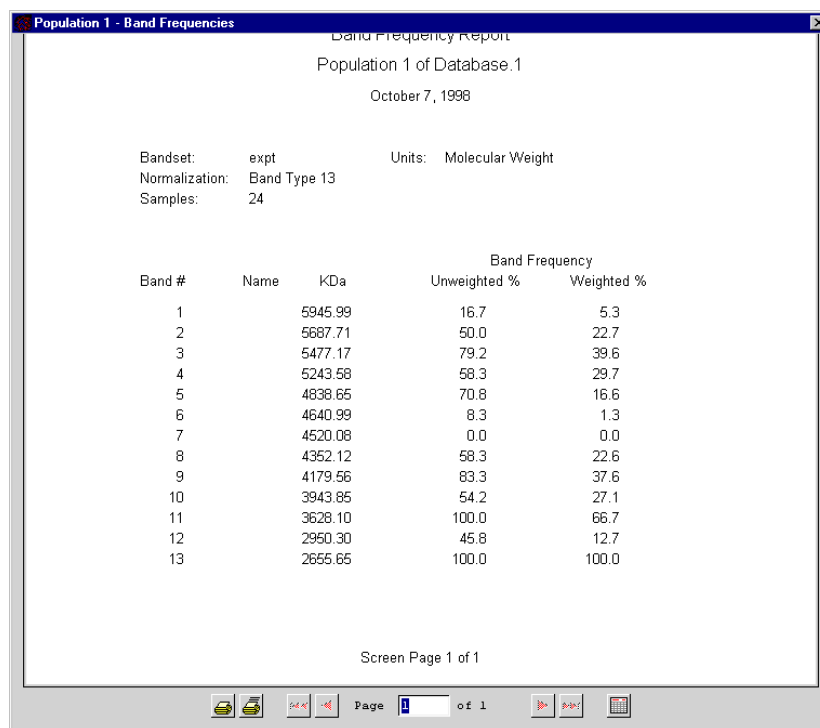
The Similarity Matrix contains all of the similarity values relating any two samples in the population. If there are N samples in the population, then the similarity matrix is an N by N matrix that is computed using one of the population comparison methods described in section 10.6.a. The matrix has the following properties:

- The diagonal elements always have values of 100. This is because a sample is always 100 percent similar to itself.

Using the Database

- The matrix is symmetrical ($M_{ij} = M_{ji}$).

When you click on the Similarity Matrix button, the report will be generated.



Band #	Name	KDa	Band Frequency	
			Unweighted %	Weighted %
1		5945.99	16.7	5.3
2		5687.71	50.0	22.7
3		5477.17	79.2	39.6
4		5243.58	58.3	29.7
5		4838.65	70.8	16.6
6		4640.99	8.3	1.3
7		4520.08	0.0	0.0
8		4352.12	58.3	22.6
9		4179.56	83.3	37.6
10		3943.85	54.2	27.1
11		3628.10	100.0	66.7
12		2950.30	45.8	12.7
13		2655.65	100.0	100.0

Fig.10-21. Similarity Matrix report.

The Similarity Matrix report is displayed in a standard report window (see section 12.1. for information about standard report window options).

10.7.f. Phylogenetic Tree

Phylogenetic trees are schematic representations of lane similarity.

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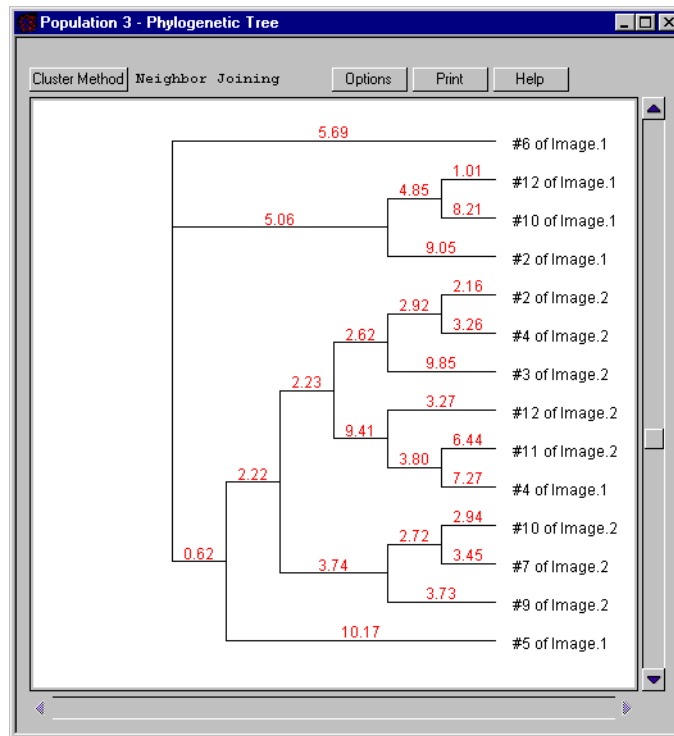


Fig.10-22. Phylogenetic Tree window.

Clicking on the Cluster Method button allows you to pick between the eight varieties of phylogenetic trees available in Diversity Database. Choosing a new method will automatically reconfigure the tree displayed in the window. Each method calculates the similarity of lanes in slightly different ways.

The report window for phylogenetic trees is slightly different than the standard report window.

The tree appears in one window with scroll bars for moving up and down and left and right.

To print the tree, click on the Print button. This will open the standard Print Report dialog box.

Using the Database

To close the window, click on the Close box in the upper right corner of the window.

Clicking on the Cluster Method button allows you to select a different clustering method for displaying the tree. This will automatically reconfigure the tree displayed in the window.

Each method calculates the similarity of samples in slightly different ways.

Neighbor Joining¹

This type of phylogenetic tree is computed is based on minimizing the total branch length at each stage of clustering. The method also finds branch lengths between nodes. The approximate distance between any two samples in this tree can be found by adding the branch lengths that connect the samples.

Other Methods^{2,3}

The following methods are based on the algorithm below:

1. Begin with n clusters—one cluster for each sample.
2. Compute the similarity matrix for the samples.
3. Convert the similarity matrix into a distance matrix d using the appropriate distance formula.
4. Join the two clusters with the minimum distance into one cluster. Compute the similarity value for this cluster.

1. For a complete explanation of the calculations and assumptions used to generate the dendrogram, please refer to Saitou and Nei, *Mol. Biol. Evol.* **4** (4): 406-425, (1987).

2. For a complete explanation of the calculations and assumptions used to generate these dendrograms, please refer to Sneath and Sokal. *Numerical Taxonomy*, San Francisco: W. H. Freeman & Company, 1973.

3. Vogt and Nagel, *Clinical Chemistry* **38** (2): 182-198, (1992).

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5. Recompute the distance matrix d using the cluster that was formed in Step 4.

Steps 4 and 5 are repeated until there is only one cluster. The difference between the methods below is based on the definition of minimum distance in Step 4, and on the method of computing the new distance matrix in step 5. In the discussion that follows, let:

p, q be indices indicating two clusters that are to be joined into a single cluster.

k be the index of the cluster formed by joining clusters p and q .

i be the index of any remaining clusters other than cluster p, q , or k .

n_p the number of samples in the p 'th cluster.

n_q the number of clusters in the q 'th cluster.

n the number of clusters in the k 'th cluster formed by joining the p 'th and q 'th cluster. $n = n_p + n_q$

d_{pq} the distance between cluster p and cluster q .

Single Linkage

This is also called Nearest Neighbor or Minimum Method.

$$d_{ki} = \min(d_{pi}, d_{qi})$$

Complete Linkage

This is also called the Furthest Neighbor or Maximum Method.

$$d_{ki} = \max(d_{pi}, d_{qi})$$

Single and Complete linkage are good algorithms for indicating outlier clusters.

Using the Database

UPGAMA

Unweighted pair group method using arithmetic averages. This is also called Weighted Average Linkage.

$$d_{ki} = \frac{\hat{p}_n}{\hat{E}_n} \diamond d_{pi} + \frac{\hat{q}_n}{\hat{E}_n} \diamond d_{qi}$$

WPGAMA

Weighted pair group method using arithmetic averages. This is also called Average Linkage.

$$d_{ki} = 0.5 \diamond d_{pi} + 0.5 \diamond d_{qi}$$

WPGAMA is a special case of UPGAMA that favors the most recent member clusters in forming new clusters.

Centroid

$$d_{ki} = \frac{\hat{p}_n}{\hat{E}_n} \diamond d_{pi} + \frac{\hat{q}_n}{\hat{E}_n} \diamond d_{qi} - \frac{(n_p \diamond n_q)}{n^2 d_{pq}}$$

Median

$$d_{ki} = 0.5 \diamond d_{pi} + 0.5 \diamond d_{qi} - 0.25 \diamond d_{pq}$$

Centroid and Median are similar to UPGAMA and WPGAMA, respectively, but the distance formula contains an additional third term. Centroid and Median methods are not monotonic hierarchical clustering algorithms. In other words, the similarity value between cluster k and any other cluster may be greater than the similarity between cluster p and cluster q. This condition occurs if the centroids (or medians) of the different clusters have approximately the same distance as the distances between the samples that make up the cluster.

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Ward's

This method attempts to minimize the information by describing a set of N samples using a fewer number of clusters.

$$d_{ki} = \frac{n_p + n_i}{n + n_i} d_{pi} + \frac{n_q + n_i}{n + n_i} d_{qi} - \frac{n_i}{n + n_i} d_{pq}$$

From our experience, we have found that Ward's method, UPGAMA, and WPGAMA give the most plausible clusters and are affected the least by samples that are outliers.

Options

Clicking on the Options button in the Phylogenetic Tree window will open the Options dialog box.

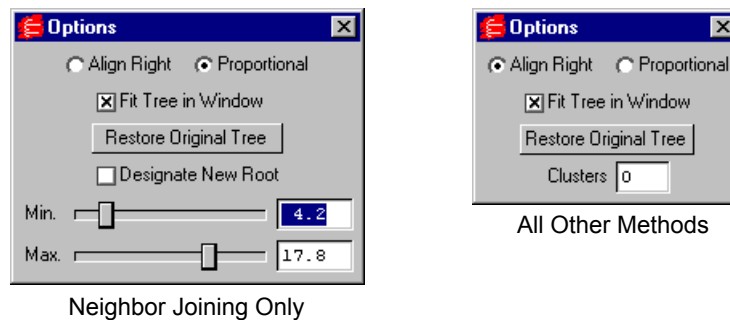


Fig.10-23. Two forms of the Options dialog box.

The box on the left has controls that are specific for the Neighbor Joining method. The box on the right is applicable to all other tree methods. Both boxes share certain features.

Each allows you to choose between aligning the tree to the right or proportionally aligning it. Right alignment uses a fixed branch length for displaying the distance between nodes; proportional alignment differs for Neighbor Joining and other tree modes.

Using the Database

In Neighbor Joining mode, proportional alignment shows branch length sizes proportional to the min./max. values (see below) set in the Options form. Other modes plot the nodes at locations determined by their similarity values.

The Fit Tree in Window option scales the tree so that the entire tree will fit onto a single printed page. With Fit Tree in Window turned off, the entire tree will appear with the correct distances between nodes preserved, and the printed tree can be tiled across multiple sheets of paper.

The Neighbor Joining Options form allows you to designate a new root node (the node at the very top of the tree), enabling you to look at the relationships between lanes from a different perspective. You do so by clicking on the Designate New Root button. This will highlight all the nodes of the tree on-screen. Choose the one you want to serve as the new root by clicking on it. The rearranged dendrogram will then be displayed. If you wish to return to the original form of the tree, click on the Restore Original Tree button.

The Min. and Max. sliders are active only for proportionally aligned Neighbor Joined trees. They allow you to adjust the Minimum and Maximum distance values between nodes on branches, so that you can adjust the display of your tree to highlight specific regions of data. Distances below the Min. or above the Max. values will be collapsed to fit on the form.

The Clusters option allows you to define the number of clusters (0 to 18) that will be identified in your tree for any non-Neighbor Joined tree. Entering a value greater than zero in this field will cause the tree to redisplay into separate clusters, identified by letters.

Populations grouped into clusters will maintain those clusters when displayed in the 3-D PCA Display form (see below).

10.7.g. 3-D PCA Display

Principal Component Analysis (PCA) is a method used for reducing the number of variables in a complex set of data. In Diversity Database, we use PCA to reduce the dimensions of the sample vectors (which generally have a high number of dimensions) to three dimensions. The samples can then be represented as points plotted on a 3-D graph. This is very useful for visually inspecting the similarity of the samples, since dissimilar samples will appear further apart on the plot than highly similar samples.

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In addition, if a phylogenetic tree was computed and the number of clusters set, the display will assign a color to each cluster and paint each sample point in the appropriate color. The display can then be used as a visual indication of the compactness of each cluster and the dissimilarity of each cluster.

Note: If you have not defined clusters in the Phylogenetic Tree form, this graph will display all its data points in the same set.

Clicking on the 3-D PCA Display button on the Populations form will display the 3-D graph.

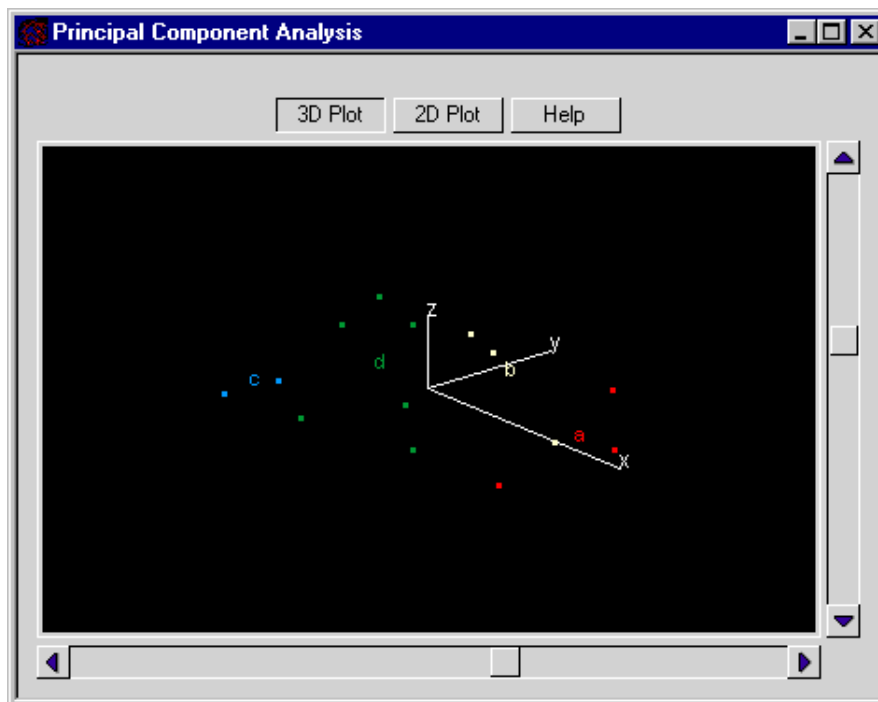


Fig.10-24. 3-D PCA Display graph, with four clusters.

Features of the 3-D PCA Display

Each cluster is displayed in a different color.

Using the Database

Each cluster's centroid is designated with a letter. Each point in the display is a sample in your population. Up to 18 clusters can be defined for a given population.

Clicking on the scroll bars on the right and bottom of the screen will rotate the display axes continuously through three dimensions to allow you to better visualize distinctions in the clusters.

If you are interested in a particular data point, click on it. The Sample Detail form (see section 10.3.) for that point will open, displaying category / attribute information for that sample and allowing you to access the samples list and open a sample image window to view the lane directly.

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11. Other Advanced Analyses

Diversity Database contains several other types of analysis tools that can be used on 1-D gels. These include volume tools, quantity calibration, differential display, and VNTR analysis. Each is explained below:

11.1. Volume Tools

You can use the Volume Tools in Diversity Database to quantitate and compare the intensity data in your images quickly and easily.

A volume is the data inside a defined boundary drawn on your image.

Volume = Sum of the intensities of the pixels within the volume
boundary x pixel area

Volume units = intensity units x mm²

Volumes are similar to contours (see section 7.7.a.), except that they are not dependent on lanes and bands.

To quantitate a particular object in your image, you first draw a box around that object using the Volume Tools. This creates a “volume object.” You can then classify your volume objects as unknowns or standards and compare them using the Volume Analysis Report and Volume Regression Curve features.

Note: The Volume Analysis Report and Volume Regression Curve are described in detail in sections 12.5 and 12.6, respectively.

To open the Volume Tools toolbar, select Analysis > Volume Tools, or click on the Volume Tools icon on your main toolbar. (The Volumes Quick Guide can be opened from the main toolbar or from the Help menu.)

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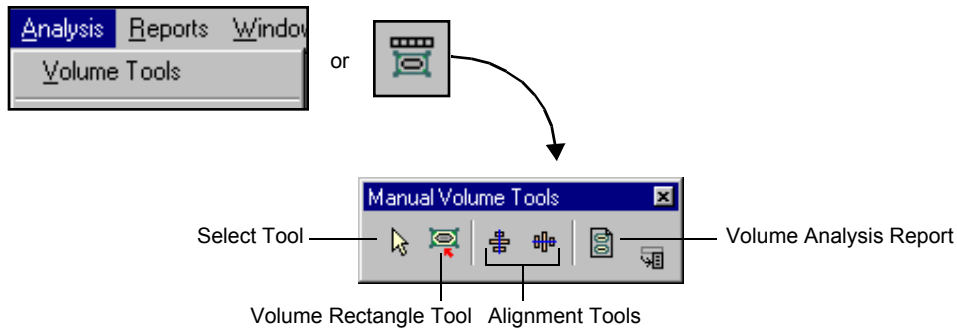


Fig.11-1. Volume Tools.

11.1.a. Creating a Volume Object

To create a volume object, click on the Volume Rectangle Tool button on the toolbar, then drag a box around the object to be quantitated.

The volume rectangle you draw should completely surround the data you want to quantitate. How big you make your box may depend on whether you are adjusting for background intensity (see section 11.1.c., Classifying a Volume Object, below).

You may want to experiment with several different rectangles drawn around the same object before selecting the volume that gives you the best quantity data.

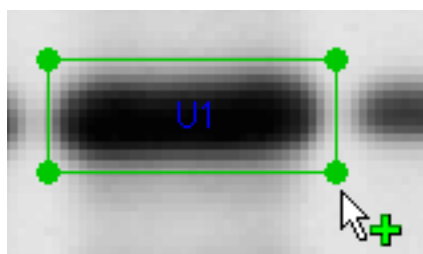


Fig.11-2. Creating a volume object.

Other Advanced Analyses

The box you draw will have a green border and anchor points (circles) at the corners. The green border indicates that the volume is selected.

If you click elsewhere on the image, the border will change to blue and the anchor points will disappear, indicating that the object is deselected.

If you move your cursor over the box—selected or unselected—the border will change to gold.

You can create as many volumes as you want.

11.1.b. Editing Volume Objects

Editing, Positioning, and Copying Individual Volume Objects

The following commands require that either the Select Tool or the Volume Rectangle Tool be assigned to your mouse. To assign one of these functions, click on the appropriate button on the Volumes Tools toolbar.

- To move a volume, position your cursor on it and drag.
- To resize a volume, click on it to select it, then position your cursor on one of the anchor points and drag.
- To copy a volume, position your cursor on it and hold down the CONTROL key while you drag.
- To rotate a volume, click on it to select it, then hold down the SHIFT key while dragging an anchor point. The volume object will pivot around its center. This is useful if the data object in your image is lying at an angle—for example, if your gel is smiling.
- To delete a volume, click on it to select it, then press the DELETE key.

Selecting and Editing Multiple Volumes

You can move, copy, or align multiple volumes. First you must first select them.

Click on the Select Tool button on your Volume Tools toolbar, then either drag a box around the objects you want to select or hold down the SHIFT key while you click on individual objects.

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Note: When dragging a box around objects to be selected, make sure that the box entirely surrounds all the objects.

Each object in the selected group will have a green border.

You can drag the selected objects to move them as a group, or click on the Alignment buttons on the Volumes Tools to align them.

Holding down the CONTROL key while you drag will copy all of the objects you have selected. Pressing the DELETE key will delete them.

11.1.c. Classifying a Volume Object

There are three types of volume objects: unknown, standard, and background. To classify a particular volume object, double-click on it with the Select Tool or Volume Rectangle Tool assigned to your mouse.

This will open the Volume Object Properties dialog box.

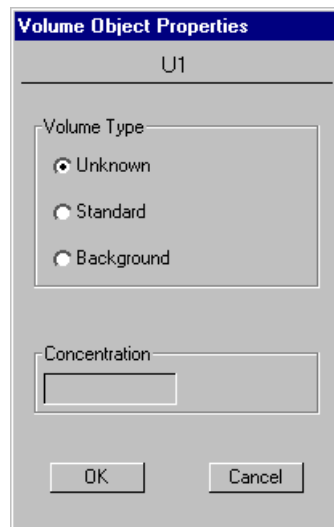


Fig.11-3. Volume Object Properties dialog box.

Other Advanced Analyses

Unknown

If you have drawn your volume around an object of unknown quantity, select the Unknown option button.

Known Standard

If you have drawn your volume around an object of known quantity, you can use it to calculate the quantities of your unknown volumes.

Select the Standard option button, then enter the quantity in the Concentration field. (Do not include units.)

After you have identified at least two standards, you can use the Volume Regression Curve feature under the Reports menu to calculate the quantities of your unknown volumes. The Volume Regression Curve and Volume Analysis Report are described in section 12.6 and 12.5, respectively.

Background

When you draw a volume object, you will probably include some background pixels around the edge of your data. These background pixels will usually have an intensity value that you do not want to include in your quantitation. Diversity Database has a mechanism for adjusting for this background intensity.

Note: You will have some residual background intensity even after you have performed background subtraction on your image. To delete background noise from your entire image, select Image > Subtract Background. To filter noise from your entire image, select Image > Filters > Filter Wizard.

To adjust for background intensity within your volumes, draw a volume object in a representative background region of your image (i.e., a region where there is no data and where the average pixel intensity appears to be the same as the background intensity surrounding your data).

Double-click on the object and select the Background option button in the Volume Object Properties dialog.

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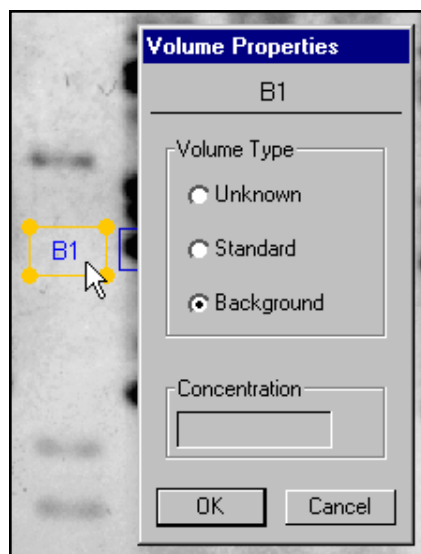


Fig.11-4. Defining a background volume object.

The average intensity value of the pixels in that background volume object will be calculated and subtracted from each pixel in your standard and unknown volumes. This will reduce the intensity of the pixels around the edge of your data in your volumes to close to zero.

The Volume Analysis Report will display both the unadjusted volume and the volume with background subtracted (adjusted volume) of your standards and unknowns, so you can see exactly how much intensity was subtracted. The Volume Analysis Report is described in section 12.5.

Note: If the region you identified as background has a higher average intensity value than your data object, then you will obtain a negative value for your adjusted volume in the Volume Analysis Report. If this happens, select a new background region that has less intensity than your data object.

Other Advanced Analyses

11.2. Calculating Quantity Standards

You can identify standards and generate a calibration curve for determining the absolute quantities of all the bands in your lanes (or cells in your arrays). (To quantitate bands outside of lanes, see section 11.1., Volume Tools.)

Note: Relative quantity is derived from band intensity, as described in previous chapters. For example, from the band intensity, you can determine that there is twice as much of Band X in Lane 4 as there is in Lane 5. However, band intensity does not tell you how many micrograms there are of Band X in the gel. Absolute quantity is calculated based on standards and a calibration curve that you create.

Using the Quantity Standards function, you can identify standards of known quantity and use these to generate a calibration curve. You can apply this curve to unknown bands in the current image as well as other images.

To create a calibration curve, the absolute quantities of at least two bands must be known. The greater the number of known bands and the wider the range of their values, the more accurate the calibration curve will be.

11.2.a. Creating and Applying a Set of Quantity Standards

Choose Quantity Standards from the Match menu. A pop-up box will appear, asking you to create a new curve or load a previously defined calibration curve (if any has been saved).

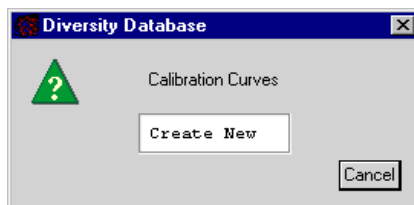
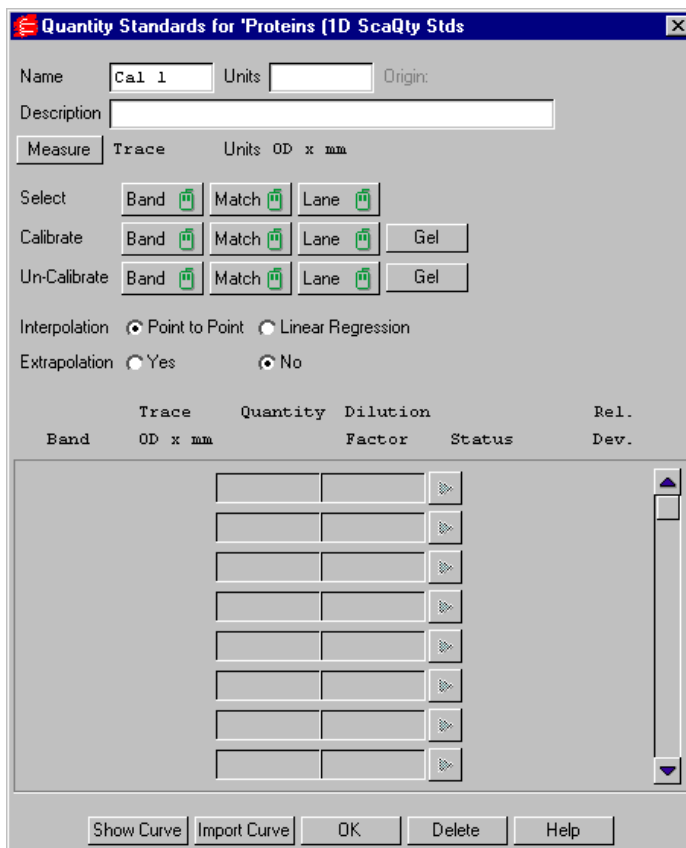


Fig.11-5. Loading a quantity calibration curve.

Selecting Create New will open the Quantity Standards dialog box.

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The dialog box is titled "Quantity Standards for 'Proteins (1D ScaQty Stds)". It contains several input fields and buttons. At the top, there are fields for "Name" (containing "Cal 1"), "Units", and "Origin:". Below these is a "Description" text area. A "Measure" button is located to the left of a "Trace" label. Below the "Trace" label is a "Units OD x mm" label. There are three rows of buttons: "Select" with "Band", "Match", and "Lane" buttons; "Calibrate" with "Band", "Match", "Lane", and "Gel" buttons; and "Un-Calibrate" with "Band", "Match", "Lane", and "Gel" buttons. Below these are radio buttons for "Interpolation" (Point to Point selected, Linear Regression) and "Extrapolation" (Yes, No selected). At the bottom, there is a table with columns: "Band", "Trace OD x mm", "Quantity", "Dilution Factor", "Status", and "Rel. Dev.". The table has 8 rows, each with input fields for the first five columns and a right-pointing arrow button in the "Status" column. At the very bottom are buttons for "Show Curve", "Import Curve", "OK", "Delete", and "Help".

Band	Trace OD x mm	Quantity	Dilution Factor	Status	Rel. Dev.

Fig.11-6. Quantity Standards dialog box.

The dialog box will open with a default name for the quantity standards (e.g., Cal 1). Enter a new name and specify the quantity value units (e.g., mg).

You can enter extra descriptive information next to the Description prompt.

The calibration curve is generated by plotting the known quantity values you provide against the intensity-based quantitation. There are several intensity-derived values that can be used for the Y axis of the plot. Click on the button labeled Measure to display a list from which to make your choice.

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Selecting Bands of Known Quantity

The dialog box includes three buttons that are used to select bands on the gel image with which to generate the calibration curve. These buttons are located next to the Select prompt.



Fig.11-7. Buttons for selecting bands of known quantity.

- To select bands one at a time, click on the Band button, then click on each of the bands of known quantity.
- If all of the bands are part of a matched group (i.e., one band type), click on the Match button, then click on one of the bands in the matched group.
- If all of the bands are in one lane, click on the **Lane** button, then click on the lane with the known quantities.

After you have selected the bands, the bands will be highlighted.

In the lower part of the Quantity Standards dialog box, the lane and band numbers of the selected bands will appear in the Band column. The values derived from their signal intensities are also displayed.

Entering the Quantity Standards

After selecting the bands to be used as quantity standards, enter their quantities in the appropriate fields in the Quantity column.

After you have entered each quantity, the band status in the Status column changes from Unknown to Known.

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Band	Trace	Quantity	Dilution	Status	Rel.
	OD x mm	Factor			Dev.
6-12	4.042	4.04		Known	6%
10-20	3.869	3.87		Calc.	N.A.
1- 1	1.702	1.70		Known	1%
4-33	1.610	1.62		Known	1%
14- 6	0.4163			Unknown	N.A.
6-22	0.2879			Unknown	N.A.

Fig.11-8. Entering the known quantities.

After you have entered a few quantities, the status of the remaining bands in the list may change to O.R., meaning that the remaining bands are out of the current range of values (based on their intensity and what you have already entered).

Alternatively, the software may automatically calculate an unknown quantity (if it is between two known quantities) and enter a value for it; in this case the Status column will indicate that the quantity has been calculated (Calc.)

In either case, you can type a quantity directly into the Quantity column and the status will change to Known.

Relative Deviation of Known Quantities

After three values have been entered, relative deviations are automatically calculated and displayed in the Rel. Dev. column. These values are based on the difference between the known value that you entered and the values that would have been calculated from the calibration curve.

If the deviation value is too high, you may want to exclude a band from the calibration curve. Click on the arrow button next to the problem band. A pop-up box offers several different band status options from which to choose.

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To remove a band from the quantity standards and omit it from calibration curve calculations, click on the Remove option. All the information regarding that band will be deleted from the quantity standards file.

Alternatively, you can select Outlier from the pop-up list. This retains the information about the band in the calibration file but does not include it when calculating the calibration curve.

11.2.b. The Calibration Curve

Calibration Curve Settings

In the Quantity Standards dialog box, select the method with which the calibration curve should be calculated. Next to the Interpolation prompt, choose either Point to Point or Linear Regression.

- **Point to Point** produces a curve in which each point is connected to the next, regardless of the shape of the resulting curve.
- **Linear Regression** (using the method of Least Squares) produces a curve which is the “best fit” of the values you provided.

Next, indicate whether the curve should be extrapolated. If you choose not to extrapolate, the highest and lowest values you entered will be the ends of the curve and you will be unable to calculate the estimated quantity for any bands that lie outside that range. If you want to extrapolate, you can calculate a quantity value for any band on the gel. But bands whose quantities are calculated from the extrapolated region of the curve may be less accurate.

Displaying the Calibration Curve

To display the calibration curve, click on the Show Curve button at the bottom of the Quantity Standards dialog box. A graph of known quantity versus measured density will be displayed in a separate window.

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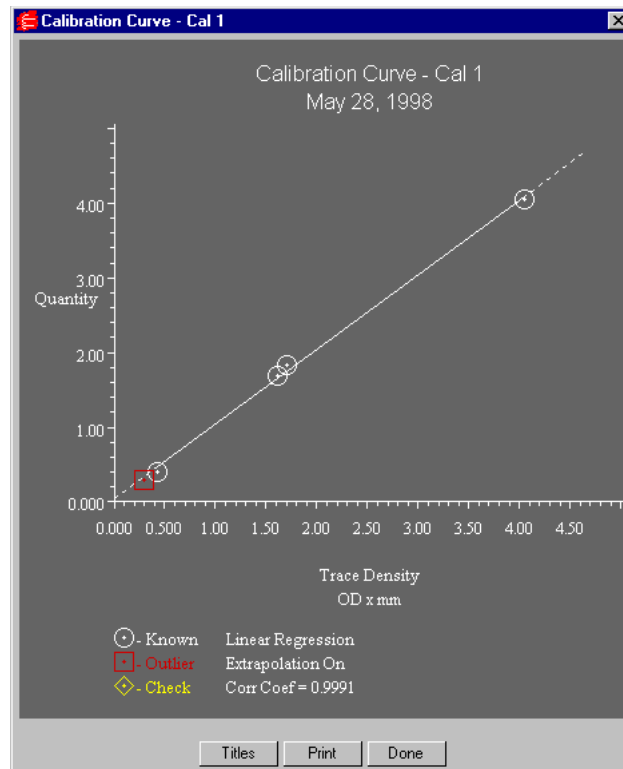


Fig.11-9. Calibration curve with titles displayed, linear regression, and extrapolation.

Points used to calculate the curve are enclosed in circles. Points that you identify as Outliers are enclosed in squares.

- To change the status of a point on the graph, click on it. The status will toggle between Known and Outlier.
- To see the legend for the different types of points on the graph, click on the Titles button.
- To print this graph, click on the Print button.
- To close the window displaying the graph, click on the Done button.

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11.2.c. Applying the Calibration Curve to Bands on the Image

Once you have selected known bands and entered values for them, you are ready to choose the bands whose quantities you want to calculate with the calibration curve. This is done using the buttons found next to the Calibrate prompt.

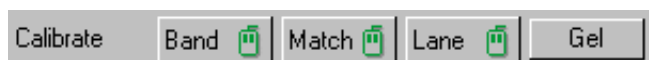


Fig.11-10. Buttons for calibrating bands of unknown quantity.

- To quantitate individual bands, click on Band, then click on the band of interest.
- To quantitate all the bands in a matched group, click on Match, then click on any of the bands in the matched group.
- To quantitate all the bands in a lane, click on Lane, then click on a lane of interest.
- To quantitate all the bands in the image, click on Gel.

Calibrated bands are highlighted after they are selected.

Unapplying the Calibration Curve

To undo the calibration of a band, match, lane, or gel, use the corresponding command next to the Un-Calibrate prompt.

11.2.d. Generating Standard Bands via a Dilution Series

One way to get a set of bands of known quantity is by starting with a stock solution and making several dilutions from it. A different dilution can be loaded onto each lane of a gel, resulting in a group of bands whose quantities can be calculated from the known quantity used to make up the stock solution and the dilution factor.

Diversity Database will perform these calculations for you if you provide it with the known quantity and the dilution factor (e.g., for a dilution factor of 1 to 10, type 1 / 10 or 0.1).

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In the Quantity Standards dialog box, next to the band that came from the undiluted stock solution, enter the known quantity.

In the Dilution Factor column, type “stock.”

Next to each of the other bands in the dialog box, enter the dilution factor in the Dilution Factor column.

Band	Trace	Quantity	Dilution	Status	Rel.
	OD x mm	Factor			Dev.
10- 3	2.060	2.06	Stock	Known	6%
8- 5	0.7845	0.72	.35	Known	5%
11- 7	0.2401	0.21	.1	Known	27%

Fig.11-11. Entering a dilution series.

The quantity of each band will be automatically calculated.

11.2.e. Quantitating Bands Using Imported Calibration Curves

A calibration curve created on one gel image can be applied to bands on other images.

Make sure that both the image you want to quantitate and the image with the calibration curve are currently open.

Click on the image you want to quantitate. Select Match > Quantity Standards and click on “create new.”

In the Quantity Standards dialog box, click on the Import Curve button. From the list displayed, choose the calibration curve you want to use to quantitate bands on the image.

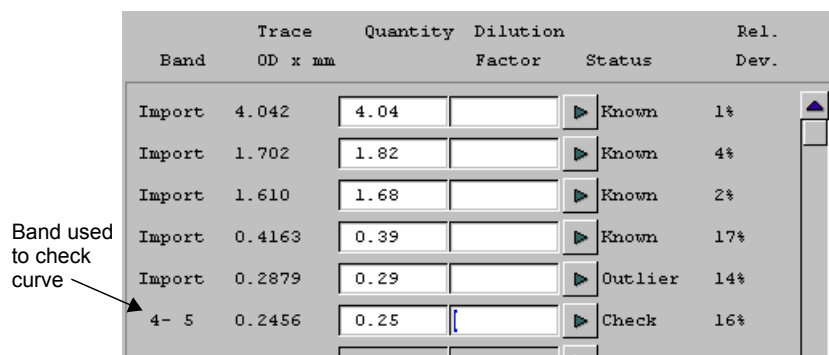
When you make your selection, the values associated with it will be displayed in the Quantity Standards dialog box. Each standard value will be labeled “Import” in the Band column.

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Checking the Imported Curve

If the quantity of one or more bands in the image is known, you can use this information to double-check the imported calibration curve to ensure that it accurately quantitates the bands on this image.

To check these known bands against the calibration curve, go to the Select prompt, click on Band, then click on the known band. Its lane number, band number, and intensity will be displayed in one of the standard values fields.



	Trace	Quantity	Dilution		Rel.
Band	OD x mm		Factor	Status	Dev.
Import	4.042	4.04		Known	1%
Import	1.702	1.82		Known	4%
Import	1.610	1.68		Known	2%
Import	0.4163	0.39		Known	17%
Import	0.2879	0.29		Outlier	14%
4- 5	0.2456	0.25		Check	16%

Fig.11-12. Checking an imported calibration curve.

When you enter the band's value, its status will change to Check. This indicates that it is a check point and is not included in the calculation of the calibration curve.

If you display the graph by clicking on the Show Curve button, these check points are enclosed in diamonds. These check points should fall on or very near the calibration curve. If that is not the case, we recommend that you *do not use* the imported standards, since the selected calibration curve will probably not give you accurate quantity data for this gel image.

11.3. Differential Display

Differential Display is a technique using mRNA and PCR amplification to study gene expression.

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Note: You must match the bands in your gel image before using this function. See section 9.1., for more information.

Differential display gels are analyzed just as any other gels in Diversity Database would be, with one important distinction. Each gel image must be normalized for quantity to a specific band type that is present in all of your lanes. (A band type that is not present in a lane is equivalent to having a zero normalized quantity band of that type.)

11.3.a. Normalization

To assign a normalized band type, select Normalize from the Match menu and then click on the band you want to normalize against. That band type will be used as the normalizing type in all subsequent calculations and reports.

11.3.b. The Differential Display Dialog Box

After you have assigned a normalized band type, you can open the Differential Display dialog box

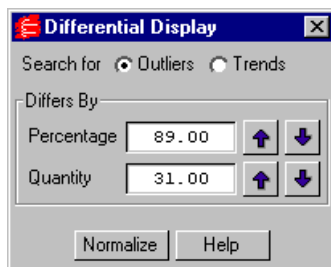


Fig.11-13. The Differential Display dialog box.

You can perform two types of searches on your gel image: an Outlier search or a Trend search. Both are determined by the Percentage and Quantity values that you specify in the Differs By field.

Select the Outlier button to search each band type for bands with normalized quantity that differ from the mean normalized quantity for its Band Type. An

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outlier is defined as a band with normalized quantity (n_q) that satisfies one of the following criteria:

$$\text{ratio} = 1.0 + (\text{percentage} / 100)$$

1. $n_q > \text{ratio} \times \text{mean}$ and
 $n_q > \text{mean} + \text{quantity}$
2. $n_q < (1 / \text{ratio}) \times \text{mean}$ and
 $n_q < \text{mean} - \text{difference}$

Select the Trend button to search for increasing or decreasing expression, represented by trends in normalized quantity across a Band Type. In Trend mode, a linear regression of (n_q) versus lane # is computed for each Band Type. The left most and right most lanes of a Band Set are determined. The normalized quantities for these lanes are calculated from the regression model. If:

$\text{abs}(\text{left most}_{n_q} - \text{right most}_{n_q}) \geq \text{difference}$ and
 $\text{Max}(\text{left most}_{n_q}, \text{right most}_{n_q}) / \text{Min}(\text{left most}_{n_q}, \text{right most}_{n_q}) \geq \text{ratio}$
then the Band Type is flagged as a trend.

Displaying the Results

Bands identified either as outliers or as belonging to a trend are displayed in white.

If a lane does not have a band assigned to a band type that is flagged as an outlier or a trend, an empty box will be drawn at the expected location of the band.

The Normalized Quantity command can be used to display a histogram of normalized quantities and the mean for any Band Type that you select.

11.4. Variable Number Tandem Repeats

If your experiments involve the use of microsatellites, VNTRs or other repeated elements, you can calculate the number of times a repeated element occurs in a band using Diversity Database.

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To calculate the VNTRs on your gel image, select VNTR Calculations from the Analysis menu. (You can also select the VNTR Quick Guide from the Help menu for guidance on analyzing gels with VNTRs.)

The Tandem Repeat Calculation dialog box will open.

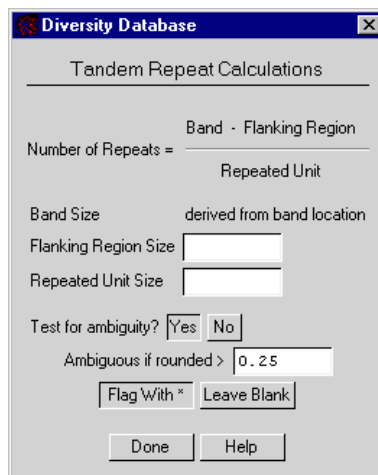


Fig.11-14. Tandem Repeat Calculations dialog box.

At the top of the dialog box is the equation used to calculate the number of times an element is repeated in a band.

The band size is determined by the position of the band on the gel image.

In the text box next to the Flanking Region Size prompt, enter the size in base pairs of the part of the fragment that does not include any repeated elements. This would include primer length and the length of any sequences that fall between the end of the primer region and the beginning of the stretch of repeats.

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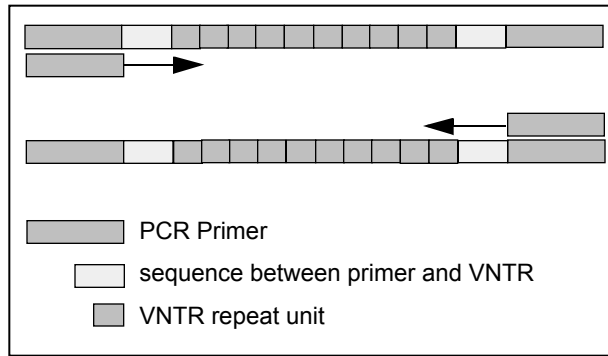


Fig.11-15. Diagram of components of a DNA fragment.

Next to the Repeated Unit Size prompt, enter the size in base pairs of the repeated element (e.g., if you are working with (CA) repeats, enter “2”).

Testing for Ambiguity

The number of repeats calculated by the software may not be a whole number, due to the limitations of exact band size determination. Since the number of times an element is repeated must be a whole number, the calculated value is rounded to the nearest whole number.

In the Tandem Repeat Calculations dialog box you have the option to “flag” those numbers that deviated from the nearest whole number. This flag serves to draw your attention to those values that may warrant further review.

Click on the Yes or No button next to the Test for ambiguity? prompt to indicate whether you want to test for ambiguity.

Next, specify what constitutes ambiguity. How much should the calculated value deviate from the nearest whole number for it to be flagged? Enter this value in the box next to the “Ambiguous if rounded >” prompt.

Finally, specify whether the tandem repeat numbers that meet the ambiguity criteria should be marked with an asterisk (*) or whether they should be left unmarked.

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When you have entered all the information in the dialog box, click on the Done button. The number of tandem repeats will be displayed next to the bands.

If the numbers have been concealed (e.g., by the Hide Overlays command), you can redisplay them by selecting VNTR Display.

11.5. Colony Counting

You can use Diversity Database to automatically count the number of white, blue, or plaque colonies in a Petri dish image.

Note: For best results, when capturing the image of a Petri dish with an imaging device, the dish should fill the imaging window. Also, images with colonies should not have asymmetric pixels. (Asymmetric pixels can be generated by densitometers and the Reduce File Size command.) The colony counting function will not work properly on images with asymmetric pixels.

Select Colony Counting from the Analysis menu to open the Colony Counting dialog box.

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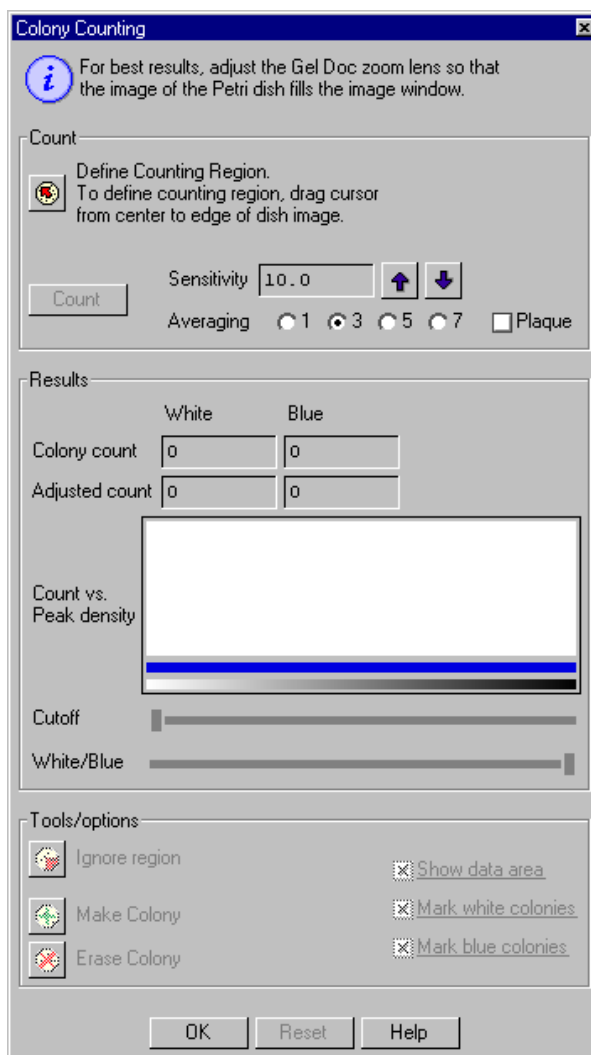


Fig.11-16. Colony Counting dialog box.

The Colony Counting dialog box has been arranged from top to bottom to guide you through the procedure.

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11.5.a. Defining the Counting Region

First, you must define the region you want to count in the Petri dish image.

Click on the Define Counting Region button in the dialog box and position your cursor at the center of dish image. Drag the cursor outward. As you drag, a blue circle will expand on the image—this defines the border of the counting region.

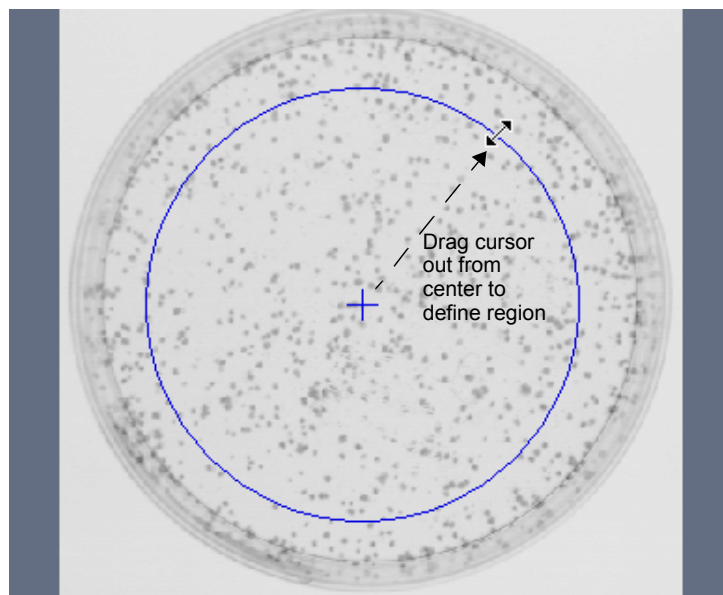


Fig.11-17. Defining a counting region.

If you make a mistake in defining your counting region, click on the Reset button at the bottom of the dialog box to start over.

Position the blue border until it is just inside the interior edge of the Petri dish.

Note: If your border disappears when you release the mouse button after dragging, check to make sure that the Show Data Area checkbox is checked. This checkbox is located at the bottom of the dialog box.

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If your circle is slightly off-center, you can reposition it by positioning your cursor on the center “target” of the circle. The cursor will change to a multidirectional arrow, and you can drag the entire circle.

To resize your counting region circle, position your cursor on the outer edge of the circle. The cursor will change to a bidirectional arrow and you can drag the border in or out.

11.5.b. Counting the Colonies

After you have positioned your circle, you are ready to automatically detect your colonies.

If you are counting plaques, click on the Plaque checkbox. (Because plaques appear as clear circles on a darker background, this checkbox must be selected for proper detection.)

Before counting, you may want to adjust the Sensitivity and Averaging parameters described below.

When you are ready to count, click on the Count button.

Sensitivity

The sensitivity setting determines the minimum signal intensity in the image that will be counted as a colony. The higher the sensitivity, the more colonies will be detected.

If the sensitivity is set too high, background noise will be erroneously detected as colonies. If the setting is too low, real colonies may be missed.

The default sensitivity setting is 10.00. If your image has faint colonies (e.g., O.D. < 0.05, counts < 2,000), you may want to increase this value to 20.00.

Averaging

Averaging affects how well the software distinguishes individual colonies from each other as well as background noise. You should select the highest value that still results in good separation of colonies (default = 3).

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A low averaging value may result in background noise being detected as colonies. A high averaging value may result in two closely spaced colonies being counted as one.

11.5.c. Displaying the Results

After you click on Count, the number of detected colonies will appear in the Results section of the dialog box in the White column.

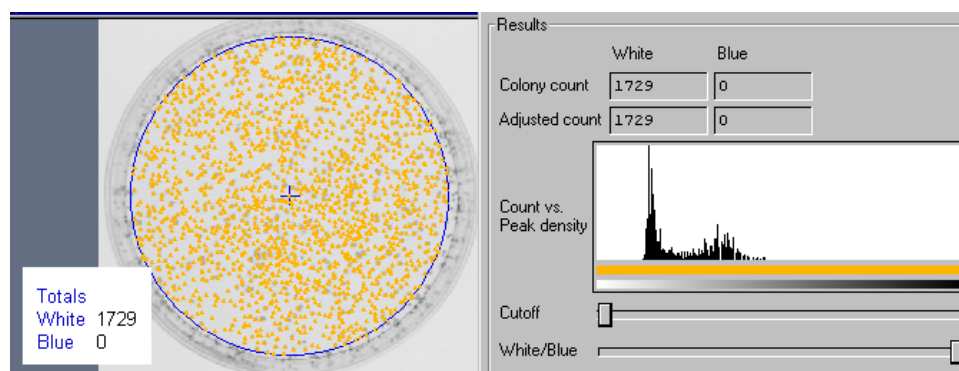


Fig.11-18. Example of a dish with white colonies.

The colonies will also appear marked as gold triangles on the image itself.

Note: If your colonies are not marked on the image, check to make sure that the Mark White Colonies checkbox at the bottom of the dialog is checked.

A text box on the image will indicate how many colonies were detected on the image.

Doing a Recount

If you want to recount using different parameters, simply change the Sensitivity and / or Averaging settings. This will erase your old count. Then click on Count again to recount.

Redrawing the counting region circle or clicking on the Reset button will also erase the count.

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11.5.d. Making and Erasing Individual Colonies

If automatic colony detection has missed or erroneously detected some colonies, you can manually mark or unmark them directly on the image using the buttons under Tools/Options.

To mark a colony, click on the Make Colony button, then click on the spot on the image that you want to identify as a colony.

To unmark a colony, click on the Erase Colony button, then click on the colony on the image that you want to unmark.

Your colony count will change accordingly.

11.5.e. Using the Histogram to Distinguish Colonies

The histogram in the Colony Counting dialog box is a graphical representation of the signal data in the image. You can use the histogram and associated sliders to reduce the number of incorrectly identified colonies and/or distinguish between white and blue colonies in the image.

Colonies Versus Background Noise

If there is a clear peak on the left end of the colony counting histogram, it is probably due to background noise in the image.

If this background noise is being detected as colonies, you can use the histogram and Cutoff slider to correct this.

Drag the Cutoff slider to the right until it is centered on the right edge of the background peak.

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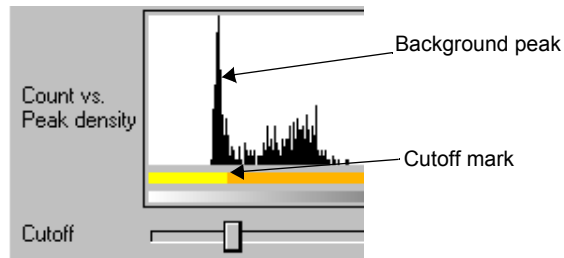


Fig.11-19. Using the Cutoff slider.

The yellow portion of the bar beneath the histogram marks the range of image data has been designated as background noise, and is not being considered for colony counting purposes. The gold portion of the bar marks white colony data range.

The colony count displayed in the dialog box and on the image should decrease. On the image, you should also see the incorrectly identified colonies disappear as you drag the slider.

White and Blue Colonies

If you know you have white and blue colonies in your image, and there are two clear peaks on the histogram to the right of the background peak, you can use the histogram to distinguish between these types of colonies.

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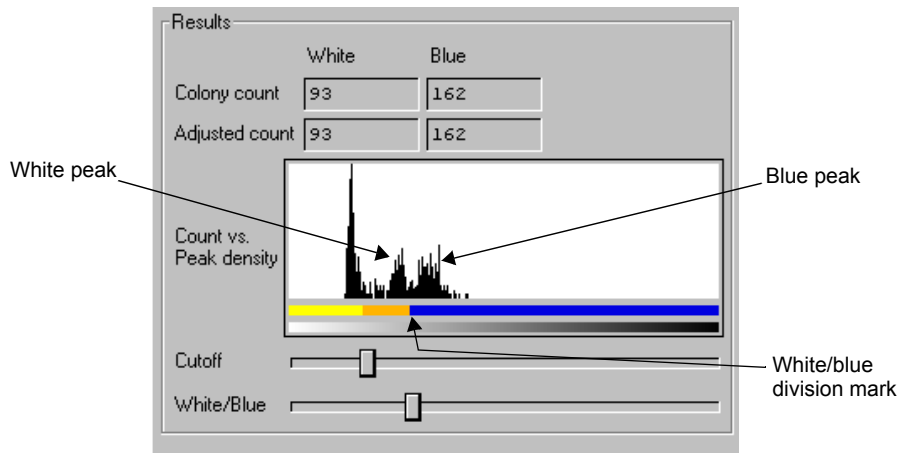


Fig.11-20. Using the White/Blue slider.

Drag the White/Blue slider to the left until it is positioned between the two peaks. The white colony data range is indicated by gold on the bar beneath the histogram, and the blue colony data range is marked with blue.

As you drag the slider, the numbers of white and blue colonies will change in the dialog box and in the text box on the image. Also on the image, you should see the marked white colonies (gold triangles) change to blue colonies (blue squares).

Note: If your blue colonies are not marked on the image, check to make sure that the Mark Blue Colonies checkbox at the bottom of the dialog is checked.

11.5.f. Ignoring a Region of the Dish

If a particular region of your Petri dish is damaged and you do not want to consider the colonies (if any) that appear there in your final count, you can use the Ignore Region function.

Click on the Ignore Region button, then position your cursor on one edge of the region you want to ignore. Drag your cursor on the image, defining the full region you want to ignore.

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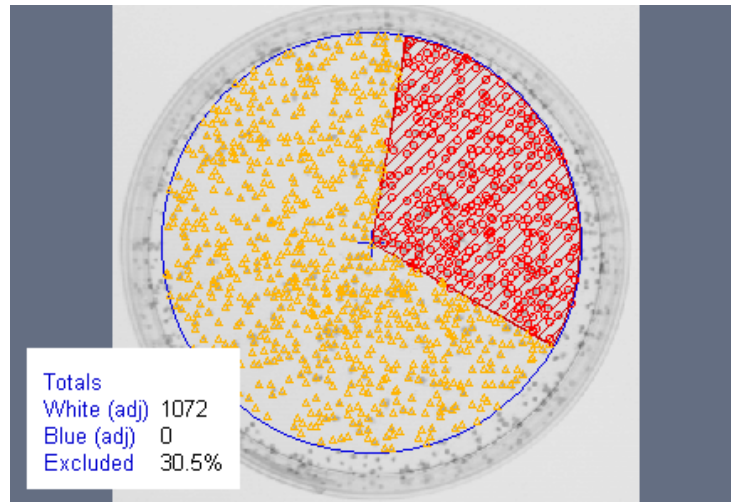


Fig.11-21. Marking a region to ignore.

As you drag, you will create a “pie slice” marked with red cross-hatching. Any colonies in this region will not be considered in the final count.

When you have defined the region, release the mouse button. If you want to change the size of the ignored region, position your cursor on the edge of the pie slice near the rim of the blue circle. Your cursor will change to a bidirectional arrow, and you can drag the edge of the pie slice.

Colony Count and Adjusted Count

After you have defined a region to ignore, two different counts will appear in the dialog box: the colony count and the adjusted count.

The colony count is the number of colonies that appear in the defined circle minus those in the ignored region.

The adjusted count is an estimate of the total colony count in the Petri dish; it uses the known colonies to extrapolate the number of colonies that might have appeared in the ignored region if it had not been damaged. The adjusted

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count is calculated based on the area of the ignored region and the density distribution of colonies in the rest of the circle.

11.5.g. Saving/Resetting Your Count

To save the count you have performed on an image, click on the OK button in the dialog box. This will dismiss the dialog box and any image overlays. Then save the image as you normally would, using the save commands in the File menu.

To view your colonies and colony counting data again, open the image and open the Colony Counting dialog box.

The Reset button will clear the Colony Counting dialog box and any changes you have made to the image. This command cannot be undone.

11.6. Arrays

The array functions in Diversity Database allow you to frame the cells in your arrays and specify their dimensions. You can then quantitate them using the Quantity Standards function described earlier in this chapter.

11.6.a. Defining an Array

The first step in defining an array is specifying the number of columns and rows in your array and creating an array frame.

Frame Arrays (on the Lane menu or Array Tools toolbar) lets you create such a frame in a single step.

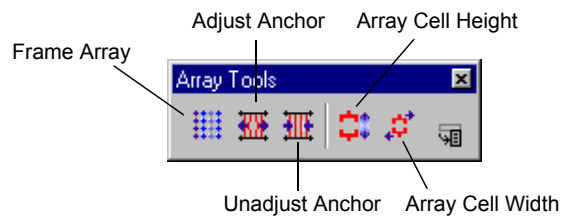


Fig.11-22. The Array Tools toolbar.

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First, select Frame Array from the Lane menu or Array Tools toolbar.

A pop-up box will ask you to enter the number of columns in the array.

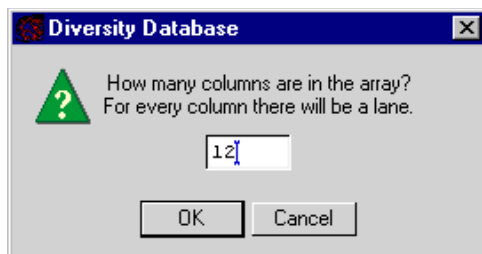


Fig.11-23. Setting number of array columns.

After you have entered the number of columns, click on OK. The next box will ask you the number of rows in your array.

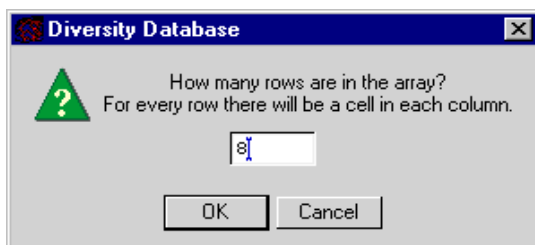


Fig.11-24. Setting the number of array rows.

Enter the number of rows and click on OK. Another pop-up box will ask you to confirm the creation of a array with the settings you specified.

If you choose to continue, an array matrix will appear on the image.

11.6.b. Adjusting the Array Matrix

Next, you will need to adjust the position of the array matrix so that it lines up with the regularly spaced cells on your dot/slot array. Select Adjust Anchors from the Array Tools toolbar or Lane menu to adjust the array frame (see section 6.1.a., Editing the Frame, for guidance on adjusting frames).

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If necessary, the Adjust Lane (see section 5.1.c) and Adjust Band (see section 6.3.b) commands can be used to “fine-tune” the placement of columns and cells within the matrix.

11.6.c. Setting Array Cell Height and Width

You may want to adjust the width and /or height of the cells in your array to maximize the area being sampled.

The sample width and height should be set to encompass as much of the cells as possible without including background region. Including background will decrease the average pixel values in the quantitation.

Use the functions Array Cell Height and Array Cell Width (on the Array Tools toolbar or Lane menu) to change the height and width of the array cells.

Note: When adjusting the cell height and width, it is useful to view your cells in “brackets mode.” To do so, select View > Band Attributes and select Brackets in the dialog box.

Array Cell Height sets the vertical distance between the brackets for every defined cell. When you select this function, a dialog box will ask you to specify the height in millimeters that should be used for all cells.

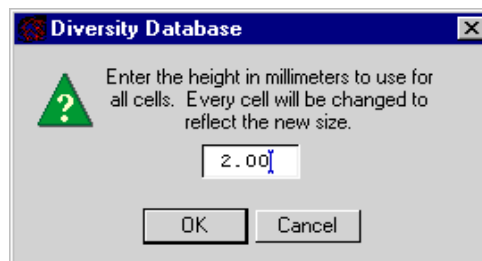


Fig.11-25. Setting the array cell height.

Changing this number and clicking on OK with your cells displayed as brackets will change your brackets in the vertical dimension.

Array Cell Width opens a dialog box in which you can set the width (in millimeters) of all the cells in your array.

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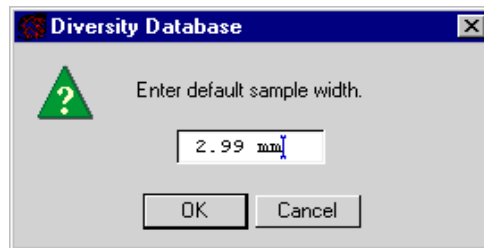


Fig.11-26. Setting the array cell width.

Changing this number and clicking on OK will change your brackets in the horizontal dimension.

To change the width of individual columns in the array, use the Lane Sampling Width function. Select Lane Sampling Width from the Lane menu and click on the particular column in the array that you want to change. A dialog box will appear, asking you to enter the new sample width for the column you selected (see section 5.1.e for more details).

You can repeat this procedure with different sample widths to find the optimal width that includes the maximum area of the cells in the column without including background noise.

12. Other Reports

Diversity Database can display and print a variety of different analysis reports in addition to the reports displayed from the Populations form (see section 10.7. for information about Populations reports). You can format these reports to include different kinds of data.

The non-Populations reports are listed on the Reports menu.



Fig.12-1. Reports menu.

12.1. The Report Window

All of the reports except Phylogenetic Tree and 3-D PCA share the same basic report window.

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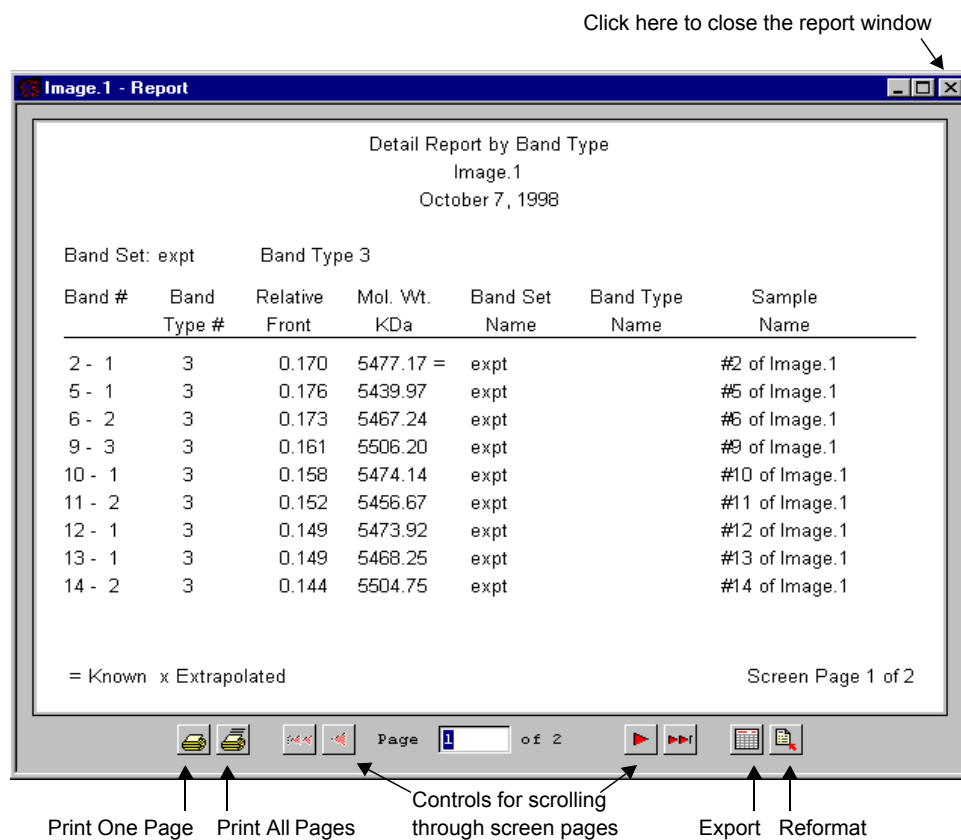


Fig.12-2. Example of a report window.

The standard report window has buttons for printing the report, scrolling through the screen pages of the report, and exporting the report to a spreadsheet application. Some report windows also have a Reformat button for displaying different report data.

To close a report window, click on the Close box in the upper right corner of the window.

Reports

Scrolling Through the Screen Pages

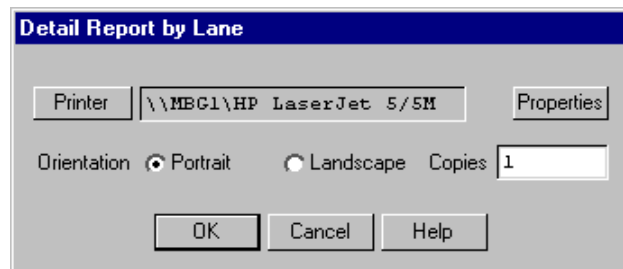
If your report has multiple screen pages, the scroll buttons in the report window will become active. You can use them to scroll to the next page, previous page, first page, or last page of the report. You can also enter a specific page number in the text field.

Printing Your Report

You can print your report using the Print One Page or Print All Pages commands in the report window.

Clicking on either of these buttons will open a smaller version of the standard print dialog (described in the next chapter).

Windows version:



Macintosh version:

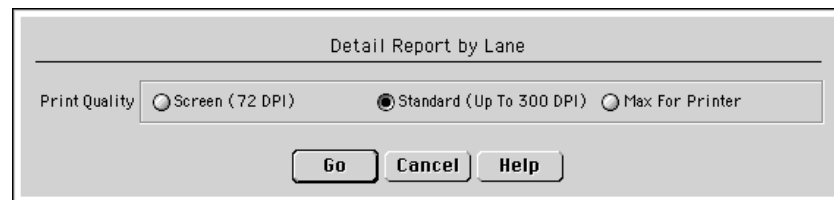


Fig.12-3. Print Report dialog box.

Print One Page sends only the current screen page to the printer. Print All Pages sends all the pages in the report to the printer.

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Note: If you select Print All Pages, the number of pages printed may be less than the number of screen pages listed in the report window. This is because Print All Pages reformats the data to make maximum use of the paper size.

Exporting Your Report Data

You can export your data to spreadsheet applications for further computation and analysis.

Click on the Export button to open the Export Report dialog box.



Fig.12-4. Export Report dialog box.

Different spreadsheet applications use different field separators when they import data. In the Export Report dialog, select Commas (for Lotus 1-2-3™) or Tabs (for Excel™) as your field separators.

You can save your data to a text file or to the clipboard by selecting the appropriate option button.

Click on the OK button to export the data.

Reports

Reformatting Your Data

If you change your mind about the data to display in your report window, click on the Reformat button (not available in all reports). This will reopen the options dialog box for your particular report.

12.2. Lane and Match Reports

There are four different lane and match reports: Lane Report, All Lanes Report, Match Report, and All Matches Report.

Lane Report generates a report on any one lane you select. First select Lane Report from the Reports menu, then click on the particular lane.

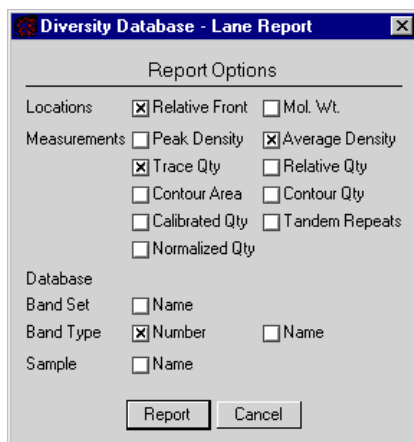
All Lanes Report generates a report on all the lanes in the current gel image.

Match Report generates a report on any band type that you select. First select Match Report from the menu, then click on a matched band.

All Matches Report generates a report on all the band types in the current gel image.

Selecting any of these commands will open the Report Options dialog box, where you can specify the type of data to be displayed in the report.

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Diversity Database - Lane Report

Report Options

Locations ☒ Relative Front ☐ Mol. Wt.

Measurements ☐ Peak Density ☒ Average Density
☒ Trace Qty ☐ Relative Qty
☐ Contour Area ☐ Contour Qty
☐ Calibrated Qty ☐ Tandem Repeats
☐ Normalized Qty

Database

Band Set ☐ Name

Band Type ☒ Number ☐ Name

Sample ☐ Name

Fig.12-5. Report Options form.

Select the data you want to display in the report by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

The report will be displayed in a standard report window (see section 13.1 above for information about standard report window options).

12.3. 1-D Analysis Report

The 1-D Analysis Report will display all the advanced analysis data (including band types, normalized quantities, amount of sample loaded, etc.) for all the lanes on your gel image. The lanes will also be ranked in similarity to the lane you initially select to generate the report.

Note: You must match the bands in your gel image before you can generate this report (see section 9.1. for more information).

Reports

Select 1-D Analysis Report from the Reports menu, then click on any experimental lane in your gel image. A dialog box will pop up, allowing you to select the report data to display.

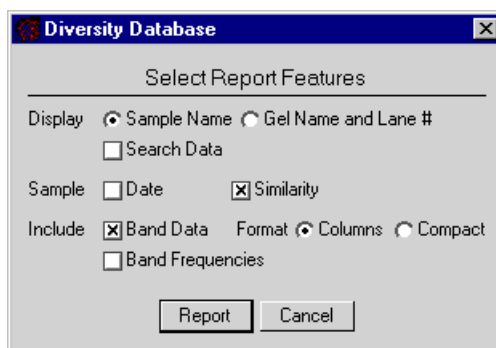


Fig.12-6. 1-D Analysis Report options.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

To display the selected data, click on the Report button.

The report will be displayed in a standard report window (see section 14.1 above for information about standard report window options).

12.4. Volume Analysis Report

The Volume Analysis Report can display all available information about your volumes.

To open this report, select Volume Analysis Report from the Reports menu or Volume Tools toolbar. The Volume Report Options dialog box will pop up, allowing you to specify the data that will appear in your report.

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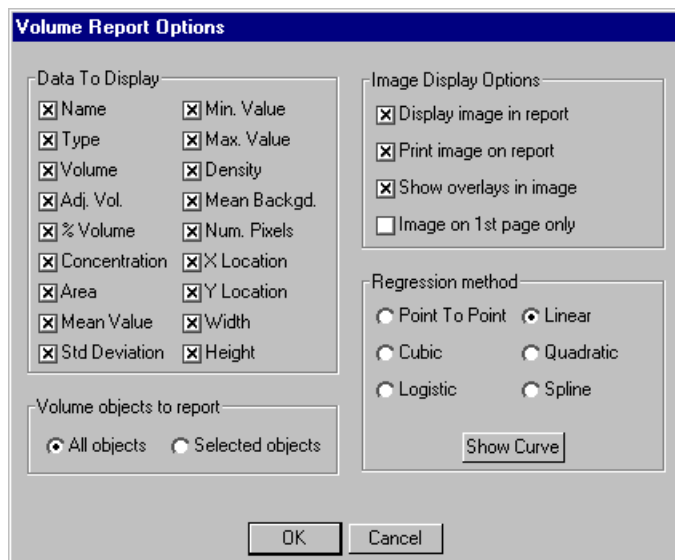


Fig.12-7. Volume Report Options dialog box.

Select the data you want to display and the image display option by clicking on the appropriate checkboxes.

Note: If you change your mind about the data to display, you can change the report options from within the report itself by clicking on the Reformat button.

In addition to the Data to Display and Image Display Options, you can choose whether to report on all your volume objects or only those objects you have selected using the Select tool. (See section 11.1.b. for information on selecting volumes.)

You can also select the regression method for displaying the Volume Regression Curve. See the following section for details. To display the curve, click on the Show Curve button.

When you have selected your report options, click on the OK button.

Reports

The report will be displayed in a standard report window (see section 14.1 above for information about standard report window options).

12.5. Volume Regression Curve

If you have defined at least two standard volumes on your image, you can display a regression curve for your volumes.

Select Volume Regression Curve from the Reports menu. This will open the Volume Regression Curve window.

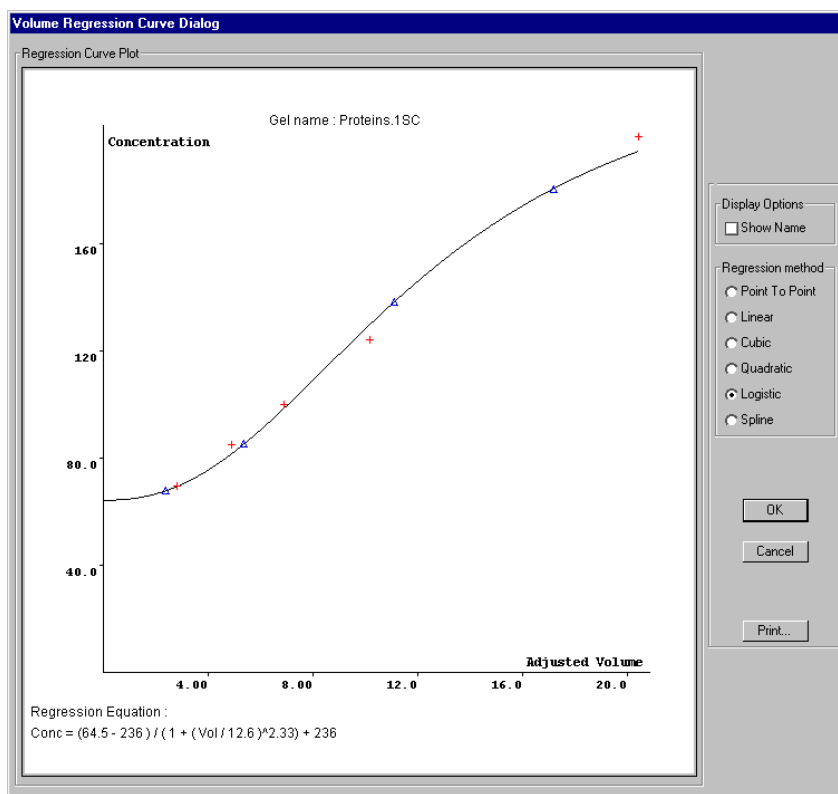


Fig.12-8. Volume Regression Curve window.

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Each standard volume is marked by a red X and each unknown is marked by a blue triangle.

The X axis is the adjusted volume and the Y axis is the concentration, based on the standards you have marked on the image.

To display the numbers and names (if any) of your volumes, click on the Show Name checkbox.

Select your preferred regression method from the list of option buttons. The regression equation for the selected method is displayed in the lower left of the window.

To print your curve, click on the Print button. This will open the standard Print Report dialog box.

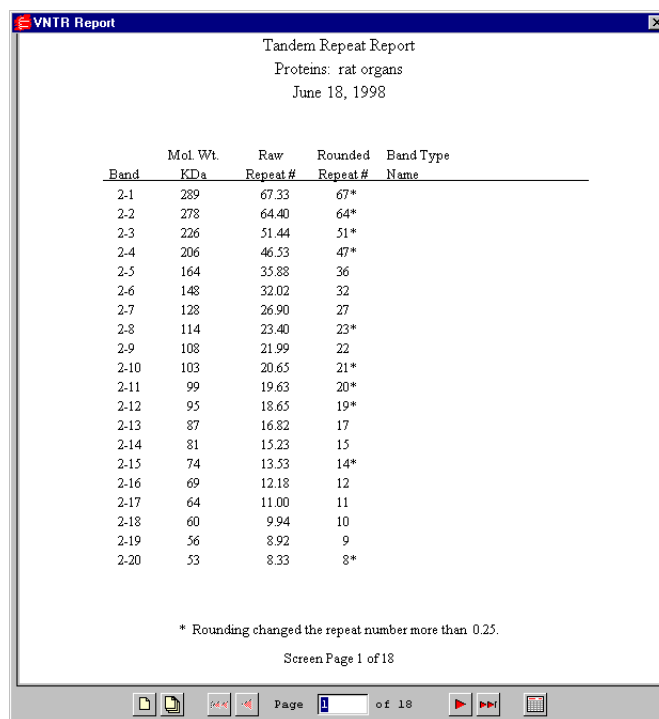
To close the Volume Regression Curve window, click on the OK button.

12.6. VNTR Report

The VNTR Report displays your VNTR data.

Select VNTR Report from the Reports menu. A pop-up box will ask if you want to display the gel image in the report.

Reports



Tandem Repeat Report
Proteins: rat organs
June 18, 1998

Band	Mol. Wt. KDa	Raw Repeat #	Rounded Repeat #	Band Type Name
2-1	289	67.33	67*	
2-2	278	64.40	64*	
2-3	226	51.44	51*	
2-4	206	46.53	47*	
2-5	164	35.88	36	
2-6	148	32.02	32	
2-7	128	26.90	27	
2-8	114	23.40	23*	
2-9	108	21.99	22	
2-10	103	20.65	21*	
2-11	99	19.63	20*	
2-12	95	18.65	19*	
2-13	87	16.82	17	
2-14	81	15.23	15	
2-15	74	13.53	14*	
2-16	69	12.18	12	
2-17	64	11.00	11	
2-18	60	9.94	10	
2-19	56	8.92	9	
2-20	53	8.33	8*	

* Rounding changed the repeat number more than 0.25.

Screen Page 1 of 18

Fig.12-9. Example of a VNTR Report.

The report is displayed in a standard report window (see section 14.1 above for information about standard report window options).

The report displays the molecular weight, the raw repeat number, and the rounded repeat number for each band in the gel image.

The raw repeat number is the number of repeats calculated using the information that you provided in the Tandem Repeat Calculation dialog box, and is likely to include fractional values. The rounded repeat number is the raw repeat number rounded to the nearest whole number.

An asterisk (*) will appear next to some of the rounded repeat numbers if you selected Test for Ambiguity and Flag with * in the Tandem Repeat Calculations dialog box. The asterisk will appear next to those numbers that

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vary from the raw repeat number by more than the ambiguity value that you specified.

13. Printing and Exporting

The commands for printing and exporting images are located on the File menu.

Reports (Sample, Phylogenetic Tree, etc.) are printed from within the individual report windows.

13.1. Printing

Diversity Database offers four different printing options under File > Print:

1. Print Image prints a copy of the image and any overlays that are currently displayed.
2. Print Actual Size prints an actual-size copy of the image.
3. Scan Report prints the image and information about its scan history, number of pixels, data range, etc.
4. Video Print prints images and reports to a video printer.*

*Video printing requires installation of the video board and cable that come with the Gel Doc 1000 / 2000 gel documentation system. The video board and cable can also be ordered separately.

Macintosh Only

On the Macintosh, specify your printer settings—including paper size, page orientation, etc.—using the File > Print > Page Setup command. This will open the standard Macintosh Page Setup dialog box.

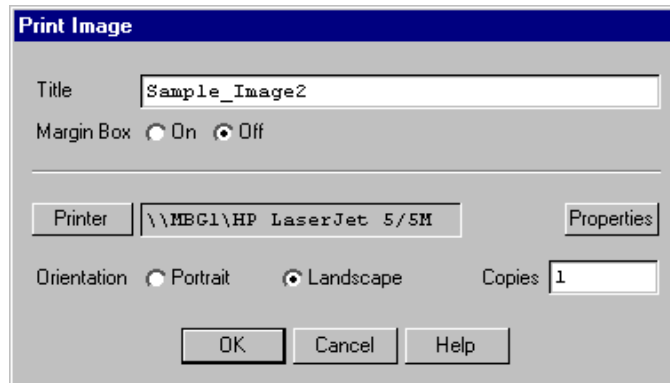
13.1.a. Print Image

Print Image prints a copy of the active image window and any image overlays that are displayed.

File > Print > Print Image opens the Print Image dialog box.

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Windows version:



Macintosh version:

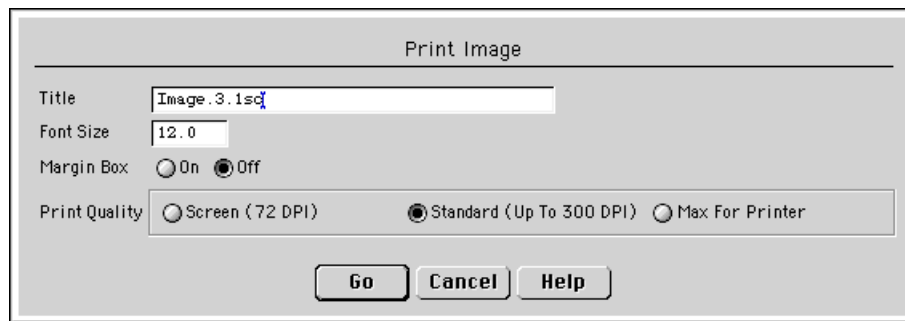


Fig.13-1. Print Image dialog box.

At the top of the dialog box, enter the title that you want to appear above the image when you print.

Choose whether or not a margin border should appear around the edge of the image by clicking either the On or Off button next to the Margin Box prompt.

Print Settings: Windows Only

Click on Printer to display a list of your available printers. After you have selected a printer, you can click on Properties to select the standard Windows print settings for that printer.

Printing and Exporting

Note: If you are using Windows 95 with certain printer drivers, you may experience difficulty printing some overlays and overlay symbols on images. While there is no single solution to this problem, you should try adjusting some settings in the Printer Properties dialog box, under the Graphics and Advanced tabs—specifically, the Dithering, DPI, Rasterizing, and RET settings.

Select the paper orientation by clicking on Portrait or Landscape next to the Layout prompt. (This can also be set in the Properties dialog box.)

Indicate how many copies should be printed in the text box next to the Copies prompt.

When you are satisfied with all the print parameters, click on the OK button to send the image to the printer.

Print Settings: Macintosh Only

Specify a font size for the image title in the text box next to the Font Size prompt. You can use font sizes ranging from 6 to 64, but we recommend you use the default setting of 12.0.

Select the printing resolution by clicking on one of the buttons next to the Print Quality prompt.

- Screen produces a printed image at the resolution of your computer screen (72 dpi).
- Standard produces a printed image at the standard resolution of a typical laser jet printer (300 dpi, unless your printer prints at a lower resolution).
- Max for Printer prints at the maximum resolution of your printer.

When you are satisfied with all the print parameters, click on the Go button to send the image to the printer.

13.1.b. Print Actual Size

You can print your images at their actual size using the Print Actual Size command.

Note: If you are using the Gel Doc or the Fluor-S, you must specify the correct image area size when capturing your images to ensure accurate 1:1

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printing. You can specify the image area size in the acquisition window for the instrument. See the chapter on each imaging device for more information.

Select Print Actual Size from the File > Print menu. The standard Print Image dialog box will open (see previous section).

13.1.c. Scan Report

Scan Report allows you to print out a single-page report of an image and its associated information. The format of the report is designed to provide a concise yet thorough summary of the most relevant features of an image for documentation purposes.

The report includes the following information:

- The image.
- The report date.
- The image title and a brief description.
- The directory location.
- The date the image was scanned.
- The type of imaging device.
- The imaging area and number of pixels.
- Pixel size.
- The intensity range, image color, and memory size.
- Image background information.
- Relevant lane and band information.

To print a scan report of a particular gel, select Scan Report from the File > Print submenu. This will open a smaller version of the standard print dialog box (see Print Image, above).

Note: TIFF images do not contain all the tagged information that would normally be included in an image file (for example, imaging device, scan date, image color, etc.). For this reason, the Scan Report may list this information as “Unknown” for imported TIFF files.

Printing and Exporting

13.1.d. Video Printing

The Video Print command allows you to print images and reports on a video printer. To create a video printout of the active window, select Video Print from the File > Print submenu.

Note: Video printing requires installation of the video board and cable that come with the Gel Doc 1000/2000 gel documentation system. The video board and cable can also be ordered separately.

Settings for the Mitsubishi P90W/P91W Video Printer

There are three settings for the Mitsubishi P90W / P91W video printer. Set Contrast to zero, Brightness to zero, and Gamma to 5.

The dip switches should stay in the orientation in which they are shipped: Pin 1 is up (on), and Pins 2–10 are down (off).

13.2. Exporting an Image in TIFF Format

To analyze your scanned image using Bio-Rad's Multi-Analyst or Molecular Analyst software, you must first save the image in TIFF format.

File > Export toTIFF Image opens a dialog box in which you can specify your export parameters.

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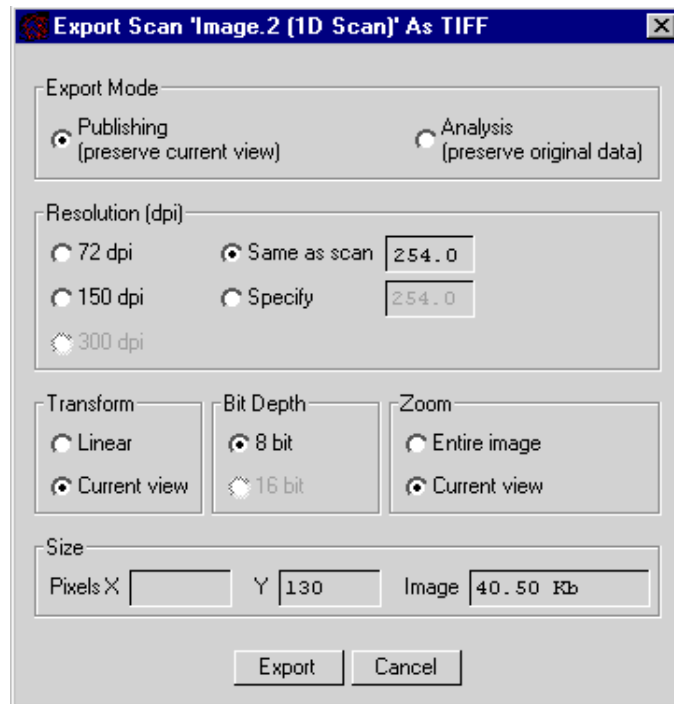


Fig.13-2. Export to TIFF Image dialog box.

Analysis Export Mode

Analysis mode exports the scan data, unmodified by Diversity Database's viewing controls.

If you select Analysis mode, the other controls in this dialog box will become inactive.

Publishing Export Mode

Publishing mode exports a TIFF image that looks like the image as it is currently displayed in Diversity Database (with whatever Transform adjustments you may have made).

Printing and Exporting

In Publishing mode, you can specify a resolution for the TIFF image by selecting 72 dpi (typical computer screen resolution), 150 or 300 dpi (standard printing resolutions), the same resolution as the scan file, or any resolution you want to specify (up to the resolution of the scan).

If you have log transformed the image, you can specify a linear transform, or preserve the current view.

If your image is 16 bits, you can compress it to 8 bits for export to TIFF.

Finally, if you are only displaying part of the image due to magnification or repositioning, you can preserve the current view or export the entire image.

Exporting the Image

The size of the pixels in the image and the file size of the image are listed at the bottom of the dialog. When you are ready to export, click on the Export button.

A Save As dialog box will open. The default file name will have a .tif extension, and the file type will indicate that this is a TIFF image. You can change the file name or select a different directory to save in.

Diversity Database will generate a grayscale TIFF image compatible with Multi-Analyst and Molecular Analyst software.

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Appendix A. Gel Doc 1000/2000



Fig.A-1. Gel Doc 1000/2000.

Before you can begin acquiring images using the Gel Doc[®] 1000/2000 gel documentation system, the system must be properly installed and connected with the host computer. See the Gel Doc hardware manual for installation, startup, and operating instructions.

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To use the Gel Doc, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install Diversity Database.

Make sure that your Gel Doc camera is turned on. If the camera is not turned on, the Gel Doc acquisition window will open but the screen will be black, and you will be unable to Video Print.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

A.1. Opening the Gel Doc Acquisition Window

To acquire images using the Gel Doc, go to the File menu and select Gel Doc.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

Appendix A. Gel Doc

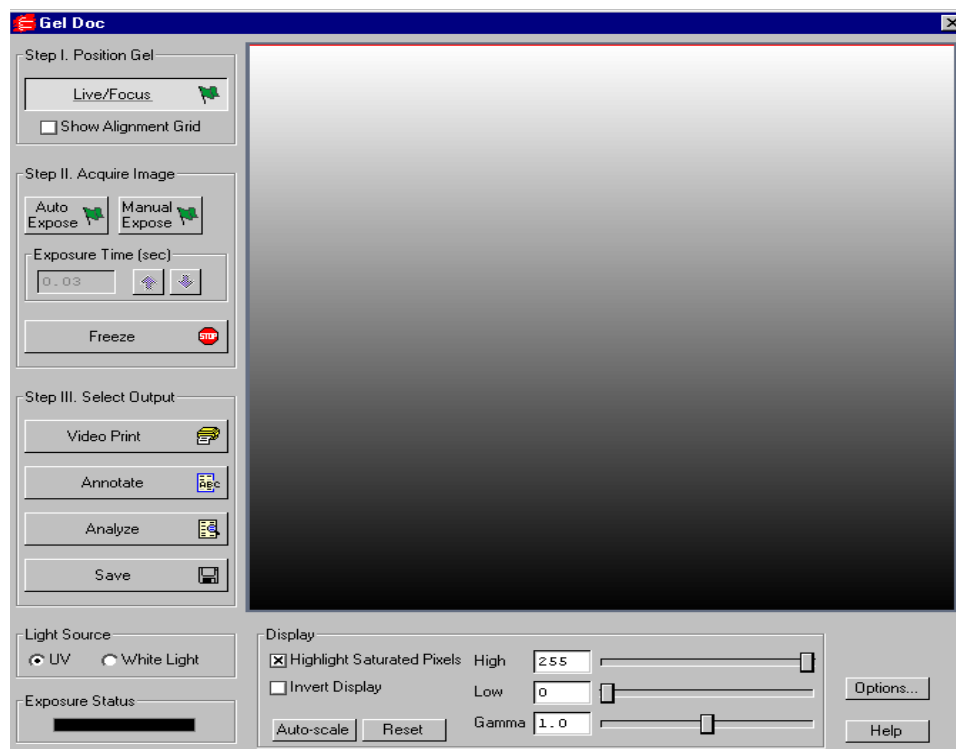


Fig.A-2. Gel Doc acquisition window

The Gel Doc video display window will open in “live” mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Gel Doc:

1. Position and focus the gel or other object to be imaged.
2. Acquire the image.

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3. Select the output.

A.2. Step I. Position Gel

The Gel Doc window will open in “live” mode, giving you a live video display of your sample. (Frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.) In this mode, the Live/Focus button will appear selected.

You can use live mode to zoom and focus the camera on your sample area and position your sample within the area. You can also select the Show Alignment Grid checkbox to facilitate positioning.

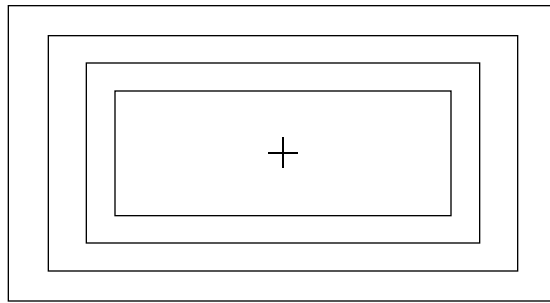


Fig. A-3. Gel Doc alignment grid.

The image should stay in focus while zooming. If focus is not maintained, refer to your Gel Doc hardware manual.

Note: After positioning your sample, you should check the Imaging Area dimensions under Options (see below) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Gel Doc box so that it is visible by the camera.

A.3. Step II. Acquire Image

For many white light applications, you can skip this step and save and print images directly from live mode.

For UV light or chemiluminescent applications, or if your aperture is closed down in white light applications, this mode will probably not give you the best possible image. To illuminate faint areas of your sample, you will need to select an exposure time for the image.

“Exposure” refers to the integration of the image on the camera CCD over a set period of time. The effect is analogous to exposing photographic film to light over a period of time. The Gel Doc has two exposure modes: Auto and Manual.

To acquire an image, you will typically perform the following steps:

1. Select Auto Expose to determine an approximate exposure time.
2. After Auto Expose is finished, “fine tune” that exposure time using Manual Expose. You can also change the camera aperture at this time.
3. Click on Freeze or any of the other buttons in the window to stop the exposure process.

Note: If you know the approximate exposure time you want (± 3 seconds), you can skip the Auto Expose step and go directly to Manual Expose.

Auto Expose

Clicking on the Auto Expose button will cancel live mode and begin an automatic exposure. (The Auto Expose button will appear selected throughout the exposure.) If you know the approximate exposure time you want (± 3 seconds), you can skip this step and go directly to Manual Expose.

In Auto Expose mode, the image is continuously integrated on the camera CCD until it has reached a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See Options below.)

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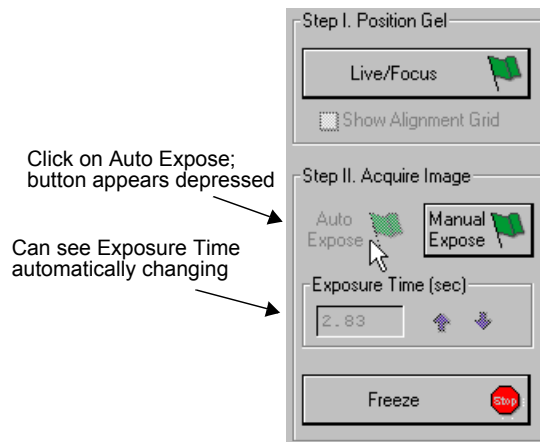


Fig. A-4. Acquiring an image, step 1: Select Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

Note: If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

Manual Expose

Manual Expose is automatically activated after Auto Expose has deactivated. Or, if you know the approximate exposure time you want, you can simply click on the Manual Expose button.

Appendix A. Gel Doc

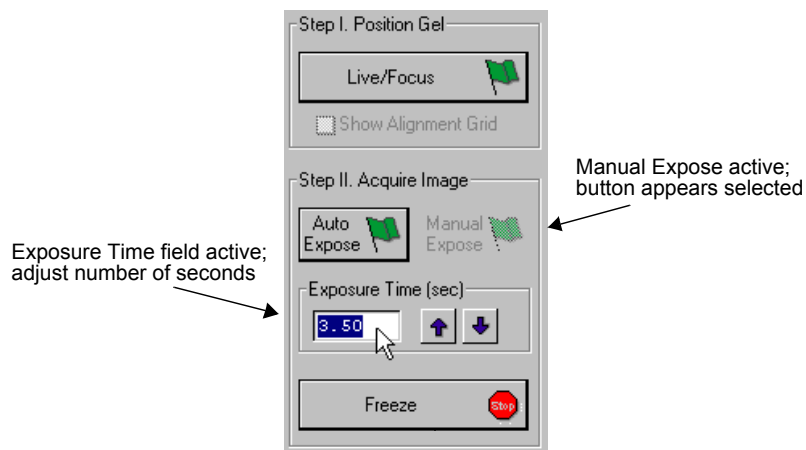


Fig. A-5. Acquiring an image, step 2: Adjust exposure time.

When Manual Expose is active (the button will appear selected), the last captured image will be displayed in the Gel Doc image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

You can adjust the exposure time by changing the number of seconds in the Exposure Time field. To do so, enter a number directly or use the arrow buttons next to the field.

Freeze

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

Note: Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Light Source, Display controls, etc.).

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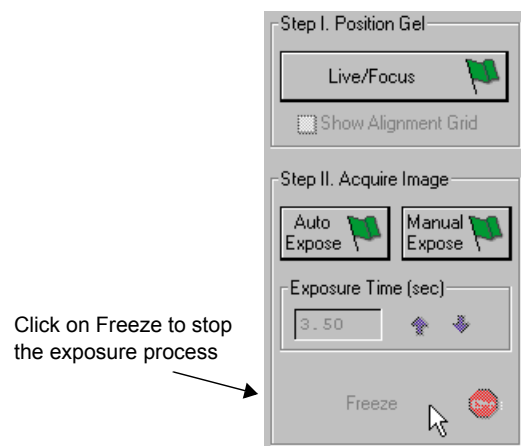


Fig. A-6. Acquiring an image, Step 3: Freeze the exposure.

A.4. Step III. Select Output

The Gel Doc window has several output options.

Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See Options, below.)

Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Annotation Tools toolbar will also pop up, allowing you to annotate your image.

Appendix A. Gel Doc

The image will not be saved until you select Save or Save As from the File menu.

Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in Diversity Database.

The image will not be saved until you select Save or Save As from the File menu.

Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also save your image as a TIFF image for export to other applications. (See Save As in section 2.4. for more information.)

A.5. Light Source

The Light Source option buttons allow you to set the type and scale of your Gel Doc data.

UV

Select this light source for fluorescent and chemiluminescent samples. The data will be measured in linear intensity units.

White Light

Select this light source for reflective and transmissive samples. The data will be measured in uncalibrated optical density (uOD) units.

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A.6. Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

A.7. Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Gel Doc window.

These controls are similar to those in the Transform dialog box.

Note: The Display controls will only change the appearance of the image. They will not change the underlying data.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa.

Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

Appendix A. Gel Doc

High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

Reset

Reset will return the image to its original, unmodified appearance.

A.8. Options

Clicking on the Options button opens the Options dialog box. Here you can specify certain settings for your Gel Doc system.

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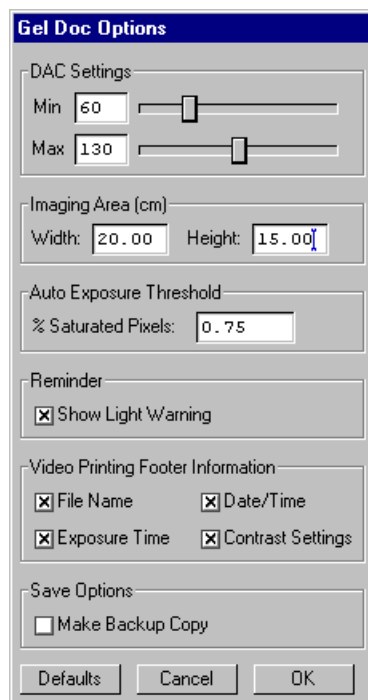


Fig. A-7. Available options in the Gel Doc acquisition window.

Clicking on OK implements any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

DAC Settings

Note: The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

Appendix A. Gel Doc

Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc acquisition window or when your system is “idle” for more than 5 minutes.

Note: If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

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With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Appendix B. GS-700 Imaging Densitometer



Fig. B-1. GS-700 Imaging Densitometer

Before you can begin scanning images with the GS-700 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-700 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-700, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open

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and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

B.1. Opening the GS-700 Acquisition Window

To acquire images using the GS-700, go to the File menu and select GS-700.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

Appendix B. GS-700

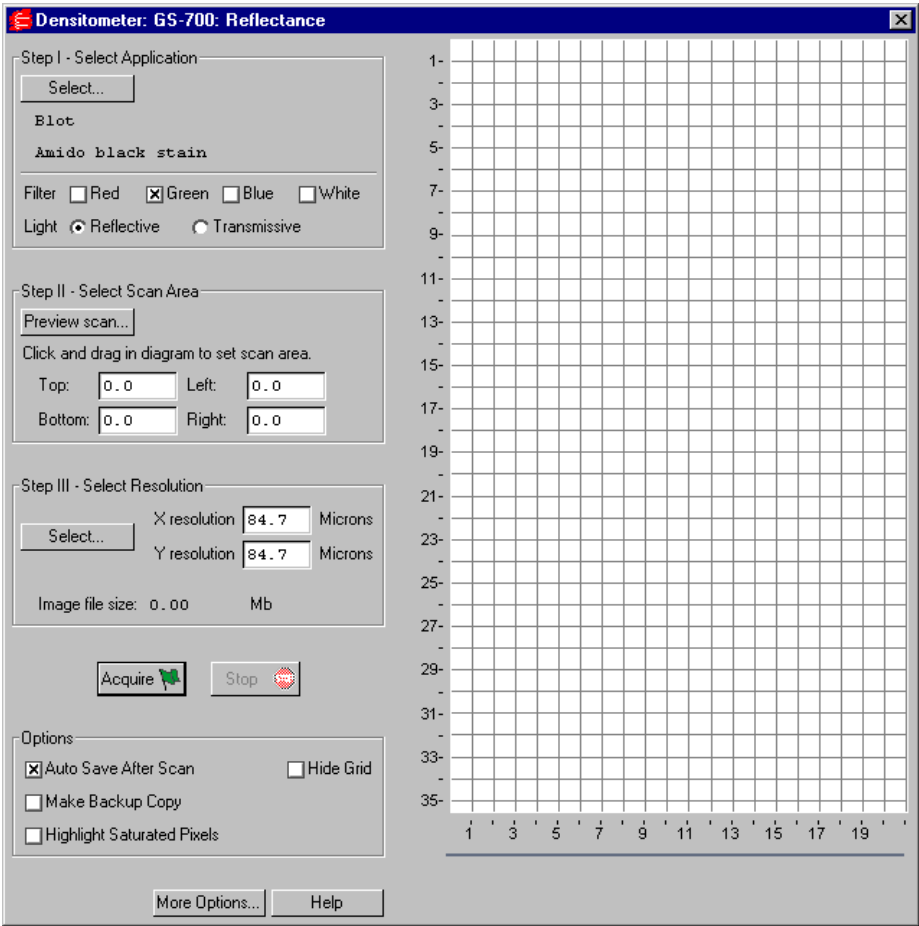


Fig. B-2. GS-700 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–35 top to bottom and 1–21 left to right if the light source is uncalibrated reflective, and 1–25 top to bottom and 1–20 left to right if the light source is uncalibrated transmissive (see below for details).

To hide the gridlines, click on the Hide Grid checkbox under Options.

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The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the GS-700:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Acquire the image.

B.2. Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

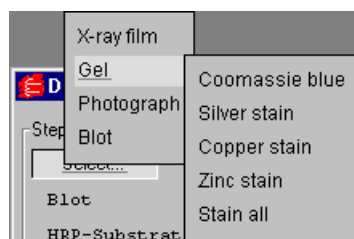


Fig. B-3. Example of the application tree in the GS-700 dialog box.

Appendix B. GS-700

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

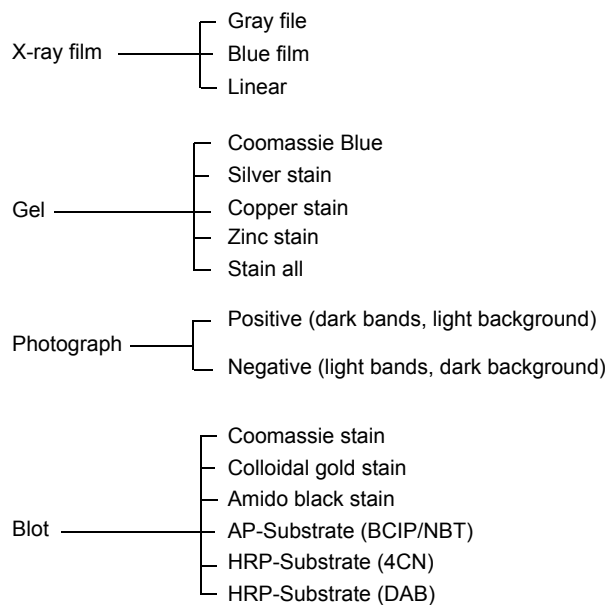


Fig. B-4. Applications available in the GS-700 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

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Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. B-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, TLC plates, and photographs. The scanning dimensions in this mode are 21 cm x 35 cm (uncalibrated).
- Transmissive mode is for scanning transparent mediums such as films, gels, and slides. The scanning dimensions in this mode are 20 cm x 25 cm (uncalibrated).

B.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Appendix B. GS-700

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

B.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

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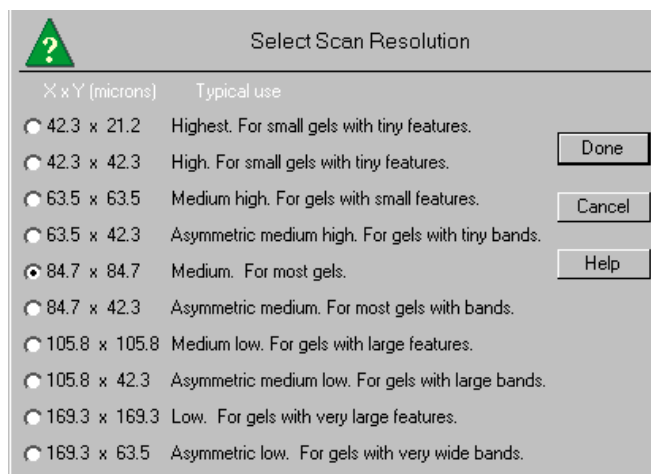


Fig. B-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section B.7. for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Appendix B. GS-700

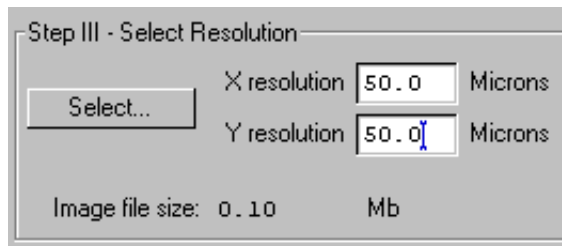


Fig. B-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

B.5. Acquire the Image

If you want to calibrate your scans, read the following section on calibration before scanning.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a Diversity Database image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

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Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

B.6. Calibration

If you have installed a calibration strip on the GS-700 scanner plate, you can automatically calibrate your scans. To order a calibration strip from Bio-Rad, contact your Bio-Rad sales representative.

See the calibration manual for details on installing the calibration strip.

To set the automatic calibration settings, click on the More Options button in the GS-700 acquisition window. This will open the Densitometer Options dialog box.

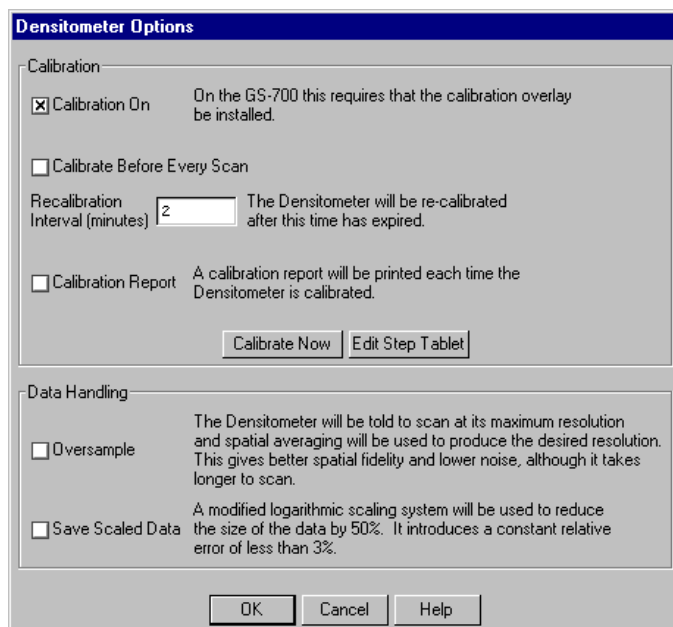


Fig. B-8. More Options in the GS-700 acquisition window.

To enable automatic calibration, click on the Calibration On checkbox.

Appendix B. GS-700

Note: Your calibration strip must be installed for this feature to work!

With Calibration On selected, the other calibration settings become active.

Calibration Strip Window

When calibration is turned on, a calibration strip window will appear below the main scanning window and the length of the main scanning window will be reduced to 29 cm reflective and 23 cm transmissive.

Every time the densitometer calibrates, an image of the calibration strip will appear in the calibration strip window.

B.6.a. Editing the Step Tablet

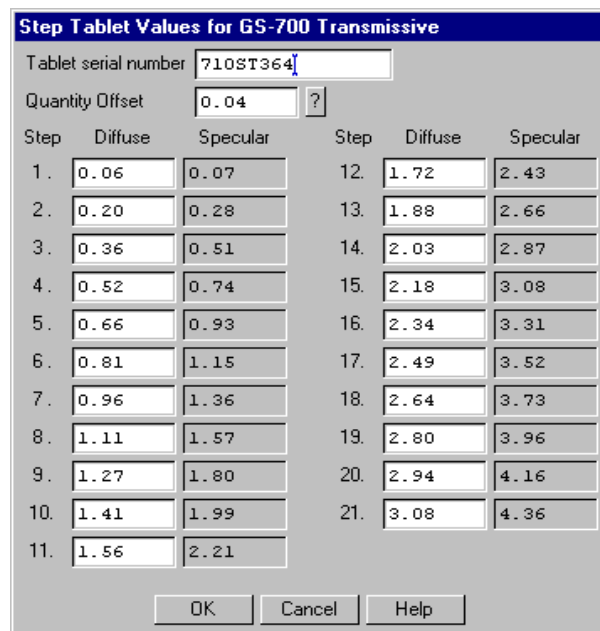
Before you can calibrate, you must enter the step tablet values for your calibration strip into the Step Tablet Values form.

In the package with your calibration strip, you will find the manufacturer's printout of the diffuse density values for each segment of your step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Scanning with incorrect step tablet values entered into the form can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-700 will open.

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The dialog box is titled "Step Tablet Values for GS-700 Transmissive". It contains a "Tablet serial number" field with the value "710ST364" and a "Quantity Offset" field with the value "0.04" and a question mark icon. Below these fields is a table with two columns of data, each with three sub-columns: Step, Diffuse, and Specular. The table contains 21 rows of data. At the bottom of the dialog box are three buttons: "OK", "Cancel", and "Help".

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.06	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig. B-9. Step Tablet dialog box.

The dialog box will indicate whether the displayed form is for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

Note: In transmissive scanning, the scanner is calibrated against a transmissive step tablet. In reflective scanning, the scanner is calibrated against a reflective step tablet, unless the reflective calibration strip is uncalibrated. If the reflective calibration strip does not have step tablet values associated with it, you can use the default values in the form.

In the Tablet Serial Number field, you can type in the serial number for the step tablet values you are entering. (The reflective step tablet will probably not have a serial number.)

Your calibration strip will have a clear plastic cover. If the quantity offset value for the clear cover is included in your calibration strip package, you can

Appendix B. GS-700

enter this number in the Quantity Offset field. Otherwise, use the default value.

Finally, enter the values in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned. You only need to enter new values if you use a new step tablet.

When you finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

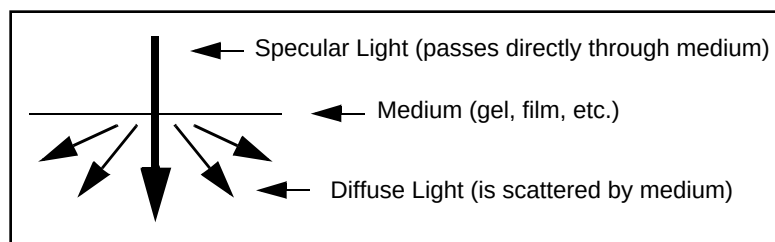


Fig. B-10. Specular and diffuse density

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Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \diamond \text{Diffuse OD}$$

B.6.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the densitometer to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: With calibration turned on, the scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

B.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-700 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

Appendix B. GS-700

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

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Appendix C. GS-710 Imaging Densitometer



Fig. C-1. GS-710 Imaging Densitometer

Before you can begin scanning images with the GS-710 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-710 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-710, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open

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and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

C.1. Opening the GS-710 Acquisition Window

To acquire images using the GS-710, go to the File menu and select GS-710.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

Appendix C. GS-710

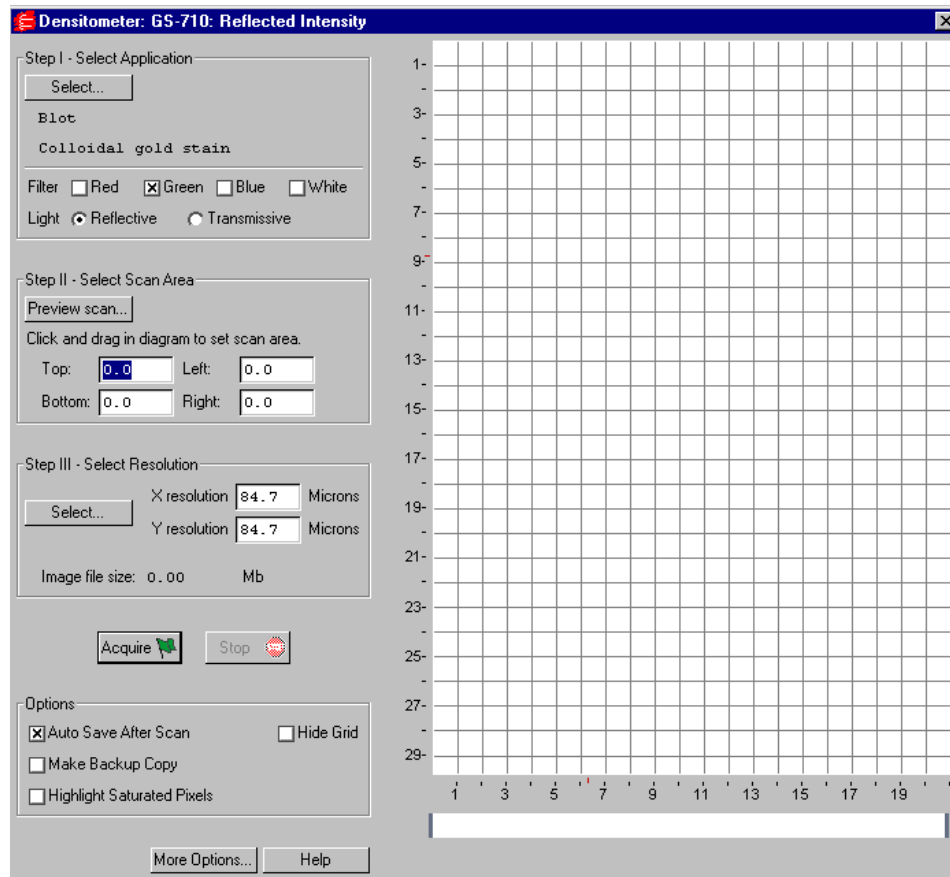


Fig. C-2. GS-710 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–29 top to bottom and 1–21 left to right if the light source is reflective, and 1–23 top to bottom and 1–20 left to right if the light source is transmissive (see below).

To hide the gridlines, click on the Hide Grid checkbox under Options.

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Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-710:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

C.2. Step I. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

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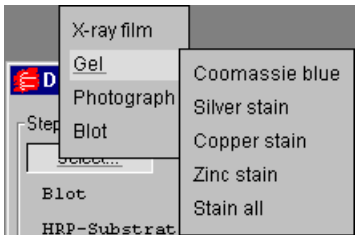


Fig. C-3. Example of the application tree in the GS-710 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

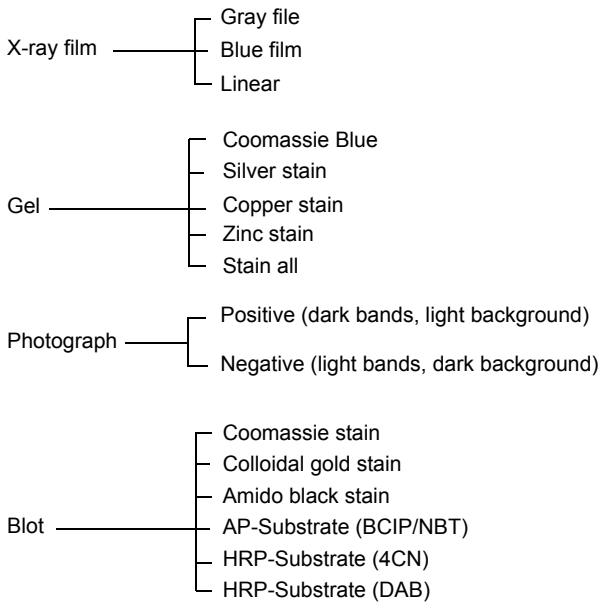


Fig. C-4. Applications available in the GS-710 acquisition window.

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Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. C-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 21 cm x 29 cm.
- Transmissive mode is for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 20 cm x 23 cm.

C.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Appendix C. GS-710

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

C.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

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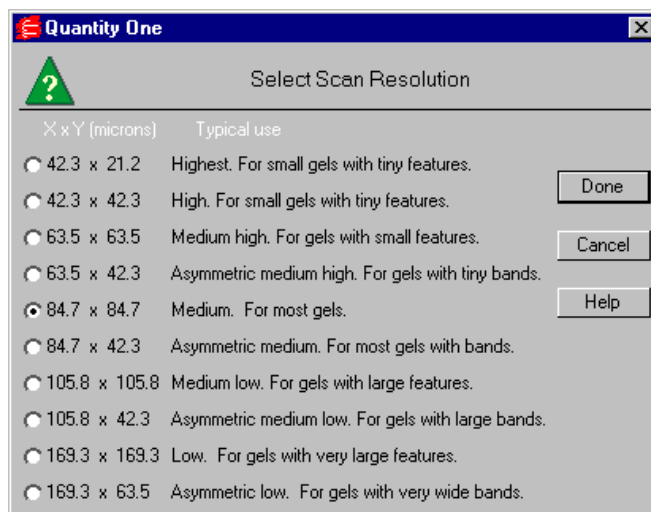


Fig. C-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section C.7., Other Options, for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

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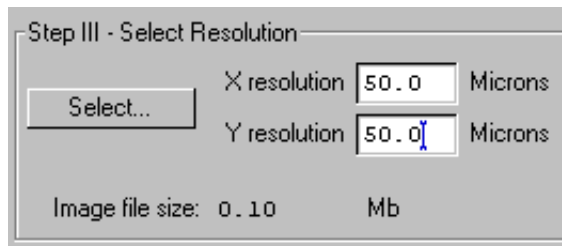


Fig. C-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

C.5. Calibration

Images scanned using the GS-710 are automatically calibrated. The scanner plate has a built-in calibration strip for both transmissive and reflective scanning.

- The transmissive calibration strip calibrates the densitometer from 0 to 3.0 O.D.
- The reflective calibration strip calibrates the densitometer to 2.0 O.D.

The first time you use the GS-710, you must select some calibration settings to ensure that your calibration is accurate.

To set the automatic calibration settings, click on the More Options button in the GS-710 acquisition window. This will open the Densitometer Options dialog box.

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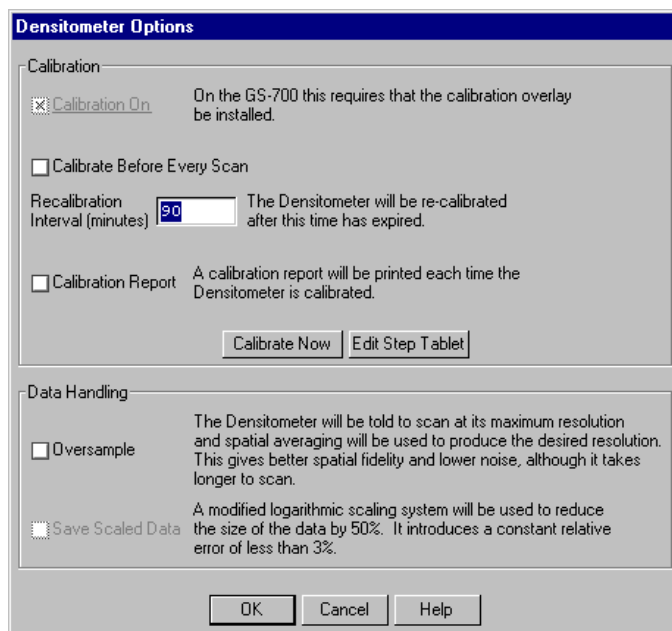


Fig. C-8. Densitometer Options dialog box.

C.5.a. Editing the Step Tablet

Before you scan for the first time, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

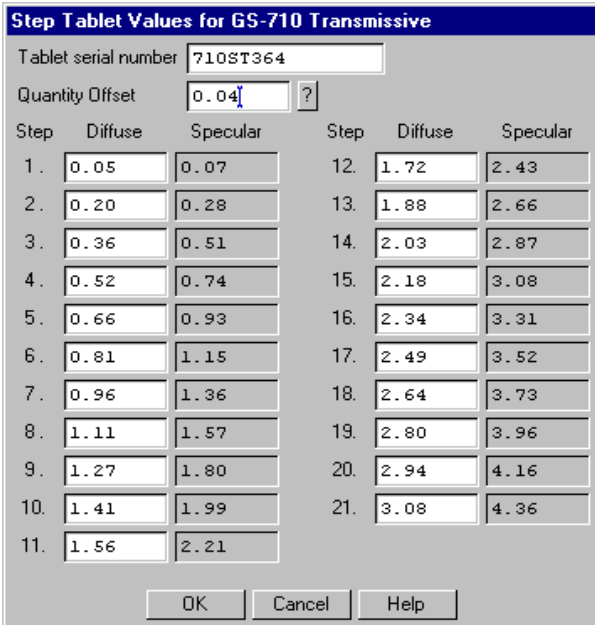
Note: In reflective scanning, the scanner is calibrated against the reflective step tablet. However, you do not need to enter the values for this step tablet; you can use the default values in the Step Tablet Values form.

Attached to the outside of your GS-710, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Appendix C. GS-710

Note: Scanning in transmissive mode with incorrect step tablet values entered into the computer can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-710 will open.



The dialog box is titled "Step Tablet Values for GS-710 Transmissive". It contains a "Tablet serial number" field with the value "710ST364" and a "Quantity Offset" field with the value "0.04" and a question mark icon. Below these fields is a table with two columns of data. The first column has steps 1 through 11, and the second column has steps 12 through 21. Each step has two input fields: "Diffuse" and "Specular". The values for the Diffuse field range from 0.05 to 3.08, and the values for the Specular field range from 0.07 to 4.36. At the bottom of the dialog box are three buttons: "OK", "Cancel", and "Help".

Step	Diffuse	Specular	Step	Diffuse	Specular
1.	0.05	0.07	12.	1.72	2.43
2.	0.20	0.28	13.	1.88	2.66
3.	0.36	0.51	14.	2.03	2.87
4.	0.52	0.74	15.	2.18	3.08
5.	0.66	0.93	16.	2.34	3.31
6.	0.81	1.15	17.	2.49	3.52
7.	0.96	1.36	18.	2.64	3.73
8.	1.11	1.57	19.	2.80	3.96
9.	1.27	1.80	20.	2.94	4.16
10.	1.41	1.99	21.	3.08	4.36
11.	1.56	2.21			

Fig. C-9. Step Tablet dialog box.

The dialog box will indicate whether the values are for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

If you are in reflective mode, you do not need to change the default step tablet values.

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If you are in transmissive mode, you do need to enter the correct step tablet values that were shipped with your GS-710. You can type the serial number for these values into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-710. This value should be set to zero.

Next, enter the values for the transmissive step tablet in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned.

When you are finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

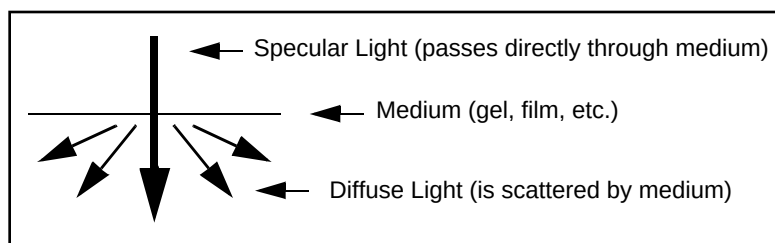


Fig. C-10. Specular and diffuse density

Appendix C. GS-710

Diffuse density values are converted to specular optical density units according to the following formula:

$$\text{Specular OD} = 1.4 \diamond \text{Diffuse OD}$$

C.5.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the GS-710 to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: The scanner will automatically recalibrate each time you change your filter or your reflective/transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

C.6. Acquire the Image

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a Diversity Database image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

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Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

C.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-710 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

Appendix C. GS-710

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

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Appendix D. Fluor-S Multimager

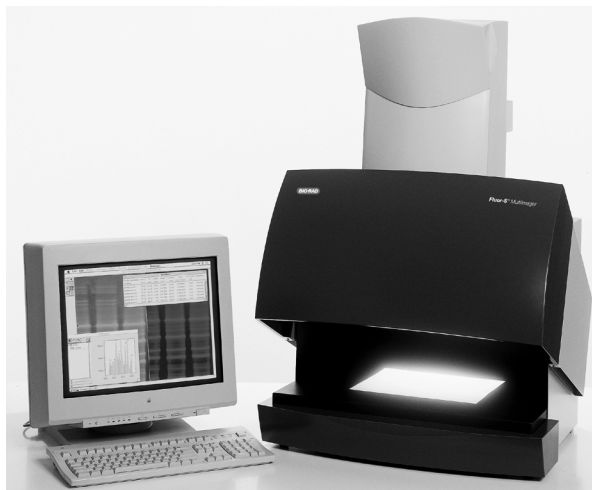


Fig. D-1. Fluor-S Multimager.

Before you can begin acquiring images using the Fluor-S[®] MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S is green before attempting to capture an image. If you are using a PC, the Fluor-S should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

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Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

D.1. Opening the Fluor-S Acquisition Window

To acquire images using the Fluor-S, go to the File menu and select Fluor-S.... The acquisition window for the imager will open, displaying a control panel and a video display window.

Appendix D. Fluor-S

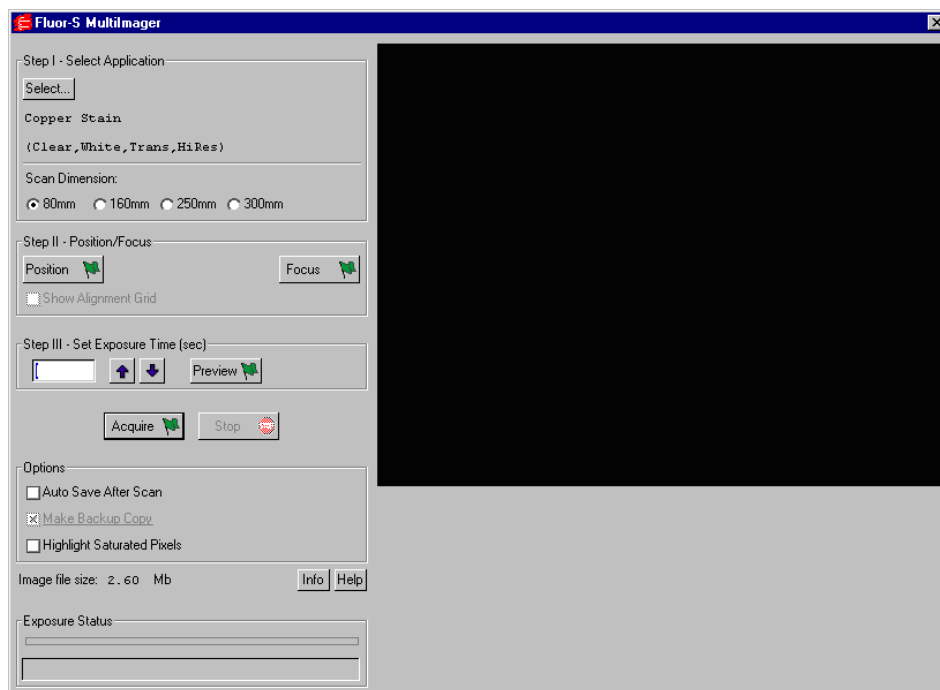


Fig. D-2. Fluor-S acquisition window

When the Fluor-S window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

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D.2. Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

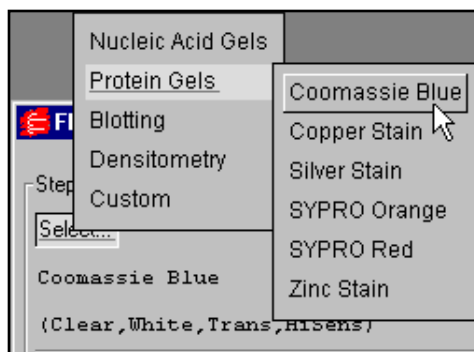


Fig. D-3. The application tree in the Fluor-S acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under “Blotting,” you also have the option of selecting High Resolution or High Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Appendix D. Fluor-S

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

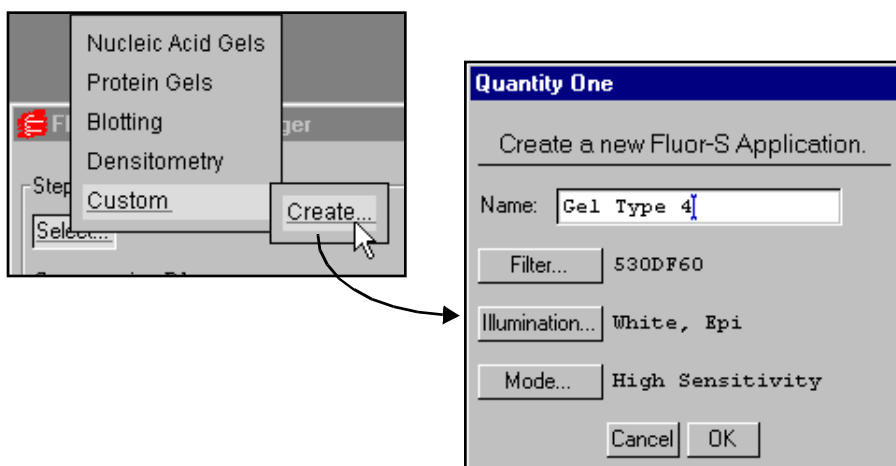


Fig. D-4. Creating a new custom application.

To select the filter (including user-defined), type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

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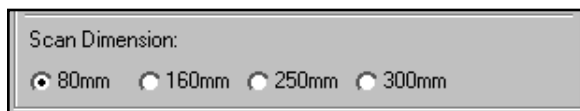


Fig. D-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Resolution Versus High Sensitivity

High Resolution and High Sensitivity are mutually exclusive options. High Resolution is the normal operating mode for the Fluor-S. High Sensitivity provides optimal sensitivity for low-light applications such as chemiluminescence.

In High Resolution mode, images are captured at the maximum resolution of the camera.

In High Sensitivity mode, the pixels in the camera are “binned” (i.e., four pixels are combined into one) to increase the amount of signal per pixel. However, combining the pixels results in a reduction in the resolution of the image.

D.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S box automatically turns on. To turn the light off while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The first step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S will begin capturing a “live” image and updating it every second.

Appendix D. Fluor-S

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

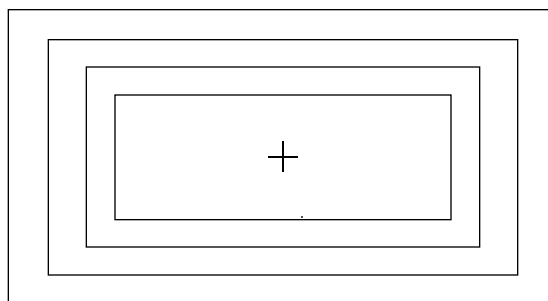


Fig. D-6. Fluor-S alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting. (See the following table on recommended f-stops.)

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

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D.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of the image on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application. The table provides recommended exposure times for various applications.

Table 1: Recommended f-stops and Exposure Times

Application	f-stop ¹ Zoom lens	f-stop ¹ 50mm lens	Exposure time ²
Ethidium bromide	f-4	f-2.8	5–30 sec.
SYBR Green	f-4	f-2.8	5–30 sec.
SYPRO Orange	f-4	f-2.8	5–30 sec.
Radiant Red	f-4	f-2.8	5–30 sec.
Fluorescein	f-4	f-2.8	30 sec.–2 min.
Texas Red	f-4	f-2.8	30 sec.–2 min.
Colorimetric Stains (e.g., Coomassie or Silver)	f-11	f-11	1–5 sec.
X-ray Film	f-11	f-11	1–5 sec.
Photographs	f-11	f-11	1–5 sec.
Chemi ³		f-1.8	1–10 min.
¹ For sharper focusing, close the f-stop down 1–2 stops from full open while focusing. ² Increase exposure time two fold for each step increase in f-stop. ³ For chemi applications, the 50mm lens is recommended. Always remove the 660 filter.			

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For

Appendix D. Fluor-S

chemiluminescent samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig. D-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

D.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

Diversity Database will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

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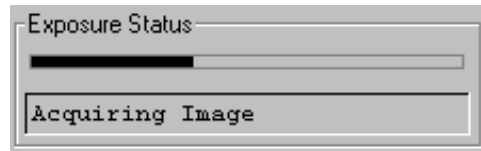


Fig. D-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database's menu and toolbar functions.

D.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Appendix D. Fluor-S

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

To specify the size of your imaging area, click on the Info button and enter each dimension in the appropriate field.

The dimensions of your imaging area are determined by how much you are zooming in or out on your sample. The dimensions determine the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. It is also necessary if you are comparing the quantities of objects (e.g., using the Volume Tools) between images.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create. This size is determined by whether you selected High Resolution or High Sensitivity when you selected an application.

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If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix E. Fluor-S MAX Multimager

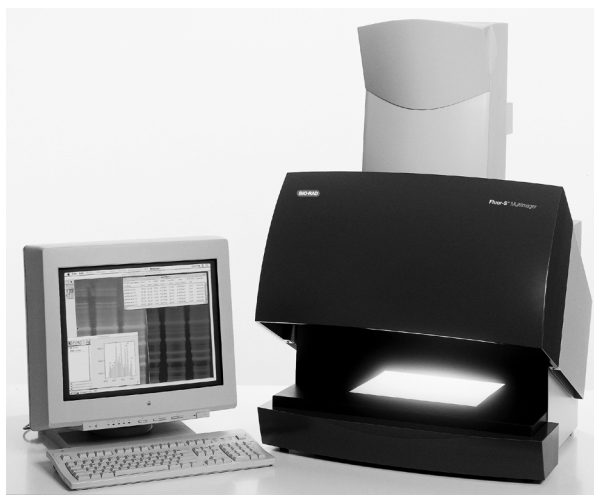


Fig. E-1. Fluor-S MAX Multimager.

Before you can begin acquiring images using the Fluor-S[®] MAX MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S MAX hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S MAX is green before attempting to capture an image. If you are using a PC, the Fluor-S MAX should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S MAX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S MAX, you must have a SCSI

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card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

E.1. Opening the Fluor-S MAX Acquisition Window

To acquire images using the Fluor-S MAX, go to the File menu and select Fluor-S MAX.... The acquisition window for the imager will open, displaying a control panel and a video display window.

Appendix E. Fluor-S MAX

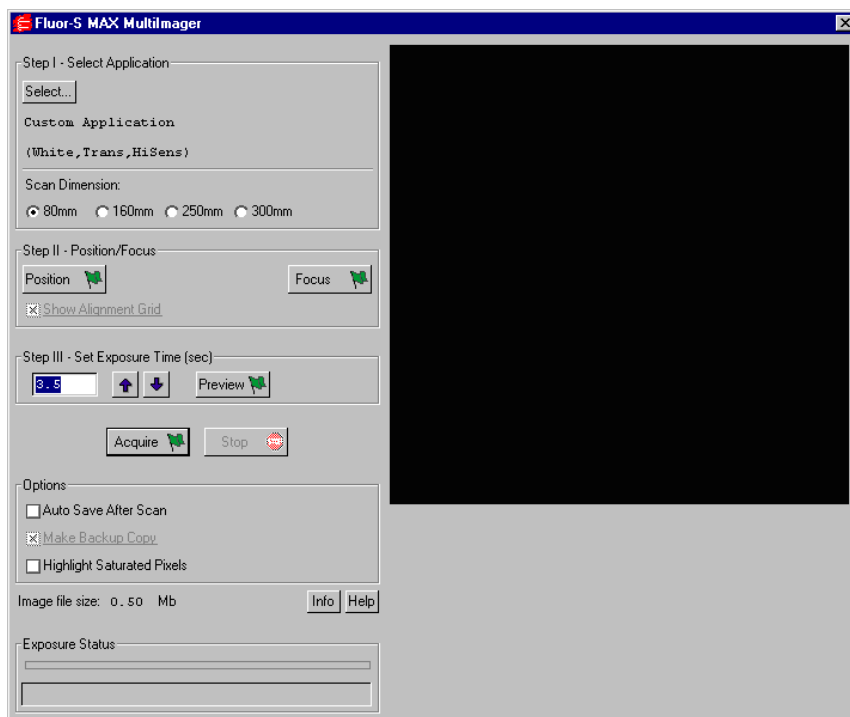


Fig. E-2. Fluor-S MAX acquisition window.

When the Fluor-S MAX window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S MAX:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

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E.2. Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

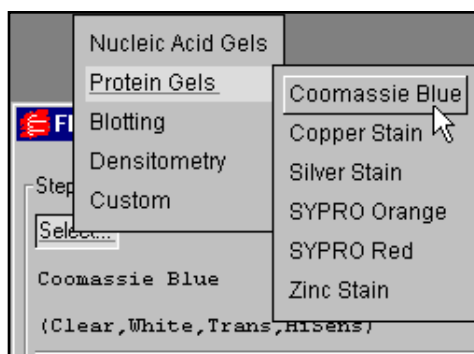


Fig. E-3. The application tree in the Fluor-S MAX acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under “Blotting,” you also have the option of selecting High Sensitivity or Ultra Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S MAX for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

For applications involving trans illumination, you must also specify a scan dimension (see below).

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Appendix E. Fluor-S MAX

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

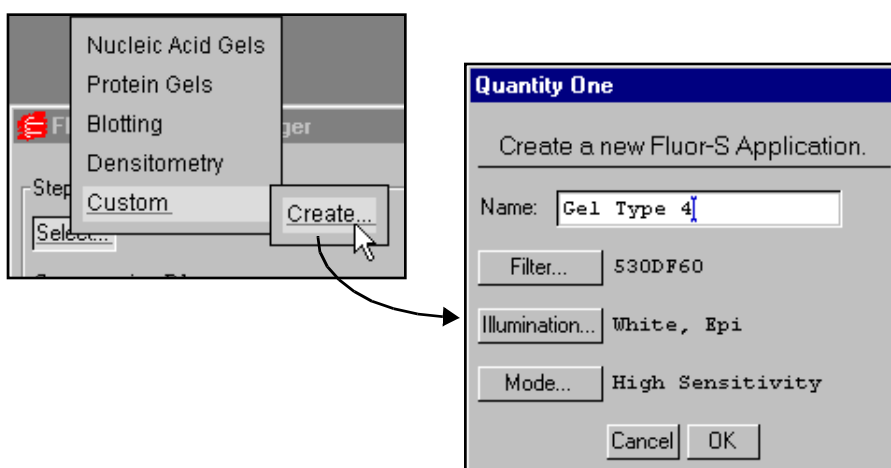


Fig. E-4. Creating a new custom application.

To select the filter, type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

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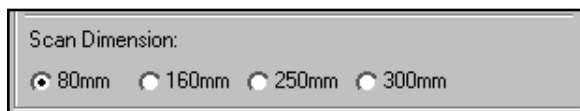


Fig. E-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Sensitivity and Ultra Sensitivity

High Sensitivity and Ultra Sensitivity are different camera modes. In Ultra Sensitivity mode, the Fluor-S MAX camera is cooled to -35.0 degrees C. In High Sensitivity, the camera is cooled to -20.0 degrees C.

High Sensitivity is the normal operating mode of the Fluor-S MAX. Ultra Sensitivity provides optimal sensitivity for low-light applications. It may be selected for the chemiluminescence and chemifluorescence applications, or for a custom application.

Note: When you change from High to Ultra Sensitivity or visa versa, there will be a delay of several minutes while the Fluor-S MAX camera cools down or warms up. If you attempt to acquire an image during this period, you will be notified of the changing temperature. If you do not want to wait, you can cancel the mode change.

E.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S MAX box automatically turns off. This is because if you focus with the camera at maximum aperture (see note below), leaving the light on would make it difficult to view the image. To turn the light on while positioning or focusing, hold down the SHIFT key when clicking on the button.

Appendix E. Fluor-S MAX

Position

The next step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S MAX will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

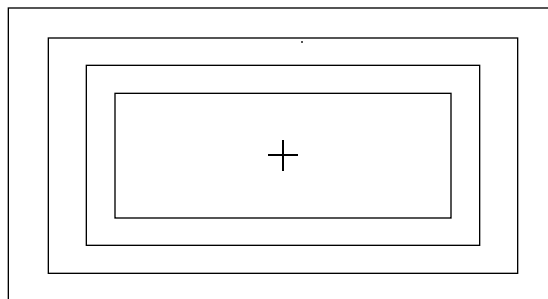


Fig. E-6. Fluor-S MAX alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting.

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

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When you are finished focusing, click on the Stop button.

E.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of image captures on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application and your “best guess” as to what exposure will give you the best image.

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemi samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.



Fig. E-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

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You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

E.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

Diversity Database will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

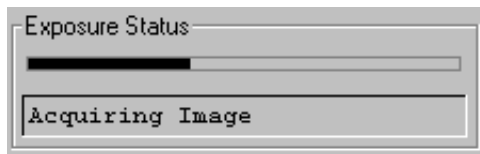


Fig. E-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

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You can then analyze the image using Diversity Database's menu and toolbar functions.

E.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

The imaging area is the area of a sample that is captured by the camera and displayed in the video display window. To specify the size of your imaging

Appendix E. Fluor-S MAX

area, click on the Info button and enter each dimension in the appropriate field. When you adjust one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a sample that is 4.5 x 4.5 cm, then you would enter 4.5 for the width and height of your imaging area.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) between images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

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Appendix F. Personal Molecular Imager FX



Fig. F-1. Personal Molecular Imager FX

Before you can begin acquiring images using the Personal Molecular Imager[®] FX, the instrument must be properly installed and connected with the host computer. See the Personal FX hardware manual for installation, startup, and operating instructions.

Note: The Personal FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

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PC Only: A Note About SCSI Cards

The Personal FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Personal FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

F.1. Opening the Personal FX Acquisition Window

To acquire images using the Personal FX, go to the File menu and select Personal FX.... The acquisition window for the imager will open, displaying a control panel and the scanning area window.

Appendix F. Personal FX

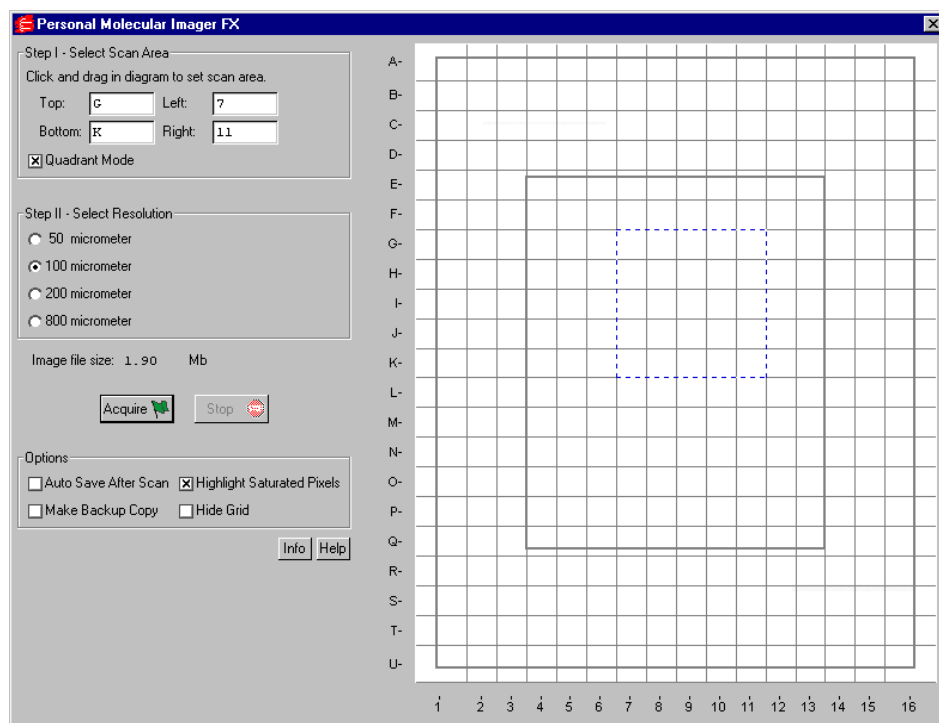


Fig. F-2. Personal FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the Personal FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Personal FX:

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1. Select the scan area
2. Select the resolution
3. Acquire the image

F.2. Step I. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

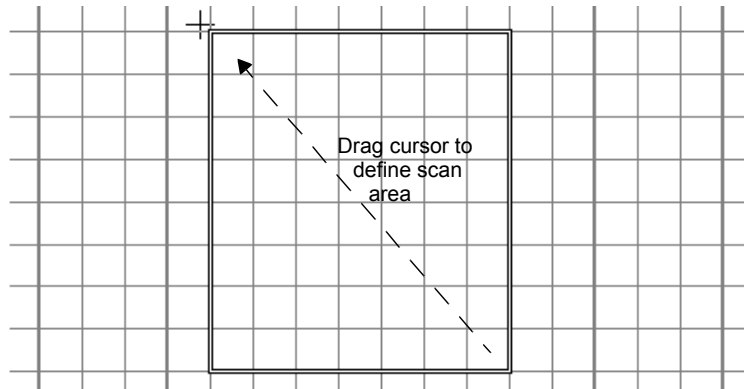


Fig. F-3. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

Appendix F. Personal FX

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

F.3. Step II. Select Resolution

The Personal FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

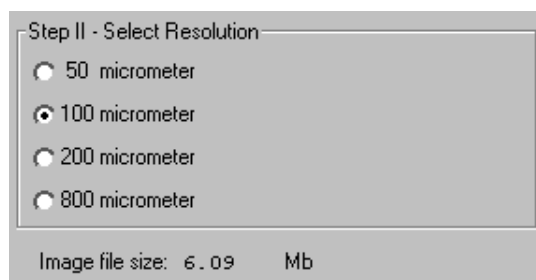


Fig. F-4. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution should be used for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

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File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

F.4. Acquire the Image

Once you have selected your scan area and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database's menu and toolbar functions.

Appendix F. Personal FX

F.5. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

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Appendix G. Molecular Imager FX



Fig. G-1. Molecular Imager FX

Before you can begin acquiring images using the Molecular Imager[®] FX, the instrument must be properly installed and connected with the host computer. See the FX hardware manual for installation, startup, and operating instructions.

Note: The FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the FX, you must have a SCSI card installed in your PC. If

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you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in Diversity Database can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

G.1. Opening the FX Control Panel

To acquire images using the FX, go to the File menu and select FX.... The acquisition window for the imager will open, displaying the control panel for the imager and the scanning area window.

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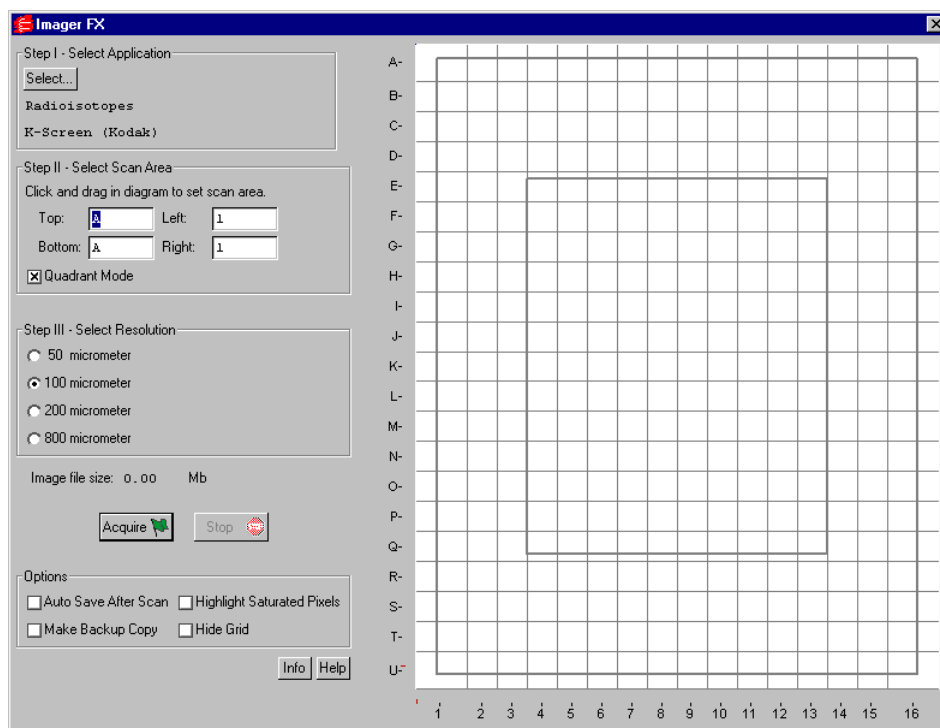


Fig. G-2. FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the FX:

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1. Select the application.
2. Select the scan area.
3. Select the resolution.
4. Acquire the image.

G.2. Step I. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application.

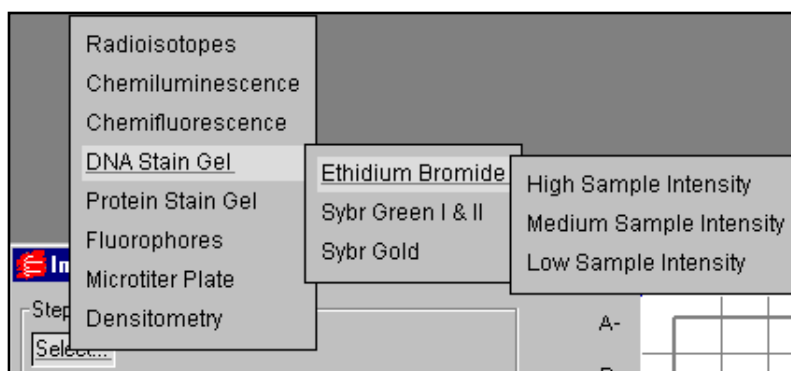


Fig. G-3. Example of an application tree: Ethidium Bromide gel.

The applications and associated settings are listed in a tree that expands from left to right. When you select an application, the software automatically sets the appropriate filter(s) in the FX for that particular application.

First you select the category of your application, next you select your particular application, and finally (if appropriate) you select the intensity of your samples.

To exit the tree without selecting, press the ESC key.

Your selection will be displayed below the Select button.

Appendix G. Molecular Imager FX

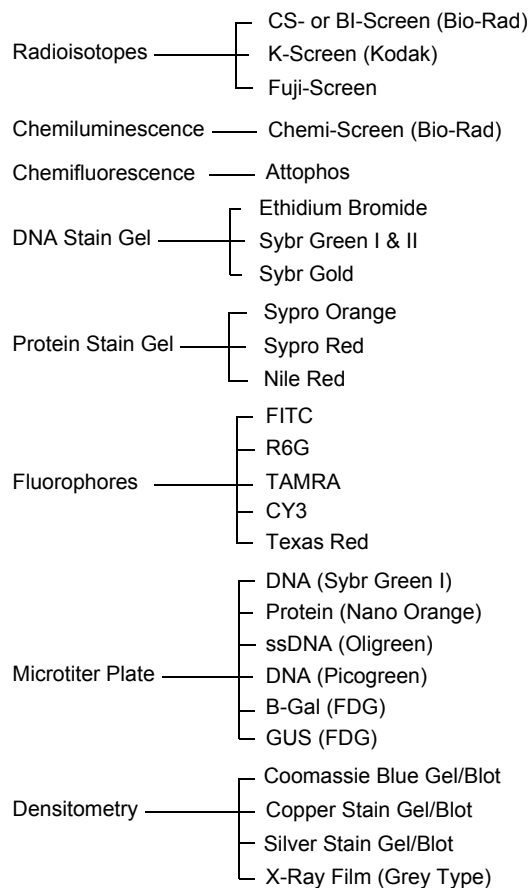


Fig. G-4. Applications available in the FX acquisition window.

Sample Intensity

When selecting any application except radioisotopes, you will need to specify a sample intensity. This is simply a rough estimate of how much sample is apparent in your gel or other object.

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Do not worry if you are not sure about the level of intensity of your sample—if you make a mistake, you can always change the setting and capture another image.

For example, if you specify a low sample intensity and the resulting image has too many saturated pixels, you will receive a warning message. Simply change the setting to medium sample intensity and rescan. If you specify a high sample intensity and the resulting image is too faint, select medium or low sample intensity and rescan.

G.3. Step II. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

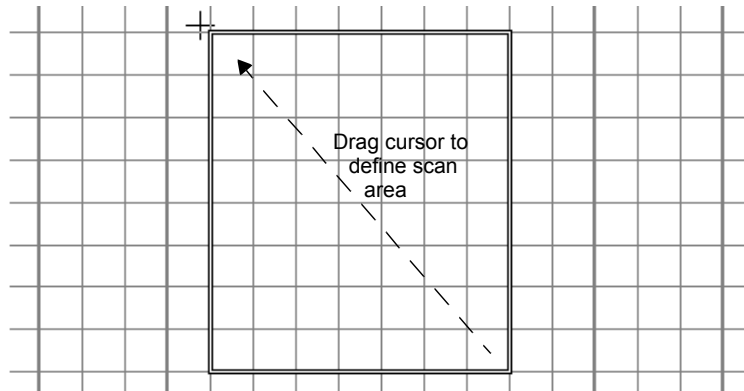


Fig. G-5. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.

Appendix G. Molecular Imager FX

- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

G.4. Step III. Select Resolution

The FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

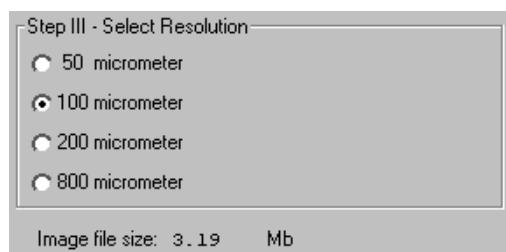


Fig. G-6. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution is useful for typical gels and arrays.

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- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

G.5. Acquire the Image

Once you have selected your application, scan area, and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message. If this happens, you can go back and select a higher sample intensity in the application tree.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date,

Appendix G. Molecular Imager FX

time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using Diversity Database's menu and toolbar functions.

G.6. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

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Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix H. Keyboard Functions

Key	Normal	Shift	Command (Mac)	Control (PC)
Left	Pan Left			
Up	Pan Up			
Right	Pan Right			
Down	Pan Down			
F1	View Entire Image			Transform
F2	Zoom box	Grab		Grab
F3	Center image on cursor	Density Tools		Density Tools
F4	Hide Overlays	New Window		New Window
F5	Show Lanes and Bands	Show Band Types		Show Band Models
F6	Band Attributes			
F7		Image Tools		Image Tools
F8	Gel Layout			
F9	Create Band	Lane Tools		Lane Tools
F10	Remove Band	Contour Tools		Contour Tools
F11	Adjust Band	Show Sample		Show Sample
F12	Save	Save All		Save All
b			Transform	Transform
d				Delete
e				Density Tools
g			Grab	Grab
h			Help	Help
i			File Info	Help
o			Open	Open
p			Print Image	Print Image

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Key	Normal	Shift	Command (Mac)	Control (PC)
q			Exit	Exit
s			Save	Save
t			Plot Cross-Section	Plot Cross-Sect.ion
w			Close	Close
y				Save All

Appendix I. Cross-Platform File Exchange

It is possible to move image data between Discovery Series software applications on different platforms. There are different protocols depending on the platform (PC or Macintosh) you are transferring from and to.

I.1. Macintosh to PC

To transfer a file from a Discovery Series application running on a Macintosh to a Discovery Series application running on a PC, you need to tag the file name with the suffix appropriate to the image file type (i.e., 1-D scan or DNA scan).

For 1-D gels (Diversity Database, Quantity One) use the suffix: .1sc

For database files (Diversity Database) use the suffix: .ddb

For DNA Scans (DNA Code) use the suffix: .dsc

I.2. PC to Macintosh

PC versions of Discovery Series files can be read directly by Macintosh applications, with no required modifications.

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